# OECD ECONOMIC SURVEYS

1999

*SPECIAL FEATURE*

**Tax reform**

# SWITZERLAND

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# OECD ECONOMIC SURVEYS

## 1998-1999

SWITZERLAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

# ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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# BASIC STATISTICS OF SWITZERLAND

## THE LAND

| | |
|---|---|
| Area (1 000 sq. km) | 41.3 |
| Cultivated land, grassland and pastures (1 000 sq. km, 01.01.98) | 15.5 |
| Forest (1 000 sq. km, 01.01.98) | 12.6 |

| Major cities, 1997 estimates (1 000 inhabitants): | |
|---|---|
| Zurich | 353.2 |
| Basle | 175.2 |
| Geneva | 174.6 |
| Bern | 130.3 |

## THE PEOPLE

| | |
|---|---|
| Total population, December 31, 1998, estimates (thousand) | 7 115 |
| Number of persons per sq. km | 172 |
| Net annual rate of population increase average 1990-98 (per 1 000 inhabitants) | 6.6 |
| Civilian employment, in 1998 (thousand) | 3 850 |
| Percentage distribution: | |
| Agriculture | 4.6 |
| Industry and construction | 26.3 |
| Other activities | 69.1 |

| | |
|---|---|
| Number of foreign workers (thousand), end of August 1998 | 863.1 |
| Average increase in the number of foreign workers census, end of August (thousand): | |
| 1962-72 | 16.8 |
| 1973-98 | 0.6 |

## PRODUCTION

| | |
|---|---|
| Gross domestic product (current prices) in 1998 (billion Swiss francs) | 383.4 |
| Growth of real GDP, 1990-98 average (annual rate, per cent) | 0.5 |
| Real gross fixed investment in 1998 (in per cent of GDP) | 25.7 |
| Growth of real investment, 1990-98 average (annual rate, per cent) | –0.2 |

## THE GOVERNMENT[1]

| | |
|---|---|
| Public consumption in 1998 (in per cent of GDP) | 14.7 |
| Revenue of the Confederation in 1998 (in per cent of GDP) | 11.5 |
| Total debt in 1998 (in per cent of GDP) | 53.4 |

| Composition of Parliament | National Council | State Council |
|---|---|---|
| Radical Democrats | 44 | 17 |
| Christian Democrats | 34 | 15 |
| Socialists | 57 | 5 |
| Central Democratic Union | 31 | 7 |
| Other | 34 | 2 |

Last elections: 1995
Next elections: October 1999

## FOREIGN TRADE

| | |
|---|---|
| Exports of goods and services (% of GDP) annual average 1992-98 | 36.8 |
| Commodity exports, fob million of Swiss francs, 1998) | 109 113 |
| Percentage distribution: | |
| By area in 1998 | |
| To OECD countries | 87.3 |
| To EU countries | 62.3 |
| To OPEC countries | 2.5 |
| By categories in 1998 | |
| Raw materials and semi-finished goods | 27.9 |
| Capital goods | 35.9 |
| Consumer goods | 36.0 |
| Energy | 0.2 |

| | |
|---|---|
| Imports of goods and services (% of GDP, annual average 1992-98) | 32.7 |
| Commodity imports, cif million Swiss francs, 1998 | 106 866 |
| Percentage distribution: | |
| By area in 1998 | |
| From OECD countries | 90.0 |
| From EU countries | 76.7 |
| From OPEC countries | 1.2 |
| By categories in 1998 | |
| Raw materials and semi-finished goods | 29.2 |
| Capital goods | 28.9 |
| Consumer goods | 38.7 |
| Energy | 3.1 |

## THE CURRENCY

Monetary unit: Swiss franc

| Currency unit per US$, average of daily figures: | |
|---|---|
| Year 1998 | 1.4497 |
| May 1999 | 1.5084 |

1. Confederation, cantons and communes.

*Note:* An international comparison of certain statistics is given in an annex table.

*This Survey is based on the Secretariat's study prepared for the annual review of Switzerland by the Economic and Development Review Committee on 19 May 1999.*

•

*After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 15 June 1999.*

•

*The previous Survey of Switzerland was issued in August 1997.*

# Assessment and recommendations

***Following export-led growth in the early phase of the upswing, domestic demand became the main engine of the recovery in 1998***

The Swiss economy recovered in 1997 after six consecutive years of broad economic stagnation. Initially, exports surged, supported by vigorous market growth and by the reversal of most of the sharp appreciation of the Swiss franc during the 1993-95 period. Export growth then lost momentum during 1998 as foreign demand slowed markedly in response to the Asian crisis. But its dampening effect on activity was more than offset by the continued buoyancy of domestic demand: private consumption was boosted by rising labour incomes and improved consumer confidence, while the acceleration of machinery and equipment investment was stimulated by stronger demand, falling prices of imported investment goods and the low cost of capital. However, construction investment has remained rather subdued so far, in spite of the federal government's infrastructure maintenance programme, suffering from the persistently high numbers of vacant dwellings and office space, and falling office rents. Altogether, real GDP grew by 2.1 per cent in 1998, the best performance in eight years. Current indicators suggest, however, that economic growth has slowed in the first half of 1999.

***Unemployment fell steeply...***

The unemployment rate peaked at 5.4 per cent in the spring of 1997. It declined gradually through the remainder of the year, but fell quickly thereafter to 2.8 per cent in May 1999. However, only about half of this decline is attributable to the recovery. The remainder is in part due to the tightening of the conditions for unemployment benefit eligibility in 1996-97, which has reduced the incentive to register. In addition, a comprehensive system of active labour market programmes (ALMP) became operational in early 1998; participants in ALMP measures which last for more

than two weeks are not counted as unemployed. Finally, the increase in long-term unemployment has raised the number of persons who have reached the end of benefit eligibility; many of them – about one third in 1998 – cease to register with the labour office.

***... while prices remained stable***

The recovery was accompanied by a further decline in inflation, reflecting the large output gap, which had built up during the slump from 1991 to 1996, and flat unit labour costs which resulted from substantial productivity gains and wage moderation in 1997-98. The process of disinflation was further underpinned by falling import prices, stable rents due to declining mortgage interest rates and intensified competition from deregulation. Consumer prices inflation averaged zero in 1998 and has remained very low since, in spite of the VAT rate increase by 1 percentage point to 7.5 per cent at the beginning of 1999.

***After a temporary slowdown in 1999, economic growth will pick up again in 2000***

The projected continuation of weak export markets in the near term is likely to exert a dampening effect on exports in 1999, followed by a pick-up in 2000, when foreign demand is projected to recover. The ongoing restructuring of the exposed sector will support the growth of machinery and equipment investment, helped by improved company profitability, low capital cost and falling prices of investment goods. Increased public infrastructure expenditure should allow a mild upswing of construction investment. Private consumption is likely to remain a mainstay of domestic demand, supported by further improving labour incomes and growing optimism about the labour market. The unemployment rate should continue to decline, partly as a result of labour market reform. With economic growth slowing this year and disinflationary pressures from abroad persisting, inflation is likely to remain very low. Given Switzerland's traditional role as "safe haven" for international investors at times of financial market unrest, which repeatedly caused wide exchange rate fluctuations, changes in the franc could move prospects considerably away from those presented here, which are based on the assumption of unchanged exchange rates. A weaker franc could impart an additional stimulus to exports, with little risk of a surge in inflation, while renewed exchange rate appreciation would entail a

dampening effect on the exposed sector of the economy with likely deflationary consequences.

***The scope for further monetary easing is limited***

The continuation of the upswing is likely to be supported by the prevailing relatively easy monetary conditions, all the more so after the Swiss National Bank's (SNB) monetary easing in the wake of the refinancing rate cut by the European Central Bank in April 1999. Real bond yields which are still somewhat above the long-term average for Switzerland, the slowdown of M3 growth, some tightening as indicated by a monetary conditions index, and actual short-term interest rates above those suggested by a Taylor rule, all indicated that there was scope for this latest relaxation of monetary policy. This is particularly so given the expected slowing of activity in 1999. Room for any further relaxation is limited.

***A new concept for monetary policy is needed***

There is also a need to define a new concept for monetary policy in Switzerland. A number of structural shifts have made it increasingly difficult to interpret changes in the monetary base, the medium-term target variable. Apart from its reduced informational value for monetary policy decisions and as anchor for expectations, the sizeable overshooting of base money with respect to its medium-term target also risks confusing financial markets. But replacing the narrow monetary base by a broader monetary aggregate as intermediate target would be similarly problematic as the inherent endogeneity of broader money tends to reduce its quality as a short-run indicator, even when its long-term stability is statistically well established. However, the long-term stability of M3 makes this aggregate an important long-run indicator of inflation. The SNB should choose a monetary policy strategy based on a clear definition of price stability and a strong commitment to it. This strategy should rely on the use of several indicators, in particular M3, and inflation forecasts. In fact, such a choice would require relatively few changes to the current operation of monetary policy as the National Bank already closely monitors a broad set of indicators, although it does not currently provide an explicit conditional inflation forecast.

***Fiscal consolidation resumed in 1998...***

Budget consolidation resumed last year, following a pause in 1995-97. Abstracting from the partial privatisation of Swisscom, the consolidated budget deficit for all levels of government fell by 1.2 percentage points to 1.1 per cent of GDP in 1998. Consolidation was timely as it coincided with the pick up in activity and reversed the deterioration in 1997. The improvement overstates the underlying trend as federal receipts were boosted by a surge in net withholding tax, but associated requests for withholding tax refunds will lead to a higher federal deficit in 1999. This will, however, be largely compensated by improving budget positions of the cantons and of the public social insurance sector, resulting in a rise in the consolidated deficit to around 2 per cent of GDP in 1999. Approximately three-quarters of this deficit is estimated to be structural.

***... and Budget Objective 2001 will take it further...***

The federal government is rightly focusing on putting its finances back on to a sound footing while economic conditions remain favourable. In view of the difficulties encountered in lowering the federal deficit in recent years, a constitutional amendment – *Budget Objective 2001* – was adopted last year aimed at bringing the federal budget into broad balance by 2001. Intermediate deficit objectives of SF 5 billion (1¼ per cent of GDP) in 1999 and SF 2½ billion (½ per cent of GDP) in 2000 were also fixed. In the event that the final objective is not met, expenditure cuts will be required to ensure it is attained within the following two years. If there is a recession, this delay can be extended for another two years. In deciding whether or not to accord this extension, Parliament is required to take into account not only current economic growth, but also a variety of forward looking indicators and the stance of monetary policy. The fact that the budget objective for 2001 is incorporated in the Constitution, together with the requirement for expenditure cuts if it is not met, considerably reinforces fiscal discipline. While the possibility of delaying attainment of the objective if the economic outlook is weak and deteriorating reduces the risk of fiscal policy aggravating a downturn, it will be important to ensure that this clause does not undermine fiscal discipline. The *1998 Stabilisation Programme* contains measures that give effect to *Budget Objective 2001*. The Medium-Term Financial Plan, which incorporates these

measures, suggests that budget consolidation will be achieved mainly through expenditure restraint. This increases the likelihood that budget consolidation proves to be durable. Taking all levels of government together, the budget deficit is projected to decline to some 1½ per cent in 2000 and to remain broadly unchanged in the following year.

***... contributing to the first declines in government debt ratios in a decade***

Budget consolidation efforts have slowed the build-up in general government debt, which reached 53 per cent of GDP in 1998, and are projected to result in a gradual decline in the debt ratio from next year onwards. The authorities have rightly been concerned about the rapid rise in debt as it limits the scope to respond to shocks, leaves a debt-servicing burden to future generations and could make Switzerland a less attractive economic site, because tax pressures would ultimately rise. Reducing debt will increase the options for dealing with the future budget pressures associated with population ageing. If the longer-term objective of amending the Constitution to require the budget to be balanced over the cycle is realised, federal debt will trend downwards swiftly.

***Unemployment insurance reform tightened benefit eligibility conditions and emphasised active labour market programmes...***

Over the past two years or so, a number of structural reforms have been implemented which aim at the "revitalisation" of the Swiss economy. Among them, the rather radical reform of the unemployment insurance system stands out. It addresses the problems of rapidly increasing unemployment in general and long-term unemployment in particular between 1992 and 1997 – developments which resulted in a large deficit of the unemployment insurance fund. In a first step, a number of conditions for access to unemployment benefits were tightened in 1996-97 which, *inter alia*, introduced a more stringent definition of "suitable work" and tightened the sanctions for uncooperative behaviour of the unemployed. In a second step, which became fully operational in early 1998, the reform made further receipt of insurance benefits conditional on participation in active labour market programmes (ALMPs) already at an early stage of unemployment, in line with recommendations of the *OECD Jobs Strategy*. The introduction of ALMPs has

been accompanied by a substantial increase in the maximum length of unemployment benefit entitlement.

***... which are to be evaluated***

Providing a large number of ALMP slots which offer a marketable work-experience has been complicated by the legal requirement that activities in subsidised employment programmes must not compete with the primary labour market. As a consequence, about one-half of the temporary employment slots are currently in the public sector. To promote the rapid integration of the unemployed into gainful work, recent measures aim at putting more emphasis on training in these programmes. However, Swiss evidence suggests that increases in benefit entitlement periods tend to reduce search efforts of the unemployed and lower the willingness to accept job offers. Recent research also concludes that the possibility of prolonged job-search through increased maximum benefit duration has raised the individual risk of becoming long-term unemployed. It would thus seem better to reallocate the substantial resources currently spent on extending maximum benefit eligibility to active labour market measures which focus on employability and education and training. An evaluation of the effectiveness of ALMPs will be provided by the end of 1999.

***A new educational pathway and higher work-time flexibility should improve labour market performance***

A further step fostering labour market flexibility is the creation of the Universities of Applied Sciences ("*Hautes Écoles Spécialisées*" or "*Fachhochschulen*") in 1998, which is a new type of specialised university. In combination with the vocational baccalaureate ("*maturité professionnelle*") – introduced in the early 1990s – it establishes a new educational pathway, which integrates applied research and technology transfer, theoretical training, work experience and graduate education; it is designed to diversify the Swiss system of higher education. Another recent reform, the amendment of the labour law in 1998, aims at increased work-time flexibility. *Inter alia*, the amendment makes the treatment of men and women equal with respect to working time, and broadens the definition of "normal" working hours, which makes two-shift working easier. However, the new law falls short of initial plans of Parliament to relax the rigid and restrictive rules for shop opening hours on Sundays. Cantonal and local authorities should relax their rules for shop opening

hours to enhance competition in the retail sector and to help Swiss retailers in border regions to compete with the more flexible regulations in neighbouring countries.

***A better competition legislation...***

Since the beginning of 1997, the new Cartel Act explicitly considers agreements setting prices, production volume or territorial distribution as "elimination of effective competition". This marks a watershed in Swiss competition policy, because previously the practice of cartel agreements was widespread. However, enforcement of the new provisions on cartels and dominant positions is a relatively slow process as it requires a great number of time-consuming investigations by the Competition Commission. The new Act notably introduced merger control, but in spite of the recent surge in merger activity, in no case was a planned merger prohibited. However, in four cases the Commission set conditions for the mergers to be accepted, among them the merger of two very large banks. The Domestic Market Act – operational since mid-1998 – is another key instrument for eliminating cantonal and communal restrictions, in particular with regard to public procurement and professional qualifications. The Act gives all persons in Switzerland the right to offer goods and services, including labour, throughout the entire country. However, enforcement of the Act is rather slow, as it led to a number of legal conflicts. Implementation of the reformed competition policy could be speeded up by providing the Competition Commission with more staff.

***... deregulated infrastructure services...***

Product market competition will also be enhanced by the actual and planned liberalisation of heavily regulated markets. At the beginning of 1998, the telecommunications market was opened up to competition. New operators have entered the market, bringing previously high telecommunication prices for households and businesses below the OECD average on a purchasing-power-parity basis. International comparisons suggest, however, that there is still scope for further price declines. A major obstacle for further price cuts is still existing barriers to market entry. The most important is the dominant position of Swisscom – especially for local calls – which is partly due to the fact that Swisscom owns the local telephone network and imposes high

interconnection charges. Hence, to encourage market entry, the setting of interconnection charges should draw on international best practice. Swisscom should also divest its stake in the leading cable television company to avoid any impediment to competition by new telecommunication suppliers via alternative fixed networks. Substantial efficiency gains could also be reaped by allowing competition in the electricity market (and, to a lesser extent, gas), for which draft legislation envisages contractually regulated "third-party access" and the separation of production, transport and distribution ("unbundling"). The recent reforms of the Post Office and the railway systems also introduced elements of competition by opening parts of the respective markets to third party access. However, important market segments remain public monopolies which leaves substantial scope for further reforms.

***... agricultural reform and participation in European integration should boost Switzerland's growth potential***

Record-high producer support estimates reveal an exceptionally high degree of protection of the agricultural sector and a vast potential for efficiency gains. A new law on agriculture came into effect at the beginning of 1999, reforming the law of 1953. It puts more emphasis on market mechanisms in determining food prices and makes use of a more comprehensive system of direct income payments, largely de-coupled from farm output and linked more closely than hitherto to the provision of ecological services. However, the reform fails to reduce subsidies in the medium term. Further reforms are likely in the context of the process of European integration, in which Switzerland will participate once the bilateral agreement with the EU of December 1998 is ratified. The latter agreement, focusing on the removal of barriers to trade and to the free movement of labour, should give an impetus for further structural reforms.

***The tax system is shaped by the decentralised federal structure***

Switzerland's tax system, which is the special topic for this year's Survey, has evolved in the context of a highly decentralised federal structure. Lower levels of government have considerable tax and spending powers and have accorded high priority to ensuring that their own revenue sources are adequate. This has contributed to the introduction or maintenance of taxes (such as the net worth tax on corporations) that score more highly on their revenue-raising capacity

than their efficiency and equity characteristics. The concern to maintain independent sources of revenue hampers tax reform efforts because reforms would most likely generate winners and losers between and across levels of government. For example, a shift of taxation from income tax to VAT would re-distribute revenues to the federal government.

***Direct democracy can slow reforms but constrains overall levels of taxation***

Direct democracy can also slow tax reform, as demonstrated by the time it took to introduce VAT (18 years and four referenda). On the other hand, reforms approved in a referendum enjoy great legitimacy and the risk of policy reversals is low. This contributes to stability of tax policies, reducing the economic costs of taxation. Direct democracy also limits the scope to spend public money in ways that serve vested interests more than the public (*i.e.* it reduces principal-agent problems). This enhances the value of public expenditures and reduces the tax burden and the associated distortions. Consistent with this feature, the overall tax burden in Switzerland is moderate by international comparison and it is difficult to point to serious under-funding of public activities.

***Having many direct tax systems is costly***

The independence of Swiss cantons is manifest not only in the priority accorded to maintaining independent sources of revenue, but also by the fact that there are 26 cantonal income and wealth tax systems. Administration and compliance costs are reduced by the fact that cantons collect the federal income tax and that municipalities levy their income and wealth taxes as a surcharge on the cantonal taxes. The 1993 tax harmonisation law should contribute to lower administrative and compliance costs as well as increasing transparency in tax competition between cantons. It sets out the direct taxes that cantons can levy and the principles that tax legislation in the cantons must respect. Cantons must comply by 2001. The law harmonises the tax bases for personal income and wealth taxes and for corporate income and net worth taxes. Neither tax-rate schedules nor personal allowances are being harmonised, leaving ample scope for cantons to compete for tax bases and/or to adhere to different social preferences for public services. An important element of the reform is that five different tax-timing

rules are to give way to a much simpler system. In view of the predominance of annual taxation already planned, it would appear that the days of the biennial system are numbered.

***Tax reform must be considered as a package***

The need to ensure that a majority of voters gains from a tax reform suggests that there are advantages in grouping reforms. A voting group's losses from one reform may well be at least partially compensated by gains from another reform. Such grouping is also preferable from an economic perspective because taxes interact. The Behnisch Commission on tax lacunae in the federal income tax system recognised this point when it recommended the introduction of a personal capital gains tax provided that there were reforms to other aspects of the system that contribute to high taxation of capital income not receiving preferential treatment. These aspects are wealth taxes and the absence of relief on distributed profits. The need to renew the constitutional authority to levy federal taxes before the end of 2006 provides an opportunity to introduce a package of tax reforms. These would need to be developed in close collaboration with the cantons. In contrast to most European countries, reforms to reduce adverse effects of taxation on labour supply and demand are not amongst the highest priorities given that unemployment in Switzerland is low and labour-force participation high.

***Tax preferences for savings should be reduced and personal interest deductibility be limited***

One priority for reform could be the reduction of tax privileges accorded to savings with pension funds and life insurance policies. Earnings-related contributions to pension funds (*i.e.* second pillar) are tax deductible, the fund's earnings accumulate tax-free and while distributions in the form of annuities are taxed at normal income tax rates, lump-sum payments are taxed at lower rates. In the case of life insurance policies, contributions are not tax deductible, but earnings on the funds invested are not taxed. There is no absolute upper limit on earnings-related contributions to pension funds and there are no ceilings on the amounts that can be placed into life insurance policies. Compared to the standard treatment under a comprehensive income tax, these arrangements are very favourable. They reduce economic efficiency by channelling savings through the

privileged institutions and could violate tax equity. Ideally, all savings should receive the same tax treatment. If all other forms of savings were taxed according to regular income tax principles, as are bank deposits, this would mean taxing fund earnings as they accumulate and either taxing fully contributions to pension funds or distributions but not both. In the event that it is not feasible to tax fully other sources of capital income, such as owner-occupied housing (in aggregate, owner-occupied housing accounts are in deficit), there is a second-best argument for retaining some preferential treatment, but then for all savings instruments. This would involve reducing tax preferences for savings with pension funds and life insurance companies and lowering taxation on other savings instruments, such as bank deposits. Considering that Parliament rejected the more modest proposals of the Behnisch Commission, there is little prospect of such reforms being implemented. This makes it vitally important that personal interest deductions be limited to declared capital income. Otherwise, households can reduce their taxable income by borrowing to obtain assets that do not generate taxable income, such as life insurance policies. Parliament recently took a step in this direction, by limiting interest deductions to SF 50 000 more than gross capital income. This reform does not, however, go far enough.

***Cantons should implement a flat-rate tax on corporate profits***

Some cantons apply a flat-rate tax on corporate profits, others have a progressive rate on corporate income and net worth. The latter ones should replace their progressive taxes by a flat-rate tax on corporate profits, as occurred at the federal level in 1998, because the existing system discourages risk taking. It tends to tax favourable profit outcomes disproportionately, while taxes are also imposed when profit outcomes are unfavourable. These arrangements are particularly disadvantageous for business start-ups, as they tend to be more risky than established businesses, and have arbitrary effects on tax equity. Such a reform is all the more necessary because of the lack of relief for the double taxation of dividends. This compounds the disadvantage faced by business start-ups by raising the pre-tax required rate of return on investments financed by new equity.

***Greater use should be made of environmental taxes***

As in other countries, the Swiss authorities have recognised that many environmental objectives could be achieved more cheaply through greater use of economic instruments. Progress in introducing such instruments has been rather modest to date. However, the Federal Council announced that an "ecological tax reform" will be included in the new legal authority for the federal government to raise tax revenues after 2006. This would introduce an energy tax and reserve the proceeds mainly for financing a reduction in social security contributions. A law on reducing $CO_2$ emissions, which would introduce a tax in the event that other measures prove inadequate, is also pending. While these reforms should enhance economic efficiency by discouraging economic activities with harmful environmental effects and/or by reducing the costs of economic regulation, there are a number of pitfalls to avoid. Special treatment should not be accorded to the greatest polluters or most intensive users of non-renewable resources to preserve their international competitiveness as this undermines the efficiency of the tax. Requirements that tax revenues be spent on environmental projects should also be avoided. If there are worthwhile environmental projects, they should be funded irrespective of whether or not environmental taxes are introduced. Ideally, revenues from environmental taxes should be used to reduce the most inefficient and/or inequitable taxes. While social security taxes may not be the worst in these respects, there should nevertheless be benefits from reducing them with the revenues from the proposed federal energy tax.

***The Swiss tax system is fairly robust to international competition...***

The Swiss tax system appears to be fairly robust in the face of increasing globalisation and financial mobility. In contrast to many other European countries, Switzerland has not been constrained to shift the weight of taxation from capital towards labour and consumption, which are relatively immobile bases. One factor contributing to this outcome is that tax incentives for saving through pension funds and life insurance companies are very attractive, dimming households' interest in shifting financial assets abroad to evade taxation. Low marginal income tax rates in certain cantons go in the same direction. Another factor contributing to this outcome is that a relatively high proportion of capital

taxation is focused on immobile bases, notably physical property and the self-employed.

***... and may also attract investments and international business activities***

The Swiss tax system and legal environment contain features that make Switzerland an attractive destination for foreign investments, both portfolio and direct. A positive feature of the Swiss system is that its tax burden is moderate by OECD standards and low in comparison to most other European countries. However, Switzerland could well come under increased pressure from other countries to modify certain features of its system:

- Swiss bank secrecy makes the use of foreign investments mandates (so-called fiduciary investments) attractive to non-resident individuals desiring to evade taxes in their country of residence and avoid all withholding taxes on interest income. Swiss law does not treat tax evasion as grounds for lifting bank secrecy in Switzerland; it may only be lifted in cases of tax fraud as defined by Swiss law. While the general rule in Switzerland is that Swiss-source capital income is subject to a 35 per cent withholding tax, the Swiss withholding tax can be avoided by setting up a Swiss fiduciary relationship by which the funds of non-resident individuals are invested abroad. The fiduciary will – as would any individual investing directly – take advantage of the absence of a withholding tax on capital income in certain other jurisdictions (source countries), thereby enabling non-resident individuals to escape all taxation on their capital income from these jurisdictions. Thus, the combination of bank secrecy, fiduciary mandates and the lack of withholding tax on capital income in certain source countries enables non-resident individuals desiring to evade taxes in their country of residence to be reasonably certain that a failure to declare the income in their country of residence will go undetected. The desire to evade taxes is, however, by far not the only reason to have recourse to a fiduciary relationship.

- Two cantonal corporate tax regimes, notably the administration company regime, may encourage some foreign companies, the functions of which are predominantly performed abroad, to locate in Switzerland.

### *Summing up*

In summary, the recovery of the Swiss economy now seems more firmly established, notwithstanding the temporary slowdown of output growth expected this year. It is supported by an accommodating monetary policy, while fiscal consolidation is on track. The upswing is accompanied by a stable price level and falling unemployment. The latter is also a result of the shift in the emphasis of labour market policies away from passive income support to the rapid reintegration of the unemployed in the work process. Labour market flexibility should be further improved by the amended work-time regulations and by the creation of new options of secondary and tertiary education. The recovery is also underpinned by a variety of other structural reforms, which have enhanced competition in factor and product markets, including the markets for infrastructure services and agriculture. Tax reforms in recent years – covering notably, VAT, federal company taxes and tax harmonisation – should also contribute to improved economic performance, while leaving ample scope for further reforms. More wide-ranging and deeper structural reforms should raise the economy's growth potential and improve Switzerland's high standard of living.

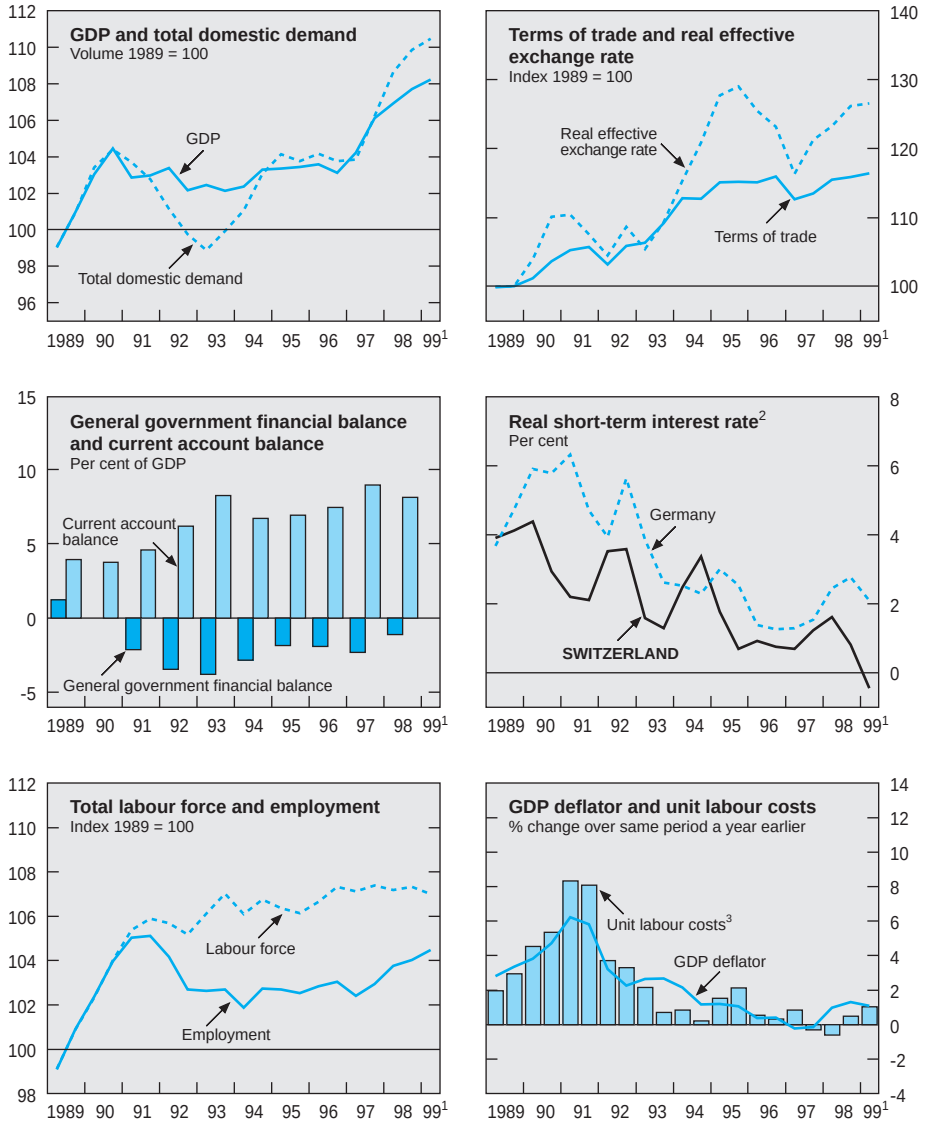
# I. Recent trends and short-term prospects

## The economy has resumed growth

After six years of broad economic stagnation, the Swiss economy recovered in 1997, with output exceeding its 1990 level for the first time (Figure 1). Initially, the upswing was led by exports, which benefited from buoyant export markets and from the reversal of most of the strong real effective appreciation of the Swiss franc that had occurred between 1992 and 1995. Part of the exchange rate correction was due to lower interest rates, induced both by easier monetary policy and sluggish demand for money and credit. Household consumption also recovered, in line with the pick-up of labour incomes[1] and improving consumer confidence, which reflected the gradual improvement in employment prospects. The cyclical turnaround was accompanied by further disinflation. This is consistent with estimates of a large output gap, which had built up during the slump from 1991 to 1996. Moreover, unit labour costs declined due to the combination of marked productivity gains and wage moderation.

Economic activity lost momentum during the first half of 1998 as the growth of exports of goods and services slowed significantly in response to the Asian crisis, following two-digit growth in the second half of 1997. The weakening of exports intensified in the second half of 1998, when exports to EU markets, Central and Eastern European economies and developing countries shrank. However, the contractionary effect from exports on economic activity was more than offset by the continued buoyancy of domestic demand (Figure 2). For 1998 as a whole, real GDP grew by 2.1 per cent, the best growth performance in eight years, although growth was stronger in most other European OECD countries. With this pick-up the (production-function based) estimate of the output gap narrowed somewhat (Figure 3). Economic growth slowed substantially in the first quarter of 1999, to an annual rate of 0.4 per cent, falling short of estimates of potential output growth.

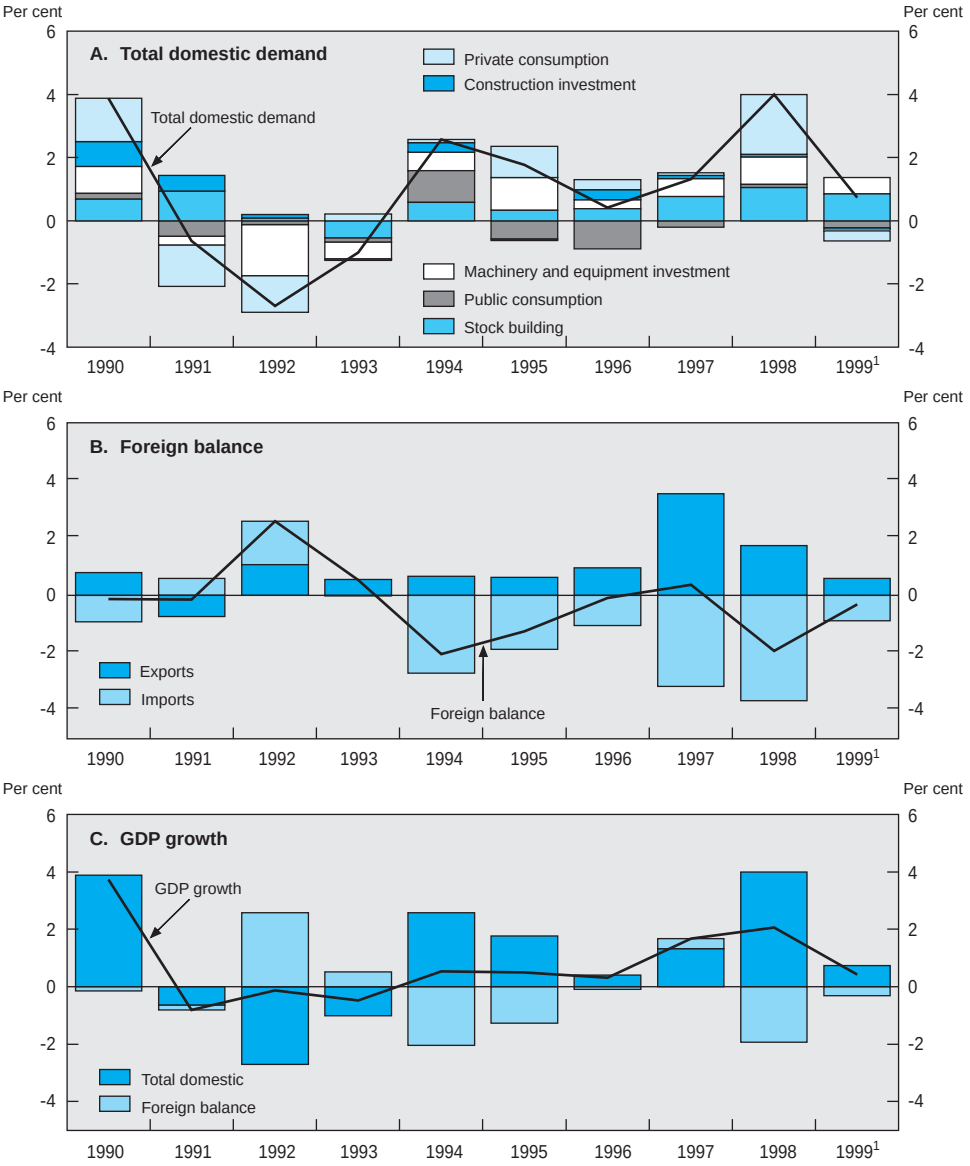
Figure 1. **Key aspects of economic activity**




1. Figures for the first half of 1999 are OECD estimates.
2. 90-day Euro-swiss franc rate and German 3-month interbank rate deflated by the private consumption deflator.
3. Total economy.

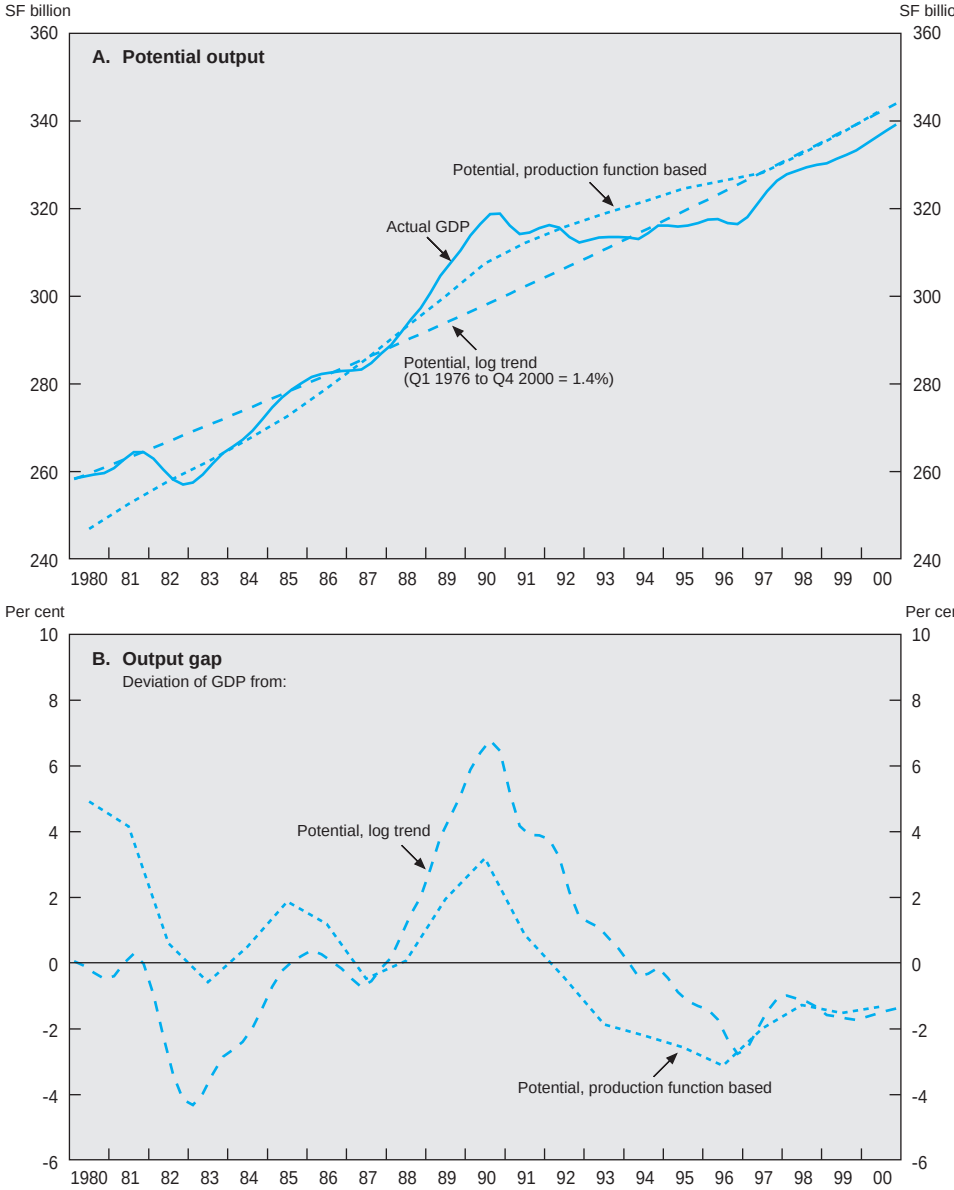
*Source:* OECD, *National Accounts, Main Economic Indicators* and OECD Secretariat.

Figure 2. **Contribution to real GDP growth**
As a percentage change of GDP in previous year




1. 1999 Q1 figures, seasonally adjusted annual rates.
*Source:* OECD, *National Accounts.*

Figure 3. **Potential output and output gap**[1]




1. Data for 1999 and 2000 are OECD projections.
*Source:* OECD Secretariat.

## Robust domestic demand

Domestic demand was supported by brisk private consumption growth in 1998 (Table 1), boosted by the recovery in employment. Preliminary estimates suggest growth of gross labour income by some 2 per cent in 1998, with property income likely to have grown somewhat faster. Disposable income rose less, however, because the biennial collection cycle of the federal and most cantonal direct taxes implies higher tax payments in even-numbered years than in odd-numbered years. Real household incomes may have grown by only about 1½ per cent. However, consumer sentiment continued to improve and turned positive[2] by mid-1998, for the first time in more than eight years (see Figure 9 below). The survey indicates that households increasingly judge job security less pessimistically and expect an improving labour market over the coming twelve months. Accordingly, preliminary estimates indicate a small fall in the household saving ratio in 1998 (by about ¼ per cent to 9 per cent). Household consumption weakened somewhat in the first quarter of 1999, growing at a seasonally-adjusted annual rate of 1.5 per cent.

Public consumption, in contrast, remained subdued in 1998 and in the first quarter of 1999, reflecting continuing efforts to bring budgets of all levels of government closer to balance. The small increase in real government expenditure is mainly due to rising government employment, primarily in the field of education and research, but is also a result of a shift of posts from the privatised Swisscom to the federal government. Defence expenditures continued to shrink.

Fixed investment picked up further due to accelerating growth of machinery and equipment investment and a mild recovery of construction activity. The latter[3] was primarily due to the take-off of major railway projects and the effects of the government investment programme of 1997 (*programme de relance conjoncturel*) which brought forward maintenance of public infrastructure. The decline in construction investment in the first quarter of 1999 may be partly due to unusually bad weather. In general, construction activity continued to suffer from the high and rising number of vacant dwellings, which not only induced rents and house and apartment prices to continue to decline, but also contributed to a further, albeit milder, contraction of residential construction. On the other hand, the fall in business construction volumes probably came to a halt in 1998, in spite of evidence of continued excess capacity of commercial structures and office space, and falling office rents. Machinery and equipment investment accelerated further in 1998, probably stimulated by the surge in exports in 1997, falling prices of imported investment goods and low cost of capital, all of which seem to have encouraged capital deepening investment. It remained buoyant in the first quarter of 1999, albeit less so than in 1998.

A major contribution to GDP growth in 1998 – close to 2 percentage points – is attributable to stockbuilding, which could give rise to fears of an

Table 1. **Supply and use of current output**

| | 1991 at current prices | | Percentage volume[1] change from previous period, annual rates, s.a. | | | | | |
|---|---|---|---|---|---|---|---|---|
| | SF billion | % share of GDP | Average 1991-96 | 1997 | 1998 | 1998 I | 1998 II | 1999 Q1 |
| Private consumption | 193.4 | 58.0 | 0.5 | 1.3 | 1.8 | 1.4 | 2.6 | 1.5 |
| Public consumption | 50.9 | 15.2 | 1.4 | 0.6 | 0.6 | 0.5 | 1.2 | –0.7 |
| Gross fixed capital formation | 85.0 | 25.5 | –1.0 | 1.5 | 3.8 | 4.2 | 2.0 | 1.2 |
| Construction | 45.8 | 13.7 | –1.4 | –1.5 | 0.7 | 1.9 | –1.8 | –1.7 |
| Machinery and equipment | 39.2 | 11.7 | –0.4 | 4.9 | 7.1 | 6.5 | 6.1 | 4.1 |
| Final domestic demand | 329.3 | 98.7 | 0.2 | 1.2 | 2.1 | 2.0 | 2.3 | 1.0 |
| Changes in stocks[2, 3] | –0.2 | –0.1 | –0.1 | 0.1 | 1.9 | 2.0 | 0.4 | –0.3 |
| Total domestic demand | 329.1 | 98.6 | 0.1 | 1.3 | 4.0 | 3.9 | 2.6 | 0.7 |
| Exports of goods and services | 116.7 | 35.0 | 1.4 | 9.0 | 4.1 | 2.7 | 0.0 | 1.4 |
| Exports of goods | 91.2 | 27.3 | 1.8 | 7.6 | 4.6 | 3.0 | –0.7 | 1.9 |
| Exports of services | 25.7 | 7.7 | –0.1 | 13.1 | 3.6 | 1.8 | 3.3 | –0.5 |
| Imports of goods and services | 112.1 | 33.6 | 1.7 | 8.1 | 8.8 | 7.5 | 3.8 | 2.0 |
| Imports of goods | 99.5 | 29.8 | 1.2 | 9.5 | 8.5 | 7.6 | 4.0 | 2.2 |
| Imports of services | 12.6 | 3.8 | 4.6 | 3.5 | 4.8 | 6.9 | 2.6 | 0.7 |
| Change in foreign balance[2] | 4.6 | 1.4 | –0.1 | 0.4 | –1.9 | –2.0 | –1.7 | –0.3 |
| Gross domestic product | 333.7 | 100.0 | 0.0 | 1.7 | 2.1 | 1.9 | 0.9 | 0.4 |
| GDP deflator | | | 2.4 | –0.1 | 1.1 | 1.3 | 0.6 | 2.1 |
| Private consumption deflator | | | 2.9 | 0.6 | 0.2 | 0.3 | –0.1 | 0.3 |
| Index of industrial production[4] | | | 0.7 | 4.7 | 3.6 | 6.0 | –1.9 | .. |

1. 1990 prices.
2. As a percentage of GDP in the previous period.
3. This item is partly a residual value; it also contains statistical errors.
4. Excluding mining and quarrying.

*Source:* Office fédéral du développement économique et de l'emploi; Office fédéral de la statistique and OECD, *Main Economic Indicators.*

imminent downward adjustment of inventories, echoing the experience in 1991-92. However, a large part of the recorded inventory accumulation is not of a cyclical nature but is due to a surge in imports of precious stones and metals,[4] to be processed and exported at a later stage. The Federal Office for Economic Development and Labour estimates that SF 4.8 billion out of the recorded "stockbuilding and statistical discrepancy" of SF 7.1 billion in 1998 were due to these imports. This leaves SF 2.3 billion as cyclical inventory accumulation (including a statistical discrepancy), which is equivalent to only 0.4 per cent of GDP, and which is broadly in line with survey measures of changes in inventories compiled by the KOF/ETH Institute in Zürich. This also implies that the adjusted growth of total domestic demand and imports of goods and services amounted to 2½ and 5¼ per cent, respectively, in 1998, instead of the recorded growth of 4 per cent and 9 per cent.

## Rapidly falling unemployment

In the current upswing, the lag between the recovery of output growth and its positive effects on employment was unusually short. This is in line with the observation that during the stagnation of the 1990s firms abstained from the previous practice of labour hoarding once they realised that the economic slump would persist longer than usual. Hence, the number of "persons in employment"[5] (*"population active occupée"*) already picked up in the spring of 1997, shortly after the turnaround in output growth, and has risen since. Employment gains were above average for foreign workers (+2¾ per cent) and for women (+1¾ per cent), compared with total employment growth of 1¼ per cent in 1998 (Table 2). These trends continued into 1999. The differences reflect, on the one hand, the above-average increase in employment in the service sector, where the share of female workers is high and, on the other hand, the pronounced – and long-awaited – pick-up in employment in the construction industry from its depressed level, where a high share of the labour force is of foreign origin. In line with past trends, the growth of part-time employment was stronger than that of full-time jobs. The improving labour market situation is also reflected in the sharp rise in vacancies in 1997 and 1998 and the falling number of persons working short-time. However, the growth of employment slowed somewhat in the second half of 1998 and in the first quarter of 1999, in line with weakened output growth. Moreover, the number of vacancies (s.a.) peaked in January 1999 and declined in the following four months.

The rate of registered unemployment peaked at 5.4 per cent (s.a.) of the labour force in March/April 1997 and then declined gradually. From the beginning of 1998 on, however, it fell sharply, to 2.8 per cent in May 1999. Unemployment of all major categories of workers declined (Table 3), while the relative incidence

Table 2. **Employment trends**

Per cent change from previous period

| | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|
| **Persons in employment** | 3.2 | 2.0 | –1.7 | –0.8 | –0.3 | 0.3 | 0.3 | –0.3 | 1.3 |
| Agriculture | –3.6 | 2.5 | –2.4 | 1.9 | –4.8 | 3.8 | 5.5 | 2.3 | 1.7 |
| Industry | 0.6 | –1.7 | –5.7 | –4.0 | –0.1 | 1.6 | –3.9 | –4.7 | –0.5 |
| *of which:* | | | | | | | | | |
| Manufacturing | 1.2 | – | –6.3 | –3.5 | –1.0 | 1.6 | –3.0 | –4.5 | –0.9 |
| Construction | 0.9 | 3.5 | –4.3 | –5.0 | 2.2 | 1.2 | –6.0 | –4.8 | 1.4 |
| Services | 5.0 | 4.0 | 0.2 | 0.5 | –0.2 | –0.4 | 1.9 | 1.4 | 2.0 |
| *Memorandum item:* | | | | | | | | | |
| Registered vacancies | –1.7 | –38.9 | –22.6 | –29.7 | –9.0 | 3.4 | 3.8 | 68.4 | 40.5 |
| Labour productivity[1] | 1.1 | –0.9 | 2.2 | 0.3 | 1.0 | 0.5 | 0.1 | 1.9 | 0.8 |

1. Real GDP per person in employment.

*Source:* Office fédéral de la statistique, *La statistique de la population active occupée;* KOF/ETH, *Konjunktur* and OECD Secretariat.

remained broadly unchanged: the French and Italian speaking parts of the country remained more affected by unemployment than the German speaking regions, foreign workers more than Swiss nationals, and women more than men. Of all age groups, prime aged workers suffer most from joblessness, and youth unemployment is lower than the national average. The relative reduction in unemployment was greatest among construction workers.

However, the recent fall in the number of registered unemployed persons – by some 35 per cent since the beginning of 1998 (Figure 4) – also reflects the tightening of the conditions for unemployment benefit eligibility in two steps in 1996 and 1997, which reduced the number of registrations with labour offices. Along with the new eligibility rules, a comprehensive system of active labour market programmes (ALMP) was introduced in 1997, which became fully operational in early 1998 (see Chapter III); participants in ALMP measures which last for more than two weeks are not counted as unemployed. Moreover, the increase in long-term unemployment (more than one year) over recent years has raised the number of those who have reached the end of benefit eligibility; many of them – about one-third in 1998 – cease to register at the labour office.[6]

To account for the effect of the unemployment insurance reform, the OFDE[7] introduced a new statistics for job-seekers in October 1997. In addition to the registered unemployed, it includes those persons who are searching for a job but who are occupied (in a broad sense) and as a result are not available for work within 30 days. This ensures that the new measure covers people who are in employment programmes, who attend training and education courses financed by

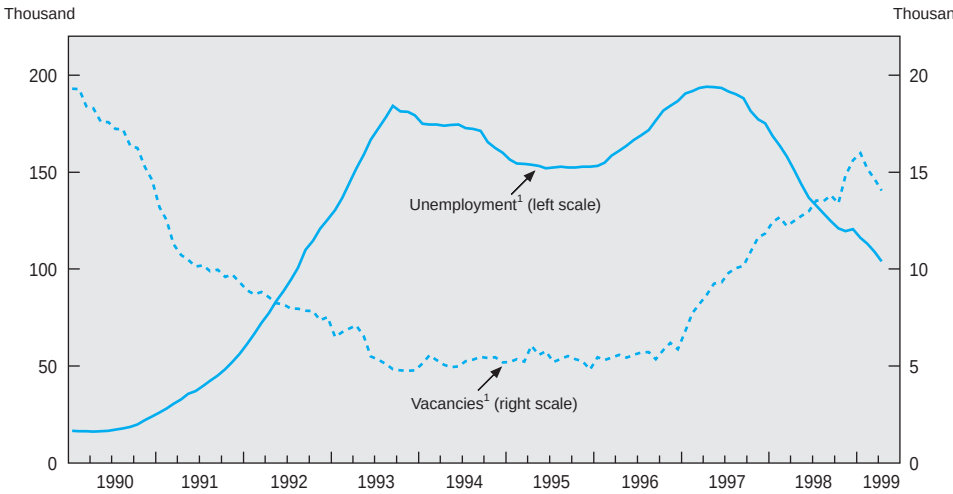
Table 3. **Level and structure of unemployment**

| | 1995 | | 1996 | | 1997 | | 1998 | | 1999 Q1 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Number | Rate[1] | Number | Rate[1] | Number | Rate[1] | Number | Rate[1] | Number | Rate[1] |
| **Total** | 153 316 | 4.2 | 168 630 | 4.7 | 188 304 | 5.2 | 139 660 | 3.9 | 120 661 | 3.3 |
| **According to region** | | | | | | | | | | |
| German speaking | 86 656 | 3.3 | 101 064 | 3.9 | 117 438 | 4.5 | 84 822 | 3.2 | 70 552 | 2.7 |
| French and Italian speaking | 66 661 | 6.6 | 67 566 | 6.7 | 70 867 | 7.0 | 54 838 | 5.4 | 50 109 | 4.9 |
| **According to gender** | | | | | | | | | | |
| Women | 67 843 | 4.8 | 71 801 | 5.1 | 79 626 | 5.7 | 62 557 | 4.4 | 52 371 | 3.7 |
| Men | 85 473 | 3.9 | 96 829 | 4.4 | 108 679 | 4.9 | 77 103 | 3.5 | 68 289 | 3.1 |
| **According to citizenship** | | | | | | | | | | |
| Swiss | 88 662 | 3.2 | 92 735 | 3.3 | 101 700 | 3.6 | 73 785 | 2.6 | 62 171 | 2.2 |
| Foreign | 64 655 | 8.0 | 75 895 | 9.3 | 86 604 | 10.7 | 65 875 | 8.1 | 58 490 | 7.2 |
| **According to age** | | | | | | | | | | |
| 15-24 years | 26 058 | 3.9 | 28 168 | 4.3 | 31 180 | 4.7 | 21 577 | 3.3 | 17 907 | 2.7 |
| 25-49 years | 96 479 | 4.4 | 107 788 | 5.0 | 120 024 | 5.5 | 88 998 | 4.1 | 77 188 | 3.6 |
| 50 years and more | 30 779 | 3.9 | 32 674 | 4.2 | 37 101 | 4.7 | 29 085 | 3.7 | 25 566 | 3.3 |
| **According to sectors** | | | | | | | | | | |
| Agriculture | 1 500 | 0.9 | 1 851 | 1.1 | 2 108 | 1.2 | 1 606 | 0.9 | 1 720 | 1.2 |
| Energy, mining | 495 | 1.5 | 552 | 1.8 | 635 | 2.1 | 486 | 1.6 | 392 | 1.6 |
| Arts and crafts, industry | 31 009 | 4.0 | 32 128 | 4.4 | 34 590 | 5.0 | 22 652 | 3.3 | 18 964 | 2.7 |
| Construction | 13 918 | 4.0 | 18 050 | 5.8 | 21 791 | 7.4 | 14 152 | 4.7 | 13 411 | 4.3 |
| Distributive trade, catering, repairs | 44 877 | 4.9 | 49 525 | 5.6 | 53 504 | 6.1 | 40 423 | 4.5 | 34 665 | 5.3 |
| Transport and communication | 4 497 | 1.8 | 5 057 | 2.0 | 5 727 | 2.3 | 4 204 | 1.7 | 3 705 | 1.7 |
| Banking, insurance, consultancy | 18 009 | 3.3 | 20 187 | 3.7 | 21 590 | 3.9 | 15 687 | 2.7 | 14 004 | 2.8 |
| Other services | 21 078 | 2.8 | 21 717 | 2.9 | 24 079 | 3.1 | 19 456 | 2.5 | 16 736 | 3.4 |
| Government | 5 909 | 4.0 | 7 078 | 4.9 | 8 445 | 5.6 | 6 174 | 4.0 | 5 049 | 3.3 |
| Not specified[2] | 12 025 | | 12 486 | – | 15 837 | – | 14 820 | – | 12 014 | .. |
| Total | 153 316 | 3.9 | 168 630 | 4.4 | 188 304 | 4.9 | 139 660 | 3.6 | 120 661 | 3.3 |
| | | Share in % | | Share in % | | Share in % | | Share in % | | Share in % |
| **According to duration** | | | | | | | | | | |
| 0-6 months | 71 078 | 41.6 | 81 417 | 47.6 | 81 210 | 47.5 | 62 244 | 36.4 | 64 137 | 53.2 |
| 7-12 months | 38 287 | 22.4 | 43 167 | 25.2 | 49 638 | 29.0 | 32 069 | 18.7 | 24 219 | 20.1 |
| More than one year | 43 951 | 25.7 | 44 046 | 25.8 | 57 456 | 33.6 | 45 347 | 26.5 | 32 305 | 26.8 |
| **According to function** | | | | | | | | | | |
| Specialists | 73 093 | 42.7 | 80 390 | 47.0 | 91 461 | 53.5 | 68 788 | 40.2 | 59 613 | 49.4 |
| Auxiliaries | 60 188 | 35.2 | 67 890 | 39.7 | 74 755 | 43.7 | 53 955 | 31.5 | 47 434 | 39.3 |
| Apprentices/students | 10 839 | 6.3 | 10 588 | 6.2 | 10 969 | 6.4 | 7 978 | 4.7 | 5 780 | 4.8 |
| Other | 9 196 | 5.4 | 9 762 | 5.7 | 11 119 | 6.5 | 8 939 | 5.2 | 7 833 | 6.5 |

1. Official unemployment statistics include those who are partially unemployed. With the exception of sectoral unemployment rates, official unemployment rates are expressed as a percentage of the workforce in the 1990 population census; sectoral unemployment rates are expressed as a percentage of the workforce in the corresponding period.
2. Persons seeking employment for the first time or re-entering the workforce.

*Source:* Office fédéral du développement économique et de l'emploi.

Figure 4. **Unemployment and vacancies**




1. Seasonally adjusted.
*Source:* OECD Secretariat.

the unemployment insurance fund or who benefit from temporary pay supplements from unemployment insurance. This measure showed an average of 217 500 job-seekers (5½ per cent of the labour force) in 1998, in contrast to 139 700 registered unemployed. It also reveals that during the twelve months to December 1998 the statistics of job-seekers declined by some 44 600 persons compared with a reduction of the number of unemployed persons by 56 900. Of the decline in the number of job-seekers, about 14 600 were persons who dropped out of the unemployment statistics because of entitlement termination. This implies that the cyclical decline in the number of job-seekers was about 30 000, which is around half the reduction of the number of registered unemployment in the course of 1998. Accordingly, half of the fall in the unemployment rate from 5.0 per cent in December 1997 to 3.4 per cent in December 1998, hence about three-quarters of a percentage point may be attributed to the economic cycle.[8]

## A stable price level

Average consumer price inflation was zero in 1998 and thus below 1 per cent for the third consecutive year[9] (Table 4). At the producer and wholesale

Table 4. **Price trends**

Percentage changes from previous year

| | May 93 Weight in % | 1995 | 1996 | 1997 | 1998 | 1998 | | 1999 |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Q3 | Q4 | Q1 |
| **Total supply index**[1] | 100.0 | 0.0 | –2.3 | 0.1 | –1.5 | –1.9 | –2.9 | –2.9 |
| **Producer price index** | 70.2 | –0.1 | –1.8 | –0.7 | –1.2 | –1.4 | –2.0 | –1.9 |
| **Import price index** | 29.8 | 0.3 | –3.7 | 2.0 | –2.2 | –3.0 | –5.1 | –5.0 |
| **Consumer price index** | 100.0 | 1.8 | 0.8 | 0.5 | 0.0 | 0.1 | –0.1 | 0.3 |
| Goods | 44.2 | 0.3 | 0.2 | 0.6 | –0.3 | –0.3 | –0.9 | –0.5 |
| Non-durables | 25.6 | 0.8 | 0.9 | 1.3 | –0.6 | –0.7 | –1.5 | –0.8 |
| *of which:* Food, beverages, tobacco | 16.3 | 0.8 | –0.1 | 0.9 | 0.9 | 0.9 | 0.2 | 0.2 |
| Energy | 3.2 | 5.5 | 5.3 | 2.2 | –6.4 | –6.7 | –8.5 | –5.0 |
| Services | 55.8 | 3.0 | 1.3 | 0.5 | 0.3 | 0.4 | 0.5 | 0.8 |
| *of which:* Rents | 22.1 | 1.1 | 1.3 | 0.5 | 0.1 | 0.1 | 0.1 | 0.2 |
| Home produced goods and services | 74.6 | 2.5 | 1.0 | 0.5 | 0.3 | 0.3 | 0.3 | 0.6 |
| Imported goods and services | 25.4 | –0.3 | 0.1 | 0.7 | –0.7 | –0.7 | –1.3 | –0.6 |
| Export prices[2] | | –1.0 | 0.7 | 2.9 | –0.0 | –0.0 | 0.3 | 0.2 |
| Import prices[2] | | –3.0 | 0.3 | 4.1 | –2.6 | –3.0 | –3.1 | –2.8 |

1. The total supply price index is the weighted average of producer and import price indices.
2. Change in the price deflator of goods and services, national account basis, 1990 = 100.

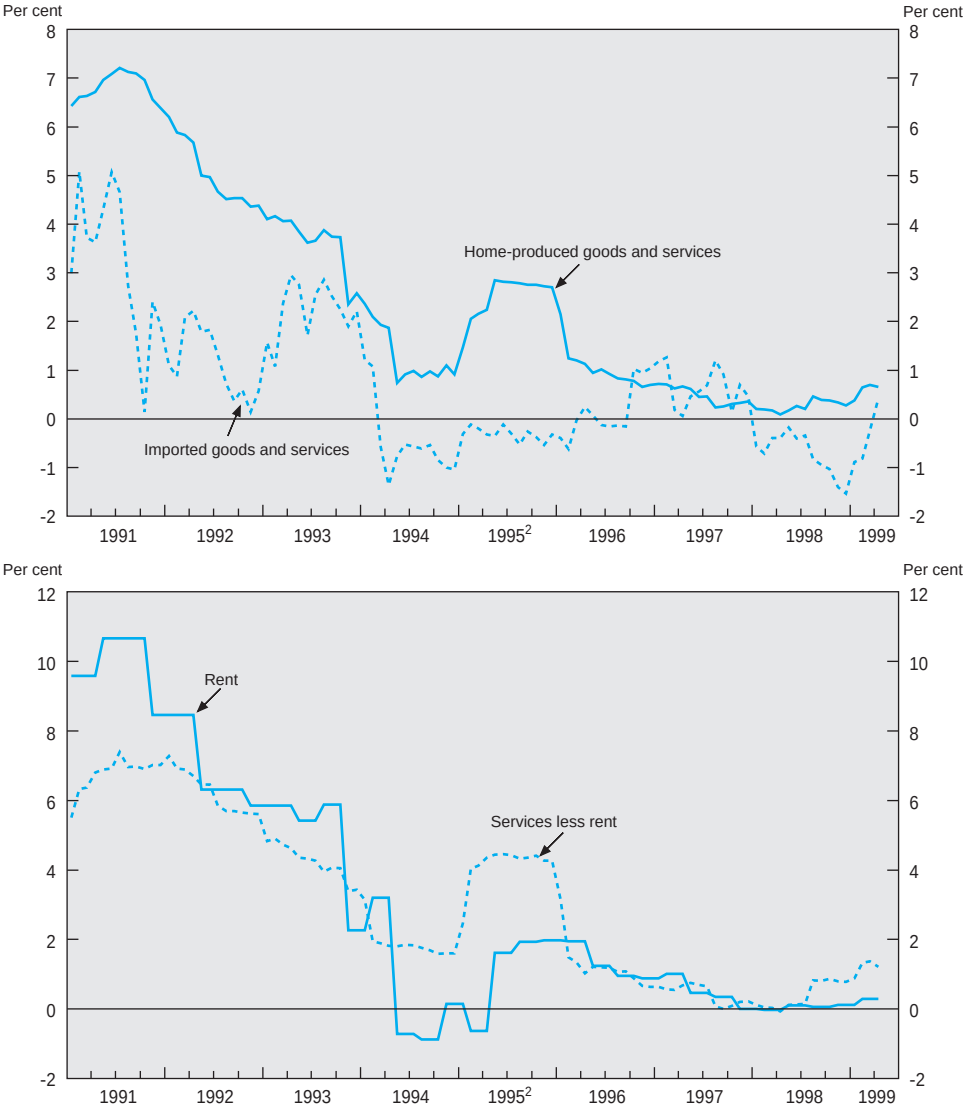
*Source:* Département fédéral de l'économie, *La vie économique;* Banque nationale suisse, *Bulletin mensuel.*

level, prices fell on average by 1.5 per cent in 1998, so that the period of roughly stable (1991-95, 1997) or falling prices (1996, 1998) now even extends to as much as eight years. Price stability was underpinned by the continuing slack in factor and product markets, which is reflected in a significant output gap. This contained labour and capital costs and curbed the pricing power of producers. The low level of mortgage interest rates also helped to keep the level of rents stable,[10] which account for 22 per cent of the consumer price index (CPI) and which were a major source of past inflation (Figure 5, lower panel). Rent increases were also kept in check by the high number of vacant dwellings, a legacy of the real estate boom in the late 1980s. Another major factor behind the persistent price stability was falling import prices,[11] (Figure 5, upper panel), notably for mineral oil; prices of imported energy at the wholesale level fell by 16 per cent in 1998. In addition, competition intensified as a consequence of deregulation and the reformed competition law (see Chapter III). A striking example was the opening of the telecommunications sector to competition, which resulted in substantial price cuts in telecommunications services.

At the beginning of 1999, the general VAT rate was raised from 6.5 to 7.5 per cent, the rate for hotel services from 3.0 to 3.5 per cent and the reduced rate for a number of goods and services[12] from 2.0 to 2.3 per cent. If the VAT rate increase were to be fully passed on to consumers, its average annual mechanical effect on the CPI would amount to about ½ per cent in 1999. But given the state of the cycle and enhanced competitive pressures, a full pass-through of the VAT rise on prices is unlikely. Up to May 1999, when the twelve-monthly rate of CPI inflation was 0.6 per cent, the actual inflation increase amounted to about one-half of the potential VAT effect estimated by the Federal Statistical Office. The continuing fall in producer and wholesale prices, which are not affected by VAT, during the first months of 1999, together with the decline in mortgage interest rates in the Spring, suggest an unchanged climate of low or slightly negative inflation. This keeps Switzerland's inflation among the lowest in the OECD (Figure 6).

The wage rate indicator rose by 0.5 per cent both in 1997 and 1998, which implies *ex-post* real wage growth of zero and ½ per cent, respectively, and broadly unchanged real wages over the past six years. The growth of effective real compensation per employee derived from the National Accounts is estimated to have been somewhat stronger in 1997-98 (see note 1), and was matched by gains in labour productivity. As a result, unit labour costs remained roughly unchanged in 1997 and 1998. Wage dispersion continued to be significant, with nominal wage changes ranging from gains of 4 to 4½ per cent in the insurance business and parts of the banking sector to losses of 1 per cent in the hotel and restaurant business in 1997.[13] On the basis of wage settlements concluded in the autumn of 1998, latest estimates indicate nominal wage increases of under 1 per cent in 1999.

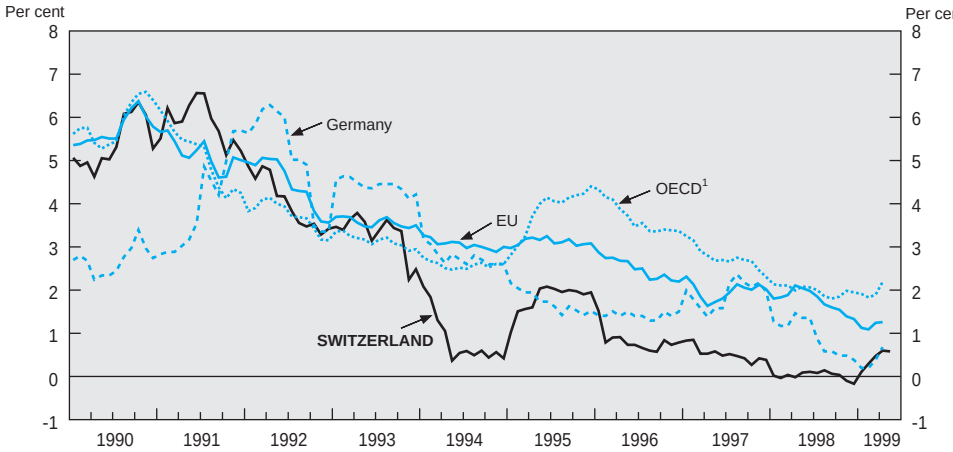
Figure 5. **Selected components of inflation**[1]
Year-on-year change




1. Weights in CPI as from May 1993 are: 0.75 for home-produced goods and services; 0.25 for imports; 0.29 for services less rent and 0.22 for rent.
2. Introduction of VAT.

*Source:* KOF/ETH, *Konjunktur* and OECD, *Main Economic Indicators.*

Figure 6. **Consumer prices: an international comparison**
Percentage change from the same period of previous year



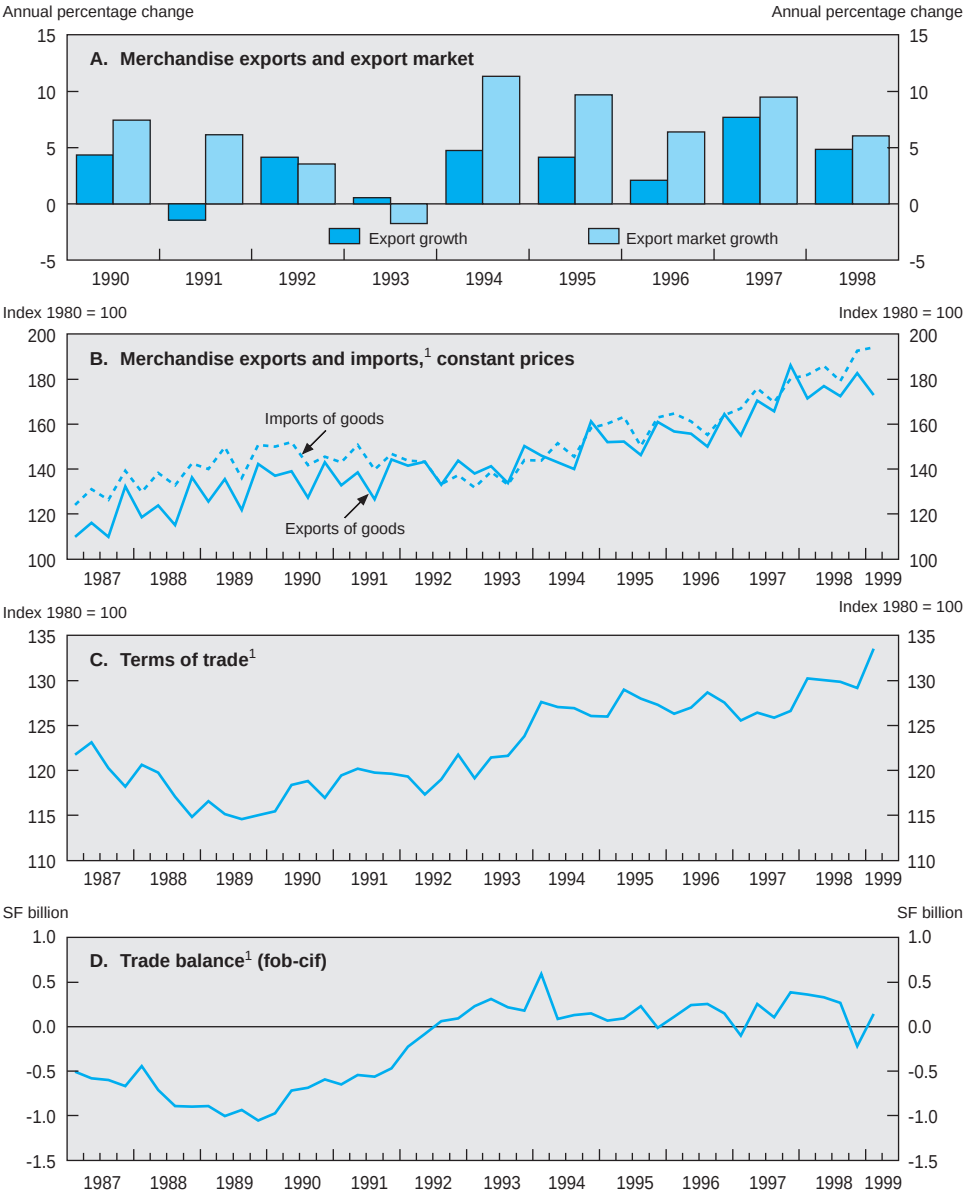

1. Excluding Turkey.
*Source:* OECD, *Main Economic Indicators.*

## A large current external surplus

Merchandise export volumes recovered vigorously in 1997 after four years of rather subdued growth, but lost some momentum in 1998, in line with decelerating export market growth (Figure 7). Rising relative unit labour costs in common currency suggest some loss in cost competitiveness in 1998, but relative export prices remained largely unchanged as exporters accepted cuts in profit margins to maintain price competitiveness (Figure 8). This helped to bring the trend deterioration in export performance to a halt in 1998. But in spite of robust average growth in 1998 (by 4.9 per cent), which was partly due to a positive carry-over from 1997, merchandise export volumes weakened during the year; in the fourth quarter of 1998, they even fell by an annual rate of nearly 11 per cent. Export volumes stabilised in the first quarter of 1999, when recovering exports of investment goods compensated for the continuing decline in exports of consumer goods, raw materials and semi-processed manufactures.

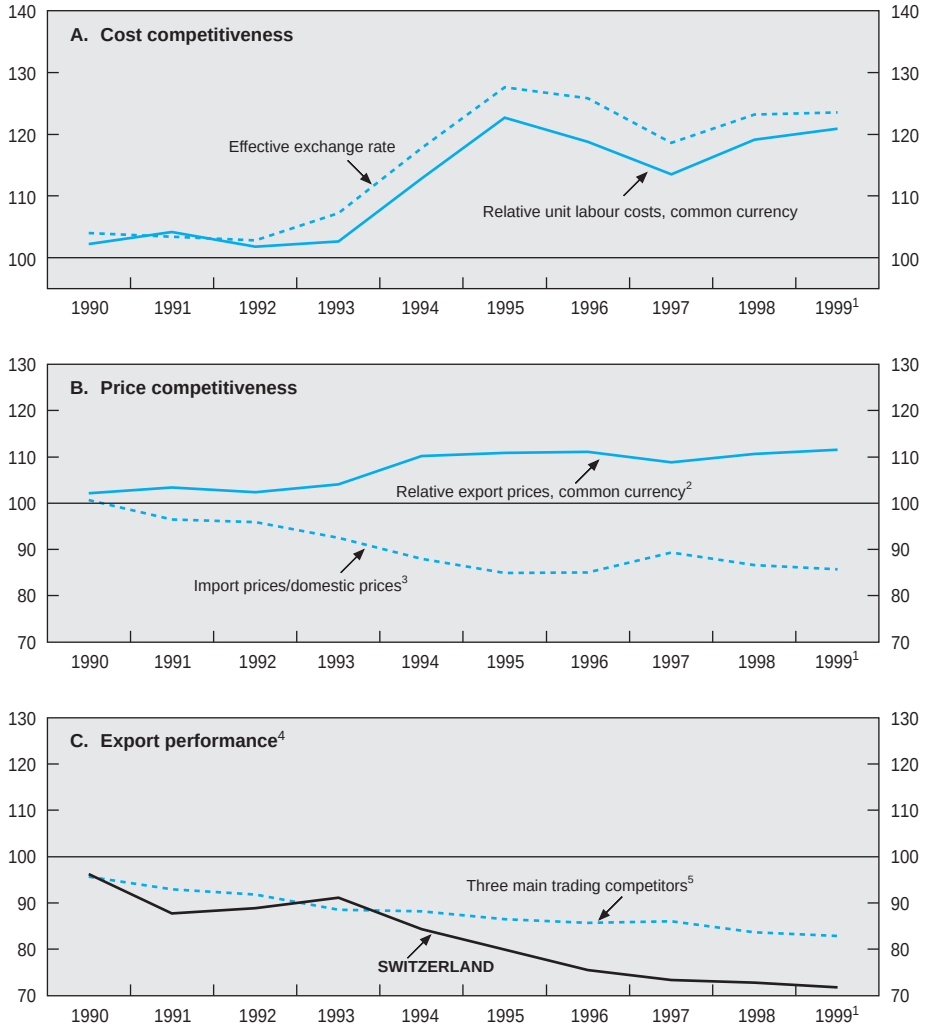
A major cause for the weakening dynamism of exports is the economic crisis in Southeast Asia and – as from mid-1998 onward – in Russia, together with the rather longer-term weakness in the Japanese economy. Apart from the direct contractionary effect on Swiss exports to these regions, these crises also weakened economic activity and overall import demand in other world regions and,

Figure 7. **Foreign trade indicators**




1. Seasonally adjusted.
*Source:* OECD Secretariat.

Figure 8. **International competitiveness and trade**
Index 1987 = 100




1. Figures for 1999 are OECD Secretariat estimates.
2. Manufactures.
3. Import prices of total goods and services divided by deflator of total domestic demand.
4. Ratio between export volumes of manufactures and export markets for manufactures.
5. Germany, France and Italy.

*Source:* OECD Secretariat.

hence, demand for Swiss products. Altogether, merchandise exports (at current prices) to troubled Southeast Asian countries and to Japan fell on average by 26 and by 4 per cent, respectively, in 1998; however, exports to these destinations improved in the first quarter of 1999. Exports to the EU and the United States – which together account for three-quarters of Swiss exports – remained relatively strong in 1998, although less so than in 1997; they contracted towards the end of 1998, in spite of generally robust domestic demand in many of these destinations, Germany, the most important export market, and Italy being exceptions. In the first quarter of 1999, sales to the United States picked up vigorously, while exports to the EU remained sluggish.

In contrast, growth of merchandise import volumes accelerated and remained strong throughout 1998, with imports of investment and consumer goods being particularly buoyant. Apart from the recovery of domestic demand this also reflects increasing import penetration, in line with the ongoing gain in the price competitiveness of imports (Figure 8, panel B). Import growth remained strong in volume terms, but fell in current prices in the first quarter of 1999. The faster growth of import than of export volumes in 1998 (Figure 8, panel B) was more than offset by a terms-of-trade gain (import prices falling by more than export prices) so that the trade surplus[14] increased from SF 2.0 billion in 1997 to SF 2.2 billion in 1998 (Figure 7, panel D). However, with imports of gold and precious stones soaring, the trade account in a broader definition turned from near balance to a deficit of SF 2.7 billion in 1998. It remained in a deficit of SF 0.6 billion in the first quarter of 1999.

With both tourism revenues and expenditures growing nearly at the same rate, the traditional surplus in the tourism account widened only little. Net revenues from banking commissions, international transport and licences and patents declined a little from their high level, so that the large surplus in non-factor services narrowed somewhat. A small decrease in the deficit on labour incomes mainly reflects the declining employment of cross-border workers (*"travailleurs frontaliers"*). The large surplus on investment income – net foreign assets were some 125 per cent of GDP by the end of 1998 – increased slightly in absolute terms, to stay a little below 7 per cent of GDP in 1998. Altogether, the current external surplus fell by about SF 2 billion, from 8.9 per cent of GDP in 1997 to 8.1 per cent in 1998 (Table 5). It is estimated to have jumped, however, to 9.6 per cent of GDP in the first quarter of 1999.

## The outlook to 2000

Current indicators are broadly consistent with an acceleration of economic growth in the second half of 1999. In the Spring of 1999, the aggregate business climate indicator showed signs of improvement and the assessment of

Table 5. **Current external account**

SF billion

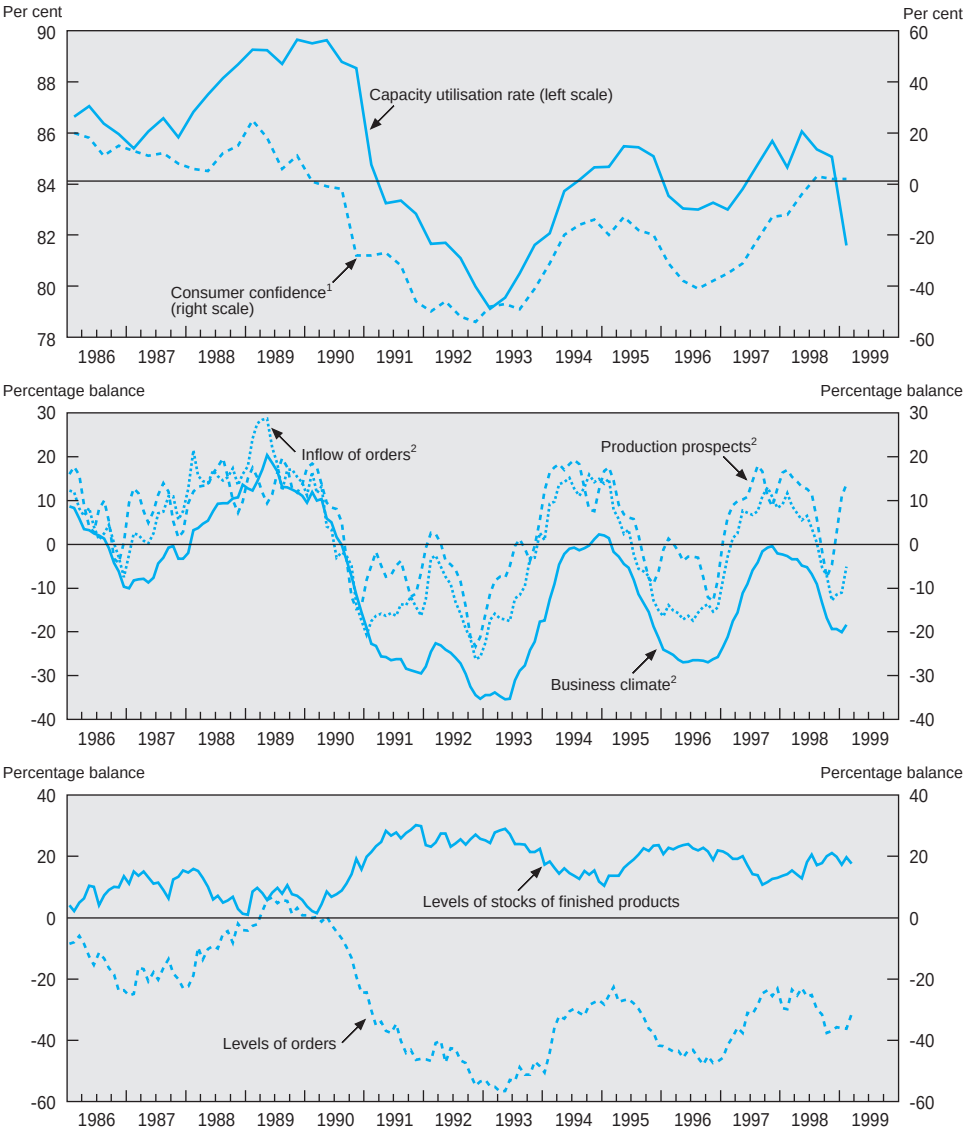
| | 1993 | 1994 | 1995 | 1996 | 1997 | 1998[1] | 1999 Q1[1, 2] |
|---|---|---|---|---|---|---|---|
| **A. Receipts** | | | | | | | |
| Exports of goods (f.o.b.) | 93.3 | 95.8 | 96.2 | 98.6 | 110.4 | 114.1 | 113.0 |
| Supplementary exports items | 3.5 | 3.6 | 3.6 | 3.6 | 3.8 | 4.0 | 4.1 |
| Services, receipts | 31.7 | 30.9 | 30.8 | 32.5 | 37.2 | 38.5 | 39.4 |
| Tourism | 11.3 | 11.4 | 11.2 | 11.0 | 11.5 | 11.8 | 12.1 |
| Other services | 20.5 | 19.5 | 19.6 | 21.5 | 25.7 | 26.7 | 27.4 |
| Factor income | 37.2 | 36.6 | 37.3 | 40.8 | 48.7 | 50.0 | 50.5 |
| Labour | 1.4 | 1.4 | 1.5 | 1.5 | 1.5 | 1.6 | 1.6 |
| Capital | 35.8 | 35.1 | 35.8 | 39.3 | 47.2 | 48.4 | 48.9 |
| Current transfers | 3.7 | 3.5 | 3.5 | 3.6 | 3.8 | 4.1 | 4.1 |
| Total receipts | 169.4 | 170.4 | 171.5 | 179.1 | 204.0 | 210.7 | 211.1 |
| **B. Expenditures** | | | | | | | |
| Imports of goods (c.i.f.) | 89.8 | 92.6 | 94.5 | 96.7 | 110.1 | 115.8 | 114.3 |
| Supplementary imports items | 4.5 | 4.5 | 4.3 | 4.4 | 4.6 | 4.9 | 5.2 |
| Services, expenditures | 14.9 | 15.3 | 15.6 | 17.1 | 18.4 | 19.8 | 16.6 |
| Tourism | 8.8 | 8.8 | 8.8 | 9.4 | 10.1 | 10.3 | 7.2 |
| Other services | 6.1 | 6.5 | 6.8 | 7.7 | 8.4 | 9.5 | 9.4 |
| Factor income | 23.7 | 25.9 | 23.4 | 25.2 | 29.1 | 30.1 | 30.4 |
| Labour | 8.5 | 8.3 | 8.3 | 8.4 | 8.1 | 8.0 | 8.1 |
| Capital | 15.1 | 17.6 | 15.0 | 16.8 | 21.0 | 22.1 | 22.3 |
| Current transfers | 7.7 | 8.1 | 8.5 | 8.6 | 8.7 | 8.9 | 9.1 |
| Total expenditures | 140.6 | 146.5 | 146.3 | 151.9 | 170.9 | 179.5 | 175.6 |
| **C. Balance** | | | | | | | |
| Trade balance | 3.5 | 3.2 | 1.8 | 1.9 | 0.3 | –1.8 | –1.3 |
| Supplementary trade items | –1.0 | –1.0 | –0.7 | –0.8 | –0.8 | –0.9 | –1.1 |
| Services balance | 16.8 | 15.6 | 15.2 | 15.4 | 18.8 | 18.7 | 22.9 |
| Tourism | 2.4 | 2.7 | 2.4 | 1.6 | 1.5 | 1.5 | 4.9 |
| Other services | 14.4 | 13.0 | 12.8 | 13.8 | 17.3 | 17.2 | 18.0 |
| Net Factor income | 13.5 | 10.7 | 13.9 | 15.6 | 19.7 | 19.9 | 20.1 |
| Labour | –7.2 | –6.9 | –6.8 | –6.8 | –6.5 | –6.5 | –6.5 |
| Capital | 20.7 | 17.6 | 20.8 | 22.4 | 26.2 | 26.3 | 26.6 |
| Net current transfers | –4.0 | –4.7 | –5.0 | –5.0 | –4.9 | –4.8 | –5.0 |
| Balance on current external account | 28.8 | 23.9 | 25.2 | 27.2 | 33.1 | 31.2 | 36.0 |
| As a percentage of GDP | 8.2 | 6.7 | 6.9 | 7.4 | 8.9 | 8.1 | 9.6 |

1. Estimates.
2. Annual rates, not seasonally adjusted.

*Source:* Banque nationale suisse, *Bulletin mensuel.*

order inflows and order stocks became less pessimistic (Figure 9). Forward looking indicators of orders and production expectations over the three months ahead suggest growing optimism, also for exporting industries. Although consumer confidence has not improved further, retailers expect a rising turnover in the near term.

Figure 9. **Indicators of economic activity**
Year-on-year change




1. Balance of positive and negative responses.
2. 3-month moving average.

*Source:* KOF/ETH, *Konjunktur ;* OECD, *Main Economic Indicators.*

The projections presented below have been made on the technical assumption of nominal exchange rates remaining broadly unchanged at their average level of 1998. It is further projected that the National Bank allows the three-month Euro-Swiss franc interest rate to stay slightly below 1 per cent this year, hence about ½ percentage point below the average level of 1998, with it remaining there in 2000. This is in line with the projected easing of monetary conditions in the Euro area and implies a broadly unchanged yield curve. It also implies an *ex-post* real short-term interest rate of close to zero and a real long-term interest rate 1 percentage point lower than in 1998. Fiscal policy is assumed to tighten its stance in general, in accordance with the objective of a balanced budget of the Confederation in 2001 (see next chapter). The Confederation's programme to bring forward maintenance expenditure on public infrastructure

Table 6. **Short-term projections**

| | Current prices SF Billion 1995 | Share in GDP % | Percentage changes | | | |
|---|---|---|---|---|---|---|
| | | | 1997 | 1998 | 1999[1] | 2000[1] |
| **Demand and output**[2] | | | | | | |
| Private consumption | 216.1 | 59.5 | 1.3 | 1.8 | 1.8 | 1.9 |
| Government consumption | 55.0 | 15.1 | 0.6 | 0.6 | 0.4 | 0.3 |
| Gross fixed investment | 77.6 | 21.4 | 1.5 | 3.8 | 3.3 | 4.0 |
| Construction | 43.8 | 12.1 | –1.5 | 0.7 | 1.2 | 2.0 |
| Machinery and equipment | 33.8 | 9.3 | 4.9 | 7.1 | 5.3 | 6.0 |
| Final domestic demand | 348.7 | 96.0 | 1.2 | 2.1 | 2.0 | 2.2 |
| Change in stocks[3, 4] | 0.0 | 0.0 | 0.1 | 1.9 | –0.5 | 0.4 |
| Total domestic demand | 348.8 | 96.0 | 1.3 | 4.0 | 1.5 | 2.6 |
| Exports of goods and services | 127.5 | 35.1 | 9.0 | 4.1 | 2.8 | 4.0 |
| Imports of goods and services | 112.9 | 31.1 | 8.1 | 8.8 | 3.4 | 5.8 |
| Change in foreign balance[3] | 14.6 | 4.0 | 0.4 | –1.9 | –0.3 | –0.9 |
| Gross domestic product | 363.3 | 100.0 | 1.7 | 2.1 | 1.2 | 1.8 |
| Industrial production[5] | | | 5.4 | 3.7 | 2.1 | 3.1 |
| **Prices** | | | | | | |
| GDP price deflator | | | –0.1 | 1.1 | 1.0 | 1.2 |
| Private consumption deflator | | | 0.6 | 0.2 | 0.6 | 0.8 |
| Unemployment rate (levels) | | | 5.2 | 3.9 | 3.0 | 2.7 |
| Current balance (US$ billion) | | | 22.8 | 21.5 | 20.3 | 20.2 |
| As a percentage of GDP | | | 8.9 | 8.1 | 7.8 | 7.6 |

1. Projections.
2. At 1990 prices.
3. As a percentage of previous year GDP.
4. Including statistical adjustments.
5. Excluding mining and quarrying.

*Source:* Office fédéral de la statistique; OECD, *Quarterly National Accounts* and OECD Secretariat.

will support construction investment during the first half of 1999, while later on activity will benefit from the start of some large new public projects (further improvement of the railway infrastructure, new alpine tunnels).

Decelerating export market growth and the recent real effective exchange rate appreciation (in terms of relative unit labour cost) will act as a drag on exports in 1999 (Table 6). But exports should be boosted by improving market growth from the second half of 1999 onwards. Solid growth of machinery and equipment investment is expected, underpinned by improved company profitability, low capital cost and falling prices of imported investment goods. Construction investment should continue its mild upswing in 1999, as a result of broadly stagnant private construction activity and the pick-up in infrastructure expenditure in 1999-2000. Meanwhile, private consumption is likely to remain a mainstay of domestic demand, underpinned by improving labour incomes and growing optimism about the labour market situation in general and job security in particular. The unemployment rate should fall a little further, reflecting rising employment and the effects of the unemployment insurance reform. With economic growth slowing this year, the slack in factor and product markets will remain significant. This should keep inflation very low, in spite of the increase in the VAT rate in early 1999.

Given Switzerland's traditional role as a "safe haven" for international investors at times of financial markets crises, and the resulting tendency of large exchange rate fluctuations, potential changes in the external value of the Swiss franc create a major uncertainty for the projections. A weaker franc could impart an additional stimulus to exports, with little risk of accelerating inflation at the current juncture. On the other hand, further exchange rate appreciation would entail a dampening effect on the exposed sector of the economy, but would also open additional scope for monetary easing.

# II. Macroeconomic policies

Switzerland's macroeconomic polices are set in a medium-term framework. A new amendment to the Constitution (*Haushaltsziel 2001/objectif budgétaire 2001*) – approved in the referendum of 7 June 1998 – requires fiscal policy to eliminate the federal budget deficit by 2001. From then on, the Confederation will aim at a balanced budget over the course of the business cycle, which implies a gradual decline in the debt/GDP ratio over the medium term. The objective of monetary policy is to ensure a stable price level over the medium term, which is viewed as a key prerequisite for the achievement of balanced and sustainable economic development. The Swiss National Bank (SNB) has recently declared annual consumer price inflation of 1 to 2 per cent as compatible with price stability.[15] In 1990, the SNB defined a medium-term growth path for the seasonally-adjusted monetary base[16] of 1 per cent per year[17] as a target conducive to price stability. In 1997, the SNB decided to downgrade the role of the monetary base as an intermediate target to allow for the instabilities in base money demand that had emerged in 1996.

Over the past two years, macroeconomic policy outcomes were broadly consistent with intentions: the Confederation's budget deficit has been reduced in accordance with the new constitutional mandate to balance the budget until 2001. Moreover, the economic recovery in 1997-98 was accompanied by a stable price level, notwithstanding an overshooting of the SNB's medium-term target for the supply of base money. At the present juncture, the main challenge of macroeconomic policy is to keep government finances on their consolidation path and to preserve the achieved price stability, while lending maximum support to a continuation of the economic upswing.

## Monetary policy

### *Easing monetary conditions*

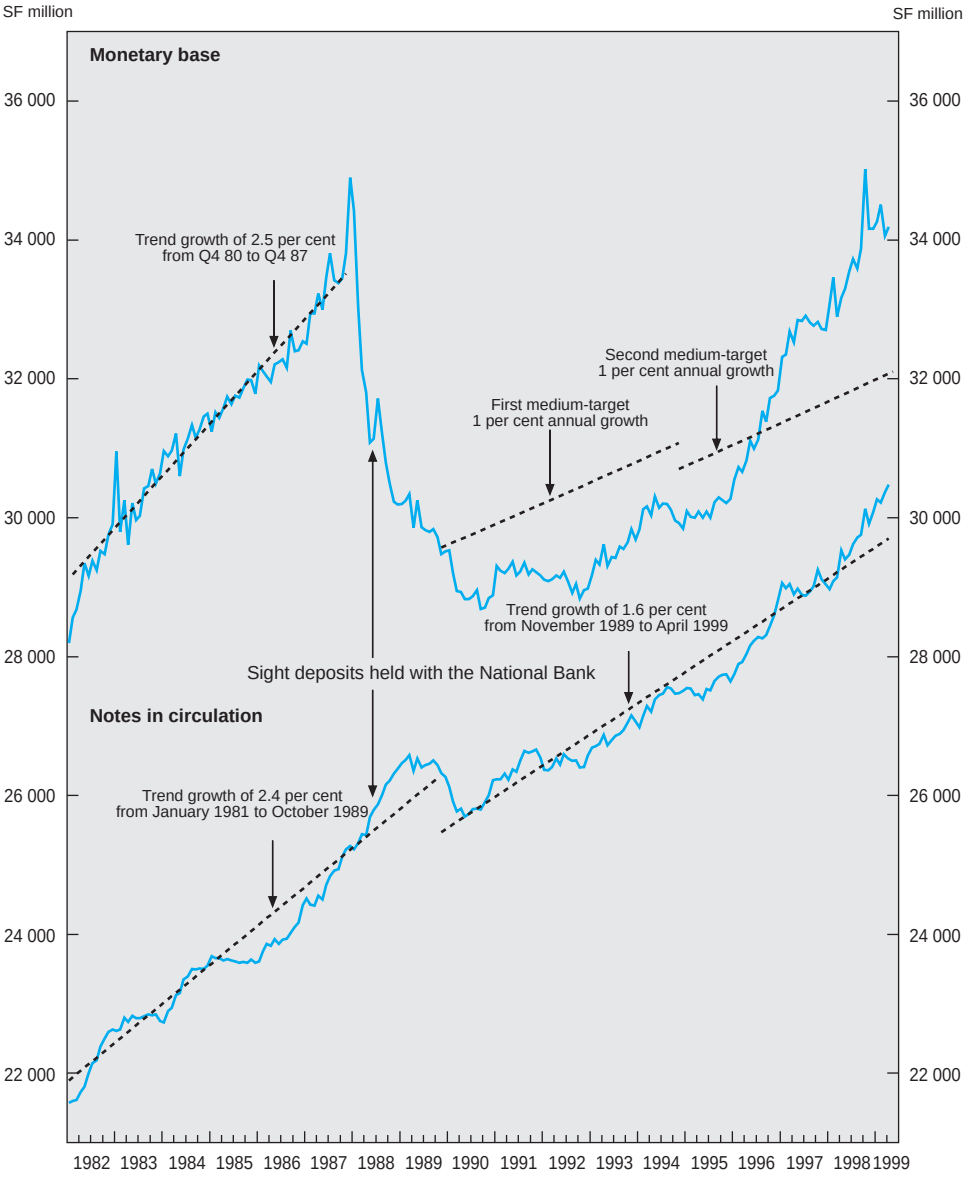
Because of the need to fight inflation, monetary conditions during most of the first half of the 1990s had been very tight. This was reflected *inter alia* in the

remarkable undershooting of the SNB's medium-term target for the supply of base money, a substantial appreciation of the real effective exchange rate and high real short- and long-term interest rates compared with their long-term averages. Given the large prevailing output gap, significant slack in the labour market and no signs of inflationary pressures, monetary policy gradually changed course during 1995 and was eased more aggressively in 1996. In consequence, the growth of the monetary base accelerated and exceeded its medium-term target from mid-1996 onward, ending more than six years[18] of substantial undershooting (Figure 10). At the end of 1997, the stock of base money was already 3½ per cent above its medium-term target. Nevertheless, in its traditional statements at year-end about its policy intentions for the twelve months ahead, the National Bank confirmed that, for both 1998 and 1999, in view of the absence of any indications of inflation pressures, it saw no need for monetary tightening or higher interest rates. Hence, the acceleration in base money supply continued during 1998, leaving the aggregate 7 per cent above its target path at the end of the year. Although the growth of base money slowed somewhat during the first five months of 1999, it remained distinctly above its medium-term target.

The more generous provision of central bank money through foreign exchange swaps and open market operations – including repo transactions[19] since April 1998 – is reflected in a further decline in the three-month euro-Swiss franc interest rate from some 2½ per cent by mid-1996 to an average of around 1½ per cent in both 1997 and 1998. However, money market interest rates fluctuated considerably over recent years, in response to developments in foreign exchange markets and subsequent interventions of the National Bank to smooth out movements in the Swiss franc exchange rate. For example, the SNB provided the commercial banks with ample liquidity injections in February and in October 1998 to counter upward pressures on the franc exchange, which were induced by turbulence in global financial markets and the subsequent flight of international investors into the Swiss franc as a "safe haven". This brought the three-month money market interest rate temporarily to below the 1 per cent mark; it returned to around 1½ per cent in the following months. Renewed upward pressure on the Swiss franc in late 1998 again induced generous liquidity provision by the SNB, which entailed another decline in the three-month Euro-Swiss franc interest rate to some 1¼ per cent in the first quarter of 1999. The National Bank's determination to maintain easy monetary conditions has also become manifest in the lowering of the discount rate[20] from 1 per cent to ½ per cent in the wake of the refinancing rate cut by the European Central Bank in April 1999. In the event, the three-month euro-Swiss franc interest rate stood at 1 per cent in the second quarter of 1999.

The recent sharp rise of base money in excess of the SNB's medium-term target has in part been the result of special factors, which accordingly gives an exaggerated impression of the degree of monetary relaxation. First, only part of

Figure 10. **The adjusted monetary base**
Seasonally adjusted




*Source:* Banque nationale suisse, *Bulletin mensuel.*

the acceleration in currency demand over the past three years is likely to be the result of the decline in interest rates on savings deposits, its key opportunity cost measure. Another substantial part of the increase seems related to the unexpectedly strong demand for one-thousand-franc bank notes, whose causes are unclear.[21] Also, the vigorous rise of the commercial banks' sight deposits held with the SNB – the second component of the monetary base – by about 50 per cent over the 1996-98 period is entirely out of line with previously observed relationships with other macroeconomic variables. Among the likely special factors which may have induced the commercial banks to raise their central bank money holdings are the rationalisation of the banks' postal accounts system, which led to the transfer of funds to accounts held with the SNB. In addition, extra central bank money demand emanated from the uncertainty linked to the merger of two large banks, the introduction of a market for repos in 1998, and possibly increased liquidity preference as a consequence of the unrest in international financial markets.[22] All this suggests that the persistent overshooting of the monetary base since mid-1996 is largely the result of a (structural) upward shift in the demand for central bank money rather than an endogenous increase in liquidity demand; it is thus unlikely to herald rising future inflation.

### *Broader monetary aggregates*

The National Bank has on a number of occasions[23] stated that the various shifts in the demand for base money in recent years make its interpretation increasingly difficult and make the aggregate too unreliable as sole guide of monetary policy. Nevertheless, the SNB has continued to comment on the development of the monetary base and to explain its deviations from the medium-term target path. In its communication with the public, the SNB has supplemented its analysis of the monetary base with commentary on the development of a broader monetary aggregate which passed econometric stability tests.[24] For a number of years, M1 had been the preferred complementary aggregate, but when it revealed features of instability, the Bank began to focus on M2 as "a reliable monetary indicator"[25] in 1997.

Both M1 and M2 grew strongly in 1996 and 1997 (Table 7), boosted by strong inflows into sight deposits and transactions accounts, which were largely induced by falling interest rates. The increase in M2 also reflected buoyant flows into savings accounts, probably caused by the flattening of the yield curve, which increased the attractiveness of savings deposits relative to bonds. This episode demonstrates the high sensitivity of M1 and M2 to portfolio reshuffling in response to interest rate movements, which complicates their use as intermediate targets of monetary policy. The growth of both M1 and M2 slowed in 1998, but it is unclear to what extent this represents a change in monetary conditions or just a normalisation after portfolio adjustments.

Table 7. **Monetary aggregates**

Percentage change from previous period, s.a.a.r.

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 | 1998 | | | | 1999 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Q1 | Q2 | Q3 | Q4 | Q1 |
| Notes in circulation | 2.2 | –0.2 | 1.8 | 1.9 | 0.7 | 2.0 | 3.1 | 1.9 | 0.6 | 4.9 | 3.3 | 3.4 | 5.1 |
| Sight deposits held with SNB[1] | –6.2 | –9.9 | 3.9 | 1.1 | –4.0 | 14.0 | 26.5 | 9.5 | 25.5 | –14.3 | 21.9 | 53.2 | –30.3 |
| Monetary base[1] | 1.1 | –0.7 | 1.5 | 1.9 | 0.2 | 3.5 | 4.8 | 3.0 | 4.9 | 2.4 | 4.8 | 8.8 | –2.0 |
| Sight deposits and transactions accounts | 1.8 | 2.3 | 13.1 | 6.6 | 8.2 | 13.5 | 11.6 | 8.9 | 19.7 | 1.1 | 7.5 | 6.1 | 6.2 |
| M1[2] | 2.0 | 2.0 | 10.5 | 5.6 | 6.8 | 11.7 | 10.1 | 7.8 | 17.4 | 0.7 | 6.5 | 6.4 | 6.7 |
| Savings deposits | 1.5 | 3.6 | 21.5 | 14.2 | 3.4 | 12.1 | 3.6 | –1.1 | –2.9 | –4.2 | –1.1 | –2.1 | –6.9 |
| M2[2] | 1.7 | 2.8 | 16.2 | 10.2 | 4.9 | 11.9 | 6.5 | 3.0 | 6.3 | –1.9 | 2.5 | 2.0 | –0.3 |
| Time deposits | 4.4 | 0.1 | –16.9 | –7.4 | –5.9 | –10.8 | 0.6 | –7.9 | –16.1 | 2.0 | –1.4 | –6.5 | 14.6 |
| M3[2] | 2.7 | 1.8 | 4.2 | 5.1 | 2.2 | 6.6 | 5.4 | 1.0 | 2.0 | –1.2 | 1.8 | 0.5 | 2.1 |
| Domestic credit expansion[3] | 4.5 | 1.7 | 0.9 | 2.4 | 2.4 | 2.3 | 1.2 | 0.5 | 0.4 | 1.0 | 0.8 | 0.5 | 2.8 |
| Nominal GDP | 5.2 | 2.6 | 2.2 | 2.2 | 1.6 | 0.7 | 1.6 | 3.2 | 3.4 | 1.4 | 1.2 | 2.2 | 2.6 |

1. Change in definition in 1995.
2. Definition of 1995.
3. Quarterly figures are not seasonally adjusted. Changes refer to the same period of the previous year.

*Source:* Banque nationale suisse, *Bulletin mensuel* and OECD, *National Accounts.*

This has motivated the SNB to attach greater importance to M3 as a complementary monetary indicator in its policy statements for 1998 and 1999. M3 was found to be less interest-sensitive than M1 and M2 and to be stable in the long run.[26] It signals changes in inflation with an average lead period of 3½ years. Given the observed trend decline of the income velocity of M3 by 1 per cent, an average annual increase by 4 per cent would be consistent with potential output growth of 2 per cent and inflation of 1 per cent. Hence, average annual growth of about 7 per cent in 1996 and 5 per cent in 1997 was consistent with the notion of monetary ease in these years. The growth of M3 slowed to 1 per cent in 1998 and remained subdued in the first quarter of 1999. This is in line with the weak expansion of domestic credits, but also seems to reflect the flow of funds out of M3 into other assets (such as mutual funds) which are about as liquid as time deposits. However, the movement of such a broad aggregate is difficult to interpret in the short run.

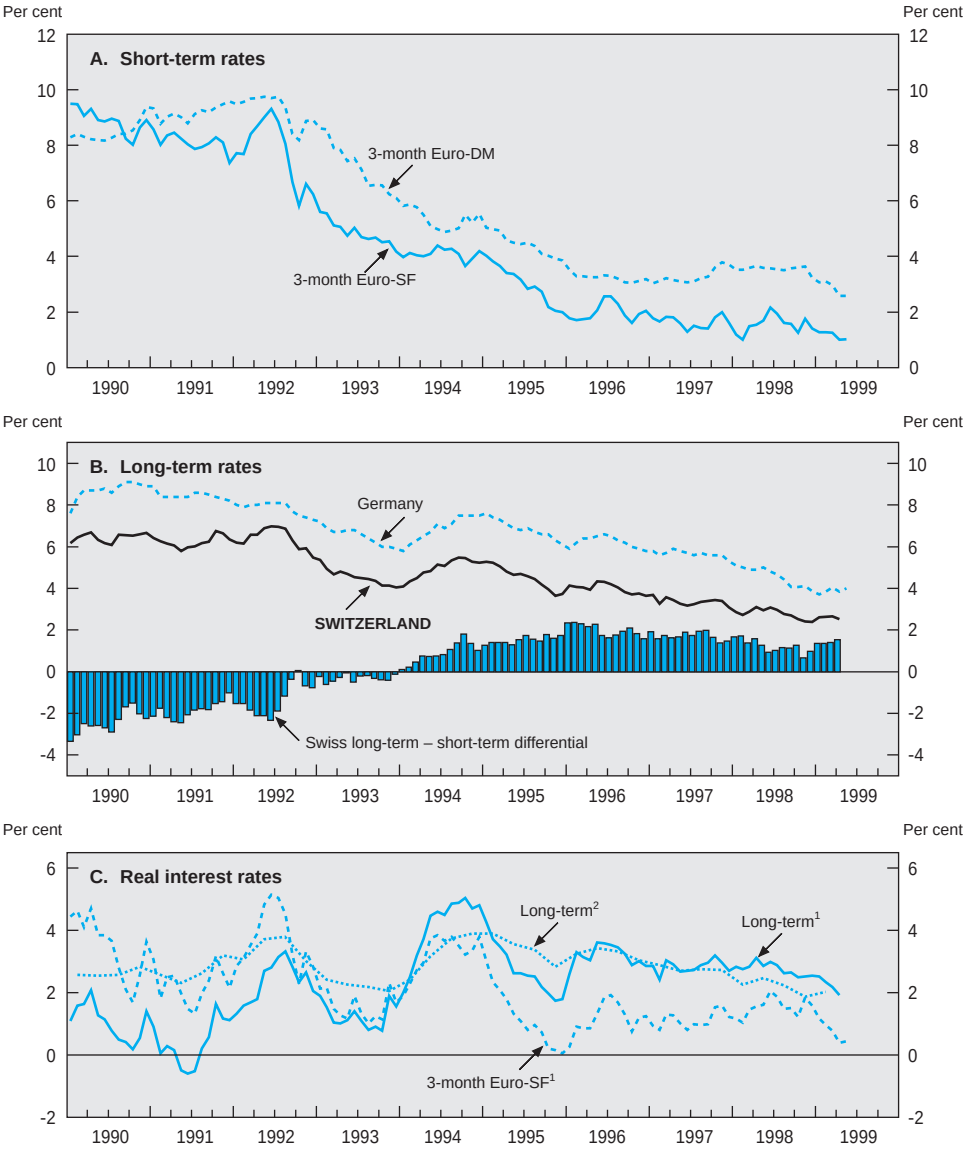
### *Other indicators of monetary conditions*

To assess the potential for inflation in the short and medium run, additional indicators need to be taken into account, in particular indicators of the business cycle and the exchange rate. Among these, short-term interest rates displayed a high degree of volatility, but showed only a minor further trend decline in 1998 and early 1999 (Figure 11). Real short-term interest rates, however, are now below their average of the past ten years. Long-term real interest rates declined further over the past two years or so, but remain above their long-term averages.

For an economy as open as Switzerland, the exchange rate is a powerful transmission mechanism of monetary policy and therefore a key determinant of overall monetary conditions. After the correction in 1996-97 of the vigorous appreciation during the three preceding years, the nominal effective Swiss franc exchange tended to trend upwards, although with marked fluctuations (Figure 12). However, thanks to inflation lower than in most trading partner countries, the *average* real (CPI-based) effective exchange rate remained broadly unchanged in 1998 and in the first five months of 1999, hence about 9½ per cent below its peak of 1995. The notion of generally easy monetary conditions in 1997-98 is confirmed by a simple monetary conditions index (MCI), which tracks the combined changes in real effective exchange rates and the (*ex-post*) real short-term interest rates (Figure 13). Nevertheless, according to this measure, monetary conditions appear to have tightened somewhat in 1998, mainly reflecting the small increase in the real three-month Euro-Swiss franc interest rate. The MCI indicates, however, further easing of monetary conditions during the first five months of 1999.

To sum up, monetary conditions remain easy, which appears appropriate in view of the remaining slack in product and labour markets and the absence of

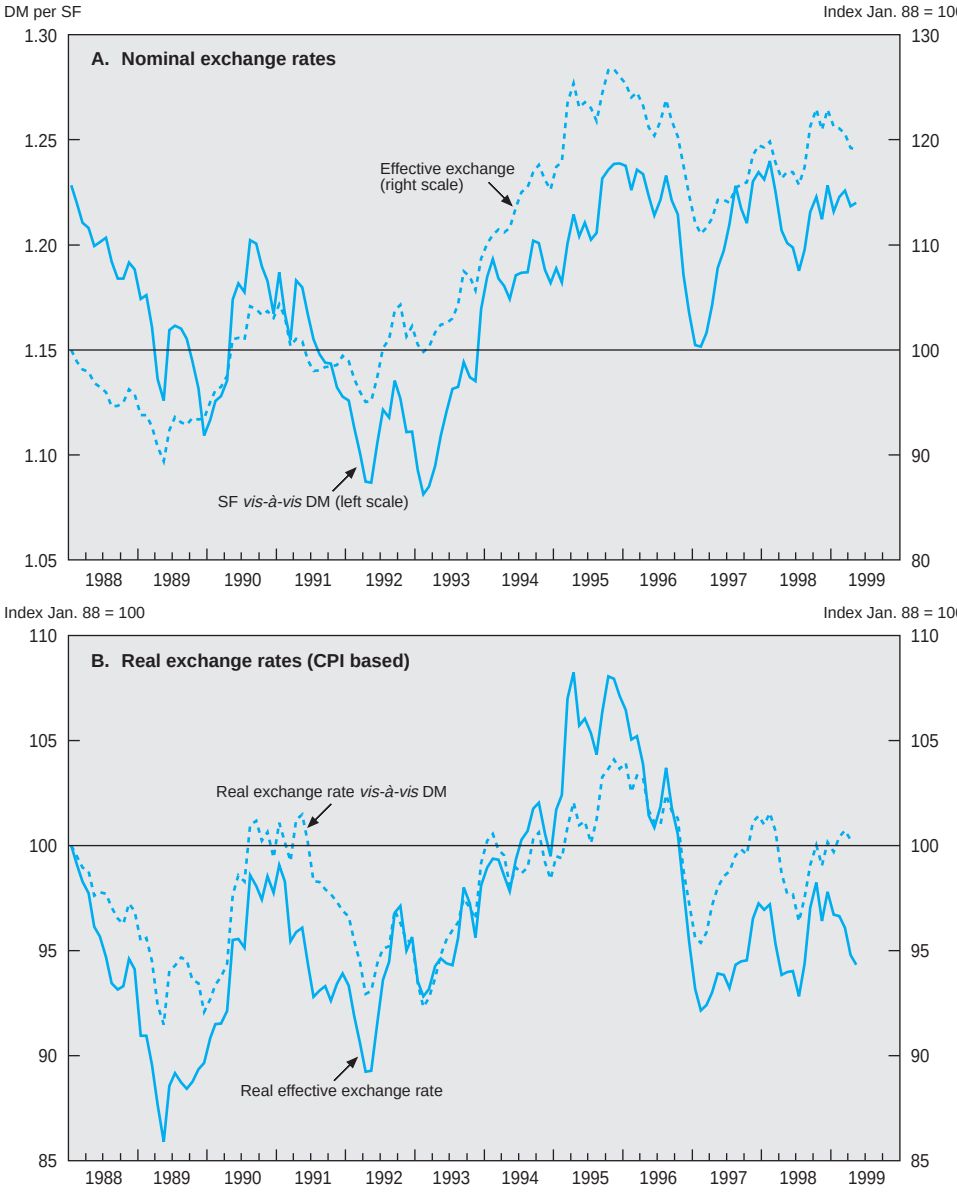
Figure 11. **Interest rates**




1. Nominal rate less the change in the consumer price index from the previous year.
2. Nominal rate less low frequency component of the annualised change in the seasonnally adjusted GDP deflator.

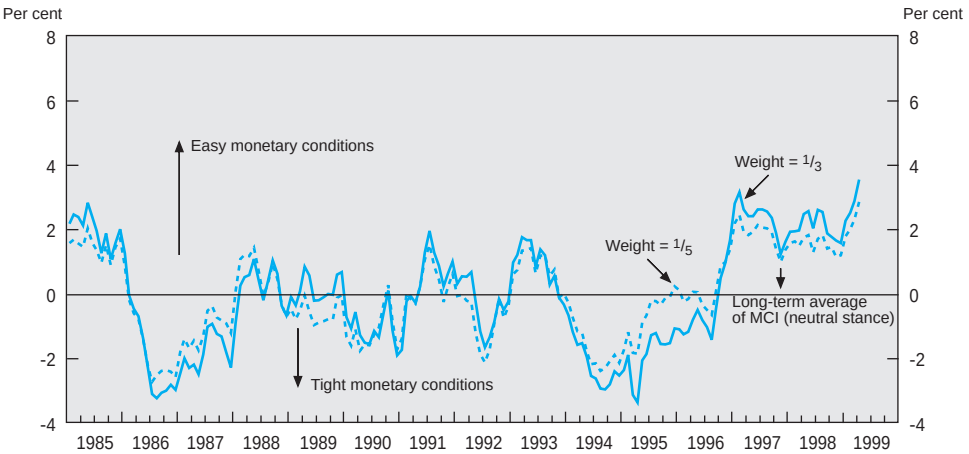
*Source:* OECD, *Main Economic Indicators* and OECD Secretariat.

Figure 12. **Exchange rates**




*Source:* Banque nationale suisse, *Bulletin mensuel;* OECD, *Main Economic Indicators,* and OECD Secretariat.

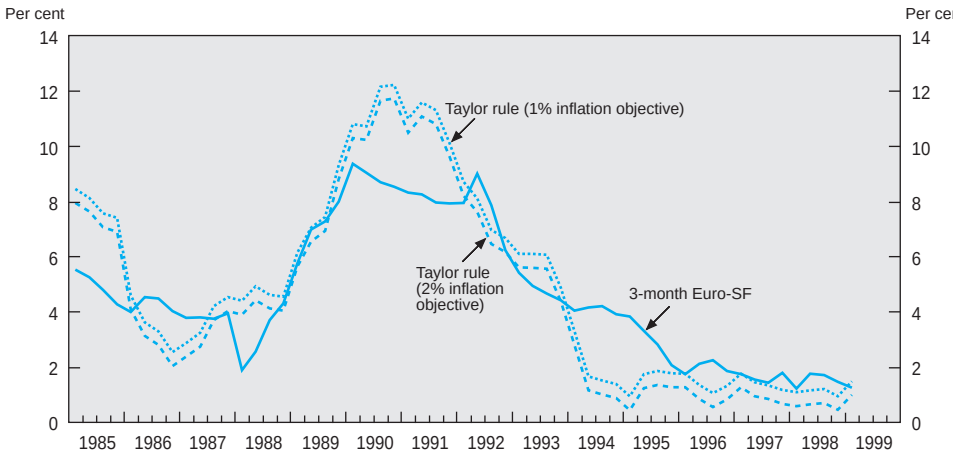
Figure 13. **Monetary conditions index (MCI)**




*Note:* The MCI is defined as MCI = -(r-r*) -(e/e* -1)/w, where r is the real (CPI deflated) three-month euro-Swiss franc interest rate, r* the long-term average of r, e the real effective exchange rate (CPI based), e* the long-term average of e, and 1/w the weight on the exchange rate deviation from its average. Since the real exchange rate in Switzerland is characterised by a significant upward trend, the MCI calculated for Switzerland uses the deviation of the real exchange rate from its (linear) trend rather than from its long-term average. A positive MCI implies easy monetary conditions (relative to long-term averages).
*Source:* OECD Secretariat.

inflation pressures, as well as the prospective shift to fiscal restraint (see next section). But with real long-term interest rates still somewhat high, the growth of M3 slowing, and the monetary conditions index signalling some (mild) tightening of monetary conditions in 1998, there appeared to be scope for some further relaxation of monetary policy in early 1999. This was particularly so given the expected weakening of economic activity in 1999. This view is further supported by a standard Taylor rule model, which explicitly assumes responsibility of monetary authorities not only for price stability, but also for full utilisation of resources.[27] The Taylor rule suggests that recent short-term interest rates were still somewhat above the "warranted rate", which takes account of output gap estimates[28] and the difference between actual inflation and the monetary authority's inflation objective (Figure 14). However, the room for further relaxation has narrowed after the SNB's monetary easing in line with the European Central Bank's rate cut in April.

Figure 14. **Taylor rule**




*Note:* The "warranted" short-term interest rate r* (Taylor rule) is defined as r* = Φr + p + (p-p*)/2 + GAP/2. r* the sum of the long-term (ten years) average real short-term interest rate Φr, the actual inflation rate p, one-half of the difference between actual inflation p and the inflation objective p* and one-half of the output gap, defined as the difference between actual and potential output as a percentage of the latter.

*Source:* OECD Secretariat.

### *The need for a new concept*

The growing difficulty in distinguishing structural shifts from endogenous changes in liquidity demand severely reduces the informational value of the monetary base and its usefulness as an anchor for expectations. Its continuing divergence from the SNB's medium-term target also risks confusing financial markets. Hence, the termination of the second five-year period of the base-money targeting exercise by the end of 1999 could be the right occasion for not only downgrading the policy relevance of base money further but for abandoning its use as an intermediate target altogether. Such decisions have been taken by central banks in a number of member countries where financial innovation caused instabilities of previously reliable relationships between targeted monetary aggregates and macroeconomic magnitudes. But, replacing the narrow base money by a broader monetary aggregate as intermediate target could also be problematic as the inherent endogeneity of broader money tends to blur the signals coming from it in the short run. However, since the long-term stability of M3 is statistically well established, it may be a useful long-run indicator for monetary policy.

A new concept of monetary policy with a clear definition of price stability and a strong commitment to it is needed in Switzerland. It should be based on several indicators, in particular the aggregate M3 and inflation forecasts. The difference to the current practice would be rather small as the National Bank already monitors a broad set of indicators.[29] The new concept should be characterised by a high degree of transparency, since motivating and explaining monetary policy to the public is crucial. A step in this direction was undertaken by the National Bank when it began the regular publication of "Conclusions for monetary policy" in the SNB's *Quarterly Bulletins* in 1998. In these "Conclusions", the SNB interprets monetary developments, explains its actions and gives an outline of its policy intentions. To enhance transparency further, the text can be consulted on the Internet.[30] The need for more accurate inflation forecasts could also give rise to a stronger effort to improve the quality, coverage and timeliness of economic statistics.

## Fiscal policy[31]

### *Background*

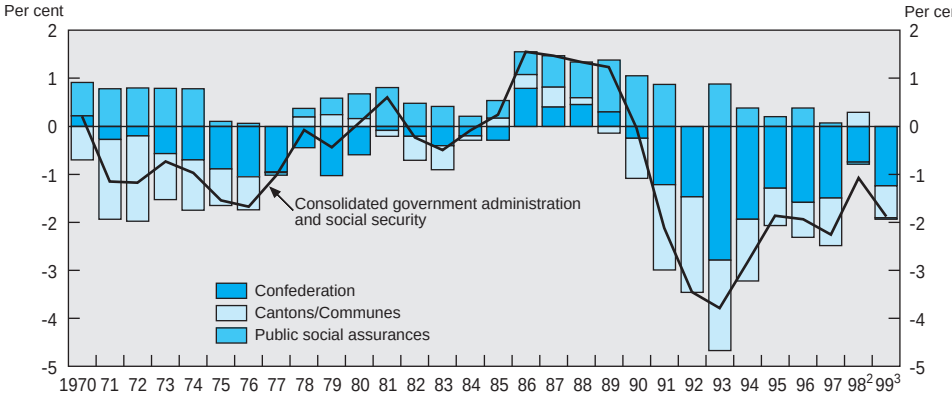
In recent years, fiscal policy has been mainly directed at reversing the large deterioration in budget balances that occurred in the recession of the early 1990s (Figure 15). Most progress was made between 1993 and 1995, when the general government deficit was cut in half, falling to 1.9 per cent of GDP in 1995. This reduction is estimated to have been structural and was achieved through an increase in revenues (Figures 16 and 17). Most of the consolidation occurred in the Confederation's budget. With the economy failing to recover as was expected in the mid-1990s, budget consolidation was put on hold over the next few years. The general government deficit rose slightly through to 1997 but fell in 1998 to the lowest level since 1990. Now that economic recovery appears to be more firmly established, the federal government has redoubled its efforts to reduce the budget deficit. It has put in place a stabilisation programme aimed at bringing the budget near to balance by 2001. In the event that this objective is not achieved, a constitutional amendment (*Budget objective 2001*), approved in a referendum last year, requires budget cuts to be made so that the objective is achieved within the following two years (or four years if there is a recession).

### *Recent developments*

*Confederation*[32]

Following two years of broadly stable deficits of around 1½ per cent of GDP, the Confederation's budget balance (Financial Accounts basis) improved

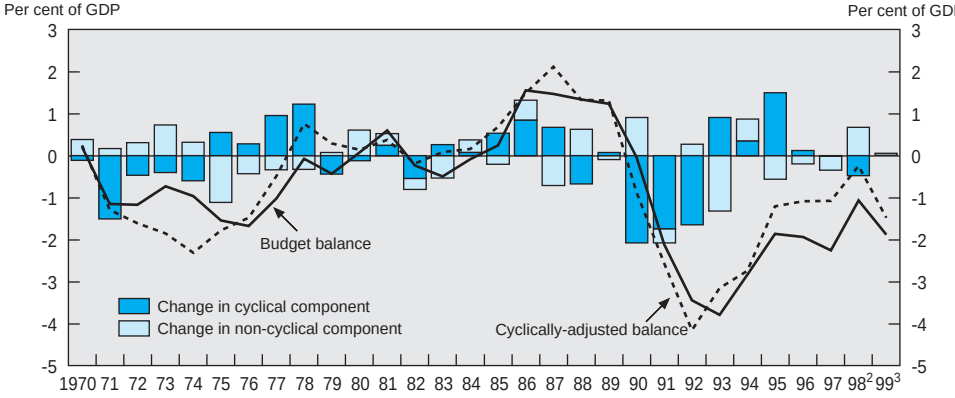
Figure 15. **Budget balances**[1]
As a per cent of GDP




1. Statistique financière révisée basis.
2. Budget estimates, except for Confederation. Excludes proceeds from the sale of shares in Swisscom (SF 2.7 billion plus SF 0.2 billion in stamp duty).
3. Budget projections.

*Source:* Administration fédérale des finances.

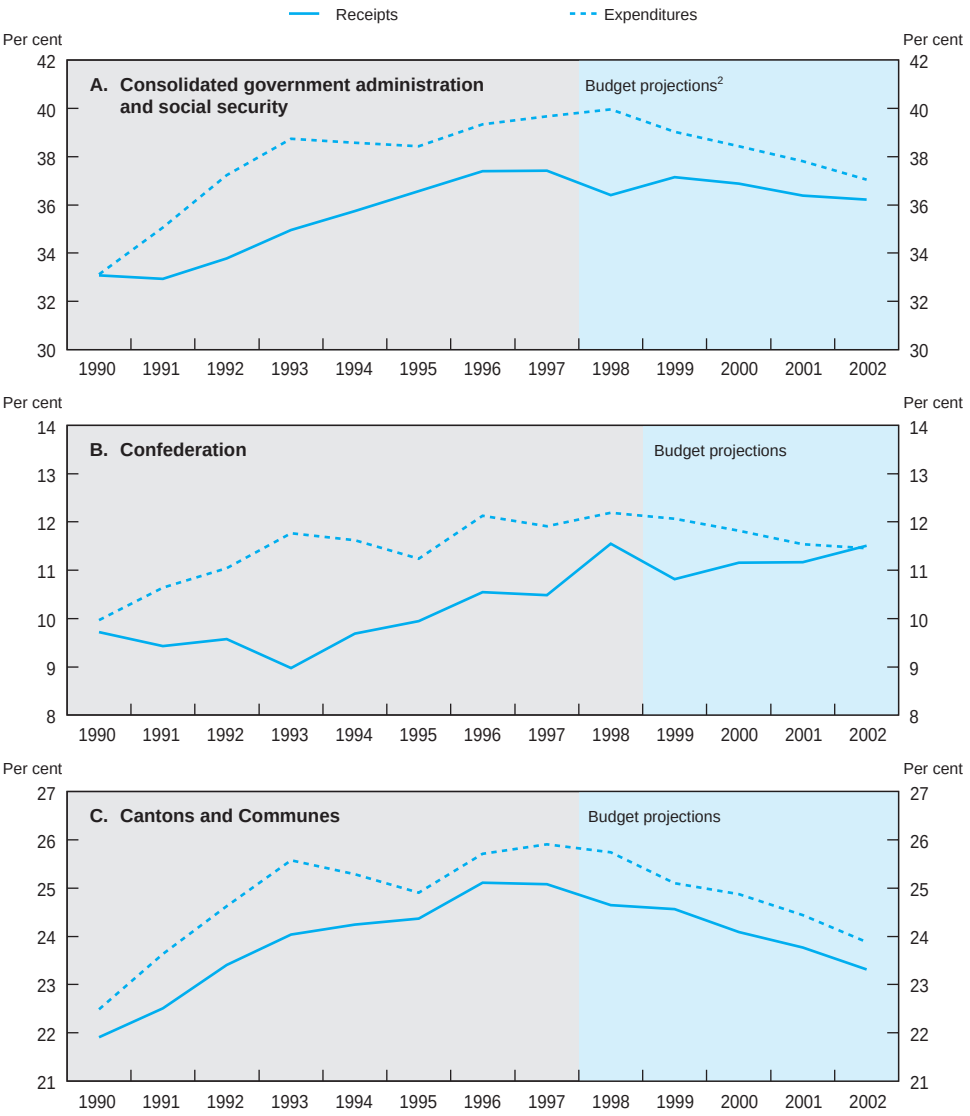
Figure 16. **Cyclically-adjusted budget balance**
General government[1]




1. Statistique financière révisée basis. Including social security.
2. Budget estimates, except for Confederation. Excludes proceeds from the sale of shares in Swisscom (SF 2.7 billion plus SF 0.2 billion in stamp duty).
3. Budget projections.

*Source:* Administration fédérale des finances.

Figure 17. **Government receipts and expenditures**[1]
As a per cent of GDP




1. Excluding proceeds from the sale of shares in Swisscom (SF 2.7 billion plus SF 0.2 billion in stamp duty).
2. Outcomes for the Confederation in 1998.

*Source:* Administration fédérale des finances.

sharply in 1998 to a surplus of SF 0.5 billion, hence 0.1 per cent of GDP (Table 8). A major factor contributing to this outcome was the partial privatisation of Swisscom. Even abstracting from this transaction (including SF 0.2 billion in stamp duty), the deficit fell significantly, to SF 2½ billion (½ per cent of GDP). Revenue was boosted by withholding tax receipts due to soaring dividend payments while, on the other hand, requests for withholding tax refunds remained weak owing to the delay with which they can be presented. As withholding tax and the federal income tax are higher in even numbered years a better indication of the underlying growth in tax revenues can be obtained by comparing receipts in 1998 and 1996. Average growth in total receipts for this period was 5.7 per cent, which is considerably higher than in earlier years. Expenditure grew rapidly, owing to a one-off charge resulting from changes in the financing of rail infrastructure projects (SF 1.9 billion). Abstracting from this charge, outlays only rose by 1.4 per cent, considerably less than the growth in nominal GDP (2.9 per cent). Most major categories of expenditure showed little or no increase, reflecting measures taken to constrain spending and the decline in unemployment. Debt interest payments, however, continued to rise, owing to the Confederation's rising debt level.

*Cantons, communes and social security*

After declining markedly through to 1995, cantons' deficits rose significantly in 1997 and are estimated to have decreased sharply in 1998 (Table 9). The deterioration in 1997 to SF 3.1 billion reflected rising transfers to cover deficits of regional hospitals and to reduce medical insurance premiums, loans to the unemployment insurance system and investment expenditure (encouraged by the programme to stimulate the economy). The estimated deficit in 1998 (SF 0.8 billion, or 0.2 per cent of GDP) was substantially lower than budgeted (SF 4.2 billion) on account of an increase in fiscal receipts by nearly SF 1 billion, the positive effects of low inflation on spending, a supplementary distribution of profits by the Swiss National Bank (SF 0.4 billion) and a reduction of SF 0.5 billion in loans to the unemployment insurance system. In addition, favourable conditions in capital markets are estimated to have resulted in lower than budgeted interest expenses.

There is a wide diversity of fiscal positions amongst the cantons. Receipts in some cantons do not even cover current expenditures while in others receipts also cover half or more of net investment. In general, the French- and Italian-speaking cantons together with the German-speaking cantons which are large agglomerations are in a weaker financial position than the other (German-speaking) cantons. The cantons in a weaker financial position were particularly adversely affected by weak activity until the mid-1990s and, in the case of large agglomerations, by the growing cost of social assistance and of supplying services that also benefit residents of neighbouring cantons. The planned reform of fiscal equalisation arrangements should reduce the financial difficulties

Table 8. **Confederation budget**[1]

| | Outturn 1997 | Outturn 1998 | Percentage change | | Budget[2] 1998 | Budget[2] 1999 | Percentage change | |
|---|---|---|---|---|---|---|---|---|
| | | | 1998/1997 | 1998/1996 average | | | Budget 99/ Outturn 98 | 1999/1997 average |
| **Total expenditure** | 44 122 | 46 589 | 5.6 | | 47 590 | 46 297 | –0.6 | |
| Social welfare | 12 864 | 12 917 | 0.4 | | 13 408 | 12 037 | –6.8 | |
| Transport | 6 651 | 8 304 | 24.9 | | 9 046 | 6 596 | –20.6 | |
| Defence | 5 395 | 5 353 | –0.8 | | 5 400 | 5 119 | –4.4 | |
| Agriculture and food | 3 922 | 3 925 | 0.1 | | 3 941 | 4 239 | 8.0 | |
| Interest | 3 079 | 3 345 | 8.6 | | 3 413 | 3 729 | 11.5 | |
| Research and training | 3 074 | 3 131 | 1.9 | | 3 145 | 3 328 | 6.3 | |
| Cantons' share of receipts | 3 179 | 3 485 | 9.6 | | 3 054 | 3 260 | –6.5 | |
| Foreign Affairs | 2 043 | 2 043 | 0.0 | | 2 103 | 2 112 | 3.4 | |
| Other | 3 915 | 4 086 | 4.4 | | 4 080 | 5 877 | 43.8 | |
| **Total receipts**[3] | 38 852 | 44 139 | 13.6 | 5.7 | 39 968 | 42 362 | –4.0 | 4.4 |
| Tax receipts | 34 694 | 39 613 | 14.2 | 7.7 | 36 077 | 38 882 | –1.8 | 5.9 |
| VAT | 12 477 | 13 255 | 6.2 | 5.3 | 13 200 | 15 150 | 14.3 | 10.2 |
| Federal income tax | 9 725 | 9 687 | –0.4 | 3.9 | 8 850 | 9 800 | 1.2 | 0.4 |
| Fuel duties | 4 131 | 4 712 | 14.1 | 3.7 | 4 567 | 4 727 | 0.3 | 7.0 |
| Stamp duty | 2 513 | 3 016 | 20.0 | 23.5 | 2 400 | 3 075 | 2.0 | 10.6 |
| Withholding tax | 2 320 | 5 388 | 132.3 | 27.4 | 3 550 | 2 700 | –49.9 | 7.9 |
| Tobacco tax | 1 440 | 1 453 | 0.9 | 2.3 | 1 520 | 1 600 | 10.1 | 5.4 |
| Customs duty | 1 094 | 1 034 | –5.5 | –12.7 | 1 025 | 888 | –14.1 | –9.9 |
| Other | 994 | 1 068 | 7.4 | 15.3 | 965 | 942 | –11.8 | –2.7 |
| Loan repayments/privatisation proceeds[3] | 1 384 | 1 463 | 5.7 | 6.0 | 1 203 | 643 | –56.0 | –31.8 |
| Other receipts[1] | 2 774 | 3 063 | 10.4 | 2.8 | 2 688 | 2 837 | –7.4 | 1.1 |
| **Balance**[3] | –5 270 | –2 450 | | | –7 622 | –3 935 | | |
| As a percentage of GDP | –1.4 | –0.6 | | | –2.0 | –1.0 | | |
| **Balance** (including Swisscom receipts) | | 484 | | | | | | |
| As a percentage of GDP | | 0.1 | | | | | | |

1. Financial Account basis.
2. Budget as adopted by the Parliament.
3. Excluding Swisscom receipts (SF 2.9 billion, including stamp duty).

*Source:* Administration fédérale des finances.

Table 9. **Government accounts**

SF million and percentage changes[1]

| | 1995 SF million | Outturns | | | Budgets | | Financial plan | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 1996 | 1997 | 1998[2] | 1998 | 1999[3] | 2000 | 2001 | 2002 |
| **Confederation**[4] | | | | | | | | | |
| Expenditure | 40 856 | 8.3 | 0.3 | 5.8 | | –1.5 | 1.4 | 1.0 | 2.9 |
| Revenue[5] | 36 162 | 6.4 | 1.0 | 13.6 | | 6.0 | 6.9 | 3.5 | 6.8 |
| Revenue (two-year average)[6] | | 5.4 | 3.7 | 7.1 | 1.9 | 4.4 | 6.4 | 5.1 | 5.1 |
| Balance[5] | –4 694 | –5 773 | –5 530 | –2 825 | –7 976 | –4 874 | –2 646 | –1 563 | 207 |
| **Cantons** | | | | | | | | | |
| Expenditure | 52 111 | 5.7 | 3.1 | 0.7 | | –1.0 | 3.1 | 1.5 | 1.0 |
| Revenue | 50 147 | 5.5 | 1.5 | 5.0 | | 2.8 | 1.3 | 2.3 | 1.6 |
| Revenue (two-year average)[6] | | 4.1 | 3.5 | 3.2 | 1.2 | 1.8 | 2.6 | 1.8 | 1.9 |
| Balance | –1 964 | –2 163 | –3 068 | –800 | –4 200 | –2 100 | –3 200 | –2 800 | –2 500 |
| **Communes** | | | | | | | | | |
| Expenditure | 38 427 | 0.7 | –0.0 | 2.3 | | 1.3 | 2.0 | 1.7 | 1.7 |
| Revenue | 37 587 | 1.6 | –0.1 | 3.2 | | 1.5 | 2.0 | 1.7 | 1.7 |
| Revenue (two-year average)[6] | | 1.6 | 0.5 | 1.3 | 1.6 | 2.6 | 1.8 | 1.9 | 1.7 |
| Balance | –840 | –496 | –600 | –300 | –600 | –500 | –500 | –500 | –500 |
| **Total, government administration**[7] | | | | | | | | | |
| Expenditure | 111 054 | 4.4 | 1.2 | | | –1.4 | 1.8 | 1.0 | 1.2 |
| Revenue | 103 555 | 3.8 | 0.6 | | | 3.3 | 2.9 | 2.3 | 3.0 |
| Revenue (two-year average)[6] | | 3.6 | 2.2 | | 1.3 | 2.0 | 3.1 | 2.6 | 2.6 |
| Balance[5] | –7 499 | –8 431 | –9 198 | –3 925 | –12 776 | –7 474 | –6 346 | –4 863 | –2 793 |
| As a percentage of GDP | –2.1 | –2.3 | –2.3 | –1.0 | –3.3 | –1.9 | –1.6 | –1.2 | –0.6 |
| **Social security** | | | | | | | | | |
| Expenditure | 40 201 | –0.8 | 4.5 | | | 5.5 | 2.4 | 5.0 | 1.9 |
| Revenue | 40 941 | 0.7 | 1.6 | | 0.1 | 7.3 | 2.4 | 2.4 | 2.5 |
| Balance | 740 | 1 378 | 250 | –200 | –800 | –100 | –100 | –1 300 | –1 000 |
| **Consolidated account of government administration and social security** | | | | | | | | | |
| Expenditure | 139 670 | 2.8 | 2.0 | | | 0.1 | 1.9 | 1.7 | 1.5 |
| Revenue | 132 911 | 2.6 | 0.7 | | | 4.4 | 2.8 | 2.0 | 3.1 |
| Revenue (two-year average)[6] | | 3.3 | 1.7 | | 1.0 | 2.8 | 3.6 | 2.4 | 2.6 |
| Balance[5] | –6 759 | –7 053 | –8 948 | –4 125 | –13 576 | –7 574 | –6 446 | –6 163 | –3 793 |
| As a percentage of GDP | –1.9 | –1.9 | –2.3 | –1.1 | –3.6 | –1.9 | –1.6 | –1.5 | –0.9 |

1. For 1996-2002, expenditures and revenues are percentage changes from the previous year unless otherwise stated. Balances in SF million.
2. Estimates for cantons, communes and social security.
3. Initial budget on initial budget.
4. These accounts have been constructed using the definitions of the "*Statistique financière révisée*". This makes the Confederation accounts comparable with those of the cantons and communes, making possible the aggregation of the accounts of all three levels of government to obtain estimates for general government finances. Includes deficits for the large rail infrastructure projects.
5. Excluding proceeds (SF 2.9 billion) in 1998 from the partial privatisation of Swisscom.
6. Average annual percentage change from two years earlier.
7. Not including social security.

*Source:* Administration fédérale des finances.

presently being faced by large agglomerations by ensuring a fairer sharing of the costs of supplying services with regional spill-over benefits.

The municipalities drastically reduced their deficits in 1993 and have since maintained small budget deficits or been in budget balance (1997). Most municipalities were obliged to reduce their deficits quickly because of legal limits on their capacity to borrow. The small decrease in municipalities' estimated deficit in 1998 (see Table 9) is attributable to an increase in fiscal receipts. Growth in spending remained subdued, helped by low inflation and interest rates. Large cities remain in deficit, partly for the reasons outlined above.

The balance of the public social insurance system deteriorated markedly in 1998, falling to a deficit of SF 0.2 billion. This deficit is accounted for by large (and growing) deficits on the first pillar of the retirement income system (*l'assurance-vieillesse et survivants* (AVS), SF 1.4 billion) and on invalidity insurance (*l'assurance-invalidité* (AI), SF 0.7 billion) offset by surpluses on the other social insurance schemes.

### *1999 Budgets*

The Confederation, cantons, municipalities and public social insurance project a combined budget deficit (SFR basis) of SF 7.6 billion (1.9 per cent of GDP) in 1999, slightly higher than the outcome in 1998 but considerably lower than the budgeted deficits for 1998 (see Table 9). Compared with 1998 budgets, expenditures are projected to be stable, reflecting spending restraint and a favourable economic environment. Receipts are projected to grow moderately. The deficit is mainly due to the Confederation (SF 4.9 billion)[33] and to the cantons (SF 2.1 billion). About three quarters of the consolidated deficit of the Confederation, cantons, municipalities and public social insurance is estimated to be structural.[34]

*Confederation*

The 1999 budget approved by Parliament provides for an increase in the deficit (Financial Account basis, excluding Swisscom receipts) of 0.4 percentage point of GDP to 1 per cent (SF 3.9 billion, see Table 8). This deterioration reflects a decline in receipts, which is attributable to a sharp fall in net withholding tax receipts. Claims for refunds of withholding tax are expected to rise sharply this year following the large increase in withholding tax paid last year. As growth in tax revenues tends to be weaker in odd-numbered years, a better indication of the underlying growth in receipts can be obtained by comparing 1999 with 1997. On this basis, total receipts are projected to grow at an annual average rate of 4.4 per cent, largely reflecting the VAT rate increase (by 1 percentage point for the standard rate and 0.3 percentage point for the reduced rate). The rise in VAT is to fund the expected rise in the financing of the first pillar of the retirement income

system. The Confederation's next most important source of revenue, the income tax, is projected to record little growth over the two years to 1999. This reflects the depressed economic conditions prevailing in 1995-96, the tax period to which receipts in 1998 and 1999 relate. Expenditures are projected to contract slightly in 1999 after having been boosted in 1998 by a one-off charge (SF 1.9 billion) associated with changes in the financing of rail infrastructure projects. Abstracting from this charge expenditures are projected to grow by only 1.2 per cent, owing to rigorous measures to restrain outlays that have been legislated in the Federal Law on the *1998 Stabilisation Programme* (see below). These measures include the prolongation of the third percentage point of unemployment social security contributions (saving SF 0.5 billion in loans to the unemployment insurance fund), the blockage of credits (saving SF 0.2 billion), and various other measures (saving SF 0.5 billion). The projected decline in transport expenditures reflects the shifting of rail infrastructure expenditures into a special fund. On a SFR basis, which includes the fund for the large rail infrastructure projects, the budget deficit is projected to rise by 0.6 percentage point of GDP to SF 4.9 billion (1.2 per cent of GDP).

Underlying the budget projections are the assumptions that real economic growth is 2 per cent, inflation is 1 per cent and that the average interest rate on Confederation bonds is 3½ per cent. The economic growth projection now appears to be too optimistic, suggesting that the budget deficit may be a little higher than projected.

*Cantons, communes and social security*

Taken together, the cantons' 1999 budgets provide for a deficit reduction of ⅓ percentage point of GDP to ½ per cent (SF 2.1 billion, see Table 9). Compared with 1998 budgets, budgeted expenditures decline by 1 per cent owing to the reduction in loans to the unemployment insurance system. Abstracting from these loans, spending rises by 2.1 per cent. While expenses on personnel, goods and services and debt servicing grow moderately, transfers are projected to grow more strongly (4 per cent). Receipts are budgeted to rise by 2.8 per cent from 1998 budgets, boosted by a dividend increase (SF 0.6 billion) from the Swiss National Bank. Adjusting for this factor, receipts only expand by 1.7 per cent, the same as the average for the two years since 1997, reflecting weak growth in the income tax base in the period 1996-97.

Communes' budgets in aggregate remain broadly in balance in 1999 (see Table 9). Both receipts and expenditure are projected to grow only modestly. With respect to expenditure, city budgets provide for stable spending on personnel and goods and services and a reduction in investment outlays. While small and medium-sized municipalities maintain a satisfactory budget position, a number of cities continue to have significant budgetary problems, despite a small

budgeted improvement. The financial situation in the cities of Zurich, Lausanne and Geneva remains particularly difficult. The deficits of their operating budgets is projected to amount to around 6 per cent of current expenditures.

Although the receipts from the increase in VAT will help finance the first pillar of the retirement income system (AVS), it is projected to remain in deficit (SF 0.3 billion) in 1999. With the finances of invalidity insurance (AI) projected to deteriorate further, the balance of the public social insurance schemes is budgeted to remain in a small deficit this year.

### *Budget Objective 2001, the 1998 Stabilisation Programme and the 1999 Medium-Term Financial Plan*

*Budget objective 2001* is a transitory constitutional article approved in a referendum in June 1998 that requires the Federal Council and Parliament to bring the Confederation's budget into equilibrium by 2001. This objective will be considered attained if the (Financial Account) deficit is less than 2 per cent of receipts, which would correspond to a deficit of less than SF 1 billion. Intermediate deficit objectives of SF 5 billion in 1999 and 2½ billion in 2000 were also fixed. In the event that the objectives are not met, the Federal Council is required to make budget cuts or propose them to Parliament (should laws need to be changed) to ensure that the objectives are attained within the following two years. If there is a recession, this delay can be extended another two years.

The Federal Council proposed a range of measures in the *1998 Stabilisation Programme* to achieve these objectives. These measures (as adopted by Parliament) reduce expenditures by SF 2 billion (½ per cent of GDP) in 2001 and raise revenues by SF 1 billion. In 2001, the expenditure savings are to be effected as follows:

- Reduction in federal transfers paid to cantons (SF 0.5 billion);
- Various measures to reduce the cost of the old age pension, invalidity benefit, and unemployment benefit (SF 0.4 billion);
- Various other measures, notably to reduce defence expenditures and transfers to the federal railways (CFF), as well as a blockage of credits (SF 0.9 billion).

Most of the increase in revenues comes from the prolongation (until the end of 2003) of the third percentage point of social security contributions for unemployment insurance. It is also intended to strengthen tax enforcement and to eliminate unfair lacunae in the tax system (based on the recommendations of the Behnisch Commission; see Chapter IV below).

The 1999 medium-term financial plan of September 1998, which incorporates the *1998 Stabilisation Programme*, shows the (Financial Account) deficit declining from SF 4 billion (1 per cent of GDP) in 1999 to SF 1 billion (¼ per cent of GDP)

Table 10. **The Financial Plan 2000-2002**

SF millions

| | Budget[1] 1999 | Outlook 2000 | Outlook 2001 | Outlook 2002 | Percentage change 1998-2002 |
|---|---|---|---|---|---|
| **Expenditure** | 45 040 | 45 335 | 45 722 | 46 997 | |
| Percentage change from previous year | –5.4 | 0.7 | 0.9 | 2.8 | –0.3 |
| **Revenues** | 41 012 | 43 460 | 44 975 | 48 099 | |
| Percentage change from previous year | 2.6 | 6.0 | 3.5 | 6.9 | 4.7 |
| **Balance** | –4 028 | –1 875 | –746 | 1 102 | |
| As a percentage of GDP | –1.0 | –0.5 | –0.2 | 0.3 | |
| **Key figures** (as a percentage of GDP) | | | | | |
| Expenditure | 11.5 | 11.2 | 10.8 | 10.7 | |
| Taxes | 9.7 | 9.9 | 9.9 | 10.2 | |
| Net interest | 6.4 | 6.3 | 6.2 | 5.9 | |
| **Economic assumptions** | | | | | |
| GDP growth | | | | | |
| Nominal | 3.0 | 4.0 | 4.0 | 4.0 | |
| Real | 2.0 | 2.0 | 2.0 | 2.0 | |
| **Interest rate on the Confederation's debt** | 3.5 | 4.0 | 4.0 | 4.0 | |

1. Draft budget of the Federal Council of 28 September 1998.
*Source: Rapport du Conseil fédéral sur le plan financier 2000-2002.*

in 2001, thus respecting *Budget objective 2001*. A small surplus is projected in the following year (Table 10). Budget consolidation is to be achieved by growing revenues while (nominal) expenditure will be held constant. As a share of GDP, expenditures are projected to decline by more than revenues rise.

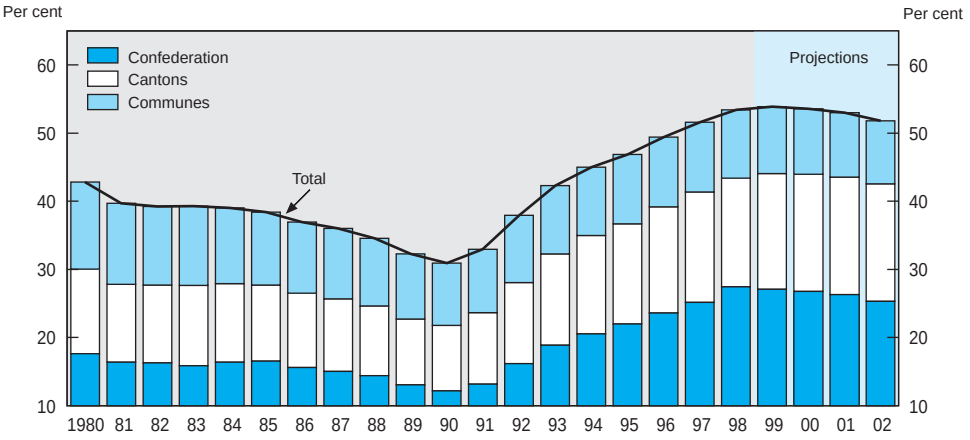
### *Government debt developments*

Gross government debt has increased sharply during the 1990s, from 31 per cent of GDP in 1990 to 53 per cent of GDP in 1998 (Figure18). Much of this rise is accounted for by the federal government's debt, which doubled to 25 per cent of GDP, with the remainder attributable to cantons. Gross government debt is projected to rise further this year to a peak of 54 per cent of GDP but to edge down thereafter, falling to 51 per cent of GDP by 2002. For the most part, this reduction is attributable to the Confederation.

### *Assessment*

Following a pause in budget consolidation in 1995-97, substantial progress was made last year. This was opportune in view of the relatively buoyant

Figure 18. **Gross government debt**
As a per cent of GDP




*Source:* Statistique financière révisée, Administration fédérale des finances.

economic conditions prevailing. The reduction (½ percentage point of GDP) in the deficit for all levels of government was attributable to the federal government and the cantons. Fiscal policy should have a broadly neutral effect on the economy this year but a somewhat contractionary effect in 2000. Budget consolidation is attributable to the federal government, the cantons and the social security system. With solid growth projected next year, this is again a good time for cutting deficits.

The projected reduction in the federal government's budget deficit next year reflects the measures of the 1998 Stabilisation Programme aimed at bringing the budget broadly into balance by 2001. The authorities are rightly focusing on putting federal finances back on to a sound footing while economic conditions remain favourable. The fact that this objective is incorporated in the constitution, along with requirements for expenditure cuts if it is not met, considerably reinforces fiscal discipline.

Consolidation efforts, together with the strengthening of growth and low interest rates, have slowed the build-up in government debt as a percentage of GDP and are projected to result in small declines from next year onwards. If the federal government's longer-term objective of a constitutional amendment requiring the budget to be balanced over the cycle is realised, the large increase in debt as a share of GDP during the 1990s should eventually be reversed.

# III. Progress in structural reform

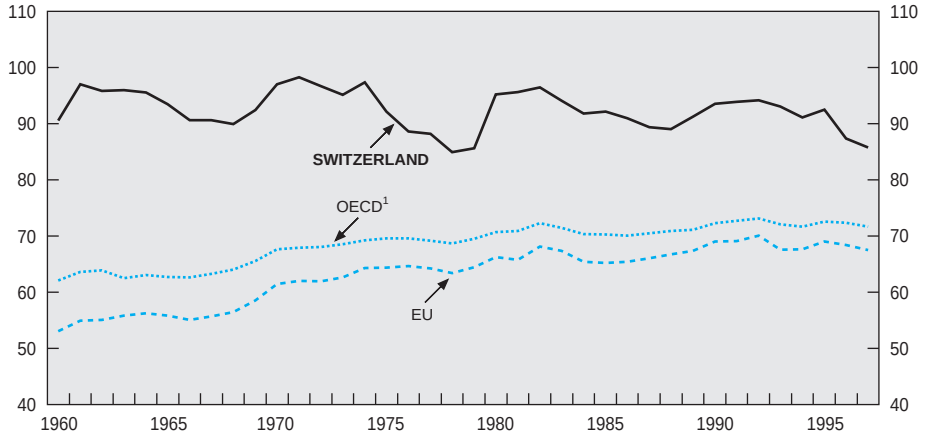
## Introduction

Following eight years of sustained growth and full employment, Switzerland's overall economic performance deteriorated substantially in the 1990s. This is reflected in the accentuation of the trend decline in per capita GDP, which narrows Switzerland's lead relative to other OECD Member countries (Figure 19). Apart from the cyclical under-utilisation of resources, the mediocre growth performance in the 1990s can be attributed to poorer total factor productivity than elsewhere (Figure 20). Although the exact causes of the productivity slowdown are still unclear, two features have marked economic conditions in Switzerland in the 1990s: first, the recession of 1991 was not followed by a quick recovery but by an agonising period of five further years of broad economic stagnation.[35] In consequence, unemployment rose to levels not seen since the 1930s, even though it has remained fairly low by international comparison. Second, in the referendum of 6 December 1992 the Swiss voters decided not to ratify the European Economic Area (EEA) Agreement. This dealt a hard blow to the federal government's policy to step up Switzerland's participation in the process of European integration, which would have implied wide-ranging microeconomic reforms. With the process of structural reform intensifying in many OECD countries, this raised widespread concern that in the absence of profound reform Switzerland's macroeconomic performance risked falling behind other Member countries over the medium term.

In order not to lose the momentum behind the efforts to prepare for the EEA and to minimise the potential damage from non-participation in the Agreement, from 1993 on, the federal government embarked on a comprehensive programme of structural reforms known as "revitalisation programme". This programme included major legislative changes in many fields of economic policy, covering product market competition, domestic market liberalisation, labour market flexibility, education, the infrastructure (electricity, gas, postal services, the railway system, and telecommunications) and agriculture. Moreover, the experience of unprecedented levels of unemployment, accompanied by a rapid rise in long-term unemployment and indications of a marked increase in structural

Figure 19. **GDP per capita**

Current PPPs, United States = 100

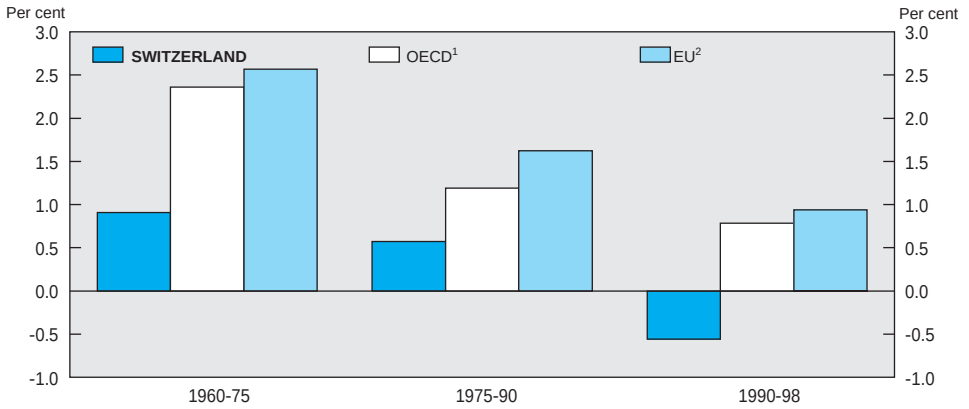



1. Excluding Czech Republic, Hungary, Korea, Poland and Switzerland.

*Source:* OECD, *National Accounts.*

Figure 20. **Total factor productivity in the business sector: an international comparison**

Average annual rate of growth




1. Excludes Czech Republic, Hungary, Iceland, Korea, Luxembourg, Mexico, Poland and Turkey.
2. Excludes Luxembourg.

*Note:* Aggregates were calculated on the basis of 1992 GDP for the business sector expressed in 1992 purchasing power parities.

*Source:* OECD Secretariat.

Table 11. **Implementing structural reform: an overview**

| Proposal | Action | Assessment/recommendations |
|---|---|---|
| **I. Redesign unemployment insurance to put more emphasis on active labour market policies** | Unemployment insurance reform of 1996 became fully operational in 1998. | Make careful evaluation as soon as possible. |
| Regional Placement Offices focus on reducing the duration of unemployment of job losers. | Ongoing training of job counsellors. | Should proceed in line with proposal. |
| Raise number of ALMP slots. | Number of ALMP slots exceeded legal requirements in 1997/98. | Improve quality of ALMP slots. |
| Lower maximum duration of unemployment benefit eligibility. | No action. | |
| Put more emphasis on skill-enhancing active measures. | Waiting for results of careful evaluation of ALMP measures. | Speed up evaluation of ALMP measures. |
| **II. Increase working-time flexibility** | | |
| Remove obstacles to equal treatment of men and women with respect to working time. | New federal labour law approved in referendum of November 1998. | Should be implemented in line with proposal. |
| **III. Improve labour force skills and competences** | | |
| Creation of Fachhochschulen/Hautes écoles spécialisées. | Creation of seven Hautes écoles spécialisées approved by Federal Council in March 1998. | |
| Tackle low literacy of new immigrants. | No action. | Should be taken up in context of active labour market programmes. |
| Make the financing of university education more efficient and fairer. | No action. | |
| Encourage further enrolment of women in scientific and technical education. | No action. | |

Table 11. **Implementing structural reform: an overview** *(cont.)*

| Proposal | Action | Assessment/recommendations |
|---|---|---|
| **IV. Enhance product market competition** | New Cartels Act of 1996. | Carefully monitor effect of new Act on degree of cartelisation and on merger activity. |
| Eliminate unlawful cartels and abuse of dominant positions. | Competition Commission undertakes numerous investigations. | Strengthen resources of Competition Commission. |
| Further liberalise telecommunications sector. | New telecommunication legislation of 1998. | Privatise larger share of Swisscom; require Swisscom to divest its holdings in Cablecom. |
| Deregulate the markets for electricity and gas "third party access" and "unbundling" of production, transport and distribution. | Draft legislation in preparation. | Accelerate deregulation of electricity and gas markets. |
| Open railway and postal services up to competition. | New legislation of 1999 and 1998, respectively. | More liberalisation needed in railway and postal services. |
| Lower agricultural protection. | New law on agriculture de-emphasises price support. | Make bolder moves to cut price support. |
| Align cantonal and communal procurement rules with Domestic Market Act. | In process, guided by Competition Commission; further liberalisation proposed by bilateral agreement with EU. | Should proceed in line with proposal. |
| Introduce more flexible shop-opening hours. | Rejected by several referenda. | Promote acceptance of shop-opening liberalisation. |
| **V. Make economic activity environmentally sustainable** | | |
| Make greater use of economic instruments to achieve environmental objectives. | Environmental tax on oil with high sulphur content came into effect in 1998. Tax on volatile organic compounds will come into effect in 2000. | |
| | The new Law on Agriculture links direct income payments to farmers to the use of integrated and organic farming methods as from January 1999. | |

*Source:* OECD.

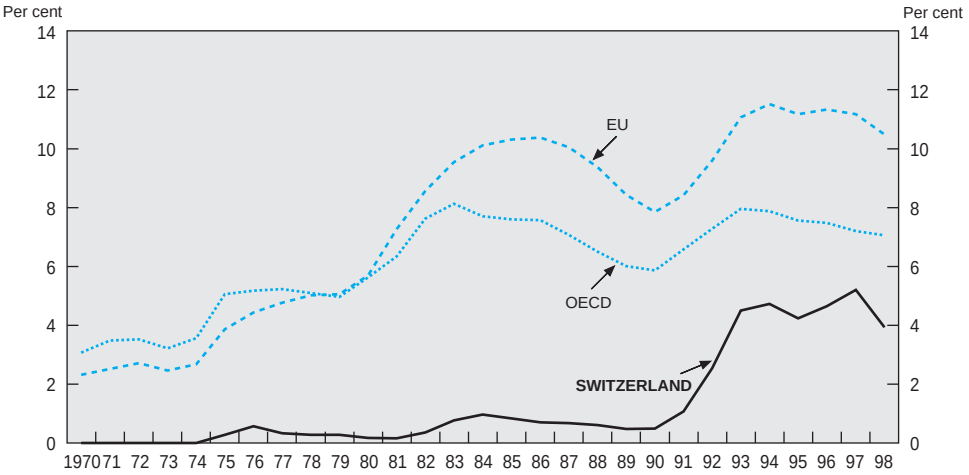
(non-cyclical) unemployment, gave rise to a rather radical reform of the unemployment insurance system, which became effective in two steps. This chapter reviews the progress made in the implementation of these structural reforms since the previous OECD *Economic Survey of Switzerland* and tries to identify the areas where more needs to be done (Table 11).

## Labour market

### *Reform of unemployment insurance*

An earlier review of the Swiss labour market[36] concluded that a number of features are in place which keep the employment-to-population ratio high and unemployment low by international comparison (Figure 21). These features are: a wage-formation system which responds sufficiently flexibly to changing economic conditions at both the macro- and micro-level; a high cyclical flexibility of the labour force; a tax and transfer system and employment protection legislation which do not appear to constitute high barriers to hiring; and the rigorous application of work tests and tight controls of benefit entitlements to assure that work

Figure 21. **International comparison of unemployment rates**
Per cent of civilian labour force




*Source:* OECD Secretariat.

disincentives or moral hazard problems are not aggravated by the provision of generous unemployment benefits.

Nevertheless, the government felt that reform of unemployment insurance was needed because of concern that in a period of rising labour market slack the generous unemployment benefit system[37] risked to lengthen the periods of job-search and to contribute to a build-up of long-term unemployment, which would further worsen the financial imbalance of the unemployment insurance fund. Accordingly, a number of conditions[38] for access to unemployment benefit were tightened in a first reform stage as from 1 January 1996. An important feature of the new unemployment insurance law[39] is that it defines work at a minimum of 70 per cent of previous earnings as "suitable" (80 per cent for job-seekers with dependents). In a second reform step, the new law introduced the "activation principle" as from 1 January 1997, which makes further receipt of insurance benefits conditional on participation in active labour market programmes (ALMPs) at an early stage of unemployment. In line with recommendations of the OECD *Jobs Study*, this shifted the emphasis of unemployment insurance from passive income support to active labour market programmes.

The new law limits payments of passive unemployment benefits to 150 working days for persons aged under 50, to 250 days for those aged between 50 and 60 years and to 400 working days for those aged 60 and over. Once passive unemployment benefits have been exhausted, benefit receipt for up to a total of two years is conditional on participation in active labour market programmes. Participation in temporary employment measures is normally limited to six months and does not generate new benefit entitlements, as was formerly the case. This avoids the so-called "carousel effect", whereby a considerable number of long-term unemployed move between spells of benefit receipt and programme participation. The minimum form of active measure is to undergo job-search counselling.

To implement the new system, the cantons created a nation-wide network of 150 regional placement offices (RPO) in the course of 1997. These are financed by the unemployment insurance fund and employ some 2 900 persons. About two-thirds of the RPO staff are placement officers, who provide job-seekers with intensive counselling, job-search assistance and – when necessary – access to ALMP slots to overcome employment handicaps, with the objective to improve the chances of reintegration of unemployed persons into gainful occupations. In addition, the RPO also have the task of limiting the scope for abuse of the unemployment benefit system.

Because the system became fully operational at the beginning of 1998, a comprehensive evaluation of the efficiency of the RPO will become available only towards the end of 1999. Hence, it is not clear yet to what extent better placement services have improved the job-seekers' employment prospects and have short-

ened the length of unemployment spells. However, a survey commissioned by the OFDE[40] suggests that a majority of clients is satisfied in general with the quality of the services of the RPO, even though the survey has been conducted at a time when not all of the RPO were fully operational. About one-half of the job-seekers who received counselling and placement services from RPO were of the opinion that it had improved their chances of re-integration into gainful employment. Most of the job-seekers' criticism focused on temporary employment programmes which were regarded as not good at all by 16 per cent of the participants. It appears that the RPO helped to enhance transparency in the labour market, as indicated by the sharp annual increase in job vacancies reported to employment offices by enterprises and job brokers: the number doubled over the preceding year in 1997 and increased by about 40 per cent in 1998, much more than what could have been expected from the mild cyclical recovery.

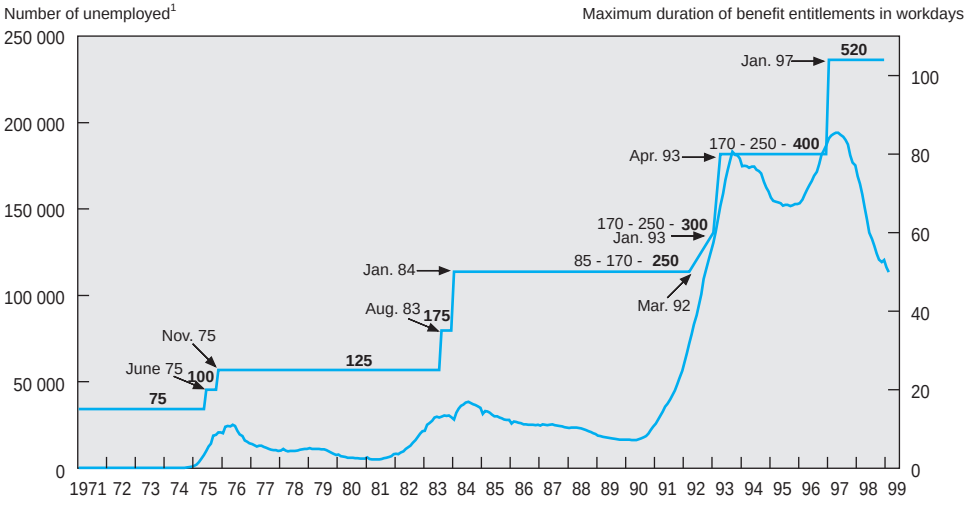
The new legislation required the cantons to raise the number of ALMP slots nation-wide from around 10 000 in 1996 to at least 25 000 in the following years. With an average number of available ALMP places of 28 000 in 1997 and of more than 30 000 in 1998, the target was exceeded by a large margin. This allowed placement of more than half of those eligible for ALMP, which is higher than the initially projected coverage of about one-third for the implementation phase of the new system. However, only about one-fifth of the around 235 000 persons in ALMP programmes in 1998 participated in education, training and other skill-enhancing measures which extended over several months. A large majority of participants only benefited from measures with a duration between one and ten days, which surely is too short a period to substantially raise the skills of the unemployed. Since around 40 per cent of the job-seekers – hence around 87 000 persons in 1998 – are likely to have low or no qualifications at all,[41] it appears that greater emphasis should be put on more thorough skill-enhancing measures.

But increasing the number of ALMP slots which provide a marketable work experience is complicated by the legal requirement that the activities of workers in subsidised temporary employment programmes must not compete with the primary labour market. This condition has been established to secure support for the reform from labour unions and private sector employers, who closely monitor that it is respected. As a consequence of the "no-competition requirement" currently about one-half of the temporary employment slots are supplied by the public sector, where they account for 8 per cent of total employment. The share of ALMP participants in total communal employment even amounts to about 15 per cent. The other half of temporary employment places is mainly with private charitable organisations. Hence, earlier fears proved founded that the "no-competition" rule favours unskilled work in the public sector with little opportunity for training, which is likely to impede the rapid integration of

the unemployed into gainful work. Responding to this problem, there is now a stronger emphasis on training in these programmes.

In the event that no ALMP place is available, an unemployed person continues to be entitled to benefit payments for up to a further 370 days, which implies a total of 520 working days (hence two years) for persons aged under 50, compared with 400 working days under the previous regime. However, the increase in the maximum length of passive unemployment benefit entitlements risks reducing search efforts of the unemployed and lowering the willingness of the job-seekers to accept job offers.[42] This view has been further supported by recent research (Sheldon, 1998) which suggests that the probability of a Swiss unemployed person finding a job tends to decline in proportion to the length of the unemployment spell. Given the observation that the average period of job-search increased over time in line with the growing maximum benefit duration, the repeated rise in the maximum duration of benefit eligibility since the mid-1970s (Figure 22) is likely to have markedly increased the individual risk of becoming long-term unemployed.[43] The research also suggests that the intensity of search efforts tends to rise towards the end of benefit eligibility. From all this it may be concluded that the substantial additional resources spent on extending maximum benefit eligibility would be used more efficiently in active labour

Figure 22. **Unemployment and maximum duration of unemployment benefit entitlements**




1. Seasonally adjusted.
*Source:* Office fédéral du développement économique et de l'emploi.

market measures which focus on raising employability and education and training. Remaining skill deficiencies should be reflected in lower pay so as to align real wages with marginal labour productivity, with intermittent pay compensation from the unemployment insurance temporarily cushioning the possible negative income effects.

### *Labour force skills and competences*

In order to meet the growing demand for workers who are flexible and possess advanced skills, the system of vocational education has been subject to continuous reform since the early 1990s. This system is known as the "dual system" because its main characteristic is the alternation between regular instruction of apprentices at vocational schools for one or two days per week and on-the-job training under the supervision of a master[44] (*Meister*). The main objective of recent reform is to enhance the attractiveness of the dual system by giving bright students better opportunities to realise their potential and to improve their earnings and employment prospects.[45] To achieve this objective, the vocational baccalaureate ("*maturité professionnelle*" (MP) or – in German – "*Berufsmaturität*") was introduced at upper secondary level between 1992 and 1995. Apprentices who prepare for an MP receive considerably more general instruction than traditional apprentices.[46] An MP qualifies its holder both to exercise the relevant profession and to enter a University of Applied Sciences ("*Haute École Spécialisée*[47]" (HES) or "*Fachhochschule*"). Since 1998, the latter is the new third pillar of higher education in Switzerland, the others being the Federal Institutes of Technology and the Universities; it constitutes a core element of the federal government's revitalisation programme. The combination of "*maturité professionnelle*" and "*Haute École Spécialisée*" establishes a new educational pathway, designed to emphasise further the integration of practical and theoretical education and to diversify the Swiss system of higher education.

Legislation creating the legal framework for the establishment of the HES came into force in the autumn of 1996. But due to various delays which in part originated in conflicting regional interests, the federal government gave its final approval for the creation of seven HES only in March 1998. The HES are to be created by regrouping the former Advanced Colleges of Engineering, of Economics and Business Administration and of Art and Design and upgrading them to university level, a process which is planned to be completed by the year 2003 at the latest. The HES will train highly-qualified professionals by offering a combination of practical and theoretical education in the framework of short-term degree programmes. It is expected that the HES integrate applied research, theoretical training and work experience substantially better than the Colleges they replace, while universities will continue to carry out basic research. The HES also aim at a high level of co-operation with the business sector, which should contribute to

improving technical and managerial skills in enterprises. It is planned to raise the number of students of the HES to 21 000 by 2003 – some 50 per cent above the number of students in Advanced Colleges in 1997. Whether this increase will materialise will depend on the success in further raising the enrolment of women in Advanced Colleges/HES. This has more than doubled in absolute terms since 1990 but still accounted for a share of only 17 per cent of the total in 1998. There is also scope for raising the number of foreign students in HES, whose share in the total was about 12 per cent in 1998, compared with a share of foreigners in the population of around one-fifth.

### *Working-time flexibility*

The federal government's project to amend the labour law of 1964 (*loi fédéral sur le travail*) with a view to increasing the flexibility of working time was rejected in the referendum of December 1996. Given the National Council's conviction that reform is desirable – an assessment shared by the OECD – the government worked out a new draft law which took account of the misgivings concerning the previous bill. In particular, the initially planned – but modest – relaxation of the prohibition of employment in the retail sector on Sundays was dropped and the introduction of time bonuses as compensation for regular night work was proposed, while the (non-contested) core parts of the previous bill were basically left unchanged. In this modified form, the reform law was accepted in the referendum of November 1998. It will become effective in 2000.

A major reform element of the new labour law is the cancellation of the ban on night work for women. In principle, this will make the treatment of men and women equal with respect to work time; preferential treatment of women will prevail for pregnant women and nursing mothers. Another important feature of the new law is the extension of "normal" working hours from 8 p.m. to 11 p.m., which will make it easier than hitherto for enterprises to schedule two-shift working. Work between 11 p.m. and 6 a.m. ("night work") will continue to require a permit (except for special categories of enterprises). Employees who regularly work at night time will get a 10 per cent time bonus for night work which they can take as special leave. Permits for overtime work will no longer be needed. The new law reduces the maximum number of overtime hours per employee per year from 220 or 260 hours (depending on the enterprise category) to 140 or 170 hours, respectively, which runs counter to the objective of increasing working time flexibility.

With the achievement of a more flexible labour law, further efforts should be made to relax the rigid and restrictive rules for shop opening hours,[48] including the prohibition of employment in the retail sector on Sundays. This not only would add to labour market flexibility but also to enhanced competition in the retail sector. Although reform is complicated by the fact that these issues fall in

the jurisdiction of cantons and that surveys suggest conflicting interests of "insiders" and "outsiders",[49] the potential benefits in terms of employment opportunities are large. This can be inferred from the impressive numbers of jobs created in the retail trade sector in the United States over the past decade and first significant signs of success of the rather modest reform of shop opening hours in Germany in 1996.[50] Less rigid opening hours would be particularly helpful for retail traders in Switzerland's border regions, who compete with the more flexible regulations in neighbouring countries such as France, Germany and Italy.

## Product market competition

### *The new Competition Law*

*Cartels and dominant positions*

Up to the mid-1980s, competition legislation regarded cartels or other agreements which create obstacles to competition as legal as long as there was no "abuse" in the exercise of their economic power. As a consequence, and given the immense practical problems to prove the abuse of economic power, cartels were widespread in Switzerland. The revision of the law in 1985 allowed some spectacular successes of the Cartel Commission to dismantle harmful cartels, for example in banking and insurance. In view of the economy-wide negative effects of restrictions to competition, which had become most visible in the high domestic price level (for a standard basket of goods and services) compared with most other OECD countries, the federal government introduced new legislation in 1996 (Federal Act on cartels and other restrictions to competition and preventive merger control) in the context of its revitalisation programme. Outright prohibition of cartels as in other countries was not considered as an option since it would have required a time-consuming amendment to the Constitution, with a risk of failing in the compulsory referendum. Instead, the new Cartels Act contains a presumption that horizontal agreements setting prices, production volume or territorial distribution have damaging effects on competition; accordingly the Act prohibits such agreements. The reformed Competition Law also includes a (non-exhaustive) list of potential abuses of dominant positions and prohibits mergers if they would eliminate effective competition. The new legislation came into force on 1 July 1996, but since cartel agreements were given six months to be brought into compliance with the new legislation, the Act became fully operational only on 1 January 1997.

Given that cartels were a common feature in many production and distribution activities at the national and local level, enforcement of the new law is a relatively slow and continuing process. To accelerate the elimination of unlawful cartels and unlawful practices of firms which dominate their market, the new

Competition Commission, the successor of the Cartel Commission, embarked on a substantial number of preliminary investigations in 1997 and 1998. In most cases either no evidence of unlawful restraint of competition was uncovered, or firms could be induced to eliminate existing restraint on competition, which kept the number of full investigations rather low. Of the full investigations of cartel agreements and dominant positions carried out so far, the Commission arrived at a formal conclusion of unlawfulness only in two cases. A significant intervention of the Competition Commission concerned the telecommunications market which is dominated by Swisscom. The Commission uncovered that Swisscom charged higher sales prices to those retailers who also offer telecommunication equipment and services of firms competing with Swisscom than to retailers who exclusively supply Swisscom products. The Commission concluded that this was an abuse of a dominant position and therefore unlawful, which eventually induced Swisscom to end this practice.

*Mergers and acquisitions*

The new Competition Law of 1996 stipulates that the Competition Commission must be notified in advance of planned mergers when the turnover of the enterprises concerned exceeds certain limits.[51] If a prior investigation has discovered a dominant position of a firm in a particular market, this firm has to notify any further merger or acquisition it intends to undertake. The Commission may prohibit the concentration or allow it subject to conditions or obligations if its review ascertains that the concentration would create or reinforce a dominant position such that it might eliminate effective competition, and that it would not cause an improvement of competitive conditions in another market which would outweigh the drawbacks of that dominant position. The creation or strengthening of a dominant position is also the decisive criterion of merger control legislation in many other OECD countries. The number of notified planned mergers were 23 in 1997 and 26 in 1998, many more than the 10 to 15 operations per year expected during the legislative procedure. In no case was the planned merger prohibited, but in four cases the Competition Commission set conditions for the mergers to be accepted.

The biggest merger case in terms of company size and potential implications for competition in the Swiss market for financial services was the merger of Schweizerische Bankgesellschaft (UBS) and Schweizerischer Bankverein (SBV) to the new UBS, which became the second largest bank in the world in terms of its balance sheet. The merger plans were notified to the Competition Commission in early 1998. In its subsequent investigation,[52] the Commission focused on the market for mortgage credit and for loans to small and medium-sized enterprises (SMEs) up to SF 2 million. The Commission concluded that the new UBS alone would not achieve a dominant position sufficient to eliminate effective competition in these markets. It argued, however, that the merger would create a situation

of "collective dominance" in the market for loans to SMEs as in almost any regional market the three largest banks together – the new UBS, Credit Suisse and the respective (publicly-owned) cantonal bank – would hold market shares of the order of 85 to 90 per cent. The Commission argued that a number of features were in place which were conducive to collusion in the case of an unconditional merger.[53] Accordingly, the Commission decided to authorise the merger contingent upon a number of conditions, the most important being that the new UBS:

- Had to sell about 25 branch offices, if possible to a single buyer engaged in retail banking in Switzerland;
- Had to sell two subsidiary banks in the two most heavily concentrated regional markets;
- Had to pledge not to drop out of several joint ventures with smaller banks;
- Had to leave unchanged until the year 2004 the conditions (especially risk classification) of the loans to those SMEs which had been customers of both banks before the merger.

The new cartels Act is clearly a great step forward in the direction of enhanced product market competition. However, since the law only prohibits the elimination of effective competition but not cartels as such, its application is a relatively slow process, hinging largely upon the investigations of the Competition Commission. Providing the Commission with more human resources should help speeding up the enforcement of the law.

### *The Domestic Market Act*

Together with the revised competition legislation and the law on the removal of technical barriers to trade,[54] the Domestic Market Act came into force on 1 July 1996. The Act lays down the basic principles for creating a level playing field throughout the Confederation. Its purpose is to eliminate remaining restrictions on competition arising from cantonal and communal regulations, in particular with regard to public procurement and the recognition of professional qualifications. The Act formally confirms the right of all persons in Switzerland to offer goods and services, including labour, throughout the entire country. To make this right effective, mutual recognition of laws and qualifications can be enforced by going to court. Along the lines of the principle of mutual recognition established by the "*Cassis de Dijon*" ruling by the European Court of Justice, market access is – in principle – regulated by the laws in force in the supplier's (cantonal or communal) home market. The Competition Commission surveys the compliance of the federal government, cantons, communes and other bodies with this Act. Since it required the alignment of cantonal and communal law with the new federal legislation, the Act provided for a transition period of two years, so that it became fully operational only on 1 July 1998.

Since mid-1996, a number of cantonal and communal laws, rules and practices have been brought into line with the new Act, guided by recommendations of the Competition Commission. In particular, substantial cantonal and communal legislation concerning public procurement has been enacted in response to the non-discrimination principles of the Domestic Market Act. Nevertheless, alignment to the Act has not been completed yet. Also, recent legal conflicts with respect to the mutual recognition of professional qualifications bear witness to frictions in the implementation of the principles of free market access in practice.[55] However, an important advantage of the Act lies in the legal means it gives to individuals to challenge before the Federal Court decisions of lower levels of government which violate the principles of free market access.

### *Infrastructure*

#### *Telecommunication*

As another cornerstone of the federal government's revitalisation programme, new legislation[56] came into effect on 1 January 1998, which ended the public monopoly on telephone networks and opened the telecommunications market up to competition. As part of the government's strategy, the telecommunications business has been separated from the former PTT (postal and telecommunications services) and operates as an independent enterprise under the name of Swisscom since the beginning of 1998. It is organised as a joint-stock company under a special telecommunications law, which allows partial privatisation but stipulates that the majority stake in the capital and the majority of the voting rights have to remain with the federal government. Profits from the telecommunications business are no longer accruing to the Confederation. Rather, Swisscom distributes dividends to its shareholders and is also liable to taxation. During the first five years of its existence, Swisscom is required to cover the whole country with a basic supply of telephone and data services. Thereafter, this public service will be put up for tender as a regional or nation-wide licence. As a first step of partial privatisation, about 35 per cent of the Swisscom shares have been sold to the public in October 1998.

Since the federal government's decision to liberalise the telecommunications sector, a number of new operators entered the market and various commercial alliances have been formed to compete with Swisscom, which held some 80 per cent of the market in 1998. New companies now offer national and international calls and have extended the diversity of the product and service range, which puts Swisscom under increasing pressure to bring down the prices for telecommunications services, which had gone up substantially during the first half of the 1990s (Table 12). Thanks to competition in the sector, telecommunication prices for households and businesses are now below the OECD average on a PPP

Table 12. **Business telecommunications charges**

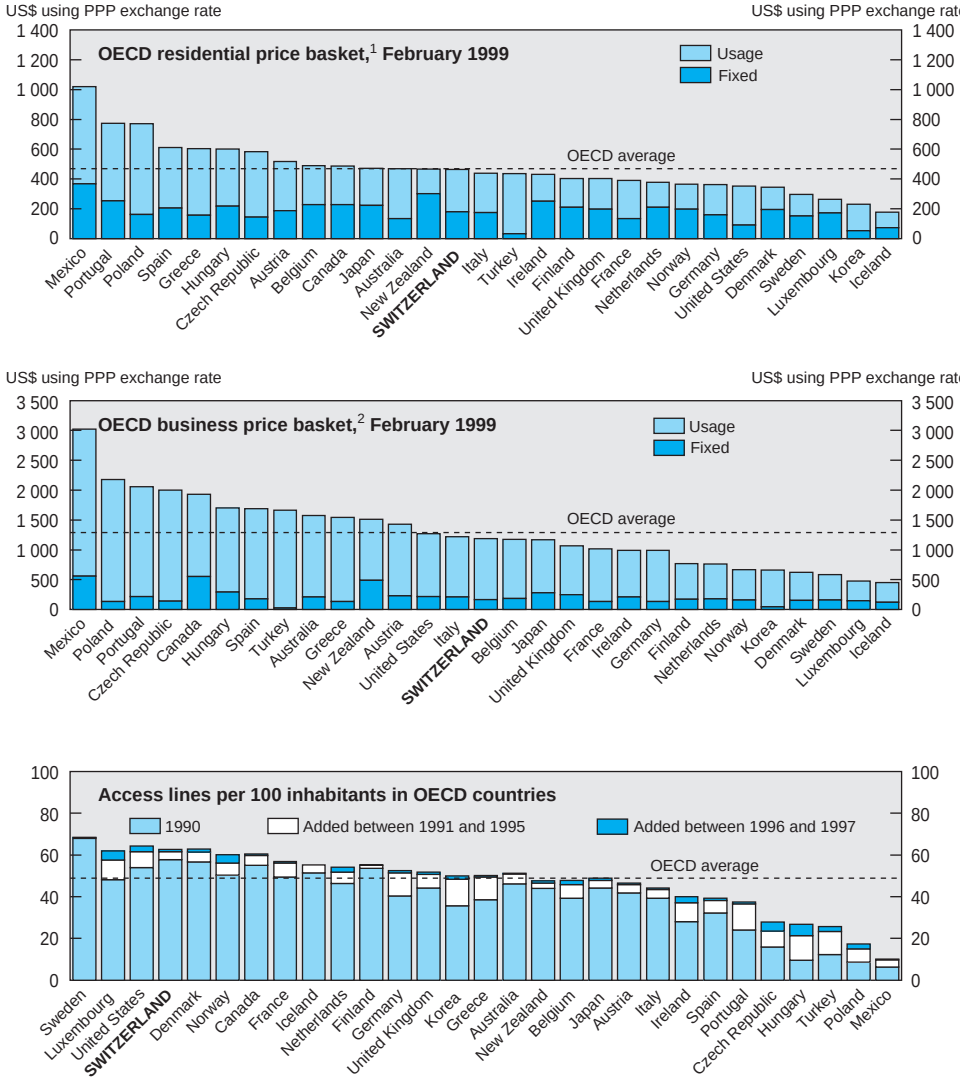
Index, basket of services,[1] 1990 = 100

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[2] | 1997[2] | 1998[2] | Nov. 98 |
|---|---|---|---|---|---|---|---|---|---|
| **Switzerland** | **103.13** | **115.64** | **116.86** | **119.41** | **132.46** | **126.80** | **114.99** | **116.38** | **110.02** |
| Australia (Telstra) | 109.28 | 112.29 | 111.85 | 108.49 | 106.04 | 107.15 | 100.79 | 111.74 | 98.16 |
| Austria | 100.05 | 91.39 | 88.20 | 86.84 | 86.93 | 86.26 | 72.51 | 65.54 | 65.54 |
| Belgium | 98.72 | 98.98 | 98.99 | 100.98 | 107.90 | 110.44 | 93.87 | 111.35 | 111.44 |
| Canada (Bell Canada) | 95.56 | 93.95 | 96.17 | 98.41 | 97.92 | 97.88 | 125.72 | 133.64 | 139.37 |
| Denmark | 108.18 | 107.04 | 108.09 | 107.27 | 103.37 | 112.47 | 121.43 | 119.41 | 114.46 |
| Finland (HTC) | 115.15 | 106.04 | 105.23 | 87.83 | 93.05 | 84.44 | 108.82 | 115.36 | 122.27 |
| France | 101.54 | 102.46 | 102.30 | 96.59 | 86.75 | 69.21 | 70.23 | 83.97 | 66.09 |
| Germany | 85.96 | 84.44 | 82.86 | 82.47 | 83.65 | 67.73 | 66.25 | 68.56 | 68.29 |
| Greece | 106.47 | 97.35 | 92.31 | 113.18 | 121.13 | 119.61 | 123.68 | 109.50 | 109.50 |
| Iceland | 85.24 | 85.71 | 91.11 | 92.30 | 98.22 | 92.42 | 97.86 | 100.28 | 99.38 |
| Ireland | 104.09 | 84.95 | 86.74 | 83.03 | 83.25 | 77.41 | 72.44 | 61.44 | 53.55 |
| Italy | 108.64 | 106.09 | 103.41 | 102.51 | 95.46 | 94.54 | 97.64 | 73.36 | 73.36 |
| Japan (NTT) | 91.25 | 91.73 | 93.17 | 79.15 | 82.58 | 86.70 | 94.63 | 98.58 | 97.00 |
| Netherlands | 100.24 | 104.23 | 101.85 | 109.56 | 111.62 | 102.70 | 97.63 | 127.18 | 123.90 |
| New Zealand (TCNZ) | 113.05 | 98.24 | 90.86 | 88.91 | 90.65 | 83.64 | 91.26 | 101.34 | 101.44 |
| Norway | 68.26 | 64.45 | 65.77 | 53.89 | 52.42 | 47.65 | 54.96 | 52.52 | 52.79 |
| Portugal (TP/TLP) | 102.85 | 93.29 | 91.25 | 92.55 | 107.73 | 97.83 | 102.89 | 101.24 | 85.47 |
| Spain | 116.45 | 113.33 | 115.36 | 114.54 | 135.27 | 105.81 | 110.44 | 106.90 | 103.43 |
| Sweden (Telia) | 114.79 | 102.68 | 101.10 | 86.71 | 87.76 | 89.37 | 91.05 | 90.12 | 86.53 |
| Turkey | 133.11 | 122.83 | 109.72 | 107.14 | 89.67 | 51.12 | 94.05 | 82.44 | 109.46 |
| United Kingdom (BT) | 98.63 | 99.03 | 95.73 | 88.33 | 82.97 | 85.54 | 84.85 | 79.47 | 79.46 |
| US (AT&T/Nynex) | 100.00 | 101.00 | 95.24 | 88.68 | 88.68 | 89.18 | 73.80 | 80.32 | 79.46 |
| OECD average[3] | 102.64 | 99.01 | 97.57 | 95.16 | 96.76 | 90.69 | 93.99 | 95.25 | 93.49 |

1. Data are from business tariff basket, and refer to 1 January if not otherwise indicated. The basket is expressed in the form of an index that makes 1990 = 100. The weighted average is calculated with 1992 telecommunication mainlines.
2. 1st February.
3. Simple average.

*Source:* OECD Secretariat.

Figure 23. **An international comparison of performance indicators in telecommunications**




1. Including VAT. The usage charge includes a basket of 1 274 calls.
2. Excluding VAT. The usage charge includes a basket of 4 426 calls.
For a full description of the tariff comparison methodology for these baskets, see OECD, ICCP Series No. 22: "Performance Indicators for Public Telecommunications Operators".

*Source:* OECD, *Communications Outlook* 1999.

basis, although international comparison suggests that there is still considerable scope for further price cuts (Figure 23).

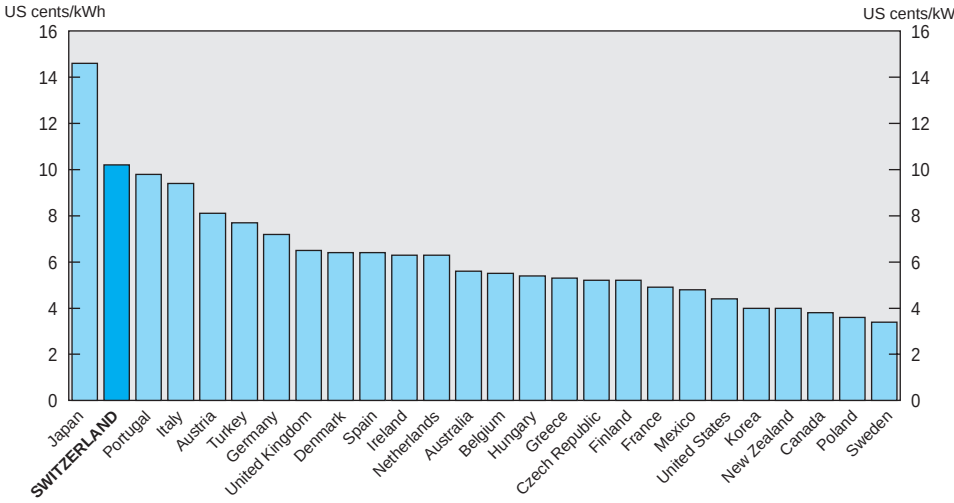
Existing barriers to market entry are a major obstacle to more rapid price cuts. The most important among them seems to be the dominant position of Swisscom – especially for local calls – which is partly due to the fact that Swisscom owns the local telephone network and that it imposes high "interconnection charges" on other operators for the use of the network. According to an investigation commissioned by the Federal Office for Communication,[57] these interconnection charges were among the highest in the world in early 1998, about five times higher than those in the United States and twice as high as in France. Accordingly, the Federal Communications Commission ordered Swisscom to cut interconnection charges – including fees for the "final mile" to subscribers – three times in 1998. Nevertheless, with interconnection charges still remaining considerably higher than EU benchmarks, competitors claim that they are above the costs which the law allows a provider in a dominant position to charge. However, on 1 January 2000, the system of calculating interconnection charges will be aligned with EU legislation, which bases charges on long-run incremental costs.

Competition is also likely to be enhanced by new suppliers who use alternative fixed line networks, such as the glass fibre networks laid along Swiss railway lines, electricity supply, or the cable television network, to which currently 82 per cent of households have access; the latter promises to become particularly competitive as it disposes of the final mile to subscribers and can be used as an alternative local telephone network. However, the cable television network is largely provided by Cablecom, of which Swisscom has a 32 per cent stake, acquired by the former PTT telecom. The Competition Commission undertook an inquiry which concluded that this stake is a major obstacle to present and potential competition in urban telephone networks, and, as such, is in contradiction to the liberalisation of telecommunications markets. Accordingly, the Competition Commission recommended that Swisscom divests its holdings in Cablecom in a manner that would be neutral from a competition standpoint, a view which the federal government as supervisory authority has not shared. Given the strategic importance of inexpensive but high quality telecommunications services for the whole economy, the Swiss authorities should dismantle existing barriers to entry to the telecommunications market.

*Electricity*

The Swiss electricity market is characterised by a complex interaction of many local distributors and a few vertically integrated, large regional suppliers, all operating under numerous legal prescriptions and affected by conflicting political goals. The cantons are primarily involved in power generation, transport and preliminary distribution, either by means of companies of their own or through

Figure 24. **Electricity prices in industry**
1997[1]




1. Average of year; 1994 for Canada.
*Source:* OECD, International Energy Agency (IEA).

participation in inter-cantonal public or mixed enterprises. The communes are primarily active in the distribution to end-users, usually through non-autonomous institutions, some of which also generate power. These features result in electricity prices for industry remaining among the highest in the OECD (Figure 24) and the highest in Europe, although electricity prices are moderate for households.

In order to reap potential efficiency gains, the so-called Cattin Report[58] of 1995 recommended opening up the market for electricity (and gas), which involves far-reaching reforms, including "third-party access" and the separation of production, transport and distribution ("unbundling"). On the basis of the results of a public consultation process in early 1998, the federal government prepared a draft Electricity Market Law, which envisages gradual liberalisation of the electricity market through contractually regulated third party access during a transition period of nine years. Another objective of the draft law is to make the sector compatible with the internal electricity market of the EU. Liberalisation is planned to begin with large power consumers (those who consume more than MWh 20 per year and account for some 12 per cent of the market) and ultimately to give all consumers a choice of supplier. For the operation of the transmission

network, it is proposed to establish a nation-wide network company. So far, it is unclear at what pace the market is to be liberalised and to what extent sunk costs will be compensated. Future work on the draft Electricity Market Law will be importantly influenced by the outcome on the parliamentary discussion during 1999 on an energy tax on non-renewable energy sources. In case of the introduction of such a tax, it is planned to use parts of the revenues to compensate investors for losses due to liberalisation. Given the various contentious issues it seems unlikely that a new Electricity Market Law could come into effect before 2001.

*Gas*

The federal government envisages reform of the gas market broadly in line with the liberalisation of telecommunication and electricity markets, with third party access being the key instrument to enhance competitive pressures. However, a somewhat lower priority has been given to the opening of the gas market, which is in line with the EU timetable. There is also less need for liberalising the gas market as Swiss gas prices are not much higher than those elsewhere, because Switzerland has no own gas resources but depends entirely on natural gas imports. Also, as heating energy, gas already faces stiff competition, especially from heating oil. Moreover, with transport and distribution carried out by different companies, unbundling is realised. In response to a request by the federal government, the Swiss Gas Industry Association is currently working on a report on the impact of the EU directive on gas market liberalisation of August 1998, which will be submitted to the Swiss authorities in the summer of 1999.

*The Post Office*

The PTT reforms came into effect on 1 January 1998, which dissolved the old PTT and separated the Post Office and Telecom business into two autonomous enterprises. This ended the traditional practice of cross-subsidisation whereby the surplus in the telecommunication business financed the recurring deficits in postal activities. The main purpose of the reform of the Post Office is to separate the economic policy from the entrepreneurial responsibilities. To this end, the Post Office became an independent institution, which gives it more entrepreneurial freedom than hitherto; it is liable to taxation and has to transfer part of its profits to the federal government. It remains, however, a fully state-owned corporation, entirely controlled by the Confederation, which sets the strategic objectives of the Post Office over a period of four years. The long-term goal of the reform is the gradual liberalisation of most of the postal market with the monopoly for the transport of addressed letters remaining. With respect to products and services which are offered in competition with other suppliers in the market, the Post Office will largely be able to determine price and performance

itself. The only condition is the exclusion of cross-subsidies from the areas covered by the monopoly, which would lead to unfair competition.

*The railway system*

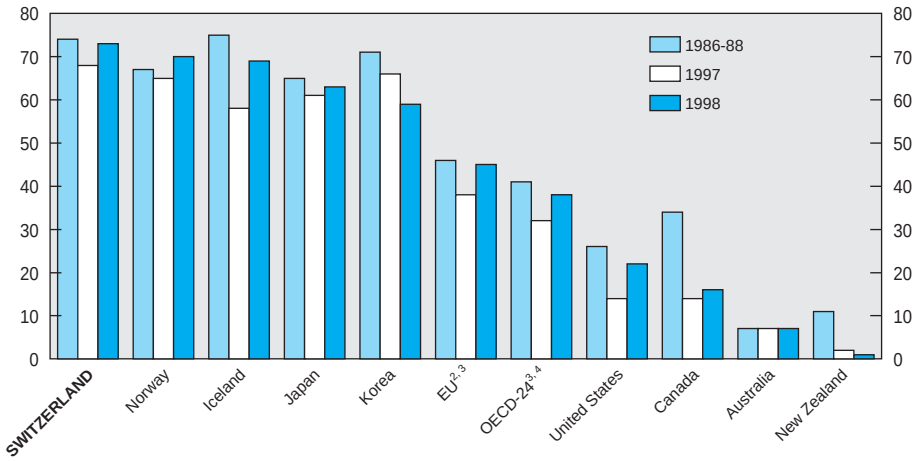
Reform of the railway system came into effect on 1 January 1999, which transformed the Swiss Federal Railways into a state-owned company with greater freedom to act as a business. The Confederation now confines itself to stating political and financial objectives, while the federal railways themselves are responsible for the operative management. Railway operations will be split into two parts, one part managing the infrastructure and the other the traffic. The railway network will be fully open to other companies against payment for merchandise transport and for non-regular transport of passengers, which will introduce elements of competition among service providers. However, for the time being, the federal railways and other companies owning their infrastructure retain the exclusive right of using their network for regular transport of persons. Regional authorities which have become financially responsible for all regional transport started to put scheduled services up for tender. This indicates the likely direction of future reforms.

### *Agriculture*

With the system of guaranteed producer prices resulting in excess production of a number of farm products, widening price differentials with abroad, rapidly rising federal government expenditures on price support – in particular for dairy products – and growing signs of environmental damage due to intensive farming, the Federal Council's Seventh Report on Agriculture of 1992 initiated a major reorientation of Swiss agricultural policy. A first phase of reform came into effect in early 1993, which aimed at shifting farm assistance gradually away from price support to a generalised system of direct transfers to farmers; formerly, direct payments were used only to a very limited extent, mainly to supplement farm incomes in mountain regions. The reform introduced two new categories of direct income payments: the first one compensates farmers for earnings shortfalls from cuts in guaranteed producer prices, while the second type of direct payments is linked to ecological forms of production or the provision of environmental services.

In spite of the reform, the level of agricultural protection in Switzerland has remained high. This is mirrored in producer support estimates[59] (PSE) as a percentage of the value of farm production which were the highest in the OECD area in recent years (Figure 25): at 73 per cent in 1998, the Swiss PSE was about double the OECD average. In line with the intentions of the reform, the rapidly increasing direct income payments to farmers since 1993 now make up for almost one-third of the PSE. Price support has fallen in recent years and now accounts for

Figure 25. **An international comparison of producer support estimates**[1]
As a percentage of value of farm production



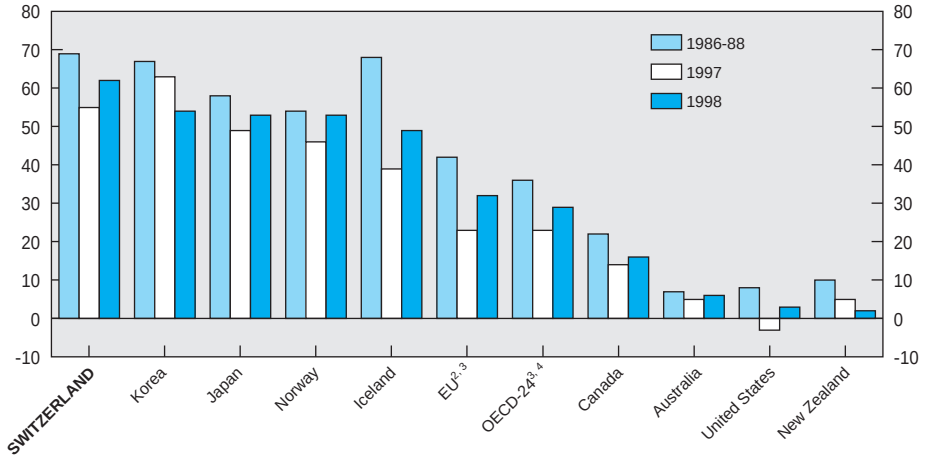

1. For detailed explanations, see OECD, *Agricultural Policies in OECD Countries,* 1999.
2. EU-12 for 1986-94, EU-15 from 1995. EU includes ex-GDR from 1990.
3. Austria, Finland and Sweden are included in the OECD totals for all years, and in the EU for 1995.
4. OECD-24 excludes Czech Republic (1995), Hungary (1996), Korea (1996), Mexico (1994) and Poland (1996).

*Source:* OECD, *Agricultural Policies in OECD Countries,* 1999.

just over 60 per cent of the PSE, which not only reflects the switch in policy instruments but also a rise in world food prices. Reflecting the increased reliance on direct payments, the consumer support estimate[60] (CSE) has decreased significantly until 1997, but bounced back in 1998 to the highest level in the OECD (Figure 26). Accordingly, domestic prices of agricultural products are still almost two and a half times as high as world prices and about double the price level in the EU. High producer prices are likely to have contributed to the declining share of the Swiss food processing industry in the domestic market as consumers increasingly buy cheaper imported foodstuff or purchase directly in neighbouring countries.

In view of the broadly unchanged high level of farm support and the concomitant severe burden on the federal budget, the Federal Council 's report of 1996 "Agricultural Policy 2002" proposed a second phase of agricultural reform. The reform required the introduction of a new article into the Constitution which was approved by a large majority of the Swiss people and the cantons in the referendum of 9 June 1996. Based on this constitutional amendment, the Federal Council proposed a new Law on Agriculture which passed the Parliament in

Figure 26. **An international comparison of consumer support estimates**[1]
As a percentage of agricultural consumption valued at the farm gate




1. For detailed explanations, see OECD, *Agricultural Policies in OECD Countries,* 1999.
2. EU-12 for 1986-94, EU-15 from 1995. EU includes ex-GDR from 1990.
3. Austria, Finland and Sweden are included in the OECD totals for all years, and in the EU for 1995.
4. OECD-24 excludes Czech Republic (1995), Hungary (1996), Korea (1996), Mexico (1994) and Poland (1996).

*Source:* OECD, *Agricultural Policies in OECD Countries,* 1999.

April 1998 and came into force[61] on 1 January 1999. The new law seeks to improve the competitiveness of the farm sector and to promote more energetically the objective of environmental "sustainability" in agriculture. These objectives are to be achieved by increased reliance on market mechanisms in determining food prices and the use of a more comprehensive system of direct income payments, linked more closely to the provision of ecological services and largely de-coupled from farm output.[62]

As probably the most important effect, the new law will substantially reduce direct government intervention in the market for agricultural products by phasing out all price guarantees. For example, since 1 May 1999, producer prices and processing margins for dairy products – by far the most important product group in Swiss agriculture – are no longer guaranteed by the government; the previously guaranteed price for milk has been replaced by a "target price" at a lower level. The subsequent income losses of farmers will be partly offset by direct payments. However, the price supplement for milk used in cheese production will be increased and paid directly to farmers. Similarly, the price guarantees for other products will also be phased out; for example, for bread cereals these

guarantees will be eliminated by 2001. As a medium-term objective of the reform, it is planned to reduce by one-half the excess of Swiss agricultural prices over EU prices by 2002. However, this target does not seem very ambitious, in view of the relatively high EU food price level.

The reform also puts more emphasis on the protection of the environment in agricultural production: during a transition period of five years, direct income payments will progressively be linked to ecological production methods. These are defined by the new law on agriculture and refer to the use of fertilisers and pesticides, soil conservation, and other criteria for "organic" farming. After the transition period, direct income payments will be made only if integrated and organic farming methods apply to the entire farm ("whole-farm approach"). However, due to the increased use of payments linked to ecological criteria, the reform fails to reduce the subsidies for the agricultural sector in the nearer future, but stabilises them at about SF 3½ billion per annum during the period from 2000 to 2003, which implies a continuing heavy burden on the taxpayer. Reflecting the goal of the new law, expenditures for price support will decline from currently SF 1 200 million to SF 800 million by 2003. Altogether, the latest reform is understood by the Swiss authorities as an additional but by no means final step in the direction of a more market oriented agricultural sector. Further reforms are to be expected in the context of the process of European integration.

## European integration

After the rejection of the agreement on the European Economic Area (EEA) in the referendum of 6 December 1992, the federal government decided to embark on negotiations with the EU with the objective of minimising the economic cost of non-participation in the EEA agreement. The negotiations began in December 1994 and covered seven key areas: land transport, air transport, free movement of labour, agriculture, research and development, public procurement, and technical trade barriers. The negotiations proved particularly contentious on the issues of free movement of persons and land transportation; major stumbling blocks with respect to the latter were the maximum weight of trucks allowed to transit Switzerland, the level of charges imposed on them and the use of railways in diverting freight traffic from road to rail. After four years of negotiations, a final agreement was concluded on 11 December 1998. The seven accords are designed as a package in the sense that if one of them ceased to apply then the whole package would be null and void and could come into force again only through new negotiations. Parliamentary discussions of the treaty are likely to terminate before the general election in October 1999. On the assumption that it passes a possible referendum, and given that all EU member countries ratify it, the treaty may come into force only in 2001.

The following are the main features of the seven bilateral accords:

- *Land transport*: as a central element of the agreement, Switzerland will raise its 28-tonne weight limit for lorries entering the country in steps to 40 tonnes in 2005 and will build two transalpine rail tunnels, in order to make rail transport more competitive. Charges will be levied on domestic and foreign heavy goods vehicles on the basis of weight, distance travelled and emission levels, which – in combination with improved rail infrastructure – is meant to induce a shift of transit traffic from road to rail. A maximum charge for the reference distance from Basle to Chiasso has been set at euro 180 or SF 297 in 2005. It will be raised to euro 200 once the first transalpine rail tunnel will be opened (in 2006 or 2007) but no later than 1 January 2008. Switzerland retains the right to ban trucks travelling during the night and to apply safeguard measures (*e.g.* raise truck charges) under certain conditions.
- *Air transport*: the air transport accord will give Swiss carriers the same rights as those of the EU within two years, enabling them to transport passengers between cities in different EU countries and pick up and drop off passengers within the EU on their way to a final destination.[63]
- *Free movement of labour:* this part of the package will facilitate the free movement across the Swiss and EU labour markets and will introduce the mutual recognition of professional qualifications and social security entitlements. The Swiss quota system for work permits granted to EU and EFTA citizens will gradually be phased out over a five-year period. However, already after the first two years, the Swiss authorities will abolish the preferential treatment of indigenous workers; at the same time, Swiss citizens will be granted full freedom of movement in the EU. During the following seven years, the Swiss authorities may monitor whether the accord leads to excessive immigration, in which case they are allowed to reinstate the old quota system for work permits. In case Switzerland decides to continue the pact, freedom of movement for EU and EFTA citizens will become permanent after this period of seven years.
- *Agriculture:* access to the EU and Swiss markets for certain agricultural goods will be facilitated through lower tariffs and the removal of other trade barriers.
- *Research and development:* Switzerland will participate under the same conditions as any EU country in EU research programmes and will allow the participation of EU researchers in Swiss programmes.
- *Public procurement:* EU companies will be granted non-discriminatory access to procurement by communes,[64] whereby Switzerland will attain the same level of liberalisation as the EU. The accord will also apply to licensed private companies supplying water, electricity, urban and regional

transport, railway and telecommunications services. This will terminate discriminatory practices such as giving preference to local bidders.

- *Technical trade barriers:* this agreement facilitates the trade of industrial products through the mutual recognition of tests, certificates and other technical rules and procedures, which will reduce compliance costs in particular in the high-technology segment of Swiss exports. However, since the agreement will apply only to technical rules and procedures, it is less comprehensive than the EEA agreement; in particular, it will not adopt the mutual recognition principle ("*Cassis de Dijon* principle").

Ratification of the bilateral agreement with the EU would help to keep the direct welfare losses from staying outside the EEA low. It would also give a fresh impetus to more structural reforms in line with policies carried out or planned in the EU. This would undoubtedly lead to better resource allocation, which in turn would entail stronger gains in productivity.

# IV. Tax reform

Switzerland's tax system has evolved in the context of a highly decentralised federal structure with extensive use of direct democracy. These features have reduced the scope for governments to provide services that are not valued by taxpayers. As a share of GDP, total tax revenue is moderate by OECD standards and low compared with most EU countries. However, these features of Switzerland's federal structure have also hampered tax reform, as this tends to redistribute revenues amongst governments. In addition, even when governments agree on reform, it can be difficult to obtain approval in a referendum. For example, the federal government first sought popular approval for the introduction of a VAT in 1977 but only gained approval in the fourth referendum held in 1995. These features, not the inherent qualities of the tax system, account for the fact that Switzerland was not amongst the OECD countries to implement comprehensive tax reforms in recent years.

An unusual feature of the Swiss Constitution is that the federal government's power to raise income taxes and VAT is time-limited. These taxes will expire by the end of 2006 unless authority to raise them is renewed in a referendum held before then. This provides an opportunity to reform the tax system. After outlining the forces shaping the tax system and its main features, this chapter discusses reforms that could reduce its economic costs (excess burden,[65] including administration and compliance costs) and/or enhance equity. Inevitably, the reforms proposed are conditioned by the decentralised federal structure that so shapes existing arrangements.

## Forces shaping tax policy

### *A highly decentralised federal structure*

The Swiss Confederation is more decentralised than any other OECD country.[66] Under the Constitution that created the Swiss Confederation in 1848, cantons retain all responsibilities unless these are explicitly ceded to the central government.[67] Insofar as tax powers are concerned, they ceded the right to levy

indirect taxes to the Confederation. But Swiss cantons have considerable autonomy in setting most types of direct tax. In addition, the cantons often allow municipalities to impose their own direct taxes by means of municipal surcharges on cantonal direct taxes. Reflecting this decentralisation, central government's share of total tax revenues (including social security) is the lowest in the OECD, only 27 per cent, and the share of state and local governments is exceeded only in Canada (Table 13). Decentralisation permeates the entire architecture of the Swiss tax system – the design of tax statutes and structure, the way redistribution

Table 13. **Tax revenues by level of government**[1]

Share of total revenues, 1996

| | Federal or central government | Social security funds | State and local governments |
|---|---|---|---|
| United States | 43.0 | 24.7 | 32.3 |
| Japan | 38.8 | 36.5 | 24.7 |
| Germany | 29.5 | 40.6 | 29.3 |
| France | 45.5 | 43.9 | 10.3 |
| Italy | 61.5 | 34.1 | 5.0 |
| United Kingdom | 76.3 | 17.3 | 3.8 |
| Canada | 40.0 | 16.3 | 43.7 |
| Australia | 77.5 | 0.0 | 22.5 |
| Austria | 50.5 | 28.2 | 20.9 |
| Belgium | 35.0 | 34.9 | 27.9 |
| Czech Republic | 46.2 | 41.9 | 13.5 |
| Denmark | 65.4 | 3.1 | 31.1 |
| Finland | 50.2 | 26.9 | 22.6 |
| Greece | 69.0 | 29.1 | 1.3 |
| Hungary | 59.6 | 30.6 | 9.8 |
| Iceland | 80.3 | 0.0 | 19.7 |
| Ireland | 84.4 | 12.0 | 2.3 |
| Korea | 71.7 | 9.1 | 19.2 |
| Luxembourg | 66.9 | 26.1 | 6.7 |
| Mexico | 81.1 | 15.6 | .. |
| Netherlands | 56.1 | 40.1 | 2.9 |
| New Zealand | 94.5 | 0.0 | 5.5 |
| Norway | 59.5 | 22.1 | 18.4 |
| Poland | 60.1 | 31.2 | 8.7 |
| Portugal | 66.7 | 27.1 | 5.7 |
| Spain | 50.7 | 35.5 | 13.4 |
| Sweden | 45.4 | 23.0 | 31.6 |
| **Switzerland**[2] | **27.4** | **37.4** | **35.2** |
| Turkey | 71.7 | 15.8 | 12.5 |

1. Tax revenues are allocated to the level of government that ultimately receives them, which need not be the level of government where they are collected.
2. Social security contributions for Switzerland include all private medical insurance premiums. Excluding such premiums that are voluntary, social security contributions were 34.6 per cent of total tax revenues.

*Source:* OECD (1998*a*), *Revenue Statistics.*

works, and measures relating to compliance and enforcement – as will be discussed in the next section.

Swiss federal arrangements reflect a broad acceptance of the effects of tax competition among cantons. The Swiss authorities consider that the positive effects of inter-cantonal tax competition outweigh the negative effects from constraining cantons' choice of tax structure. As a result, mechanisms that are used to weaken or eliminate base mobility in other federations are either not found or are very weak in Switzerland. There is no rate harmonisation whatsoever and equalisation transfers are comparatively small (see below). With primary responsibility for income and wealth taxation assigned to lower levels of government, there are few institutional constraints to prevent large differentials in effective tax burdens from developing among cantons and, indeed, this has occurred. These differences are a stable feature of the Swiss Confederation, suggesting that the cantons have entered what might be called a "tax equilibrium" with relatively little pressure for changing relative tax rates. In the future, however, institutional, technological and social change – by lowering the real and perceived costs of mobility – may create tensions in this equilibrium.

With such small government units, spatial spillovers are important and, in particular, major cities have structural budget problems. These are, at least in part, due to not receiving adequate compensation for the services they provide which benefit citizens living beyond their boundaries. More generally, the fact that there are inevitably winners and losers amongst governments associated with almost any spending or tax initiative introduces some inertia into the federal tax reform process. For example, while introducing dividend imputation might be an option in some countries, companies and shareholders are not evenly distributed amongst the cantons, with the result that cantons with many shareholders and few corporate head offices would lose tax revenue.

Cantons remain fiercely independent, but recent trends show a slight movement toward re-centralisation. The Confederation has been granted more and more reserved tax bases and the recent law on the harmonisation of definitions of direct tax bases (described below) has also considerably reduced the tax sovereignty of the cantons. As tax policy evolves in response to various pressures, the challenge for Switzerland will be to find arrangements that allow the Swiss people to express their political will for a decentralised state, while simultaneously reducing the administrative and economic costs that such a state may entail.

### *The tradition of direct democracy*

The tradition of direct democracy is also an essential element in the political economy of public finance in Switzerland. Switzerland offers the most extensive example of a functioning direct democracy in the OECD area. Anyone

can initiate a change in the federal Constitution (a new article or the modification or elimination of an existing article) by gathering 100 000 signatures within 18 months from citizens who have the right to vote.[68] When Parliament initiates changes to the Constitution or urgency laws that run counter to the Constitution, voter approval must be obtained through a referendum. Other changes to federal law are subject to optional referendum. These are held if someone gathers 50 000 signatures within 100 days of publication of the changes in the law. Increases in the rates of VAT and Federal Direct Tax are subject to compulsory referendum. During the 1990s, there has been an average of about three referenda per year. Several items are generally submitted to voter scrutiny during each referendum and these often bear directly or indirectly on government finances, spending and public management.[69]

This feature has a direct bearing on the political economy of government finances in Switzerland. Direct democracy means that electoral control of the fiscal process is effective and lowers "principal-agent" problems that can arise in the context of representative democracies. This should mean that demand-based factors are more important determinants of public expenditure than in representative democracies and that supply-based factors (as emphasised in the theory of public choice) are less important. A recent study (Kirchgässner and Pommerehne, 1997) confirms that a demand-based model of public expenditure performs much better for Switzerland than for Germany, a country with a similar federal- and economic structure except that it is a representative democracy.

By constraining the overall tax burden, these features may also ease problems of tax enforcement. In Switzerland, the moderate overall tax burden has meant relatively low marginal tax rates. As such rates are the most important determinant of tax evasion in Switzerland (Weck-Hannemann and Pommerehne, 1989), as in other countries, this limits the incentive to evade. These arrangements may also reduce evasion by lifting "tax morale". Studies of tax enforcement emphasise the fact that, at least for some tax bases, the importance of "voluntary compliance" is large and probably growing (see Slemrod, 1995). If direct democracy in Switzerland succeeds in raising the legitimacy of government spending and taxes, then this could be a positive force to limit evasion – citizens are made more responsible, having participated directly in expenditure decisions. This is reinforced by arrangements that keep expenditures and taxes at the same level of government and at the lowest possible level of government – the aim is that citizens voting expenditures should also have to pay for them.

### *Pressures on government expenditure*

Government expenditure in Switzerland (34 per cent of GDP in 1996) is below the OECD (weighted) average (38 per cent of GDP) and far below the EU (weighted) average (45 per cent of GDP). It has been growing broadly in line with

Table 14. **Trends in government social spending**

Per cent of GDP

| | Unemployment[1] | | Disability[2] | | Health[3] | | Pension[4] | | Total insurance | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 |
| United States | 0.6 | 0.0 | 1.0 | 0.3 | 6.6 | 2.1 | 6.3 | 0.1 | 14.5 | 2.5 |
| Japan | 0.5 | 0.2 | 0.5 | 0.0 | 5.9 | 1.0 | 6.3 | 1.4 | 13.2 | 2.6 |
| Germany | 3.7 | 1.5 | 1.4 | 0.1 | 9.2 | 1.2 | 10.9 | 0.0 | 25.2 | 2.8 |
| France | 3.1 | 0.0 | 1.4 | –0.4 | 9.3 | 1.6 | 12.2 | 1.3 | 26.1 | 2.5 |
| Italy | 2.0 | 0.2 | 1.9 | 0.1 | 5.7 | –0.2 | 13.6 | 2.3 | 23.2 | 2.5 |
| United Kingdom | 1.4 | –1.2 | 2.9 | 1.5 | 6.7 | 0.9 | 7.4 | 0.2 | 18.4 | 1.3 |
| Canada[5] | 1.9 | –0.7 | 1.0 | 0.1 | 6.7 | 0.4 | 4.9 | 1.0 | 14.4 | 0.9 |
| Australia | 2.1 | 0.4 | 1.2 | 0.3 | 6.1 | 0.3 | 3.3 | –0.3 | 12.8 | 0.6 |
| Austria | 1.8 | 0.6 | 1.8 | 0.0 | 6.4 | 0.9 | 13.4 | 0.8 | 23.4 | 2.3 |
| Belgium | 4.2 | –0.6 | 2.2 | –0.9 | 7.5 | 0.8 | 10.3 | 0.4 | 24.2 | –0.3 |
| Czech Republic (1990-1995) | 0.3 | .. | 1.5 | 0.1 | 8.7 | 1.7 | 6.4 | –0.5 | 16.8 | 1.6 |
| Denmark | 6.6 | 1.2 | 2.3 | 0.4 | 9.0 | –0.2 | 7.8 | 1.8 | 25.6 | 3.2 |
| Finland | 5.6 | 3.3 | 4.0 | 0.9 | 7.9 | 0.5 | 9.1 | 1.6 | 26.6 | 6.3 |
| Greece (1993) | 0.7 | 0.2 | 1.9 | 0.0 | 3.9 | 0.5 | 10.1 | 0.3 | 16.6 | 0.9 |
| Iceland | 0.8 | .. | 1.6 | .. | 9.0 | .. | 4.2 | .. | 15.6 | .. |
| Ireland | 4.4 | –0.7 | 0.9 | 0.1 | 6.6 | –1.9 | 4.6 | –1.3 | 16.4 | –3.8 |
| Korea (1990-1995) | 0.1 | 0.0 | 0.3 | 0.1 | 2.0 | 0.3 | 1.4 | 0.6 | 3.8 | 0.9 |
| Mexico (1990-1995) | 0.0 | .. | 0.0 | 0.0 | 2.9 | 1.1 | 0.4 | 0.1 | 3.3 | 1.3 |
| Netherlands | 4.1 | –0.1 | 4.1 | –0.2 | 9.4 | 0.5 | 7.8 | –0.1 | 25.4 | 0.0 |
| New Zealand | 1.9 | 0.3 | 1.7 | 0.6 | 6.6 | 1.4 | 5.8 | –2.0 | 15.9 | 0.3 |
| Norway | 2.4 | 1.3 | 2.7 | 0.6 | 11.4 | 2.9 | 6.2 | 1.0 | 22.7 | 5.9 |
| Portugal | 1.8 | 1.5 | 1.8 | 0.0 | 5.8 | 1.8 | 7.7 | 3.1 | 17.1 | 6.4 |
| Spain | 3.3 | 0.1 | 1.4 | –0.4 | 7.1 | 1.7 | 9.2 | 1.4 | 21.0 | 2.8 |
| Sweden | 4.7 | 1.6 | 2.8 | 0.6 | 10.4 | –1.6 | 9.0 | 0.9 | 26.9 | 1.5 |
| **Switzerland** | **1.6** | **1.2** | **2.5** | **0.6** | **7.4** | **2.2** | **7.1** | **0.8** | **18.6** | **4.8** |
| Turkey | 0.0 | .. | 0.1 | 0.1 | 2.4 | 1.2 | 3.7 | 1.9 | 6.3 | 3.2 |
| Total of above countries[6] | 2.7 | 0.5 | 1.9 | 0.2 | 7.5 | 0.8 | 8.2 | 0.6 | 20.2 | 2.2 |
| Total of EU countries | 3.4 | 0.5 | 2.2 | 0.1 | 7.5 | 0.5 | 9.5 | 0.9 | 22.6 | 2.0 |

*Note:* The last year of the period covered is 1995 unless otherwise stated.
1. Includes active labour market programmes.
2. Includes occupational injury and disease.
3. Includes services for the elderly and disabled people and sickness benefits.
4. Includes survivors' pension or benefits.
5. From 1990 onwards family services are included in another category of social programmes.
6. Excludes Iceland, Korea, Mexico and Turkey.

*Source:* OECD, Social expenditure database, 1980-1996.

the EU and OECD averages since the early 1980s. As in most other countries, this growth is mainly attributable to increases in social spending (Table 14). In Switzerland, most of this growth over the 1985-95 period occurred in health care and unemployment programmes. The rise in the share of health care spending in GDP was particularly strong.

Table 14. **Trends in government social spending** *(cont.)*

Per cent of GDP

| | Family services[1] | | Housing services | | Other contingencies | | Total assistance | | Total social spending[2] | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 | 1995 | Change 1985-95 |
| United States | 0.6 | 0.0 | .. | .. | 0.6 | 0.3 | 1.3 | 0.3 | 15.8 | 2.8 |
| Japan | 0.4 | 0.0 | .. | .. | 0.2 | 0.0 | 0.6 | –0.1 | 13.8 | 2.6 |
| Germany | 2.0 | 0.2 | 0.2 | 0.0 | 0.6 | 0.2 | 2.8 | 0.4 | 28.0 | 3.3 |
| France | 2.6 | –0.2 | 0.9 | 0.2 | 0.5 | .. | 4.0 | 0.5 | 30.1 | 3.0 |
| Italy | 0.5 | –0.4 | 0.0 | 0.0 | .. | .. | 0.5 | –0.4 | 23.7 | 2.1 |
| United Kingdom | 2.4 | 0.1 | 1.9 | 0.6 | 0.3 | –0.6 | 4.5 | 0.0 | 22.9 | 1.3 |
| Canada[3] | 0.8 | 0.1 | .. | .. | 3.1 | 0.7 | 3.9 | 0.8 | 18.3 | 1.7 |
| Australia | 2.5 | 1.3 | 0.2 | –0.2 | 0.1 | –0.1 | 2.8 | 1.0 | 15.6 | 1.6 |
| Austria | 2.4 | –0.4 | 0.1 | –0.1 | 0.3 | 0.0 | 2.8 | –0.5 | 26.2 | 1.8 |
| Belgium | 2.3 | –0.4 | .. | .. | 0.6 | 0.3 | 2.9 | 0.0 | 27.1 | –0.4 |
| Czech Republic (1990-1995) | 2.0 | –0.8 | 0.0 | .. | 0.4 | .. | 2.4 | –0.4 | 19.2 | 1.3 |
| Denmark | 4.0 | 1.3 | 0.8 | 0.3 | 1.6 | 0.8 | 6.5 | 2.5 | 32.1 | 5.7 |
| Finland | 4.2 | 1.6 | 0.5 | 0.3 | 0.7 | 0.3 | 5.3 | 2.2 | 32.0 | 8.5 |
| Greece (1993) | 0.1 | –0.2 | 0.1 | –0.1 | .. | .. | 0.2 | –0.4 | 16.8 | 0.6 |
| Iceland | 2.4 | .. | 0.1 | .. | 0.5 | .. | 3.1 | .. | 18.6 | .. |
| Ireland | 1.7 | 0.2 | 0.6 | –0.3 | 0.7 | 0.4 | 3.0 | 0.2 | 19.4 | –3.5 |
| Korea (1990-1995) | 0.1 | 0.0 | .. | .. | 0.1 | –0.1 | 0.2 | 0.0 | 4.0 | 0.9 |
| Mexico (1990-1995) | 0.1 | .. | 0.0 | 0.0 | 0.2 | 0.2 | 0.3 | 0.3 | 3.7 | 1.6 |
| Netherlands | 1.4 | –0.8 | 0.3 | 0.0 | 0.7 | –0.3 | 2.4 | –1.1 | 27.8 | –1.1 |
| New Zealand | 2.2 | –0.2 | 0.6 | 0.5 | 0.2 | –0.1 | 2.9 | 0.1 | 18.9 | 0.5 |
| Norway | 3.7 | 1.8 | 0.2 | 0.0 | 0.9 | 0.1 | 4.9 | 2.0 | 27.6 | 7.9 |
| Portugal | 1.0 | 0.2 | 0.0 | 0.0 | 0.1 | 0.0 | 1.1 | 0.3 | 18.3 | 6.7 |
| Spain | 0.3 | 0.1 | 0.1 | 0.1 | 0.1 | 0.0 | 0.5 | 0.2 | 21.5 | 3.0 |
| Sweden | 3.9 | –0.5 | 1.2 | 0.5 | 1.0 | 0.4 | 6.1 | 0.4 | 33.0 | 1.9 |
| **Switzerland** | **1.1** | **0.0** | **0.1** | **..** | **1.2** | **0.9** | **2.4** | **1.1** | **21.0** | **5.8** |
| Turkey | 0.3 | –0.4 | .. | .. | 0.2 | 0.1 | 0.5 | –0.2 | 6.8 | 3.0 |
| Total of above countries[4] | 1.9 | 0.1 | 0.4 | 0.1 | 0.7 | 0.2 | 2.9 | 0.4 | 23.1 | 2.6 |
| Total of EU countries | 2.1 | 0.1 | 0.5 | 0.1 | 0.6 | 0.1 | 3.0 | 0.3 | 25.6 | 2.3 |

*Note:* The last year of the period covered is 1995 unless otherwise stated.
1. Includes family cash benefits.
2. Sum of total assurance and total assistance.
3. From 1990 onwards family services are included in another category of social programmes.
4. Excludes Iceland, Korea, Mexico and Turkey.

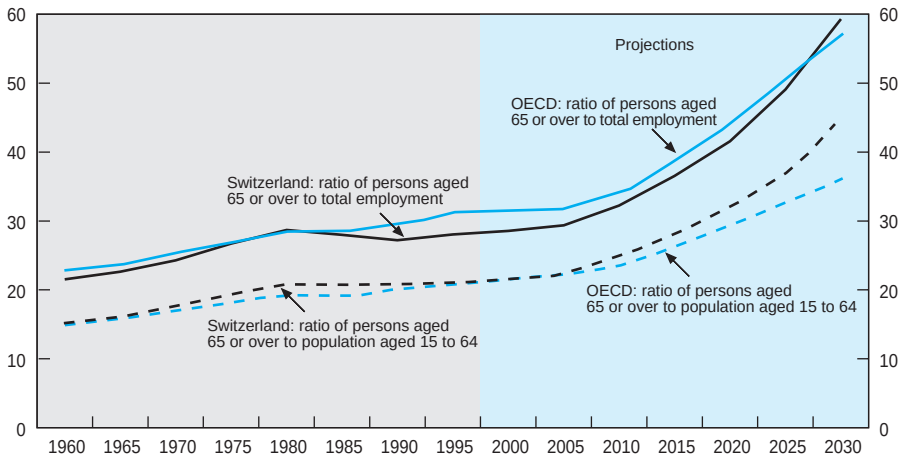
*Source:* OECD, Social expenditure database, 1980-1996.

A factor that contributes to moderate levels of government expenditure in Switzerland is that the unemployment rate (it had fallen to 3 per cent by early 1999) is one of the lowest in the OECD. Another factor is that the (dominant) second pillar of retirement income, the earnings-related component, is funded. Having a funded second-pillar that is more actuarially fair (indeed, defined contribution schemes are totally actuarially fair) than typical pay-as-you-go systems

means that contributions are more like savings than taxes. (This is one of the reasons why even compulsory contributions to funded private schemes are not classified as social security taxes in either OECD *Revenue Statistics* or the System of National Accounts.) This allows financial preparations for old age to be made in a way that avoids needlessly incurring the tax distortions that stem from using the general tax system for this purpose.[70] This feature also reduces some of the problems in the political economy of social security that other countries have encountered (*i.e.* it is politically attractive to expand social security promises, especially if future taxpayers will be paying for them). As discussed below, however, tax expenditures on savings channelled though pension programmes are very large indeed, which poses other issues for tax design. Nevertheless, the private funding of the second pillar of retirement income takes much of the pressure off public finances.

Looking further ahead, private funding of the second pillar of retirement income will also reduce the pressure on public finances from population ageing. Switzerland's elderly dependency ratio,[71] which is presently around the OECD average, is projected to more than double by 2030, a somewhat larger increase than the average for the OECD (Figure 27). A similar increase in the population

Figure 27. **Dependency ratios**
Per cent




*Source:* OECD Secretariat.

aged 65 or over as a proportion of total employment is also projected, taking this proportion to almost 60 per cent in 2030, slightly more than the OECD average. In other words, the number of people working per person aged 65 or over is projected to decline from 3.5 today to 1.7 by 2030. Despite these unfavourable demographic trends, the associated fiscal pressures are expected to be less intense than in most other OECD countries because the (dominant) second pillar of retirement income is funded.

### *Deepening globalisation and Switzerland's role in European economic integration*

Increasing international integration poses challenges for the tax systems of many countries, including Switzerland. Switzerland enjoys a major advantage in that its tax burden is moderate by OECD standards and low compared with most other European countries (see below). Reflecting this and other Swiss advantages, many multinationals have their headquarters or European headquarters in Switzerland. Deepening European economic integration also influences the development of the Swiss tax system. Such integration gives rise to tensions between high and low tax countries and brings into relief various problems of "fit" between domestic tax regimes. Switzerland, though neither a member of the EU nor of the European Economic Area, is also affected by these issues because of its close links with the EU economy.

Another force acting on Swiss tax policy stems from the possibility that it might eventually join the EU. Switzerland has already taken some steps to ensure that it can make the required modifications to its tax system should it become necessary. Various tax reforms have been made with a view to enhancing Switzerland's compatibility with EU arrangements. This was one of the considerations in introducing and then increasing the VAT (which, with the major exception of rates, is largely compatible with EU directives). But a number of other modifications to Switzerland's tax system would be required for full compatibility.[72]

Switzerland's tax rates and in some cases, cantonal practices frequently draw attention from tax authorities in other OECD countries. The importance of its private banking sector and long tradition of bank secrecy, combined with other factors described below, raise the risk of non-resident tax evasion. Moreover, two cantonal tax regimes in the area of company taxation make certain cantons an attractive place to locate certain companies, the functions of which are predominantly performed abroad.

### *International economic co-operation*

International economic co-operation also shapes Member country tax policies. OECD efforts in this area have been extensive. The Organisation's

approach to this issue recognises that there are both costs and benefits from tax competition, but also emphasises the need to improve the effectiveness of international tax administration and to strengthen the ability of OECD tax systems to adapt to emerging technological and market developments. To this end, the OECD Model Tax Convention on Income and Capital represents an attempt to improve tax co-ordination among member governments. The OECD Model and its Commentary provide a uniform basis for settling the most common problems that arise in the field of international tax treaties. In addition, the OECD has issued guidelines on "harmful" tax practices OECD (1998*b*).[73] Switzerland abstained from these guidelines.[74] The OECD is also working on related projects involving enhancing the taxation of cross-border interest flows and access to bank information for tax purposes.

## Main features of the system

### *Level and structure of taxation*

Total tax revenues (including social security contributions)[75] have increased as a share of GDP in Switzerland since the early 1980s at a broadly similar rate to that in most other OECD countries (Table 15). This increase mostly has been in social security contributions, reflecting historically high levels of unemployment in the 1990s and the growing costs of medical insurance (Table 16). While rising social security contributions have also been an important factor in the increase in taxation in other countries, those countries also tended to increase consumption taxes significantly. Despite the large increase in taxation in Switzerland, the tax ratio (33 per cent of GDP in 1996, excluding voluntary private health insurance premiums) remains significantly below the OECD average (37 per cent of GDP) and far below the EU average (42 per cent of GDP). As noted above, a factor contributing to the moderate level of taxation in Switzerland, especially compared with EU countries, is that the earnings related (*i.e.* second pillar) part of the pension system is privately funded. Such contributions amounted to 7.3 per cent of GDP in 1996.

By international comparison, the share of consumption taxes in total taxation is low in Switzerland and the shares of individual income tax and social security contributions are high (despite the fact that the second pillar of the retirement income system is not included in social security) (Table 17). Corporate taxes represent a somewhat smaller share of tax revenues than in most other countries but property and wealth taxes (which both individuals and corporations pay) are higher.

Table 15. **Total tax revenues**

Per cent of GDP

| | 1980-85 | 1986-90 | 1991-96 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|
| United States | 26.4 | 26.7 | 27.4 | 26.7 | 26.8 | 26.7 | 27.0 | 27.5 | 27.9 | 28.5 |
| Japan | 26.7 | 30.1 | 29.0 | 31.3 | 30.8 | 29.2 | 29.1 | 27.8 | 28.5 | 28.4 |
| Germany | 37.8 | 37.7 | 38.8 | 36.7 | 38.2 | 38.9 | 39.0 | 39.2 | 39.2 | 38.1 |
| France | 43.2 | 43.9 | 44.3 | 43.7 | 43.9 | 43.7 | 43.9 | 44.1 | 44.6 | 45.7 |
| Italy | 33.6 | 37.2 | 42.0 | 39.2 | 39.7 | 42.1 | 43.8 | 41.7 | 41.3 | 43.2 |
| United Kingdom | 37.1 | 36.8 | 35.1 | 36.5 | 35.6 | 35.1 | 33.5 | 34.5 | 35.6 | 36.0 |
| Canada | 33.2 | 34.9 | 36.2 | 36.0 | 36.6 | 36.2 | 35.6 | 35.9 | 36.0 | 36.8 |
| Australia | 29.2 | 30.7 | 29.5 | 30.6 | 28.9 | 28.4 | 28.6 | 29.6 | 30.4 | 31.1 |
| Austria | 41.0 | 41.7 | 42.9 | 41.0 | 41.5 | 43.0 | 43.4 | 43.3 | 42.3 | 44.0 |
| Belgium | 45.4 | 45.1 | 45.2 | 44.0 | 44.1 | 44.3 | 44.9 | 46.0 | 46.0 | 46.0 |
| Czech Republic | – | – | 42.1 | – | – | – | 43.7 | 42.8 | 41.5 | 40.5 |
| Denmark | 46.4 | 50.7 | 50.7 | 48.7 | 48.8 | 49.2 | 50.4 | 51.9 | 51.4 | 52.2 |
| Finland | 38.5 | 42.9 | 46.8 | 45.4 | 46.9 | 46.8 | 45.4 | 47.6 | 46.1 | 48.2 |
| Greece | 32.7 | 35.7 | 39.6 | 37.0 | 37.6 | 39.0 | 39.5 | 40.2 | 40.8 | 40.6 |
| Hungary | – | – | 44.3 | – | 45.9 | 45.9 | 45.8 | 44.7 | 43.3 | 40.3 |
| Iceland | 29.4 | 30.5 | 31.5 | 31.4 | 31.4 | 32.2 | 31.3 | 30.9 | 31.2 | 32.3 |
| Ireland | 35.4 | 36.6 | 34.9 | 34.8 | 35.2 | 35.5 | 35.4 | 36.1 | 33.7 | 33.7 |
| Korea | 17.4 | 17.4 | 21.1 | 19.0 | 18.8 | 19.8 | 20.7 | 21.5 | 22.3 | 23.2 |
| Luxembourg | 44.8 | 43.4 | 43.6 | 43.4 | 42.6 | 41.8 | 43.9 | 44.3 | 44.1 | 44.7 |
| Mexico | 16.7 | 16.9 | 17.1 | 17.3 | 17.3 | 17.6 | 17.7 | 17.2 | 16.6 | 16.3 |
| Netherlands | 44.8 | 45.9 | 45.5 | 44.6 | 47.2 | 46.8 | 47.5 | 44.7 | 43.8 | 43.3 |
| New Zealand | 33.3 | 36.9 | 36.9 | 38.0 | 36.6 | 36.9 | 36.9 | 37.3 | 37.9 | 35.8 |
| Norway | 43.0 | 43.1 | 41.1 | 41.8 | 41.8 | 41.0 | 40.1 | 41.3 | 41.5 | 41.1 |
| Poland | – | – | 41.0 | – | 37.2 | 38.3 | 42.5 | 43.2 | 42.7 | 42.1 |
| Portugal | 27.2 | 29.6 | 33.8 | 30.9 | 32.2 | 34.4 | 32.4 | 33.8 | 34.9 | 34.9 |
| Spain | 26.5 | 32.8 | 34.5 | 34.2 | 34.5 | 35.6 | 34.7 | 34.7 | 34.0 | 33.7 |
| Sweden | 49.6 | 54.7 | 51.2 | 55.6 | 53.7 | 51.0 | 50.1 | 50.8 | 49.5 | 52.0 |
| **Switzerland** | **30.0** | **31.4** | **32.5** | **30.9** | **30.6** | **31.2** | **32.2** | **33.0** | **33.5** | **34.7** |
| Turkey | 17.1 | 18.6 | 22.7 | 20.0 | 21.0 | 22.4 | 22.7 | 22.2 | 22.6 | 25.4 |
| Average OECD | 34.1 | 35.8 | 37.3 | 36.1 | 36.6 | 36.9 | 37.3 | 37.5 | 37.3 | 37.7 |
| Average G7 | 34.0 | 35.3 | 36.1 | 35.7 | 36.0 | 36.0 | 36.0 | 35.8 | 36.1 | 36.7 |
| Average EU (15) | 38.9 | 41.0 | 41.9 | 41.1 | 41.4 | 41.8 | 41.9 | 42.2 | 41.8 | 42.4 |

*Source:* OECD (1998*a*), *Revenue Statistics* and OECD Secretariat.

### *Main taxes by level of government*

The Constitution gives the federal government the exclusive right to raise a general consumption tax, customs duties, a withholding tax on capital income, and stamp duties. The federal government shares with lower levels of government the right to levy taxes on personal income and on firms' profits and net worth. Municipal taxes are generally levied as a surcharge on cantons' taxes.[76]

Table 16. **The level and structure of taxation by type**

Per cent of GDP

| | 1980 | | | | | | 1996 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Total | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Total |
| United States | 2.9 | 10.5 | 5.9 | 4.7 | 2.9 | 26.9 | 2.7 | 10.7 | 7.0 | 4.9 | 3.1 | 28.5 |
| Japan | 5.5 | 6.2 | 7.4 | 3.8 | 2.1 | 25.1 | 4.7 | 5.7 | 10.4 | 4.4 | 3.3 | 28.4 |
| Germany | 2.1 | 11.3 | 13.2 | 10.3 | 1.3 | 38.1 | 1.4 | 9.4 | 15.5 | 10.6 | 1.2 | 38.1 |
| France | 2.1 | 5.4 | 18.7 | 12.6 | 2.7 | 41.6 | 1.7 | 6.4 | 20.8 | 12.4 | 4.3 | 45.7 |
| Italy | 2.4 | 7.0 | 11.7 | 7.5 | 1.1 | 29.7 | 4.0 | 10.8 | 14.8 | 10.6 | 2.3 | 42.6 |
| United Kingdom | 2.9 | 10.8 | 7.4 | 10.3 | 4.2 | 35.5 | 3.8 | 9.3 | 6.2 | 12.7 | 3.9 | 35.8 |
| Canada | 3.7 | 10.9 | 3.4 | 9.9 | 3.3 | 31.2 | 3.3 | 13.9 | 6.0 | 8.7 | 4.2 | 36.0 |
| Australia | 3.4 | 12.5 | 1.4 | 8.8 | 2.2 | 28.4 | 4.7 | 12.8 | 2.1 | 8.7 | 2.8 | 31.1 |
| Austria | 1.4 | 9.3 | 15.3 | 12.6 | 1.5 | 40.2 | 2.1 | 9.2 | 18.1 | 12.5 | 1.1 | 43.0 |
| Belgium | 2.5 | 15.4 | 13.3 | 11.4 | 1.1 | 43.6 | 3.1 | 14.3 | 14.9 | 11.6 | 1.2 | 45.1 |
| Czech Republic | – | – | – | – | – | – | 4.2 | 5.3 | 17.0 | 13.5 | 0.6 | 40.5 |
| Denmark | 1.5 | 23.5 | 0.8 | 17.0 | 2.6 | 45.5 | 2.4 | 27.8 | 1.9 | 17.1 | 1.8 | 51.0 |
| Finland | 1.4 | 14.3 | 7.2 | 12.9 | 0.7 | 36.6 | 3.2 | 16.9 | 12.4 | 14.5 | 1.2 | 48.2 |
| Greece | 1.1 | 4.4 | 10.2 | 10.9 | 1.4 | 27.9 | 2.6 | 5.0 | 12.8 | 17.4 | 1.4 | 39.1 |
| Hungary | – | – | – | – | – | – | 1.9 | 7.1 | 13.7 | 16.8 | 0.8 | 40.2 |
| Iceland | 0.7 | 6.8 | 1.7 | 15.9 | 2.5 | 27.6 | 0.9 | 10.4 | 2.8 | 15.1 | 2.7 | 31.8 |
| Ireland | 1.5 | 10.4 | 4.7 | 14.1 | 1.7 | 32.4 | 3.2 | 10.5 | 4.9 | 13.3 | 1.6 | 33.6 |
| Korea | 1.9 | 2.0 | 0.3 | 9.6 | 1.8 | 15.7 | 2.7 | 4.2 | 2.2 | 10.2 | 3.9 | 23.2 |
| Luxembourg | 6.9 | 11.5 | 12.5 | 8.7 | 2.4 | 41.9 | 7.2 | 9.8 | 11.9 | 12.4 | 3.4 | 44.7 |
| Mexico | 2.7 | 2.0 | 2.4 | 4.8 | 0.6 | 12.6 | 2.3 | 1.8 | 2.7 | 4.8 | 0.6 | 12.3 |
| Netherlands | 3.0 | 11.9 | 17.2 | 11.4 | 1.7 | 45.1 | 4.1 | 7.6 | 17.1 | 12.3 | 2.1 | 43.2 |
| New Zealand | 2.6 | 20.3 | 0.0 | 7.4 | 2.6 | 32.9 | 3.5 | 15.6 | 0.3 | 12.3 | 2.0 | 33.8 |
| Norway | 5.7 | 12.1 | 9.0 | 15.0 | 0.7 | 42.5 | 4.3 | 10.7 | 9.6 | 15.5 | 0.9 | 40.9 |
| Poland | – | – | – | – | – | – | 3.1 | 9.3 | 13.4 | 15.0 | 1.3 | 42.1 |
| Portugal | – | – | – | 10.4 | 0.8 | 11.2 | 3.3 | 6.6 | 9.0 | 14.0 | 1.1 | 33.9 |
| Spain | 1.2 | 4.9 | 11.6 | 4.8 | 1.1 | 23.7 | 2.0 | 7.7 | 12.1 | 9.6 | 2.0 | 33.4 |
| Sweden | 1.2 | 20.0 | 15.3 | 11.7 | 0.5 | 48.8 | 2.9 | 18.4 | 16.8 | 11.8 | 2.0 | 51.9 |
| **Switzerland**[2] | **1.7** | **10.4** | **9.0** | **5.7** | **2.1** | **28.9** | **1.9** | **11.1** | **13.0** | **6.1** | **2.4** | **34.5** |
| Turkey | 0.7 | 7.8 | 2.5 | 4.6 | 1.5 | 17.2 | 1.5 | 5.2 | 4.0 | 9.7 | 5.0 | 25.4 |

Table 16. **The level and structure of taxation by type** *(cont.)*

Per cent of GDP

| | 1980 | | | | | | 1996 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Total | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Total |
| Average OECD | 2.5 | 10.5 | 8.1 | 9.9 | 1.8 | 32.0 | 3.1 | 10.1 | 10.1 | 11.7 | 2.2 | 37.2 |
| Average G7 | 3.1 | 8.9 | 9.7 | 8.4 | 2.5 | 32.6 | 3.1 | 9.5 | 11.5 | 9.2 | 3.2 | 36.4 |
| Average EU (15) | 2.2 | 11.4 | 11.4 | 11.1 | 1.7 | 36.1 | 3.1 | 11.3 | 12.6 | 12.9 | 2.0 | 42.0 |
| Dispersion OECD | | | | | | | | | | | | |
| Range | | | | | | | | | | | | |
| (maximum-minimum) | 6.1 | 21.5 | 18.7 | 13.2 | 3.7 | 37.5 | 6.3 | 25.9 | 20.4 | 13.0 | 4.5 | 39.6 |
| Standard Deviation | 1.6 | 5.4 | 5.7 | 3.7 | 0.9 | 10.4 | 1.3 | 5.2 | 5.8 | 3.6 | 1.2 | 8.7 |
| Coefficient of variation | 0.62 | 0.52 | 0.70 | 0.37 | 0.50 | 0.33 | 0.42 | 0.51 | 0.58 | 0.31 | 0.56 | 0.23 |

1. Consumption taxes equal total taxes on goods and services less "profits of fiscal monopolies" and "other taxes".
2. Social security contributions and payroll tax figures are unadjusted for voluntary medical insurance premiums and exclude other voluntary contributions to government (similar to payroll taxes). Adjusted figures for 1980 and 1996 are respectively 7.1 and 12.5 per cent. Compulsory contributions were introduced in 1996.

*Source:* OECD (1998*a*), *Revenue Statistics.*

Table 17. **The structure of taxation by type of tax**

Share of total tax revenues

| | 1980 | | | | | 1996 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes |
| United States | 10.8 | 9.1 | 21.9 | 17.6 | 10.7 | 9.6 | 37.6 | 24.7 | 17.1 | 11.0 |
| Japan | 21.8 | 24.3 | 29.1 | 15.0 | 8.4 | 16.4 | 20.2 | 36.5 | 15.4 | 11.6 |
| Germany | 5.5 | 29.6 | 34.5 | 27.0 | 3.3 | 3.8 | 24.7 | 40.6 | 27.8 | 3.0 |
| France | 5.1 | 12.9 | 44.9 | 30.3 | 6.6 | 3.8 | 14.1 | 45.4 | 27.2 | 9.4 |
| Italy | 7.8 | 23.1 | 38.7 | 24.7 | 3.7 | 9.2 | 25.1 | 34.3 | 24.5 | 5.4 |
| United Kingdom | 8.2 | 30.7 | 21.0 | 29.2 | 12.0 | 10.5 | 25.9 | 17.3 | 35.2 | 10.7 |
| Canada | 11.6 | 34.1 | 10.5 | 30.9 | 10.3 | 8.9 | 37.7 | 16.3 | 23.6 | 11.5 |
| Australia | 12.2 | 44.0 | 5.0 | 31.1 | 7.8 | 15.0 | 41.2 | 6.7 | 28.0 | 9.0 |
| Austria | 3.5 | 23.2 | 38.0 | 31.3 | 3.8 | 4.7 | 20.9 | 41.1 | 28.4 | 2.5 |
| Belgium | 5.7 | 35.2 | 30.4 | 26.1 | 2.4 | 6.8 | 31.0 | 32.3 | 25.3 | 2.6 |
| Czech Republic | .. | .. | .. | .. | .. | 10.3 | 13.1 | 42.0 | 33.3 | 1.4 |
| Denmark | 3.2 | 51.8 | 1.8 | 37.4 | 5.8 | 4.6 | 53.2 | 3.6 | 32.7 | 3.5 |
| Finland | 3.9 | 38.8 | 19.6 | 35.0 | 2.0 | 6.7 | 35.0 | 25.8 | 30.0 | 2.5 |
| Greece | 3.8 | 14.9 | 34.7 | 36.9 | 4.7 | 6.3 | 12.4 | 31.4 | 42.8 | 3.4 |
| Hungary | .. | .. | .. | .. | .. | 4.6 | 17.5 | 33.9 | 41.7 | 2.1 |
| Iceland | 2.5 | 23.1 | 6.0 | 54.4 | 8.5 | 2.7 | 32.2 | 8.6 | 46.6 | 8.2 |
| Ireland | 4.5 | 32.0 | 14.5 | 43.4 | 5.3 | 9.6 | 31.3 | 14.6 | 39.6 | 4.8 |
| Korea | 11.0 | 11.5 | 1.6 | 55.0 | 10.3 | 11.6 | 18.0 | 9.4 | 43.7 | 16.9 |
| Luxembourg | 16.4 | 27.3 | 29.7 | 20.7 | 5.7 | 16.0 | 22.0 | 26.6 | 27.7 | 7.6 |
| Mexico | 17.1 | 12.7 | 15.2 | 29.8 | 3.8 | 13.9 | 11.2 | 16.9 | 29.7 | 3.9 |
| Netherlands | 6.6 | 26.3 | 38.1 | 25.2 | 3.9 | 9.5 | 17.5 | 39.6 | 28.5 | 4.8 |
| New Zealand | 7.8 | 61.6 | 0.0 | 22.3 | 7.9 | 9.8 | 43.5 | 1.0 | 34.5 | 5.6 |
| Norway | 13.3 | 28.4 | 21.1 | 35.1 | 1.7 | 10.5 | 26.0 | 23.3 | 37.6 | 2.1 |
| Poland | .. | .. | .. | .. | .. | 7.4 | 22.0 | 31.8 | 35.7 | 3.0 |
| Portugal | .. | .. | 32.1 | 41.5 | 3.3 | 9.5 | 18.9 | 25.7 | 40.0 | 3.2 |
| Spain | 5.1 | 20.4 | 48.6 | 20.1 | 4.8 | 5.9 | 23.0 | 35.9 | 28.6 | 5.9 |
| Sweden | 2.5 | 41.0 | 31.4 | 23.9 | 1.1 | 5.6 | 35.3 | 32.4 | 22.8 | 3.9 |
| **Switzerland**[2] | **5.8** | **35.6** | **30.9** | **19.5** | **7.3** | **5.6** | **32.0** | **37.4** | **17.5** | **7.0** |
| Turkey | 4.1 | 43.5 | 14.0 | 25.6 | 8.5 | 5.7 | 20.5 | 15.8 | 38.3 | 19.8 |

Table 17. **The structure of taxation by type of tax** *(cont.)*

Share of total tax revenues

| | 1980 | | | | | 1996 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes | Corporate income taxes | Individual income tax | Social security and payroll taxes | Consumption taxes[1] | Other taxes, including property taxes |
| Average OECD | 8.0 | 30.6 | 23.6 | 30.3 | 5.9 | 8.4 | 26.3 | 25.9 | 31.2 | 6.4 |
| Average G7 | 10.1 | 27.7 | 28.6 | 25.0 | 7.9 | 8.9 | 26.5 | 30.7 | 24.4 | 8.9 |
| Average EU (15) | 5.8 | 29.1 | 30.5 | 30.2 | 4.6 | 7.5 | 26.0 | 29.8 | 30.7 | 4.9 |
| Dispersion OECD | | | | | | | | | | |
| Range (maximum-minimum) | 19.4 | 50.2 | 48.6 | 40.0 | 10.9 | 13.7 | 42.0 | 44.5 | 31.2 | 18.4 |
| Standard Deviation | 5.1 | 12.3 | 14.0 | 10.1 | 3.0 | 3.7 | 10.3 | 12.6 | 8.1 | 4.6 |
| Coefficient of variation | 0.64 | 0.40 | 0.59 | 0.33 | 0.52 | 0.44 | 0.39 | 0.49 | 0.26 | 0.71 |

1. Consumption taxes equal total taxes on goods and services less "profits of fiscal monopolies" and "other taxes".
2. Social security contributions and payroll tax figures are unadjusted for voluntary medical insurance premiums and exclude other voluntary contributions to government (similar to payroll taxes). Adjusted figures for 1980 and 1996 are respectively 24.3 and 36.1 per cent. Basic medical insurance became compulsory in 1996.

*Source:* OECD (1998*a*), *Revenue Statistics.*

Approximately one half of the federal government's tax revenues are derived from consumption taxes, with VAT being by far the most important such tax (Table 18). Income tax is the next most important revenue source, accounting for about one quarter of Federal revenue. The cantons and municipalities derive most of their revenue from income taxes. These account for three quarters of cantons' revenues and a somewhat higher proportion of municipalities' revenues. Most of the remainder of cantons' and municipalities' tax revenues comes from taxes on personal wealth and on corporate net worth.

Taking all three levels of government together, the personal income tax (PIT) accounts for almost half of all tax revenues (excluding social security contributions), far more than any other tax. The cantons and municipalities raise most of this revenue. The next largest category of tax is on consumption, contributing

Table 18. **Tax revenues of the Confederation, the cantons and the communes**[1]

1996, percentages

| | Confederation | 26 cantons | About 3 000 communes | Total |
|---|---|---|---|---|
| **Total tax revenues (SF million)** | 34 159.0 | 24 949.0 | 19 486.6 | 78 594.6 |
| (per cent of total tax revenues) | 43.5 | 31.7 | 24.8 | 100 |
| **Personal income tax, incl. capital gains** | 17.7 | 66.7 | 76.6 | 47.9 |
| **Taxes on personal property and wealth** | .. | 19.4 | 11.6 | 9.0 |
| Wealth taxes | .. | 6.7 | 7.6 | 4.0 |
| Inheritance and gift tax | .. | 3.9 | 0.6 | 1.4 |
| Tax on sale of real-estate property | .. | 2.2 | 1.1 | 1.0 |
| Property tax | .. | 0.6 | 2.4 | 0.8 |
| Motor vehicles | .. | 6.0 | 0.0 | 1.9 |
| **Corporate taxes** | 8.5 | 13.6 | 11.4 | 10.8 |
| On profits | 7.5 | 10.1 | 8.6 | 8.6 |
| On net worth | 1.0 | 3.5 | 2.9 | 2.3 |
| **Consumption and excise taxes** | 52.5 | 0.1 | 0.1 | 22.9 |
| VAT | 35.3 | .. | .. | 15.3 |
| Tobacco, liquors and beer | 4.4 | .. | .. | 1.9 |
| Mineral oils | 12.8 | .. | .. | 5.6 |
| **Other taxes** | 21.2 | 0.2 | 0.2 | 9.3 |
| Withholding tax | 9.7 | .. | .. | 4.2 |
| Financial and insurance stamp duty | 5.8 | .. | .. | 2.5 |
| Customs duties | 4.0 | .. | .. | 1.7 |
| Road taxes | 1.3 | .. | .. | 0.6 |
| Other | 0.4 | 0.2 | 0.2 | 0.3 |
| **Total** | 100 | 100 | 100 | 100 |

1. Excludes social security contributions. Percentages refer to total tax revenue at the top of each column.

*Source:* Administration fédérale des contributions, *Finances publiques en Suisse 1996,* Bern 1998 and OECD Secretariat.

almost one quarter of tax receipts. The remainder of taxation comes roughly equally from taxes on personal property and wealth and on corporations.

### *The main taxes*

*Personal income tax*

The personal income tax (PIT) base includes all world-wide income from paid activity, income in kind, income from wealth and income from social security and pensions. Income in kind includes not only prerequisites, such as company cars, but also imputed rentals from owner-occupied homes. Capital gains on financial wealth are not taxed, unless they relate to a business activity. However, under the federal law on tax harmonisation, capital gains on real estate (including the principal residence) are taxed in all cantons, albeit at rates which are inversely related to the holding period. There are no dividend imputation credits to counter the double taxation of dividends paid to individuals. All interest expenses are deductible.

Tax exempt thresholds (allowing for personal exemptions, standard deductions and zero-rate brackets) tend to be high. Taking all three levels of government together, average marginal rates for a single taxpayer range from 22 per cent at an annual income of SF 50 000 to 38 per cent at SF 200 000 (Table 19). There is a wide distribution of rates around these averages.

*Taxes on personal property and wealth*

The cantons and municipalities (but not the federal government) levy net wealth taxes. Net wealth includes world-wide assets, less real estate located outside the taxing government's territory, less debt. The law on the harmonisation of direct taxation (see below) sets out harmonised rules for determining the tax base as well as for the continued levying of this tax by cantons and municipalities. Wealth is assessed at its current resale value (for one or two years).[77] Tax

Table 19. **Marginal tax rates combining all levels of government, for the single taxpayer**

| Gross income ('000 francs) | Minimum | Approximate Swiss average | Maximum |
|---|---|---|---|
| 15-20 | 0 | 10 | 15 |
| 50-60 | 14 | 22 | 26 |
| 100-150 | 22 | 32 | 37 |
| 200+ | 25 | 38 | 46 |

*Source:* Administration fédérale des contributions, *Finances publiques en Suisse 1996,* Bern 1998 and OECD Secretariat.

schedules are progressive in all but three cantons, which apply proportional rates.[78] Tax rates never exceed 1 per cent of net wealth.[79]

*Corporate taxes*

The main taxes levied on corporations are the corporate income tax (CIT) and the net worth tax. The CIT accounts for about 75 per cent of corporate tax revenue. In calculating taxable income, corporations may deduct all expenses incurred in earning that income, including depreciation charges that conform to normal business practice.[80] Dividends received, and to a large extent, capital gains from holdings in other corporations are exempt from the profits tax. There is no group taxation on a consolidated basis. Under the exemption method, income from a foreign permanent establishment or from foreign immovable property is tax exempt in Switzerland. The net worth tax applies to the book value of net equity (*i.e.* including retained earnings).

An unusual feature of the CIT in about two-thirds of cantons is that rates are a progressive function of rate of return. This is calculated as profits divided by net worth. Thus, the CIT and the net worth tax are closely related. The net worth tax reduces the progressiveness of the CIT when profits are positive. The federal government also had a progressive rate CIT and a net worth tax before its business tax reforms came into effect in January 1998. Since then, the federal CIT rate has been 8.5 per cent and there has been no federal tax on net worth. Taking all three levels of government together, the (simple) average CIT rate across cantons (based on municipal rates in the capital of each canton) ranges from 13 per cent to 30 per cent.[81] Allowing for the net worth tax, average rates range from 23 per cent to 35 per cent These rates are low by international comparison,[82] although there are OECD countries with lower rates.

For companies with permanent establishments outside the canton where the head office is located, there are legal rules that determine the apportionment of the tax base between governments. These rules, which are intended to avoid double taxation, depend upon the company activity. As an example, industrial enterprises apportion income using a two-factor formula based on capitalised assets and payroll (Daly and Weiner, 1993).

*VAT*

VAT was introduced in 1995 to replace the turnover tax. There are two preferential rates and, aside from the usual exemptions, the VAT base in Switzerland is relatively broad.[83] The standard rate was raised 1 percentage point in January 1999 to the constitutional limit of 7.5 per cent and the preferential rates were increased respectively by 0.3 percentage point to 2.3 per cent and by 0.5 percentage point to 3.5 per cent. The effective rate, which equals VAT revenue divided by consumption expenditures, is among the lowest in the OECD

Table 20. **Productivity of value added taxes**[1]

Per cent, 1996

| | Value added taxes revenues over GDP | Standard rates A | Effective VAT[2] B | Effective rate over standard rate B/A |
|---|---|---|---|---|
| United States | n.a. | n.a. | n.a. | n.a. |
| Japan | 1.5 | 3.0 | 2.6 | 86.1 |
| Germany | 6.7 | 15.0 | 11.6 | 77.0 |
| France | 7.9 | 20.6 | 14.3 | 69.4 |
| Italy | 5.6 | 19.0 | 9.6 | 50.4 |
| United Kingdom | 7.0 | 17.5 | 10.6 | 60.9 |
| Canada | 2.6 | 7.0 | 4.4 | 62.3 |
| Australia | n.a. | n.a. | n.a. | n.a. |
| Austria | 8.4 | 20.0 | 16.2 | 80.9 |
| Belgium | 7.0 | 21.0 | 12.5 | 59.4 |
| Czech Republic | 7.2 | .. | .. | .. |
| Denmark | 10.2 | 25.0 | 22.1 | 88.4 |
| Finland | 8.2 | 22.0 | 17.0 | 77.2 |
| Greece | 9.2 | 18.0 | 10.9 | 60.8 |
| Hungary | 7.5 | .. | .. | .. |
| Iceland | 9.4 | 24.5 | 17.1 | 70.0 |
| Ireland | 7.0 | 21.0 | 15.5 | 73.8 |
| Korea | 4.3 | 10.0 | .. | .. |
| Luxembourg | 6.7 | 15.0 | .. | .. |
| Mexico | 2.9 | 15.0 | 4.4 | 29.2 |
| Netherlands | 7.0 | 17.5 | 13.0 | 74.2 |
| New Zealand[3] | 8.5 | 12.5 | 15.3 | 122.7 |
| Norway | 8.6 | 23.0 | 21.2 | 92.1 |
| Poland | 7.8 | .. | .. | .. |
| Portugal | 8.0 | 17.0 | 13.5 | 79.1 |
| Spain | 5.5 | 16.0 | 9.4 | 58.8 |
| Sweden | 7.0 | 25.0 | 14.1 | 56.2 |
| **Switzerland** | **3.3** | **6.5** | **5.5** | **85.2** |
| Turkey | 6.0 | 15.0 | 9.2 | 61.3 |
| OECD average[4] | 6.7 | 16.9 | 12.3 | 71.6 |
| G7 average[4] | 5.2 | 13.7 | 8.8 | 67.7 |
| EU average[4] | 7.6 | 19.5 | 13.7 | 68.8 |
| Dispersion OECD | | | | |
| Range (maximum-minimum) | 8.7 | 22.0 | 19.5 | 93.5 |
| Standard deviation | 2.2 | 5.9 | 5.2 | 18.5 |
| Coefficient of variation | 0.3 | 0.3 | 0.4 | 0.3 |

*Notes:* n.a. is not applicable; .. is not available.

1. VAT productivity is defined as effective VAT rate divided by the standard rate.
2. The effective VAT rate is VAT revenue divided by the tax base (*i.e.* consumption).
3. The general sales tax at standard rate is levied on 60 per cent of the value of the supply for long-term stay in a commercial dwelling which may partly explain why the effective VAT rate exceeds the standard rate and produces productivity figures above 100 per cent.
4. Simple average over available countries.

*Source:* OECD (1998*a*), *Revenue Statistics;* (1997), *Consumption Tax Trends* and OECD Secretariat.

(Table 20). On the other hand, VAT productivity, measured by the effective VAT rate divided by the standard VAT rate, is one of the highest in the OECD. This reflects the relatively narrow range of goods and services that benefit from reduced rates in Switzerland as well as reduced rates that are relatively high in relation to the standard rate (Table 21).

### *Tax harmonisation law (la loi sur l'harmonisation des impôts)*

The Tax Harmonisation Law (THL), which came into effect in 1993, sets out the direct taxes that cantons can levy and the principles that tax legislation in the cantons must respect. Cantons have until 2001 to bring their legislation into conformity. The law harmonises the tax bases for net personal income and wealth taxes (before social allowances) and for corporate income and net worth. Tax rates and schedules are not being harmonised. Similarly, personal allowances for personal income and wealth taxation are not being harmonised as they are considered to be a component of the schedules. A major element of harmonisation was to give up five different tax-timing rules. The law provides for only the two following timing rules: taxation every second year on the average income of the previous two years (biennial praenumerando); or taxation every year on current income (annual postnumerando). The cantons may choose either for personal direct taxation but only the latter is allowed for direct corporate taxation.

Federal direct taxation was brought into conformity with the THL in 1995 through the first ordinary law for federal direct taxes (personal income tax, corporate income and net-worth tax). Amongst the changes required to harmonise the federal tax with cantons' taxes was a change in the timing of corporate taxation from taxation every second year on the average income of the previous two years to taxation every year on current income.[84] Most cantons already have annual taxation of companies and 23 out of 26 aim also to have this system in place for personal taxation by 2001. As the cantons collect federal direct taxes, the timing they choose also applies to federal income taxation.[85] The Federal Council is required by the THL to prepare a report for the Federal Assembly on tax harmonisation and unifying the timing of taxation after 2001. In view of the predominance of annual taxation already planned, it would appear that the days of the biennial system are numbered.

### *Tax administration and enforcement*

Although all three levels of government collect direct taxes, there are not separate administration arrangements for each level of government. Cantons collect the federal income tax on behalf of the federal government (and under its supervision) and retain 30 per cent of the proceeds. (Of the federal income tax revenues retained by cantons, 17/30 remains in the canton where the tax was collected and the remainder is returned to the cantons through fiscal-equalisation

Table 21. **Trends in value added taxes – tax parameters**

| | Year VAT introduced | Initial standard rate | 1996 standard rate | Zero rated goods | Lower rated goods |
|---|---|---|---|---|---|
| United States | None | None | None | .. | .. |
| Japan | 1989 | 3 | 5 | None | None |
| Germany | 1968 | 10 | 15 | None | Books, food, newspapers, transport |
| France | 1964 | 20 | 20.6 | None | Medicine, equipment for the disabled, books, hotels, entertainment, authors' rights, museums, transport, travel, passenger travel, accommodation, agriculture, books, catering, food, newspapers and water |
| Italy | 1973 | 12 | 19 | Books, newspapers, scrap iron, recycled paper | Food, medicine, telecom, weekly publications and accommodation |
| United Kingdom | 1973 | 10 | 17.5 | Children's clothing, food, passenger transport, books, newspapers, domestic sewage and water, prescription drugs, medicine, certain supplies for the disabled | .. |
| Canada | 1991 | 7 | 7 | Medicine, basic groceries, exports, certain financial services, certain agricultural and fishing products, medical devices, international travel and transportation services, agriculture, precious metals | None |
| Australia | None | None | None | .. | .. |
| Austria | 1973 | 16 | 20 | None | Agriculture, books, food, forestry, hospitals, newspapers, art, culture, letting transport, wine |
| Belgium | 1971 | 18 | 21 | Cars for handicapped, newspapers and certain weeklies | Agriculture, original art, clothing, food, coal and coke, gold |
| Czech Republic | n.a. | n.a. | n.a. | Not available | .. |

Table 21. **Trends in value added taxes – tax parameters** *(cont.)*

| | Year VAT introduced | Initial standard rate | 1996 standard rate | Zero rated goods | Lower rated goods |
|---|---|---|---|---|---|
| Denmark | 1967 | 10 | 25 | Newspapers | None |
| Finland | 1969 | 11.1 | 22 | Newspaper and magazine subscriptions, some printing services, vessels and international transport | Food, non-alcoholic drinks, animal feed, medicine, books, passenger travel, accommodation, TV licenses, admission to cultural entertainment, sporting events and to the cinema |
| Greece | 1987 | 16 | 18 | None | Books, culture, food, medicine newspapers |
| Iceland | 1989 | 22 | 24.5 | International transport, food, fuel and equipment delivered for use in ships and aircraft engaged in international traffic, shipbuilding | Food, newspapers, books, hotels, warm water, electricity and fuel oil used for the heating of houses and swimming pools |
| Ireland | 1972 | 16.4 | 21 | Books, children's clothing and footwear, oral medicine, certain medical equipment, certain goods, seeds, fertilisers | |
| Luxembourg | 1970 | 8 | 15 | None | Agriculture, books, food, fuel, medicine, newspapers |
| Mexico | 1980 | 10 | 15 | | 10 per cent rate applied along the US border |
| Netherlands | 1969 | 12 | 17.5 | None | Accommodation, agriculture, books, equipment for the disabled, books, hotels, entertainment, authors' rights, museums, transport, travel, passenger travel, accommodation, agriculture, books, catering, food, newspapers and water |
| New Zealand | 1986 | 10 | 12.5 | Fine metals from refiner to dealer, exports | Long-term stays in a commercial building |

Table 21. **Trends in value added taxes – tax parameters** *(cont.)*

| | Year VAT introduced | Initial standard rate | 1996 standard rate | Zero rated goods | Lower rated goods |
|---|---|---|---|---|---|
| Norway | 1970 | 20 | 23 | Books, newspapers, certain aircraft and ships, transport services by ferrying vehicles, second-hand vehicles | None |
| Portugal | 1986 | 16 | 17 | None | Books, food, supplies to the disabled, medicines, entertainment, newspapers, fuel, transport, electricity, accommodation and restaurant services |
| Spain | 1986 | 12 | 16 | None | Books, social lodgings, catering, certain cultural and entertainment services, food, hotels, restaurants, supplies to the disabled, medicines, transport, newspapers, public amenities, burial services, agriculture and forestry, domestic passenger transport |
| Sweden | 1969 | 11.1 | 25 | Commercial aircraft and ships, aircraft fuel, prescription medicine, printing of certain publications | Accommodation, food, passenger transport, ski-lifts, newspapers, certain works of art, imported antiques, collectors items |
| **Switzerland** | **1995** | **6.5** | **6.5** | **None** | **Water, food, medicine, books, newspapers, non commercial television** |
| Turkey | 1985 | 10 | 15 | None | Agriculture, leasing, second hand cars, newspapers, books, magazines, basic foodstuffs, natural gas, certain entertainment and cultural services |

*Source:* Owens, J. and E. Whitehouse (1996), "Tax reform for the 21st century", *Bulletin for International Fiscal Documentation,* Vol. 50, No. 11/12; OECD (1997) *Consumption Tax Trends;* and OECD Secretariat.

arrangements, see below). Municipal direct taxes are generally levied as a surcharge on the cantons' taxes. The local cantonal tax office collects direct taxes for all three levels of government. The taxpayer (without direct tax liabilities in other cantons) files a single tax return under his canton's definition of the tax base and the canton tax administration makes the necessary adjustments to establish taxable income under the federal tax base. For example, 20 entries are necessary to convert the personal income tax base in the canton of Vaud in 1999 to the federal definition. The THL will reduce the extent of these conversions for personal income tax and eliminate them for company tax.

Administrative costs for the federal government amount to 0.4 per cent of tax revenue. For VAT at 6.5 per cent in 1998, compliance costs were about 0.55 per cent of tax revenue. For the cantons and communes, administrative costs amount to 1.5-2.5 per cent of revenues. A recent study on compliance costs in five cantons for small and medium sized enterprises shows that they spend about 100-250 hours per year on tax compliance.[86] This is less than what they spend on compliance with social security or with business regulations (construction permits, work permits, and so forth). When payments for external help are added, the total cost for tax compliance is in the neighbourhood of SF 3 500-6 500 per year in 1998. That is significantly less than in Austria and Germany. The comparison with a similar study performed in 1985 shows that tax compliance represents a much smaller share of total compliance costs in 1998. Within tax compliance, most time and money is spent on VAT compliance, followed by the withholding tax, and by stamp duties for those businesses that are concerned. There are, however, some isolated examples (such as death duties in Bern) where compliance costs are high owing to special regulatory arrangements.[87]

The Swiss tax authorities do not have many of the tools available in other countries to enforce compliance though they consider that the tools they do have are adequate. Taxes on labour income are not withheld at source (no pay-as-you-earn system) but are assessed according to the individual tax return. Many cantons require employers either to report salaries paid or to submit complete payrolls directly to the tax administration. All employers must report all salaries paid when asked to do so by the tax administration. In addition, all employers must provide their employees with a salary certificate and all employees must submit this certificate together with their tax return. If an employee does not submit the certificate, he runs the risk of being taxed on the basis of an estimation of income and of facing penalties. Banking records cannot be inspected owing to bank secrecy except where there is evidence of a criminal offence[88] having been committed. Under Swiss law, failing to report income is not a criminal offence but tax fraud is. Individuals do not have a taxpayer identification number (owing to the decentralised collection of taxes), although this exists for VAT taxpayers (the collection of this tax being centralised). The federal government and the cantons exchange information concerning direct taxes and the results of

VAT audits. Since 1995, the tax authorities have also been able to consult social security files. For income on financial assets and lottery prizes, enforcement is facilitated by a high withholding tax – 35 per cent – although this is still less than the combined marginal income and wealth tax rates faced by many individuals.[89] (Income from foreign debtors is not subject to the withholding tax.)

### *Regional differences in tax burdens*

Even after some formal harmonisation, there will still exist great differences in tax rates charged in different cantons and even municipalities. This reflects both the cantons' freedom in setting expenditures and substantial differences in wealth between jurisdictions. Income per capita in the three richest cantons (including Zurich, which has the largest population of any canton) is double that in the three poorest cantons (Table 22).[90] Total tax revenue per capita is about two-thirds higher in the three richest cantons than in the three poorest, but within each of these groups there are enormous differences in total taxation. The index of global tax burden, which is computed as a weighted average of the effective tax rates imposed on representative citizens of each canton by different taxes, shows that there are very large differences in tax burdens between cantons. This index (mean = 100) ranges from 57 in Zug, the richest canton, to 131 in Jura, the poorest canton. Income per capita is positively correlated (correlation coefficient of 0.56) with total tax revenue per capita but negatively correlated (correlation coefficient of –0.60) with the index of tax burden. There is no correlation between tax revenue and the index of tax burden. Some cantons obtain large tax revenues from high tax rates, but the majority of cantons with high tax revenues simply benefit from rich tax bases.

These differences in tax burden affect taxpayers' choice of residential location. The proportion of taxpayers with taxable income exceeding SF 100 000 in 1990 was 9.4 per cent in the most favourable canton (Zug) and 2.5 per cent in the least favourable canton (Jura), for a Swiss average of 5.5 per cent.[91] However, other factors also affect taxpayers' choice of location. The canton of Geneva, which has a tax schedule that is markedly less favourable to high-income taxpayers than that in Zug, has an even higher proportion of high income taxpayers. An econometric study taking into account a variety of influences on location decisions found that differences in income tax rates between cantons and communes can *each* account for differences in the proportion of the highest-income taxpayers (income > SF 100 000) across municipalities exceeding 4 percentage points (Kirchgässner and Pommerehne, 1996).[92] The results for the other high-income groups show that differences in municipal tax rates have a somewhat greater impact than differences in cantons' tax rates. Those results are confirmed when the categories of taxpayers in the highest-income group are broken up (retired

Table 22. **Incomes and global tax burdens by canton**[1]

Per capita

| Cantons | GDP (francs, 1995) | Total tax revenue (francs, 1996) | Global index of tax burden[2] (Switzerland = 100, 1997) |
|---|---|---|---|
| Zug | 67 452 | 6 196 | 57 |
| Basel-City | 54 769 | 9 727 | 112 |
| Zurich | 53 994 | 6 895 | 80 |
| Geneva | 48 066 | 10 342 | 113 |
| Basel-Land | 45 610 | 6 456 | 93 |
| Glarus | 45 483 | 5 227 | 112 |
| Nidwalden | 43 281 | 4 880 | 72 |
| Schaffhausen | 43 072 | 6 000 | 100 |
| Vaud | 42 628 | 6 447 | 109 |
| Switzerland | 42 514 | 6 254 | 100 |
| Aargau | 42 448 | 5 233 | 97 |
| Schwyz | 39 689 | 4 740 | 83 |
| Solothurn | 39 186 | 5 130 | 99 |
| St. Gallen | 38 668 | 5 415 | 99 |
| Ticino | 37 599 | 6 239 | 97 |
| Graubünden | 37 449 | 6 038 | 91 |
| Lucerne | 37 210 | 5 498 | 117 |
| Bern | 37 147 | 5 867 | 120 |
| Uri | 37 056 | 4 188 | 89 |
| Thurgau | 36 945 | 5 242 | 101 |
| Fribourg | 35 942 | 5 688 | 125 |
| Neuchâtel | 35 635 | 6 070 | 124 |
| Appenzell A-Rh | 33 819 | 5 104 | 106 |
| Obwalden | 33 590 | 4 576 | 116 |
| Appenzell I-Rh | 31 428 | 4 462 | 98 |
| Valais | 29 663 | 4 738 | 130 |
| Jura | 29 094 | 4 954 | 131 |

1. Each canton is represented by its capital city.
2. This is calculated as a weighted average of the effective tax rates imposed on representative citizens of each canton by different taxes.

*Source:* Federal finance administration and Federal statistical office.

people, dependent workers and self-employed).[93] (The same study found that tax variables have no significant impact on rents.)

There has not been convergence of individual tax burdens over the past decade (Table 23). However, there has been some convergence in company tax burdens, which on (a simple) average declined by 4 percentage points between 1985 and 1997. There was a marked increase in the number of cantons with company tax burdens below the Swiss average (from five to ten) and a reduction in the number with above average company tax burdens and below average individual tax burdens (from ten to six) (Figure 28). Reductions in corporate tax

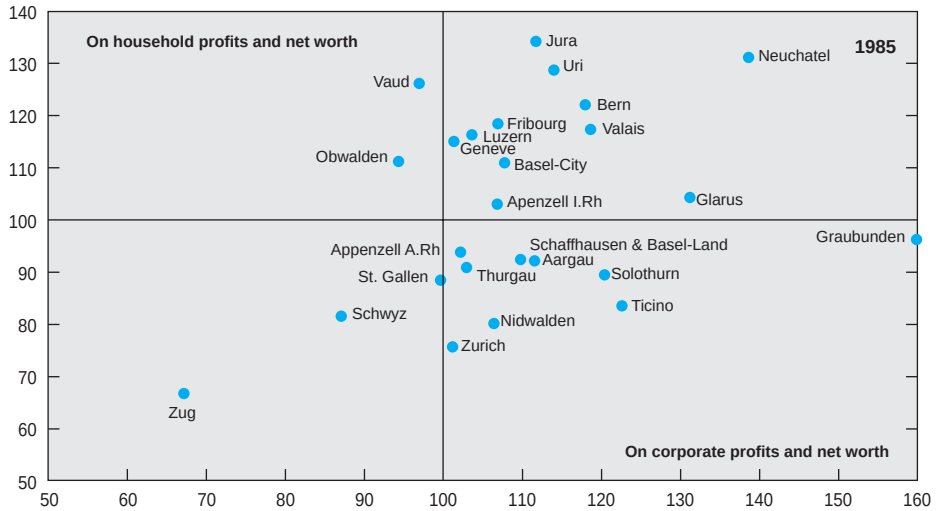
Table 23. **Fiscal burden indexes on households and companies**[1]

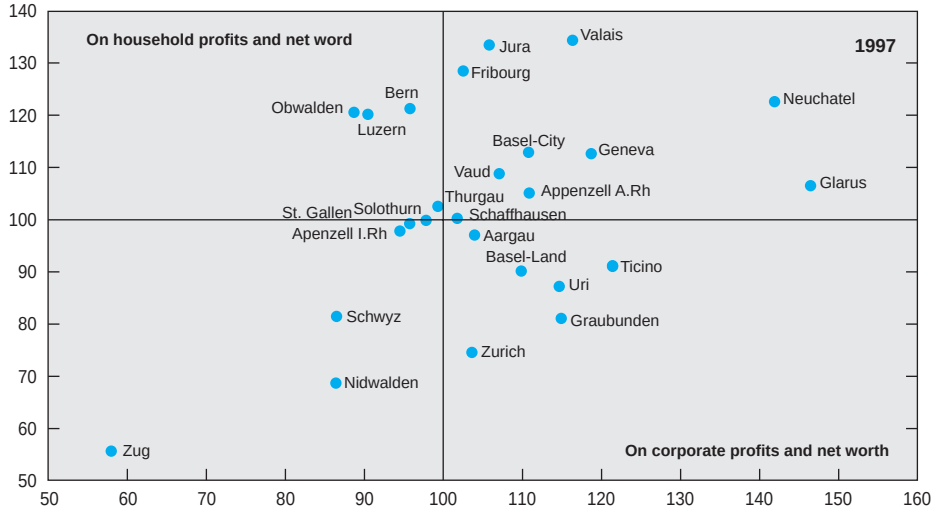
| | Company net profits and capital[2] | | Household income and net wealth[3] | | Global fiscal burden[4] | |
|---|---|---|---|---|---|---|
| | 1985 | 1997 | 1985 | 1997 | 1985 | 1997 |
| Zurich | 101.1 | 103.6 | 75.9 | 74.8 | 82.1 | 80.3 |
| Bern | 118.1 | 95.8 | 122.1 | 121.5 | 121.9 | 120.4 |
| Luzern | 103.7 | 90.4 | 116.7 | 120.4 | 114.9 | 116.6 |
| Uri | 114.1 | 114.7 | 128.8 | 87.2 | 125.7 | 88.7 |
| Schwyz | 87.3 | 86.5 | 81.5 | 81.7 | 82.9 | 82.7 |
| Obwalden | 94.3 | 88.8 | 111.3 | 120.6 | 108.1 | 116.0 |
| Nidwalden | 106.4 | 86.4 | 80.3 | 68.8 | 84.4 | 72.2 |
| Glarus | 131.3 | 146.5 | 104.0 | 106.5 | 107.0 | 111.9 |
| Zug | 67.2 | 58.0 | 66.8 | 55.6 | 68.0 | 57.3 |
| Fribourg | 106.8 | 102.5 | 118.6 | 128.5 | 116.6 | 124.8 |
| Solothurn | 120.5 | 97.8 | 89.4 | 100.0 | 93.2 | 99.4 |
| Basel-City | 107.9 | 110.9 | 110.8 | 112.5 | 109.6 | 111.9 |
| Basel-Land | 109.6 | 109.9 | 92.5 | 90.3 | 96.3 | 93.2 |
| Schaffhausen | 109.6 | 101.8 | 92.5 | 100.2 | 93.7 | 99.5 |
| Apenzell A.Rh | 102.1 | 111.0 | 94.0 | 105.1 | 95.2 | 106.2 |
| Apenzell I.Rh | 107.0 | 94.5 | 103.1 | 97.8 | 103.4 | 97.5 |
| St. Gallen | 99.6 | 95.3 | 88.6 | 99.4 | 91.8 | 99.2 |
| Graubunden | 159.8 | 115.0 | 96.3 | 81.3 | 109.6 | 90.5 |
| Aargau | 111.5 | 104.0 | 92.1 | 97.1 | 94.4 | 97.3 |
| Thurgau | 102.9 | 99.3 | 91.0 | 102.6 | 91.7 | 101.1 |
| Ticino | 122.9 | 121.4 | 83.3 | 91.2 | 92.8 | 96.9 |
| Vaud | 97.1 | 107.1 | 126.2 | 108.8 | 121.9 | 109.4 |
| Valais | 118.6 | 116.4 | 117.4 | 134.4 | 115.9 | 130.1 |
| Neuchatel | 138.6 | 142.0 | 131.0 | 122.7 | 130.7 | 124.4 |
| Geneva | 101.3 | 118.8 | 115.2 | 112.8 | 111.6 | 113.3 |
| Jura | 111.8 | 105.8 | 134.3 | 133.4 | 131.0 | 130.5 |
| Average of Canton | 109.7 | 105.5 | 102.8 | 103.1 | 103.1 | 103.1 |
| Standard deviation | 17.2 | 19.2 | 18.7 | 18.8 | 18.8 | 18.8 |
| Coefficient of variation | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Maximum | 159.8 | 146.5 | 134.3 | 134.4 | 131.0 | 130.5 |
| Minimum | 67.2 | 58.0 | 66.8 | 55.6 | 68.0 | 57.3 |
| Range | 92.6 | 88.5 | 67.5 | 78.8 | 63.0 | 73.2 |

1. Switzerland, calculated as a weighted average, has a base value of 100. Each canton is represented by its capital city.
2. Total index of the burden on net profits and capital.
3. Global index of the burden on the income and wealth of physical persons, taking into account changes in income caused by inflation.
4. Global index of tax burden taking into account changes in personal income caused by inflation. This is the weighted average of the tax burden indexes on income, net profits and capital and motor vehicles.

*Source:* Administration fédérale des Contributions, *Charge fiscale en Suisse.*

Figure 28. **Fiscal burden indexes on households and companies**[1, 2]




1. For Households: global index of the tax burden on the income and wealth of physical persons, taking account of changes in income caused by inflation.
   For companies: total index of tax burden on net profits and capital.
2. Switzerland (a weighted average) has a base value of 100.

*Source:* Administration Fédérale des Contributions, *Charge fiscale en Suisse.*

burdens have mainly occurred by way of the tax on corporate profits (as opposed to the net worth tax).

### *Fiscal equalisation*

Fiscal equalisation is intended to compensate governments for services provided that benefit citizens in other jurisdictions as well as to reduce differences in wealth across the country. Most transfers are from the federal government to cantons. These "vertical transfers" amount to around 3 per cent of GDP. About three-quarters of these transfers are linked to expenditures. Approximately one-third of vertical transfers principally depend on cantons' index of financial strength. This index is based not only on the potential resources available, but also on the tax burden (the higher the tax burden, the higher the equalisation transfer). The index ranges from 30 to 206 (mean = 100). Such transfers, which are known as "financial equalisation in the narrow sense", include the following:

- Certain federal subsidies, which are subject to ceilings;
- Sharing of part of the cantons' share of federal tax revenues (13 per cent of the federal income tax, 10 per cent of the withholding tax);
- Sharing of fuel duties and Swiss National Bank profits;
- Sharing of the cantons' contributions to the federal government's social expenditures.

The redistribution effect of the transfers that principally depend on cantons' financial strength is weak: only 15 per cent of these transfers (or 5 per cent of total vertical transfers) go to cantons with a weak or average financial capacity. The transfers between cantons, which are relatively small, mainly reflect agreements that have been made to share the costs of health and education services that benefit citizens in neighbouring jurisdictions. Fiscal equalisation arrangements also exist between municipalities.

Consultation procedures are underway to reform fiscal equalisation. The present arrangements are considered to be too dependent on expenditures, weakening cantons' incentives to spend money efficiently, and not to reduce wealth inequalities adequately. Moreover, spatial spill-over effects between jurisdictions are considered not to be adequately compensated. This is creating financial difficulties for large cities, as they provide facilities of benefit to persons living outside the city. Such facilities include universities, cultural institutions and hospitals equipped to offer sophisticated medical services. In addition, large cities bear a disproportionate share of social assistance costs because many persons in difficulty are attracted to large cities by the anonymity that they offer. Key to the revised fiscal equalisation arrangements is the redistribution of expenditure responsibilities to the level of the decider-payer subject to the principle that responsibilities remain at the lowest feasible level of government. This should give cantons more responsibility for expenditure decisions and increase

the proportion of vertical transfers not tied to expenditures. The redistribution element of the system is to be reinforced and equalisation transfers are to depend on a new "resources index" that does not take into account the tax burden. The aim is eventually to reduce the range of the tax-burden index to 85-120 (mean = 100). In addition, there will be legal obligations for cantons to negotiate co-financing arrangements for services that generate spatial spill-over effects.

### *Progressiveness of tax system*

Switzerland's tax system is relatively progressive by international comparison. The sum of the marginal income tax rate and social security contribution rate for a top rate earner is some 20 percentage points higher than for the average production worker (APW) (Table 24).[94] Over a wide range of incomes (up to three times APW earnings) which covers most taxpayers, the rate schedule (also allowing for cash social transfers) rises steadily, whereas it is much flatter in neighbouring countries and the United States (Figure 29). In line with these results, the effective distribution of personal income tax burdens in Switzerland is quite progressive.[95]

Even so, redistribution in Switzerland principally occurs through government expenditures. About 63 per cent of the reduction in the Gini-coefficient between factor income and "final incomes" (*i.e.* factor incomes augmented by public expenditure and reduced by taxes under standard incidence assumptions) is attributable to government expenditures, with the remainder being attributable to taxes (Kirchgässner and Pommerehne, 1996) (Table 25). This is comparable to the amount of redistribution through expenditures in the United States, Canada and Germany, the other federations included in the aforementioned study. What is unusual about Switzerland compared to these other federations is that most of the redistribution through taxation occurs at sub-central levels of government (despite there being no great difference in the proportion of revenues raised at lower levels of government – see Table 13). Allowing for both expenditures and taxes, the amount of redistribution in Switzerland is approximately the same as in the United States, somewhat higher than in Canada and a little lower than in Germany (*ibid.* p. 365).

### *Labour tax wedges*

The wedge between labour earnings before and after tax generally reduces employment.[96] In the adjustment phase to a new equilibrium, a higher labour-tax wedge also increases unemployment. The more flexible are real wage rates, the less an increase in the labour-tax wedge reduces employment and, during the transition phase, increases unemployment. The total labour tax

Table 24. **Progressiveness: average production worker versus top bracket marginal tax rates**[1]

Percentage points of taxation

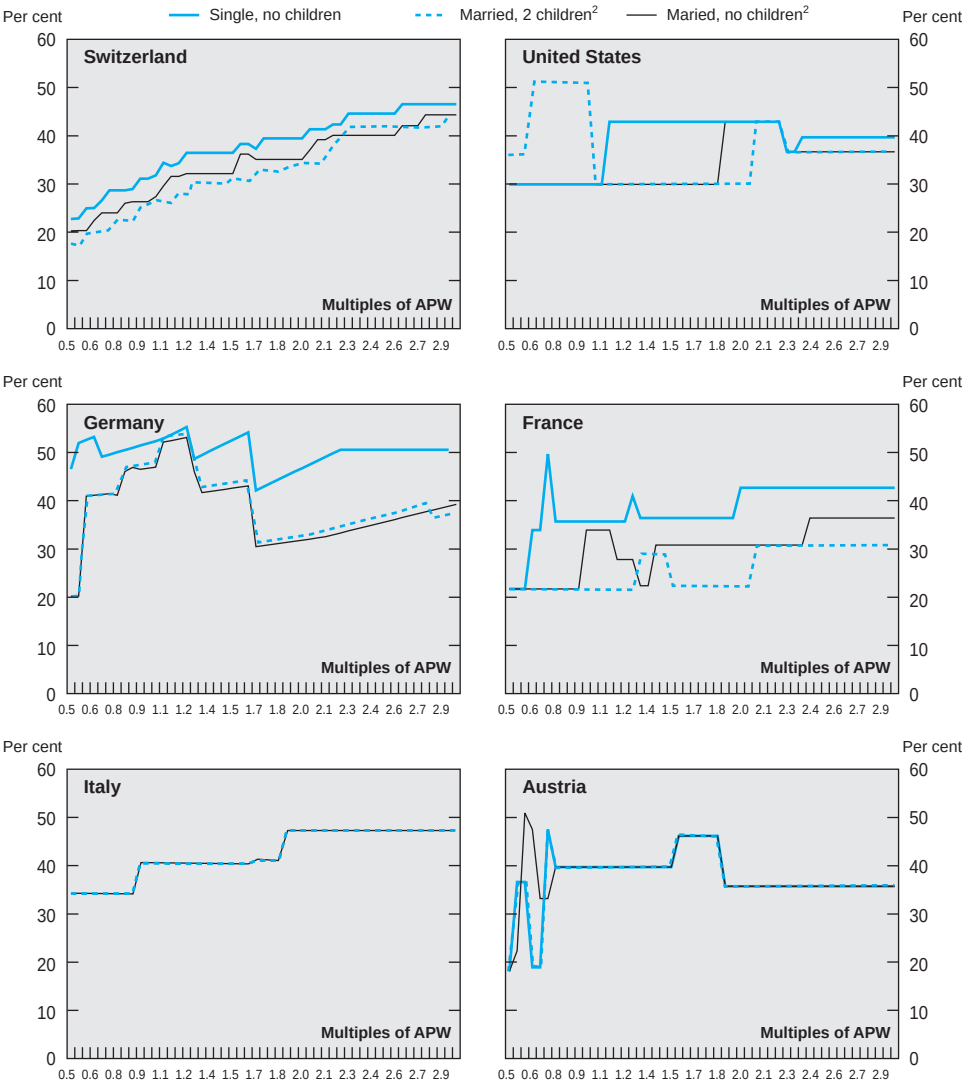
| | Average Production Worker's Rate (APW)[2] (A) | Top marginal tax rate and social security contributions for top bracket earner[3] (B) | Difference top rate and APW (B – A) |
|---|---|---|---|
| United States | 29.9 | 48.1 | 18.1 |
| Japan | 21.6 | 65.0 | 43.4 |
| Germany | 55.2 | 55.9 | 0.7 |
| France | 35.4 | 61.6 | 26.2 |
| Italy | 40.7 | 50.8 | 10.1 |
| United Kingdom | 33.0 | 40.0 | 7.0 |
| Canada | 45.0 | 54.1 | 9.1 |
| Australia | 35.5 | 48.5 | 13.0 |
| Austria | 42.5 | 50.0 | 7.5 |
| Belgium | 55.9 | 66.1 | 10.2 |
| Czech Republic | 30.0 | 47.5 | 17.5 |
| Denmark | 52.1 | 62.4 | 10.3 |
| Finland | 50.7 | 62.0 | 11.3 |
| Greece | 20.1 | .. | .. |
| Hungary | 46.5 | 43.5 | –3.0 |
| Iceland | 39.2 | 49.6 | 10.4 |
| Ireland | 54.7 | 50.2 | –4.5 |
| Korea | 8.1 | 45.9 | 37.8 |
| Luxembourg | 43.1 | 46.6 | 3.5 |
| Mexico | 17.6 | 35.0 | 17.4 |
| Netherlands | 55.1 | 60.0 | 4.9 |
| New Zealand | 33.0 | 33.0 | 0.0 |
| Norway | 45.3 | 49.5 | 4.2 |
| Poland | 20.0 | 40.0 | 20.0 |
| Portugal | 26.0 | 46.6 | 20.6 |
| Spain | 26.8 | 56.0 | 29.2 |
| Sweden | 35.7 | 59.6 | 23.9 |
| **Switzerland**[4] | **31.1** | **51.4** | **20.3** |
| Turkey | 30.5 | 61.3 | 30.8 |

1. This measure of progressivity is based on the difference between the top rate paid and the average production worker's rate. The figures for the APW are for the year 1997, while the figures for the top bracket earner are for the year 1998.
2. Marginal tax rate covering employee's social security contributions and personal income tax for single worker, no children.
3. This is the sum of the standard rate, net surcharges and net sub-central highest rates. The net rate is different from the legal rate whenever one of the taxes is deductible from the other. It includes social security paid by employees.
4. Figures for Switzerland are for the canton of Zurich (Zurich city).

*Source:* OECD, *The Tax/Benefit Position of Employees* and the OECD Tax database.

wedge[97] for a single person in Switzerland rises quickly to around 40 per cent at a little over APW earnings, and then rises slowly to around 50 per cent at three times APW earnings (Figure 30). The wedges for married persons are a little lower. These wedges are comparable to those in the United States but are below those

Figure 29. **Marginal tax rates[1] – selected countries, 1996**




1. Marginal total payments by employees (income tax plus social security contributions) less cash transfers as a percentage of their gross income from labour (this excludes employer contributions to social security). Compulsory employee contributions to private pension schemes (which should be considered as savings rather than taxes) are included as taxes.
2. Second income is zero.

*Source:* OECD, *The Tax/Benefit Position of Employees, Tax Equations* and OECD Secretariat.

Table 25. **Gini coefficients for selected definitions of income**

Switzerland, 1987

| | Gini Coefficients (× 1 000) |
|---|---|
| 1. Factor income | 334 |
| 2. Final income | 226 |
| Difference | 108 |
| Percentage of differences attributable to | |
| 1. Budget side | |
| *a)* Taxes | 37.4% |
| *b)* Expenditure | 62.6% |
| 2. Government level | |
| *i)* Central government | 37.5% |
| *a)* Taxes | 7.3% |
| *b)* Expenditure | 30.2% |
| *ii)* Sub-central government | 62.5% |
| *a)* Taxes | 30.1% |
| *b)* Expenditure | 32.4% |

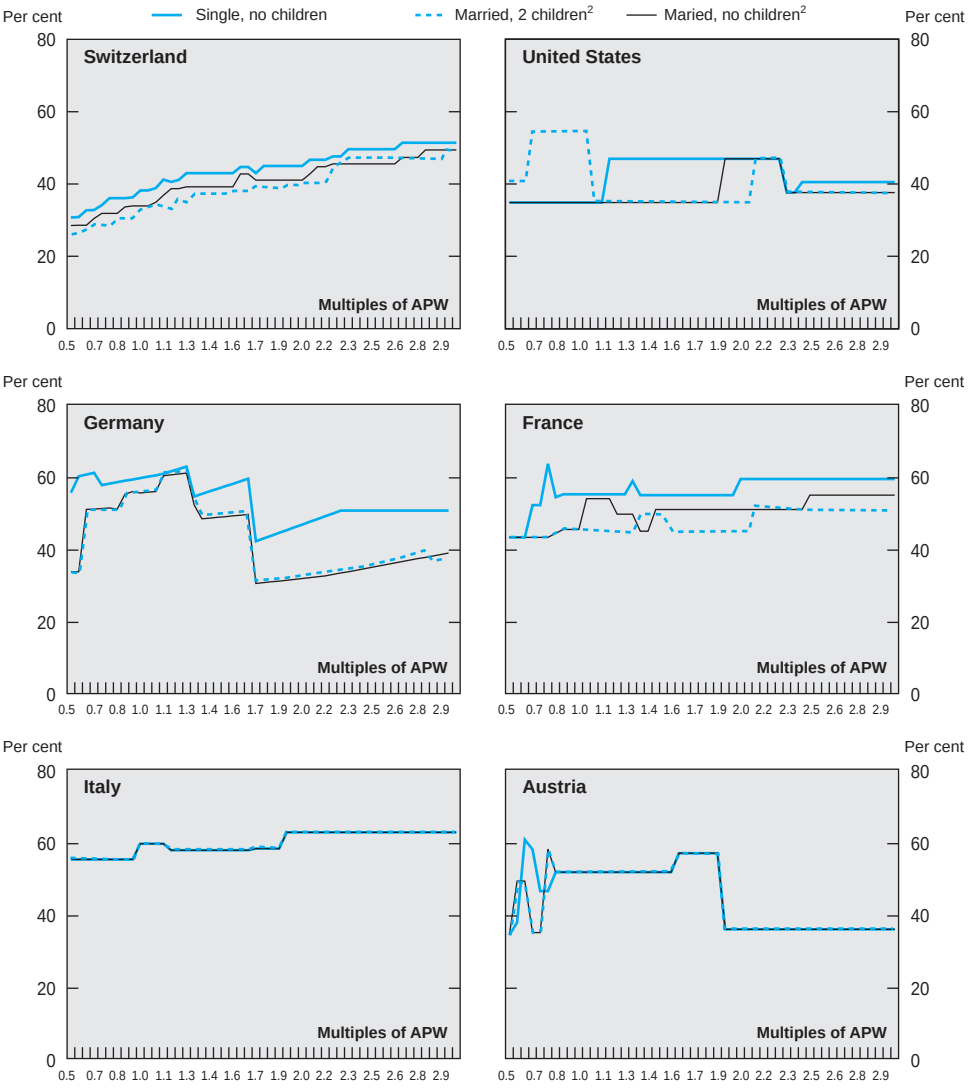
*Source:* G. Kirchgässner, W.W. Pommerehne (1996).

in Switzerland's neighbours, especially for low paid workers. The Swiss wedges are also below the OECD average and well below the EU average. If consumption taxes were also taken into account (they also affect the incentive to supply labour), the Swiss and US wedges would be even lower compared with most other OECD countries. As real wage rate flexibility in Switzerland appears to about average for an OECD country,[98] the labour tax wedge is likely to have less damaging effects on employment and unemployment than those in most other countries.

### *Capital tax wedges*

The OECD Secretariat has measured marginal effective tax rates for three types of asset – machinery, buildings and inventories – and for three funding sources – debt, new equities (*i.e.* sources that pay dividends) and retained earnings (*i.e.* sources that pay capital gains) (Table 26). These are based on statutory parameters on both personal and corporate taxes for nine cantons[99] (they also include federal taxes). The estimates reflect the complex array of instruments used in personal and corporate direct taxation in Switzerland. This includes progressive rates of return taxes on corporations (top rates are taken) and net wealth taxes on both corporations and individuals. The estimates show that, on the asset side, cantonal and federal tax practices tend to favour buildings and

Figure 30. **Marginal total labour tax wedges[1] – selected countries, 1996**




1. This is the difference between gross labour costs to the employer (including employer contributions to social security and compulsory employer contributions to pension funds) and employees' net income from labour after deduction of income taxes, social security contributions and compulsory contributions to private pension schemes (which should be considered as savings rather than taxes).
2. Second income is zero.

*Source:* OECD, *The Tax/Benefit Position of Employees, Tax Equations* and OECD Secretariat.

Table 26. **Marginal effective tax rates in manufacturing – selected cantons**[1, 2]

Wealth taxes included – per cent, 1998

| | Sources of finance | | | Physical assets | | | Overall average | Standard deviation |
|---|---|---|---|---|---|---|---|---|
| | Retained earnings | New equity | Debt | Machinery | Building | Inventories | | |
| **I. Pre-tax required real rate of return**[3] | | | | | | | | |
| Aargau | 4.2 | 6.4 | 4.5 | 4.2 | 4.6 | 5.0 | 4.5 | 0.9 |
| Basel-City | 3.8 | 6.3 | 4.5 | 4.1 | 4.4 | 4.8 | 4.3 | 0.9 |
| Bern | 3.8 | 6.2 | 4.5 | 4.0 | 4.4 | 4.7 | 4.3 | 0.9 |
| Geneva | 3.5 | 6.3 | 4.5 | 3.9 | 4.2 | 4.6 | 4.1 | 1.0 |
| Nidwalden | 4.2 | 5.9 | 4.6 | 4.3 | 4.6 | 4.9 | 4.5 | 0.6 |
| Schwyz | 4.5 | 6.1 | 4.6 | 4.4 | 4.8 | 5.1 | 4.7 | 0.6 |
| Vaud | 3.7 | 6.3 | 4.5 | 3.9 | 4.3 | 4.7 | 4.2 | 0.9 |
| Zug | 4.4 | 5.8 | 4.7 | 4.4 | 4.7 | 5.0 | 4.6 | 0.5 |
| Zurich | 3.7 | 6.2 | 4.5 | 3.9 | 4.3 | 4.7 | 4.2 | 0.9 |
| Average | 4.0 | 6.2 | 4.5 | 4.1 | 4.5 | 4.8 | 4.4 | 0.8 |
| Standard deviation | 0.4 | 0.2 | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Coefficient of variation | 0.1 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.2 |
| **II. Effective marginal tax wedges**[4] | | | | | | | | |
| Aargau | 1.6 | 3.9 | 1.9 | 1.7 | 2.1 | 2.5 | 2.0 | 0.8 |
| Basel-City | 1.0 | 3.4 | 1.6 | 1.2 | 1.6 | 1.9 | 1.5 | 0.8 |
| Bern | 8.0 | 10.4 | 8.7 | 8.2 | 8.6 | 8.9 | 8.5 | 0.8 |
| Geneva | 1.7 | 4.5 | 2.7 | 2.1 | 2.4 | 2.8 | 2.3 | 0.9 |
| Nidwalden | 0.8 | 2.4 | 1.2 | 0.9 | 1.2 | 1.5 | 1.1 | 0.5 |
| Schwyz | 1.3 | 2.9 | 1.4 | 1.2 | 1.6 | 1.9 | 1.5 | 0.5 |
| Vaud | 1.7 | 4.4 | 2.6 | 2.0 | 2.4 | 2.8 | 2.3 | 0.9 |
| Zug | 1.0 | 2.4 | 1.2 | 1.0 | 1.3 | 1.6 | 1.2 | 0.5 |
| Zurich | 1.5 | 4.0 | 2.3 | 1.8 | 2.1 | 2.5 | 2.0 | 0.8 |
| Average | 2.1 | 4.3 | 2.6 | 2.2 | 2.6 | 2.9 | 2.5 | 0.7 |
| Standard deviation | 2.2 | 2.4 | 2.4 | 2.3 | 2.3 | 2.3 | 2.3 | 0.2 |
| Coefficient of variation | 1.1 | 0.6 | 0.9 | 1.0 | 0.9 | 0.8 | 0.9 | 0.2 |

1. Top personal taxes, real rate assumed to be 5 per cent. Asset weighting is as follows: machinery 50 per cent; buildings 28 per cent; inventories 22 per cent. Finance weighting as follows: retained earnings 55 per cent; new equity 10 per cent; debt 35 per cent. Economic depreciation rates are: machinery 12.5 per cent; buildings 3.6 per cent; inventories 0 per cent. See OECD (1991) for discussion of this methodology.
2. Each canton is represented by its capital city.
3. This is the rate of return (p) on an investment that must be earned to give a household the same after-tax rate of return as a bank deposit earning 5 per cent.
4. This is the difference between the pre-tax required rate of return (p) and the rate of return after all taxes (s), assuming a 5 per cent real rate of return before personal taxes.

*Source:* OECD Secretariat.

equipment over inventories. In this respect, the cantons exhibit patterns typical of nearly all OECD countries (Table 27) (OECD, 1991 and Gordon and Tchilinguirian, 1998). However, among the cantons, this discrimination is not very

Table 27. **Marginal effective tax wedges in manufacturing: selected OECD countries**[1]

Excluding wealth taxes – per cent, 1998

| | Sources of financing[2] | | | Physical assets[3] | | | Overall weighted average[4] | Standard deviation |
|---|---|---|---|---|---|---|---|---|
| | Retained earnings | New equity | Debt | Machinery | Building | Inventories | | |
| United States | 1.9 | 5.0 | 1.7 | 1.5 | 2.8 | 2.7 | 2.1 | 1.3 |
| Japan | 4.2 | 6.4 | 0.1 | 1.5 | 4.9 | 3.9 | 3.0 | 2.3 |
| Germany | 1.5 | 1.0 | 1.4 | 1.0 | 1.7 | 2.1 | 1.4 | 0.4 |
| France | 3.7 | 6.9 | 0.8 | 2.2 | 3.8 | 3.8 | 3.0 | 2.1 |
| Italy | 2.3 | 2.6 | 0.6 | 1.0 | 1.8 | 3.3 | 1.7 | 1.0 |
| United Kingdom | 2.2 | 2.8 | 1.8 | 1.7 | 2.1 | 3.1 | 2.2 | 0.6 |
| Canada | 4.3 | 5.0 | 1.3 | 2.2 | 4.1 | 4.8 | 3.3 | 1.5 |
| Australia | 2.7 | 2.5 | 2.5 | 2.1 | 2.8 | 3.4 | 2.6 | 0.5 |
| Austria | 0.8 | 2.7 | 0.1 | –0.1 | 1.0 | 2.3 | 0.7 | 1.1 |
| Belgium | 1.4 | 2.6 | –0.6 | 0.1 | 0.8 | 2.7 | 0.9 | 1.3 |
| Denmark | 2.4 | 3.1 | 2.9 | 2.2 | 2.5 | 3.8 | 2.6 | 0.6 |
| Finland | 2.3 | 0.9 | 0.9 | 1.3 | 1.8 | 2.6 | 1.7 | 0.7 |
| Greece | 2.4 | 2.4 | –0.2 | 1.1 | 0.5 | 3.4 | 1.4 | 1.3 |
| Iceland | 2.1 | 2.6 | –0.0 | 1.0 | 1.6 | 2.2 | 1.4 | 1.0 |
| Ireland | 1.7 | 3.7 | 2.6 | 1.8 | 2.1 | 3.3 | 0.7 | 0.8 |
| Luxembourg | 3.6 | 2.4 | 1.7 | 2.2 | 2.8 | 4.2 | 2.8 | 1.0 |
| Mexico | 1.3 | 1.3 | 0.4 | 0.7 | 0.8 | 1.7 | 1.0 | 0.5 |
| Netherlands | 0.6 | 6.4 | 3.0 | 1.8 | 2.3 | 2.0 | 2.0 | 2.0 |
| New Zealand | 1.7 | 1.7 | 1.7 | 1.7 | 1.5 | 2.1 | 1.7 | 0.2 |
| Norway | 1.3 | 1.3 | 1.3 | 1.0 | 1.3 | 2.1 | 1.3 | 0.4 |
| Portugal | 2.1 | 4.7 | –0.1 | 1.6 | 1.5 | 1.9 | 1.6 | 1.6 |
| Spain | 3.9 | 3.2 | 2.6 | 3.1 | 3.6 | 3.7 | 3.4 | 0.5 |
| Sweden | 2.2 | 2.9 | 0.9 | 1.5 | 1.9 | 2.5 | 1.8 | 0.7 |
| **Switzerland**[5] | **0.8** | **3.7** | **1.5** | **0.9** | **1.4** | **2.0** | **1.3** | **1.1** |
| Turkey | .. | .. | .. | .. | .. | .. | .. | .. |
| OECD average[6] | 2.2 | 3.2 | 1.2 | 1.5 | 2.1 | 2.9 | 1.9 | 1.0 |
| G7 average[6] | 2.9 | 4.3 | 1.1 | 1.6 | 3.0 | 3.4 | 2.4 | 1.3 |
| EU average[6] | 2.2 | 3.2 | 1.2 | 1.5 | 2.0 | 3.0 | 1.9 | 1.0 |

1. These indicators show the degree to which the personal and corporate tax systems scale up (or down) the real pre-tax rate of return that must be earned on an investment, given that the household can earn a 5 per cent real rate of return on a bank deposit. Wealth taxes are excluded. See OECD (1991) for discussion of this methodology.
2. The weighted average uses the following weights: machinery 50 per cent, buildings 28 per cent, inventories 22 per cent.
3. The weighted average uses the following weights: retained earnings 55 per cent, new equity 10 per cent, debt 35 per cent.
4. The weighted average uses weights indicated in footnotes 2 and 3.
5. Figures for Switzerland are for the canton of Zurich (Zurich city). They are lower than in the preceding table because wealth taxes are excluded.
6. Averages are simple averages across available countries.

*Source:* OECD Secretariat.

pronounced. This suggests that "normal tax practice" involves a reasonably close alignment of economic and tax depreciation and that there are few other incentives for physical investment in the general tax regime.[100] However, the Swiss

personal and corporate tax codes favour financing by retained earnings and, to a lesser extent, debt. New equity is fairly strongly discriminated against due to the lack of an imputation system or other form of relief from the double taxation of dividends. According to these estimates, financing by retained earnings is slightly favoured relative to debt, reflecting the absence of a personal capital gains tax and higher personal income tax rates than corporate tax rates. These numbers also suggest that the cantons are not very different from one another when it comes to the investment and financing incentives created by their general corporate and income tax systems. The differences in their overall average tax wedges are quite small (the range is far less than 1 percentage point of effective marginal taxation, which is far less than the range for the OECD as a whole).[101] One noteworthy feature of the numbers is the degree to which the cantons use a complex range of tax instruments (indeed, one whose complexity is rivalled by no other OECD country) in order to end up with an overall investment incentive structure that is neither "best practice" nor "worst practice".

### *Average effective tax rates*

The OECD Secretariat has also calculated average effective tax rates (AETR) for capital, labour and consumption. These are based on measured revenue flows and on National Accounts statistics.[102] Compared with the averages for the EU and for the OECD, Switzerland's AETRs in 1991-96 are about the same for capital (based on gross operating surplus[103] and including all property income), at 25 per cent, low for labour, at 30 per cent (compared with 36 per cent for the EU and 33 per cent for the OECD), and extremely low for consumption, at 8 per cent (compared with 21 per cent for the EU and 17 per cent for the OECD). These estimates overstate the AETR on capital and understate the AETR on labour in countries that have favourable tax treatment for savings with pension funds and/or life insurance companies. For Switzerland, adjusting the estimates for the fact that no tax is paid on earnings as they accumulate reduces the AETR on capital in 1991-96 by 6 percentage points and raises the AETR on labour by 2 percentage points. As pension fund and life insurance assets as a share of GDP are greater in Switzerland than in most other OECD countries (see below), this adjustment makes a greater difference to the estimates for Switzerland than those for most other countries. Allowing for this factor, Switzerland has low taxation of capital income and moderate taxation of labour income by international comparison.

A much more important influence on the estimates of the AETR on labour is the fact that Switzerland's second pillar of the retirement income system is privately funded. If these contributions were instead social security contributions, Switzerland's AETR on labour in 1991-96 would have been 9 percentage points higher, bringing it into line with the rates in many European countries.

## Main issues for strengthening the system

An economic analysis of a tax system focuses on its effects on economic efficiency and equity. Efficiency calls for taxes which minimise distortions in otherwise efficient economic choices (minimise excess burden) and in administrative and compliance costs, subject to the constraints due to equity objectives. Taxes on negative externalities (such as man-made pollution) also contribute to economic efficiency. Equity generally calls for taxation according to ability to pay. The most widely accepted measure of this is comprehensive income (*i.e.* total income, regardless of its source or form). Trade-offs are often required between these objectives. Notably, society may prefer to have more redistribution at the expense of less economic efficiency or vice versa. Sometimes, reforms are possible that would enhance both efficiency and equity.

Tax systems may also reflect the pursuit of other objectives and/or the effects of political constraints. These objectives and/or constraints may be at the expense of efficiency and/or equity. In the case of Switzerland, the desire to maintain a highly decentralised political structure precludes a number of reforms that could enhance economic efficiency and/or equity from a national viewpoint. Another important point to bear in mind is that tax reforms are complementary with each other and with other reforms. If certain parts of a package are unlikely to be implemented, this must be taken into account in designing the rest of the package. For example, the desirability of introducing a personal capital gains tax in Switzerland depends on the structure of other taxes on capital income and the prospects for reforming them. Switzerland's political structure and the chances of implementing specific tax reforms condition the recommendations made in this section.

### *Reduce tax privileges for savings with pension funds and insurance companies*

Compared to the standard treatment under a comprehensive income tax, present arrangements for the taxation of savings with pension funds and life insurance companies are very favourable (Box 1). Tax incentives for these forms of long-term saving are intended to assist individuals to accumulate sufficient wealth to finance a broadly similar (but reasonable) living standard in retirement to that enjoyed while working. Although contributions to earnings-related pension schemes have only been compulsory since 1985, the Swiss have nevertheless accumulated substantial pension fund assets (75 per cent of GDP in 1996) by international comparison. Life insurance company assets (61 per cent of GDP in 1996) (Tables 28 and 29) are also high. Indeed, the combined total of pension fund and life insurance assets (136 per cent of GDP) is among the highest in the OECD. These figures certainly suggest that these incentives contribute to the accumulation of this form of long-term savings (as must do the mandatory nature

### Box 1. Tax treatment of savings with pension funds and life insurance companies

Earnings-related contributions to pension funds (second pillar) are tax deductible provided that contributions are made in accordance with the rules and conditions of the particular pension fund. There is no absolute upper limit on earnings-related contributions. Voluntary contributions by taxpayers not affiliated to a second pillar scheme and voluntary contributions by employees that are paid in addition to the contributions to their second pillar (pillar *3a*) are tax deductible as well. However, contributions to pillar *3a* are limited to SF 28 656 (for taxpayers not affiliated to a second pillar) and to SF 5 731 (for taxpayers affiliated to a second pillar fund) per year. Life insurance premiums (pillar *3b*) are not deductible. The earnings on all these funds accumulate tax-free. Distributions from deductible contributions (pillars 2 and *3a*) are taxed. Annuities are fully taxed with other income while lump sum payments are taxed separately from other income at a lower rate, partly to compensate for the effects of the progressive tax schedule, partly as compensation for the loss of interest due to the early payment of the tax. In the case of the Federal Income Tax, this rate is one-fifth the ordinary rate, while cantons tax such lump-sum payments at a variety of favourable rates according to their own legislation. Distributions from life insurance policies are exempt from tax as lump sums. If the proceeds are used to purchase an annuity, it is taxed after making an adjustment to exclude the part of the annuity that represents a return of capital (this adjustment reduces the annuity by 40 per cent under Swiss tax law). The tax treatment of each of the stages in constituting long-term savings (contributions, fund earnings and distribution of benefits) can be summarised as exempt-exempt-taxed (EET) for pension funds and taxed-exempt-exempt (TEE) for life insurance policies. Many other OECD countries have similar arrangements.

These arrangements are all more favourable than the normal treatment (as is accorded savings in a bank account, for example) under a comprehensive income tax, which is TTE. The main concession is that earnings are not taxed as they accumulate.* For life insurance policies (TEE), this means that taxation does not affect the incentive to save. The taxation of pension-fund annuities (EET) is even more favourable for two reasons. First, marginal personal income tax rates in retirement are generally lower than the weighted average of corporate and personal rates on income from which contributions were deducted. And second, taxation partly depends on investment returns, reducing risk. Pension fund savings distributed as lump sums receive still more favourable treatment, as benefits are taxed at reduced rates. In addition, all savings with pension funds are exempt from personal wealth taxes and estate duties.

* Assuming certainty and proportional income tax rates, this makes the taxation of pension-fund (*i.e.* pillars 2 and *3a*) annuities (EET) and of life insurance policies (TEE) the same as expenditure taxation. Under these assumptions, neither tax arrangement affects the incentive to save.

of most pension fund contributions). On the other hand, the rate of owner-occupied housing, the other main vehicle for long-term saving, is very low by international comparison (Figure 31). The Swiss appear to prefer (or, to some extent are obliged) to hold a larger proportion of their long-term savings in

Table 28. **Financial assets of pension funds**

As a percentage of GDP

| | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|---|---|
| Australia | 17.0 | 21.6 | 21.3 | 27.3 | 27.0 | 29.1 | 30.4 | 30.7 |
| Austria | .. | 0.5 | 0.5 | 0.6 | 0.7 | 1.0 | 1.2 | .. |
| Belgium | 2.0 | 2.8 | 2.5 | 2.9 | 2.9 | 3.7 | 4.1 | .. |
| Canada | 28.8 | 30.7 | 31.3 | 34.0 | 35.9 | 38.6 | 40.7 | 43.3 |
| Czech Republic | .. | .. | .. | .. | 0.1 | .. | .. | .. |
| Denmark | 14.6 | 15.5 | 14.4 | 16.8 | 17.2 | 16.8 | 16.9 | .. |
| Finland | .. | .. | .. | .. | .. | .. | .. | .. |
| France | .. | .. | .. | .. | .. | .. | .. | .. |
| Germany | 3.1 | 3.3 | 2.9 | 2.5 | 2.7 | 2.7 | 2.8 | 2.9 |
| Greece | 6.5 | 7.1 | 6.9 | 8.0 | 10.2 | 10.8 | 11.9 | .. |
| Hungary | .. | .. | .. | .. | .. | .. | .. | .. |
| Iceland | 37.3 | 42.1 | 40.8 | 46.8 | 55.0 | 57.6 | 62.5 | 66.5 |
| Italy | 3.5 | 4.3 | 3.1 | 3.4 | 3.5 | 3.6 | 3.2 | 2.9 |
| Japan | .. | .. | .. | .. | .. | .. | .. | .. |
| Korea | 3.1 | 2.9 | 3.2 | 3.4 | 3.3 | 3.1 | 2.8 | 1.8 |
| Luxembourg | 17.7 | 18.5 | 17.2 | 17.2 | 18.7 | 18.2 | 18.2 | .. |
| Netherlands | 81.0 | 83.5 | 76.0 | 83.0 | 87.0 | 88.3 | 93.3 | 102.0 |
| Norway | 4.4 | 4.9 | 4.5 | 5.5 | 6.4 | 6.4 | 6.5 | .. |
| Portugal | 1.6 | 2.6 | 2.9 | 5.0 | 6.6 | 8.3 | 9.1 | 10.1 |
| Spain | 2.9 | 3.2 | 2.5 | 2.6 | 2.4 | 2.2 | 2.0 | 2.0 |
| Sweden | 1.7 | 1.7 | 1.6 | 2.0 | 2.2 | 2.4 | 2.4 | .. |
| **Switzerland** | **60.3** | **..** | **60.6** | **..** | **72.5** | **..** | **75.1** | **..** |
| Turkey | .. | .. | .. | .. | .. | .. | .. | .. |
| United Kingdom | 55.0 | 59.4 | 52.7 | 72.7 | 64.7 | 68.7 | 77.5 | .. |
| United States | 44.9 | 50.6 | 51.9 | 54.4 | 53.3 | 59.4 | 64.4 | 72.5 |

*Source:* OECD (1998*c*), *Institutional Investors Statistical Yearbook.*

pension funds and life insurance companies than do persons from other countries.

From an economic perspective, the question arises as to why incentives are required to encourage individuals to save more for retirement than they otherwise would. Assuming that individuals are rational, the main argument for incentives is that they are necessary to counter the moral hazard inherent in social assistance arrangements. In other words, people may choose not to save enough for retirement so as to be able to receive social assistance. But this argument does not apply in Switzerland because participation in the first[104] and much of the second pillars[105] of the retirement income system is compulsory. Another argument for privileged tax treatment is that this reduces the disincentive to save inherent in an income tax (it reduces the return to saving, whereas an expenditure tax does not). There is considerable debate amongst economists

Table 29. **Financial assets of insurance companies**

As a percentage of GDP

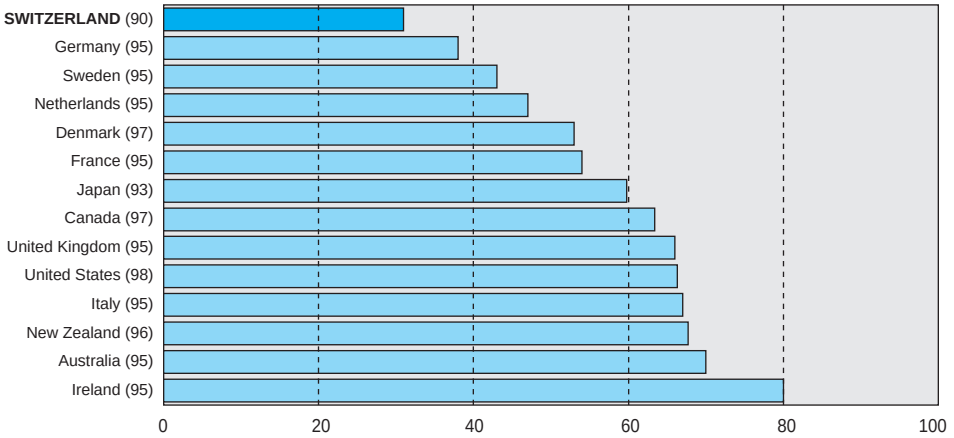
| | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|---|---|
| Australia | 24.6 | 27.7 | 30.0 | 36.2 | 35.0 | 34.6 | 38.6 | 35.7 |
| Austria | 15.4 | 16.3 | 15.7 | 17.2 | 19.3 | 20.3 | 21.0 | .. |
| Belgium | 28.3 | 29.5 | 26.5 | 27.1 | 30.4 | 29.9 | 30.9 | .. |
| Canada | 24.0 | 25.7 | 25.5 | 27.1 | 26.8 | 28.5 | 29.7 | 29.5 |
| Czech Republic | .. | .. | .. | 5.5 | 6.0 | 0.2 | 0.3 | 0.3 |
| Denmark | 38.3 | 41.0 | 36.8 | 41.0 | 44.2 | 43.7 | 45.1 | .. |
| Finland | 6.5 | 7.3 | 7.1 | 8.8 | 13.4 | 12.3 | 14.0 | .. |
| France | 22.0 | 25.0 | 24.9 | 31.9 | 34.3 | 39.5 | 45.8 | 52.6 |
| Germany | 24.4 | 24.9 | 22.1 | 23.7 | 27.1 | 28.3 | 29.5 | 31.9 |
| Greece | .. | 1.7 | 1.6 | 1.9 | 2.4 | 3.2 | 3.5 | .. |
| Hungary | .. | 2.5 | 2.4 | 2.5 | 2.9 | 2.9 | 3.8 | .. |
| Iceland | 4.2 | 4.5 | 6.8 | 7.9 | 9.3 | 9.2 | 9.4 | 9.2 |
| Italy | 6.0 | 6.8 | 6.0 | 7.7 | 9.8 | 11.1 | 12.0 | 13.4 |
| Japan | 36.2 | 37.0 | 36.2 | 37.8 | 41.0 | 38.0 | 37.9 | 38.1 |
| Korea | 18.9 | 19.6 | 21.3 | 21.7 | 22.1 | 23.5 | 24.9 | 15.9 |
| Luxembourg | .. | .. | .. | .. | .. | 44.5 | .. | .. |
| Mexico | 1.0 | 1.1 | 1.2 | 1.3 | 0.9 | 1.2 | 1.5 | 1.5 |
| Netherlands | 41.1 | 45.8 | 43.2 | 47.5 | 51.8 | 54.3 | 56.6 | 60.3 |
| New Zealand | .. | .. | .. | .. | .. | 11.1 | 12.1 | 10.8 |
| Norway | 29.3 | 30.7 | 26.3 | 29.8 | 32.4 | 31.2 | 30.4 | .. |
| Poland | .. | .. | .. | .. | 1.3 | 1.4 | 1.6 | .. |
| Portugal | 2.8 | 3.4 | 3.4 | 4.6 | 6.1 | 7.6 | 9.4 | .. |
| Spain | 9.9 | 11.2 | 10.3 | 11.8 | 15.7 | 17.5 | 18.6 | 20.4 |
| Sweden | 34.6 | 39.4 | 32.7 | 45.2 | 46.6 | 52.2 | 56.9 | .. |
| **Switzerland** | **50.5** | **52.7** | **50.8** | **55.6** | **61.4** | **62.9** | **61.3** | **71.9** |
| Turkey | 0.2 | 0.3 | 0.3 | 0.3 | 0.4 | 0.4 | 0.6 | .. |
| United Kingdom | 46.6 | 52.7 | 49.3 | 71.0 | 64.7 | 73.8 | 88.6 | .. |
| United States | 33.9 | 36.0 | 36.2 | 37.8 | 37.7 | 39.8 | 40.9 | 43.1 |

*Source:* OECD (1998*c*), *Institutional Investors Statistical Yearbook.*

about whether these schemes increase net saving or largely encourage households to switch the form in which they save.[106] Even if these arrangements do increase net saving, this is unlikely to be the most cost-effective way of doing so.

The favourable treatment accorded savings with pension funds and life insurance companies reduces economic efficiency by giving these institutions a tax-based competitive advantage over other financial intermediaries. This enables pension funds and life insurance companies to hold market share that other, more efficient financial intermediaries would otherwise hold. Economic efficiency is further undermined by the influence of tax incentives on the asset allocation choices of pension funds and life insurance companies.[107] At best, this

Figure 31. **Share of owner-occupied housing in selected OECD countries**[1]




1. Data in brackets is the census year.
*Source:* National sources and OECD Secretariat.

creates a regulation and/or tax-based clientele for different asset classes. In particular, pension funds and life insurance companies have a tax incentive to weight their portfolios towards bonds, loans and property, which generate taxable returns for other investors, at the expense of equities (as personal capital gains are not taxed). This is indeed the way that Swiss pension funds' portfolios are weighted (Table 30). This contributes to a sub-optimal allocation of savings on risk-return criteria. Long-term savings tend to be under invested in equities, which have a high return but high risk, while short-term savings are over invested in risky equities. As savings with pension funds and life insurance companies are large relative to other savings (see below), their investment choices will further undermine economic efficiency by distorting the allocation of investment resources. For example, there would tend to be over investment in property (and fewer owner-occupiers) at the expense of other forms of investment. Moreover, this narrowing of the tax base necessitates higher tax rates on remaining bases, increasing the excess burden of those taxes.

These arrangements also give rise to economic rent that is not competed away because of barriers to entry. Savers cannot switch to non-resident pension funds and life insurance companies as the tax benefits would no longer apply

Table 30. **Portfolios of pension funds**

Per cent of assets, 1994

| | Equities | Bonds and loans | Property | Liquidity and deposits | Foreign assets[1] |
|---|---|---|---|---|---|
| Australia | 48 | 27 | 8 | 17 | 16 |
| Austria | 11 | 75 | 2 | 12 | 20 |
| Belgium | 36 | 47 | 7 | 10 | 35 |
| Canada (1992) | 38 | 49 | 3 | 7 | 9 |
| Denmark | 22 | 65 | 9 | 4 | 7 |
| Finland | 5 | 73 | 12 | 10 | n.a. |
| France | 14 | 39 | 7 | 40 | 5 |
| Germany | 11 | 75 | 11 | 3 | 6 |
| Ireland | 55 | 35 | 6 | 4 | 37 |
| Italy | 9 | 62 | 23 | 6 | 5 |
| Japan (March 1994) | 27 | 61 | 2 | 3 | 7 |
| Luxembourg | 20 | 70 | 0 | 10 | n.a. |
| Netherlands | 23 | 67 | 9 | 2 | 17 |
| Portugal | 10 | 72 | 3 | 15 | n.a. |
| Spain | 4 | 82 | 1 | 13 | 5 |
| Sweden | 32 | 47 | 8 | 13 | 12 |
| **Switzerland** | **13** | **52** | **20** | **15** | **9** |
| United Kingdom | 80 | 11 | 6 | 3 | 30 |
| United States | 48 | 38 | 0 | 7 | 10 |
| Prudent man | 49 | 38 | 5 | 7 | 20 |
| Asset restrictions | 16 | 63 | 8 | 12 | 11 |

1. Foreign assets are included in the categories to the left.

*Source:* European Federation for Retirement Provision (1996), *European Pension Funds, Their Impact on Capital Markets and Competitiveness.*

while it would be costly for foreign institutions to set up permanent establishments in Switzerland. Pension funds and life insurance companies are able to retain some of the tax benefits through higher charges and fees and still remain competitive with other financial intermediaries. This reduces the incentive to long-term saving provided by the tax incentives and encourages pension funds and life insurance companies to invest resources (lobbying, for example) in obtaining these rents. While such activity may be profitable for the companies, it is not productive from a national viewpoint.

It can also be argued that the tax preferences for savings with pension funds and life insurance companies are unfair. Persons with the same comprehensive income but different proportions derived from savings with pension funds and life insurance companies will not pay the same amount of tax, violating horizontal equity. Similarly, a person with a lower comprehensive income than

another could nevertheless pay more tax if they have considerably less income derived from savings with these institutions. The tax preferences also reduce the progressiveness of the tax system because deductions and exemptions are worth more to persons on high marginal income tax rates.

The Behnisch Commission in Switzerland recently examined the tax treatment of savings with pension funds and life insurance companies in the context of an investigation into major lacuna in the Federal Income Tax on individuals. The Commission recommended that earnings-related contributions to pension funds be limited to a salary (SF 214 920, corresponding to a deduction of SF 36 536 at the average private sector contribution rate of 17 per cent) of three times the salary up to which contributions are compulsory. They noted that the relevant article (No. 34) in the Constitution calls for benefits under the first and second pillars of the retirement income system that enable beneficiaries to maintain their living standards in an appropriate measure, not to enjoy a luxurious or ostentatious living standard. Noting that Swiss tax law applies the principle that contributions to pension funds should be deductible, fund earnings should be tax free as they accumulate and distributions should be fully taxed, the Commission also recommended that earnings-related benefits taken as lump sums should be taxed along with other income. This would entail adding the life annuity that could be bought with the lump sum to other income to determine the average tax rate on such an annuity and applying this rate to the lump sum, as was done up until 1986. The Commission added that the current treatment was particularly inappropriate because it encouraged people to take their benefits as lump sums rather than as pensions.[108] The Federal Council adopted the Behnisch Commission's recommendations with one main modification, that earnings-related benefits taken as lump sums be taxed at half the normal rate. In the event, Parliament rejected the Commission's main proposals. Hence, the major features of long-term savings arrangements remain unchanged.

Economic efficiency could be enhanced by abolishing all tax privileges for this form of saving and using the revenue to reduce taxes that impose high economic costs. This could include lower taxes on saving in general. The removal of tax privileges would entail the taxation of earnings of pension fund and life insurance policies as they accumulate. If such a reform were to be made, it would be vital to maintain the compulsory elements of the retirement income system so as to avoid the moral hazard problem of people not providing adequately for their retirement in order to obtain social assistance benefits.

### *Limit personal interest deductions to declared capital income*

Households in all but five cantons are allowed unlimited deductions for interest expenses. If income flows from all assets were fully taxed, this treatment would be consistent with a comprehensive income tax. However, income flows

from all assets are not fully taxed. The most important case in point is assets held in insurance companies. Households are able to reduce their tax liability by borrowing to acquire such assets. This gives rise to a deduction for interest expenses but no extra taxable income from the investments, compounding the attractiveness of such investments. However, there are a number of factors that restrain households from unduly reducing their income tax liability:

- The courts have established rules defining what constitutes tax avoidance in this area. If a given situation in this field is considered to constitute tax avoidance, interest is not deductible;
- The fees charged by life insurance companies (these include entry costs, management fees and risk premiums, which have a negative influence on returns);
- The 2½ per cent stamp duty on single premium insurance policies;
- The lack of liquidity of the assets;
- The desire to maintain prudent gearing ratios, which may also be imposed through credit rationing; and
- The risk that government withdraws some or all of the tax privileges that create this tax avoidance opportunity.

Owner-occupied housing also generates income that is not fully taxed.

The Behnisch Commission recommended that household interest deductions should be limited to the amount of declared capital income or 50 per cent of gross income, whichever is greater. This is the model followed in Geneva. The Federal Council took up this issue in its 1998 Stabilisation Programme, proposing that household interest expenses should be deductible up to SF 20 000 more than declared capital income. The possibility of declaring a loss on the capital account up to SF 20 000 was intended to help taxpayers over-burdened with debts but with little capital income and to enable new home owners to declare a loss on their housing accounts. This amount was raised to SF 50 000 in the final reform approved by Parliament. While this reform goes in the right direction, it does not go far enough. So long as it is possible to invest in assets that generate a non-taxable income, households should not be permitted to claim more interest expenses than the capital income they declare. However, the possibility of financing assets that potentially generate non-taxable capital gains through credit finance is limited. By virtue of practice and jurisprudence capital gains are often taxed as income from professional activity. Capital gains on immovable property are always subject to cantonal and/or municipal taxation.

### *Reform taxation of owner-occupied housing*

In relation to housing, the Swiss tax system aims to treat the owner-occupier in the same way as the pair formed by the landlord and tenant. This means fully taxing imputed rents after deduction of expenses. In practice,

imputed rents are below market rates, although not by so much as in some other countries where imputed rents are taxed. The imputed rental may not be less than 70 per cent of the market rental for the federal income tax and 60 per cent of the market rental for the income tax of the cantons and municipalities. Presently, imputed rentals are less than 70 per cent of market rentals in ten cantons. A supplement is added to the imputed rental as calculated by these cantons to bring it up to 70 per cent of the market rental for the purposes of the federal income tax. In four of these cantons, imputed rentals are less than 60 per cent of market rentals. These cantons are obliged to give a deduction to tenants in order to re-establish equality between owner-occupiers and tenants. In the remaining 16 cantons, imputed rentals are mostly between 70 and 90 per cent of market rentals.

The main problem with the taxation of imputed rentals appears to be excessive expense claims. Indeed, housing accounts in aggregate are in deficit. Abolition of the taxation of imputed rentals (along with the deductibility of interest payments and maintenance expenses) actually would increase tax revenue. One source of the high expense claims is that homeowners have an incentive to maintain large mortgages and to invest in life insurance policies. This generates a tax-deductible expense and no extra revenue. The other major source is capital improvements claimed as maintenance expenses.

If the tax incentives for long-term saving with pension funds and life insurance companies were significantly reduced, there would be a strong case for raising imputed rentals closer to market values and more carefully controlling maintenance expenses. One solution for controlling maintenance expenditures could be to allow a fixed charge as a percentage of the value of the building and to treat additional expenditures as capital outlays, to be depreciated over time. A solution along these lines was adopted in the Netherlands in 1971. (Even so, owner-occupied housing accounts there remain in deficit by about 2 per cent of GDP.) In the event that housing accounts in aggregate remained in deficit, consideration should be given to phasing out the taxation of imputed rentals and all of the associated expense claims (including mortgage interest payments). However, if the tax incentives for long-term saving with pension funds and life insurance companies remain unchanged, there is not a strong case for reducing the tax advantages that owner-occupiers presently enjoy. Doing so would only serve to encourage an even more unbalanced allocation of long-term savings between pension funds and life insurance companies on the one hand and owner-occupied housing on the other.

### *Phase out personal net wealth taxes*

The cantons and municipalities are obliged by the Law on Tax Harmonisation to levy personal net wealth taxes (as well as corporate net worth

taxes). These taxes, which do not exist in most other OECD countries,[109] are an important factor contributing to the relatively high taxation of capital income in Switzerland not receiving preferential treatment. A personal net wealth tax represents a surcharge on capital income from assets included in the tax base. If comprehensive income is accepted as a good measure of ability to pay, this surcharge undermines equity by making tax liability also depend on the proportion of income derived from taxed assets. For example, horizontal equity is violated when an individual with the same comprehensive income as another pays more tax because they have a smaller proportion of their capital in forms exempt from the wealth tax, such as pension fund assets or human capital. Consideration should be given to terminating personal net wealth taxes, as has already occurred at the Federal level (and, in recent years, in Germany, Austria and Denmark). The enforcement benefits of net wealth taxation could be retained by continuing to require households to file a declaration of net worth. The phasing out of personal net worth taxes would lower one of the barriers identified by the Behnisch Commission to introducing a personal capital gains tax.

### *Replace existing corporate taxes with a flat-rate tax on corporate profits*

Other unusual features of capital taxation in Switzerland are the progressive rate of return tax on corporate profits and the corporate net worth tax levied by the cantons and municipalities. No other OECD country has a progressive rate of return tax on corporate profits and only Finland (on companies in which the shareholders are not subject to wealth tax), Iceland and Mexico have corporate net worth taxes. This tax structure discourages risk taking. If a company makes a high rate of return it faces a high tax rate (from the progressive rate of return tax, although this disadvantage is attenuated by the possibility of carrying forward losses)[110] while even if the company makes a loss it must still pay the net worth tax. Thus, the tax system tends to tax disproportionately favourable profit outcomes, while continuing to impose taxes (via the wealth tax) when profit outcomes are unfavourable. These arrangements are particularly disadvantageous for business start-ups as they tend to be more risky than established businesses. This disadvantage is compounded by the lack of relief to shareholders for taxation of distributed profits under both the corporate- and personal income tax systems.[111] These arrangements also have arbitrary effects on tax equity. Looking through the corporate veil, there is no reason to expect these taxes to be in any way related to shareholders' personal ability to pay.

It would be fairer and more efficient to replace the progressive rate of return tax on corporate profits with a flat-rate tax, abolish the net worth tax and introduce shareholder relief for the double taxation of dividends.[112] This would also bring corporate taxation into line with practice in most other OECD countries. The latter reform would also contribute to establishing the Behnisch

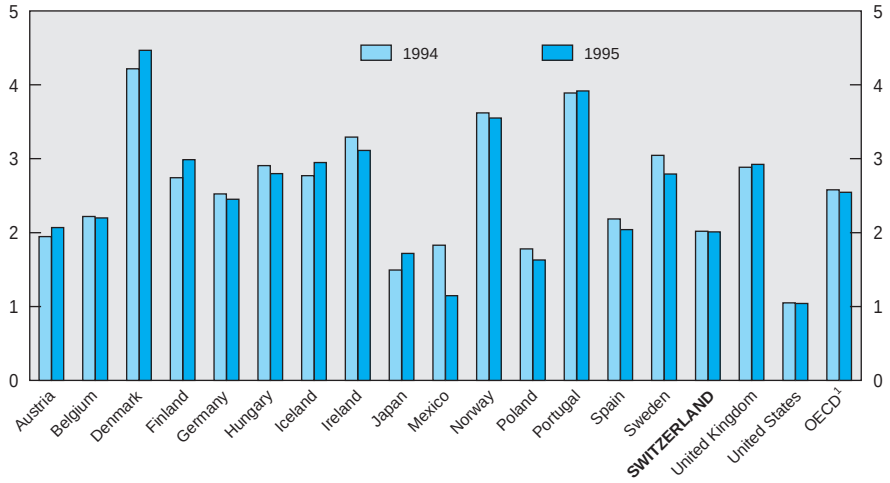
Commission's pre-conditions for introducing a personal capital gains tax – namely, lower taxation of capital income. Unfortunately, there are serious obstacles to introducing these reforms in Switzerland. When the federal government initially proposed replacing its progressive rate-of-return tax on corporate profits with a flat-rate tax that would generate the same amount of revenue it encountered considerable opposition. It seems that most companies had a lower rate of return than the (weighted) average and would consequently have faced a higher tax bill. (A minority of companies, with above average rates of return, pay most of the tax.) In order to gain political acceptance for the reform, the federal government was obliged also to abolish the net worth tax (a desirable reform in its own right), which falls most heavily on companies with a low rate of return, and to concede tax revenue. If the cantons wished to implement a similar reform, they too would probably have to concede tax revenue to gain popular approval. Introducing shareholder relief for the double taxation of dividends would be even more difficult because some cantons would lose tax revenue. Switzerland's highly decentralised federal structure is probably an insurmountable barrier to such a reform.

### *Environmental tax reforms*

Although Switzerland has an enviable record in protecting its environment, it has largely done so through directive regulation. As in other countries, the authorities have recognised that many environmental objectives could be attained more cheaply through greater use of economic instruments. (This also makes the costs of environmental regulation more transparent.) However, progress in substituting economic instruments for environmental regulation has been rather modest to date. At the federal level, the first pure environmental tax came into effect in June 1998, on light oil with high sulphur content. The introduction of a second one, on volatile organic compounds, was delayed from 1998 to 2000. [The revenue from these taxes is to be returned to the population through lump sum subsidies for medical insurance premiums on the grounds that this benefits the whole population and compensates for the possibly small regressive effects of these taxes. The disadvantage of this approach is that there is no efficiency gain from reducing taxes that distort economic choices (lump sum subsidies are effectively negative poll taxes).] There are also some specific environmental taxes (*e.g.* on waste) at the levels of the cantons and municipalities. In addition, there is a range of taxes mainly aimed at raising revenue but which also have positive environmental effects. Environmentally related taxes amount to about 2 per cent of GDP, somewhat less than the average of the OECD countries for which data are available (Figure 32). As in other countries, most such revenue is derived from taxes on energy used in transportation (such taxes have other objectives as well, such as paying for roads and reducing congestion).

Figure 32. **Revenues from environmentally related tax-bases – selected OECD countries**

Share of GDP, per cent




1. Simple average of above countries.

*Source:* OECD, *Revenue Statistics* and Database on environmentally related taxes.

The Federal Council announced in 1998 that an "ecological tax reform" would be included in the new financial order (*i.e.* the legal authority for the federal government to raise tax revenues), which is to replace the existing order at the end of 2006 at the latest. This would introduce an energy tax and reserve the proceeds (SF 2-3 billion per year) mainly for financing a reduction in social security contributions. A project is to be submitted to the official consultation procedure in 1999 and a proposal is to be submitted to Parliament in 2001. The law on reductions in $CO_2$ emissions, which would introduce a tax in the event that other measures prove insufficient, has not yet been approved by Parliament because agreement has not yet been reached on whether Parliament or the Federal Council should have the authority to introduce the tax.

The move towards greater use of environmental taxes should enhance economic efficiency by discouraging economic activities with harmful environmental effects and/or by reducing the costs of environmental regulation. The extent to which these benefits are realised, however, depends on the nature of the reforms. One feature to avoid is special treatment for the greatest polluters or most intensive users of non-renewable resources. This is often done to protect the international competitiveness of the industries concerned. Unfortunately, it

undermines the efficiency of the tax. This is especially so for a local pollution problem, where shifting the most polluting activities abroad may be part of an efficient solution. But even for a global pollution problem (such as $CO_2$ emissions), where shifting polluting activities abroad does not help to solve the problem, such crude exemptions may not be necessary to enable these industries to remain internationally competitive. For example, it would be possible to levy the tax in proportion to the consumption or emissions that are to be discouraged and to refund the revenues in proportion to sales or production. The proportion of sales or production to be refunded would be set for a whole industrial sector at a level such that the average polluter in that sector is subject to the net burden that is considered to be bearable.[113] It would also be possible to apply the tax to imports of polluting products. Article XX of the GATT permits a WTO member to place its national environmental goals ahead of its general obligation not to raise trade restrictions or to apply discriminatory trade measures, provided that the steps are transparent and non-discriminatory between domestic and foreign producers.[114] However, these rules probably would not permit the taxation of (non-polluting) imports manufactured by industries that are taxed domestically but not abroad.

Another feature to avoid is requirements that the tax revenues be spent on environmental projects. If there are worthwhile environmental projects, they should be funded irrespective of whether or not environmental taxes are introduced. The risk is that public funds are spent on projects of such low value that the authorities would be unwilling to finance them out of general tax revenue. Ideally, revenues from environmental taxes should be used to reduce the most inefficient and/or inequitable taxes. While social security taxes may not be the worst in these respects, there should nevertheless be some economic benefit from reducing them with the revenues from the proposed federal energy tax. This would represent an attempt to exploit the so-called "double-dividend" from environmental taxes, the second dividend being increased employment as a result of lower social security charges. Nevertheless, the available evidence from OECD countries suggests that any such double dividend is small.

## Swiss tax arrangements in an international setting

This section addresses two questions relating to the international dimensions of Swiss tax arrangements. How robust will these arrangements prove to be in the face of increasing globalisation and financial mobility? Do Swiss tax practices attract mobile tax bases from abroad?

### *Vulnerability of Swiss tax bases to international base mobility*

Switzerland's tax structure appears to be fairly robust to international competition. In attempting to deal with such competition, Switzerland has foregone the approach – adopted by a number of OECD countries (see Carey *et al.* 1999) – of shifting the tax burden on to the relatively immobile bases, labour and consumption, and away from capital. Compared with many Continental European countries, Switzerland taxes capital rather heavily in relation to the burdens imposed on labour and consumption, and this overall structure seems to be fairly stable across time. Normally, this approach could make the Swiss tax system vulnerable to base erosion in the more mobile segments of its capital tax base. However, the particular structure of capital taxation makes it less vulnerable to competition. The effective tax burden on household financial capital is rather low, principally owing to the favourable tax treatment of the earnings of pension funds and of funds invested in life insurance. As a result, much of the household tax burden is focused, directly or indirectly, on physical property. Indeed, over a third of capital taxation comes from various forms of property taxation (inheritance, real estate, various types of wealth tax, etc.) (Table 31). The (estimated) taxes on the capital income of the self-employed sector also account for a significant share of capital taxation, as do the taxes on corporations (including income, wealth and other business and property taxes). The corporate tax base might be subject to some vulnerability due to international mobility, while the self-employed sector's mobility does not, as yet, appear to be very high. Thus, some elements of the Swiss tax structure seem fairly robust to international competition (in particular, the emphasis on direct and indirect taxation of property), while others seem less so (the large share of corporate taxes in overall revenues).[115]

Table 31. **Corporate, property and financial taxation**

As a share of total capital taxes, per cent

| | Average 1980-85 | Average 1986-90 | Average 1991-96 |
|---|---|---|---|
| Corporate income taxes | 28.8 | 30.6 | 31.6 |
| Property taxes | 36.9 | 38.7 | 38.2 |
| Recurrent taxes on immovable property | 2.5 | 2.1 | 2.5 |
| Taxes on financial and capital income | 11.3 | 14.9 | 10.9 |
| Capital income of the self employed[1] | 34.2 | 30.7 | 30.2 |

1. Taxes on capital income of the self employed were estimated by applying an average rate of household taxation to property and entrepreneurial income (the latter adjusted for an imputed wage component).

*Source:* OECD (1998*a*), *Revenue Statistics* and OECD Secretariat.

Switzerland avoids double taxation not by granting tax credits but by applying the exemption method according to Article 23A OECD-Model. This means that profits realised by a foreign subsidiary or attributed to a foreign permanent establishment are not subject to corporate tax in Switzerland, whether or not they are repatriated by the Swiss parent company or head office. However, should Swiss companies take advantage of foreign tax havens by transferring to them profits that would otherwise normally be taxable in Switzerland, then a profit adjustment would normally need to be made in the assessment procedure for the relevant commercial year. If – for any reason – such an adjustment has not been made, there still remains the possibility for the tax authorities to refuse the participation exemption when those profits are distributed from the tax haven subsidiary to its Swiss parent company or in case the participation is sold and gives rise to a capital gain.

There is nevertheless a risk that income is unduly reinvested abroad rather than being reinvested at home. Therefore, appropriate rules on compulsory distribution might be envisaged to impede such abuses. Although it is generally thought in Switzerland that there is no need for Controlled Foreign Corporation (CFC) legislation with respect to active enterprises, whether located in a tax haven or not, it may be worth considering whether the requirements that entitle companies to the participation exemption should be made more restrictive in respect of tax haven subsidiaries.

### *Switzerland as a destination for some internationally mobile tax bases*

Switzerland's "*régime forfaitaire*" may make it an attractive destination for some foreign individuals. Expatriates not exercising any gainful activity in Switzerland can apply to be taxed on their assumed living expenses (including the cost of living of their families). This tax replaces the ordinary income and net wealth taxes (at all levels of government) and is subject to certain limits.[116] The canton authorities also have some discretion in determining the annual living expenses of the taxpayer. In the whole country, 2 730 foreign taxpayers have presently applied for the "*régime forfaitaire*"; this regime is often the last resort for tax authorities to overcome practical difficulties due to the lack of information on possibly wealthy foreign taxpayers. The regime may explain why some cantons manage to attract wealthy foreign individuals; three cantons exceptionally tax at special rates capital income from Swiss- or foreign sources for which benefits under a double taxation treaty have been claimed (these special rates rely on clear legal bases). Other factors (quality of life, security, access to transport facilities, local language and climate) are also important influences on expatriates' location decisions.

Another issue for international taxation arises from the interaction of Swiss tax law with other countries' tax practices and with the long-standing Swiss

tradition of bank secrecy. The general rule in Switzerland is that Swiss-source capital income is subject to a 35 per cent withholding tax, which is refunded to resident taxpayers if they report the respective income and investments in their tax returns. Non-resident investors have to submit a claim for refund, duly certified by the tax authorities of their countries of residence. Depending on the relevant Double Taxation Treaty, the Swiss withholding tax will be refunded. Moreover, as to the non-refundable part of the Swiss withholding tax, the beneficial owners can claim a credit against the income tax in their residence country.[117] If a non-resident decides not to file a claim for refund so as to avoid declaring income to his residence country, he loses the 35 per cent withholding tax.[118] As noted above, in the discussion of enforcement practices, tax evasion is not a crime in Switzerland and the right of tax and law enforcement officials to extract bank information, for their own tax purposes or for a treaty partner, on matters that do not relate to a suspected crime is extremely limited.

Non-residents can avoid all withholding taxes by setting up fiduciary accounts in Switzerland through which their funds are invested abroad. This may be particularly advantageous for non-residents unwilling to be taxed on capital income in their countries of residence for the following reasons. First, their funds generally are invested in countries that do not levy a withholding tax on bank deposits belonging to non-residents.[119] Second, Switzerland does not impose a withholding tax on the income because it is not Swiss-source. Third, Switzerland's tradition of strict privacy – which is laid down in the Constitution – and bank secrecy protect the identity of the beneficial owner of the account from disclosure to the tax authorities of the beneficial owner's country of residence. It is the combination of these three factors that makes tax evasion through Swiss fiduciary accounts attractive to non-residents. This scheme allows a non-resident desiring to evade taxes to be reasonably certain that a failure to declare the invested capital and/or the interest thereon to his country of residence will go undetected.

Finally, there are two cantonal (not federal) corporate tax regimes that can function as tax shelters. In particular, in these cantons, administration companies are not normally taxed on capital income that does not qualify as participation income, except if the country of source requires – by virtue of the respective Double Taxation Treaty – full taxation in Switzerland. (As noted above, capital income that does qualify as participation income is not taxed in Switzerland as it is an exemption country.) Administration companies may also benefit from the cantonal global exemption method as regards foreign permanent establishments. Consequently, profits that are attributable to foreign permanent establishments are not taxable in Switzerland. Cantons retain some discretion in determining exactly what these companies will be taxed on. This makes certain cantons attractive places for such companies to locate.[120] The tax status of holding companies is similarly advantageous.[121]

Thus, under certain circumstances, the tax policies of Switzerland (or of certain cantons) do attract mobile tax bases from abroad. The interaction of bank secrecy and international tax enforcement is now being discussed in several forums, including the OECD. What is clear is that, in Switzerland, this practice stems from a long legal tradition that places high value on individual privacy. It turns out that, for reasons discussed elsewhere in this chapter[122] the legal and information apparatus available to the Swiss tax authorities does not appear to give rise to major compliance problems in Switzerland. On the other hand, it is also clear that access to information is an essential ingredient for effective tax enforcement and that, as globalisation and technology continue to advance, it will become increasingly important. Thus, Switzerland may find itself under continued pressure to provide more complete access to information to foreign tax authorities. The other major base mobility issue discussed here – special tax regimes for holding and administration companies – is closely linked to fiscal control of cantons. While harmonisation has the merit that domestic tax competition between cantons and – within cantons – between municipalities just depends on their respective tax rates, some cantons take a competitive approach at the international level. The practices of some cantons are sometimes considered to be not only competitive but to go beyond the practices of some comparable countries.

## Concluding remarks

The decentralised federal structure and the widespread use of direct democracy shape Switzerland's tax system. Direct democracy, together with private funding of the second pillar of the retirement income system, have constrained the overall tax burden to a moderate level compared with other OECD countries. However, decentralisation has circumscribed the scope for tax reform, notably because there would be winners and losers amongst governments. For example, a shift from income taxation to VAT would be difficult to implement because the federal government would receive the VAT revenue while lower levels of government would lose income tax revenue and face increased expenditures for social assistance. While, in principle, it should be possible to negotiate a deal that would be satisfactory to all governments, it would, in practice, probably be difficult in view of the large number of parties involved. This has created a situation in which governments in some instances have maintained taxes for their capacity to generate revenues, with the effects on economic efficiency and/or equity being a secondary consideration. This is particularly so in the area of capital taxation, notable examples being the taxes on personal net wealth and on corporate net worth.

Bearing in mind these constraints, the tax system generally measures up quite well in terms of efficiency and equity. Income tax bases are broad, labour tax wedges do not present a serious obstacle to employment and VAT is highly productive by international standards. Significant progress has been made towards harmonising direct tax bases and administration arrangements are stream lined so that taxpayers are not required to submit different tax returns to each of the three levels of government that levies income tax. Moreover, the tax system is quite progressive by international comparison. Nevertheless, there is substantial scope to improve the tax system, even within the constraints of the decentralised federal structure. Priorities for tax reform include: reducing the privileges accorded to individuals for their savings, for example with life insurance companies (pillar 3b); greater limitation of personal interest deductions than recently approved by Parliament; the replacement of cantons' existing corporate taxes with a flat-rate tax on corporate profits; and a greater use of environmental taxes. Switzerland should further consider whether additional clauses should be included either in its internal legislation or in its double tax treaties in order to be more in line with international standards.

# Notes

1. Preliminary statistical evidence of a recovery in labour incomes suggests that both in 1997 and 1998 the increase in average wages derived from the National Accounts was significantly higher than the rise in the official wage rate indicator of the Federal Statistical Office – by about ½ percentage point per year. One possible source of the discrepancy could be the rising importance of labour compensation components which are not included in the official wage rate statistics (*e.g.* bonuses, redundancy payments). Another possible source is compositional changes in the labour force which result in a growing share of better qualified and better paid workers in the total. There is also the possibility that the official wage statistics, which are based on data on accident insurance payments to blue collar workers, are downward biased, in spite of the adjustments made by the Federal Statistical Office.
2. The balance of positive and negative survey responses.
3. A breakdown of construction investment in 1998 is not available yet.
4. Most of these imports origin from Southeast Asia. Since they have no immediate counterpart in exports, consumption or investment, they were attributed to the residual item "stockbuilding and statistical discrepancy"; the latter is the statistical discrepancy between production- and demand-based GDP measures.
5. These are persons – including seasonal and frontier workers – who have worked for at least six hours per week.
6. However, this does not necessarily mean that all "benefit exhaustees" remain unregistered unemployed, because some of them may withdraw from the labour force while some may continue job search without the assistance of placement offices and eventually find employment. A recent survey found that of those who reached the end of their entitlement period in 1995 and 1996, about 43 per cent had found a job before March 1997. Around 44 per cent of the sample were still searching for a job and 13 per cent not. Three-quarters of those still seeking a job were registered at regional placement offices and were accordingly registered as unemployed. Hence, only about 11 per cent of the entire sample were both looking for a job and were not registered as unemployed (see Aeppli, Hoffmann and Theiss, 1998).
7. Office fédéral du développement économique et de l'emploi.
8. If all participants of active labour market programmes were counted as unemployed, the average number of unemployed would have amounted to 195 600 instead of 139 700 persons in 1998 and the average unemployment rate would have been 5.4 instead of 3.9 per cent. Such calculations would treat employed people who receive intermittent pay supplements from the unemployment insurance fund as unemployed. This is justifiable for those benefit recipients – a relatively small group – who would not accept work at market wages. It seems, however, not justified for the larger number of persons who are employed but wish to work longer hours.

9. If adjusted for the introduction of VAT in 1995, which caused an estimated one-time increase in the consumer price level of 1 to 1¼ per cent in 1995, then consumer price inflation was under 1 per cent in 1998 for the fifth year in a row. The inherent upward bias in the CPI stemming from the omission of product substitution, quality improvements, new goods and outlet substitution could amount to about ½ per cent a year (Office fédéral de la statistique, 1999). If allowance is made for this bias, the true rate of inflation may have been zero or even negative during the past five years.

10. A federal decree authorises an increase in rents by 2 to 3 per cent for each quarter of a percentage point increase in the mortgage rate, depending on the actual level of the mortgage rate. A recent proposal of the federal government aims to break this link, which also complicates monetary policy as it causes an initial inflationary effect of tightening monetary policy.

11. The weight of import prices in the CPI is 25.4 per cent.

12. These are primarily food, non-alcoholic beverages and medication. Exempt from VAT are housing rents and various services in the field of health, education, post, insurance and banking.

13. At the time of writing, data for 1998 were not yet available.

14. Based on the Index II definition of exports and imports, which excludes the – primarily transit – trade in precious stones, precious metals and antiques, which is very volatile and tends to blur the trends in foreign trade.

15. See Schweizerische Nationalbank (December 1998). However, in a recent speech the Chairman of the Governing Board of the SNB, Hans Meyer, offered a more flexible definition of price stability, which "... tolerates inflation rates up to 2 per cent" (Meyer, 1999).

16. The monetary base comprises notes in circulation and sight deposits held with the National Bank. In addition to seasonal adjustment, the series is corrected for the biennial peak in note circulation at the end of even years, when property and income taxes are assessed, because of evidence that taxpayers try to lower their tax base by converting parts of their financial assets into currency.

17. When the "1 per cent money supply rule" was formulated in the early 1990s, it was based on the assumption of average potential output growth of about 2 per cent, an inflation objective of 1 per cent and a trend increase in the income velocity of base money by 2 per cent per year; the latter was expected to be a consequence of the rising use of non-cash means of payment.

18. The SNB adopted medium-term money supply targeting in 1990 (see Meyer, 1998).

19. Until the beginning of 1997, repurchase (repo) transactions – the sale (purchase) of securities with a simultaneous agreement to repurchase (resell) securities of the same type and quantity at a later date – were subject to stamp duty; this prevented the development of repo instruments in Switzerland. With the abolition of the stamp duty on repurchase transactions a repo market has been established in Switzerland and began its trade in April 1998. The use of repos as a monetary policy instrument widened the National Bank's scope for implementing monetary policy. Technical details on repo transactions as a monetary policy instrument can be found in Schweizerische Nationalbank (September 1998).

20. Although the SNB no longer discounts bills, changes to the discount rate are registered by financial markets as indicators of the National Bank's policy intentions.

21. Possible explanations are changes in fees by banks for the acceptance of banknotes, currency substitution effects in the run-up to the introduction of the euro and hoarding of Swiss francs abroad.
22. The SNB reckons that these special factors in total led to an upward shift in the monetary base by around SF 800 million by 1998. Base money adjusted for this shift would thus exceed the medium-term target trajectory by 2½ per cent instead of about 7 per cent at the end of 1998.
23. See, for example, Schweizerische Nationalbank, December 1997, p. 258, and December 1998, p. 10.
24. See Peytrignet (1996).
25. Schweizerische Nationalbank (December 1996), p. 296.
26. See Peytrignet and Stahel (1998).
27. The Constitution requires the Swiss National Bank to serve the interests of the country as a whole. The federal government proposed a constitutional amendment in May 1998, which defines the maintenance of price stability as the SNB's primary objective. Other major objectives of the amendment are to formally dissociate the Swiss franc from gold and to institute the independence of the National Bank in the Constitution. After parliamentary debate of the proposal during 1999, a referendum is expected for early 2000.
28. The level of the "desirable" short-term interest rate as calculated by the Taylor rule is, of course, affected by the uncertainty around estimates of the output gap. Sensitivity analysis shows that under the assumption of a zero output gap in 1998, the actual interest rate would have been about right, being located just between the two normative interest rates (based on inflation objectives of either 1 or 2 per cent). There is, however, little doubt that six years of economic stagnation in Switzerland resulted in substantial unused physical and human resources, which have not been entirely absorbed by the recovery during 1997 and 1998.
29. See Schweizerische Nationalbank (December 1998), p. 10, and Meyer (1998), p. 39-40.
30. http://www.snb.ch.
31. Data used in this section are on a Statistique Financière Révisée (SFR) basis unless otherwise indicated. This is the basis on which recent data on government accounts for different levels of government may be aggregated.
32. Data discussed in this subsection are on a Financial Accounts basis.
33. This includes SF 1 billion of expenditures for the large rail infrastructure projects.
34. The cyclically-adjusted budget balances discussed in this Survey were provided by the Office fédéral du développement économique et de l'emploi. The methodology used to calculate them is described in Ammann (1995).
35. For a summary of factors which contributed to the 1991 to 1996 period of economic slump see Chapter II of the 1997 OECD *Economic Survey of Switzerland*.
36. See OECD (1996), *Economic Surveys, Switzerland*, Chapter III.
37. Switzerland is among the most generous OECD countries with regard to benefit levels as well as benefit duration. *Ibid.*, Table 21.
38. *Ibid.*, Chapter III.

39. Loi fédérale sur l'assurance-chômage obligatoire et l'indemnité en cas d'insolvabilité.
40. Office Fédéral du Développement Économique et de l'Emploi (1998).
41. Based on data of the OFDE.
42. For earlier empirical evidence of the unemployment-increasing effect of longer duration of benefit eligibility, see Scarpetta (1996) and Sheldon (1997).
43. Sheldon reckons that the two increases in benefit duration in 1992 and 1993 increased the individual risk of becoming long-term unemployed by about 27 per cent and raised the share of long-term job-seekers in the total number of unemployed by more than 70 per cent. The latter is estimated to be roughly equivalent to the effect of an average cyclical downturn of two years.
44. For details see the 1997 OECD *Economic Survey of Switzerland*, Chapter IV: Education and Training.
45. The recent reform also gives less gifted students the opportunity to prolong their apprenticeship in order to obtain a qualification recognised in the labour market.
46. For example, during a four-year apprenticeship, they must spend 4 500 hours in on-the-job training and do 1 440 lessons in vocational subjects and as much again in general subjects. This is about 600 to 1 000 lessons more than a traditional apprentice.
47. High school graduates (baccalaureates or A levels) can also enter a HES provided that they have at least one year of professional experience.
48. Opinion polls reveal strong preferences of consumers for an extension of shop-opening hours from 6 p.m. to 8 p.m. from Monday to Friday and until 5 p.m. on Saturdays. The need for shopping on Sundays seems somewhat less pressing. See Inderbitzin and Hoch (1998).
49. See Inderbitzin, Hoch and Wolter (1998).
50. German bakers are reported to have created some 5 000 new jobs during the first seven months of the new law on shop opening hours of 1996, which now allows sales of bakery products on Sunday mornings (see Deutscher Bundestag, 1997).
51. Notification is required if, during the most recent financial year prior to the proposed transaction: *i)* the participating firms achieved combined turnover of at least SF 2 billion or turnover in Switzerland of at least SF 500 million; and *ii)* at least two of the participating firms individually achieved at least SF 100 million in turnover in Switzerland.
52. Reported in Autorités suisses de la concurrence (1998).
53. These features were market shares of the two largest banks of roughly the same magnitude, similar cost structures, high barriers to entry for new competitors, a high degree of market transparency among the banks involved but limited transparency for clients, relatively homogeneous products, and low actual and prospective market growth and profitability.
54. The new legislation aims at aligning Swiss technical standards as much as possible with those of main trading partner countries in order to minimise technical barriers to trade.
55. A case where the principles of free market access remain to be implemented is the market for certain notarial services in the canton of Bern, which is sheltered from competition by notaries from other cantons. For example, cantonal law makes the

intermediation of a notary compulsory when determining taxable wealth as a basis of the cantonal inheritance tax. As a consequence, notaries are in a position to extract substantial monopoly rents from their clients, which can raise the compliance costs of taxation to above 50 per cent of the payable inheritance tax.

56. Fernmeldegesetz and Telekommunikationsunternehmungsgesetz.

57. Federal Office for Communication (1998).

58. Office fédéral de l'énergie (1995).

59. The PSE is an indicator of the value of the monetary transfers from consumers to support agricultural producers, measured at the farm gate level, resulting from agricultural policies in a given year. The PSE includes transfers from consumers of agricultural products (through domestic market prices) and transfers from taxpayers (through budgetary expenditures or tax exemptions). PSEs and CSEs (see next note) are calculated from a standard set of 13 commodities.

60. The CSE is a measure of the value of monetary transfers to consumers resulting from agricultural policies in a given year. The CSE is usually negative because the transfers from consumers as a result of market price support policies exceed budgetary-financed consumer subsidies.

61. In September 1998, a referendum was held on the proposition of an alternative approach to agricultural policy, which has been rejected.

62. The federal Constitution mandates the Confederation to ensure that agricultural production is both sustainable and market-oriented and contributes substantially to:
    - A safe supply of food;
    - The preservation of national resources;
    - The maintenance of the usual landscape;
    - A decentralised use of land.

63. Swissair has estimated that it currently loses about SF 200 million per year because of discrimination in the EU air transport market.

64. The public procurement agreement of the WTO already opens federal and cantonal contracts up to competition with foreign companies.

65. Excess burden is the cost of taxation in excess of the revenue raised. Taxes generally impose costs greater than the revenue they raise because they distort economic choices, diverting resources from their most valued uses. Administration and compliance costs also contribute to excess burden.

66. According to some measures, federal arrangements in the United States are almost as decentralised as those in Switzerland, but effective base harmonisation is fairly complete in the US federation (though *de jure* States have the right to define their own bases). Switzerland uses "equalisation transfers" that effectively weaken base mobility, but these are relatively small in that country. They also have complete autonomy in setting rates. In this respect the systems are similar.

67. In this respect, the Swiss confederation is often mentioned as a good empirical model for understanding how fiscal relations are likely to evolve in the European Union. See Kirchgässner and Pommerehne (1996) for a discussion of this issue.

68. The initiative either sets out the wording of the proposed change in the Constitution or the principle of the proposed modification, in which case Parliament modifies the Constitution. If the initiative is approved in a referendum, consequent changes in the relevant law(s) must also be made.

69. The following is a selection of issues submitted to votes in 1997 and 1998: a federal law concerning the realisation and finance of public transport infrastructure; a law concerning a charge for heavy trucks use of the road system; a popular initiative for a revision of the "AVS" without raising the retirement age; a federal law containing measures to balance the budget; a federal law on unemployment insurance.
70. Nevertheless, the Swiss use of compulsory private insurance probably involves some regulatory excess burden.
71. Defined as population aged 65 or over as a per cent of the total working-age population.
72. These include: elimination of the withholding tax on dividends paid by a subsidiary to its parent company; other modifications to the system of collecting the current withholding tax regime; integration of the clause regarding co-operation among EU tax authorities.
73. The report is intended to develop a better understanding of how tax havens and harmful preferential tax regimes, collectively referred to as harmful tax practices, affect the location of financial and other service activities, erode the tax bases of other countries, distort trade and investment patterns and undermine the fairness, neutrality and broad social acceptance of tax systems generally. Such harmful tax competition diminishes global welfare and undermines taxpayer confidence in the integrity of tax systems (OECD 1998*b*, paragraph 4).
74. The reasons for this abstention are set out in OECD (1998*b*), pp. 76-78.
75. The social security data are as reported in *OECD Revenue Statistics*. This includes all health insurance premiums, even though it is difficult to see why the voluntary part of health insurance premiums to private funds should be considered as social security contributions. This treatment, which is not consistently applied to other countries (including the United States) tends to overstate social security contributions in Switzerland by international comparison. The social security data (correctly) do not include contributions to pension funds.
76. Each canton sets its basic tax rates in law. Most apply a coefficient to these rates to determine the current rates. Municipalities apply their own coefficients to the basic rates.
77. This is so even for financial assets that are rarely or never traded; their value is estimated nation-wide every year.
78. In two cantons, the communes do not apply a coefficient to the canton tax, as in the other cantons, but have their own schedules.
79. In addition to exhaustive wealth taxation, two cantons and the communes in a dozen cantons levy a tax on real-estate property. Debt is generally not deductible and tax rates are proportional.
80. All direct taxes paid to the Confederation, cantons and the municipalities on income and net worth are also deductible in the year in which they are paid.
81. Tax rates may be higher than this in the sixteen cantons that apply a minimum income tax to corporations that realise little profit compared to their economic importance. This tax is a fixed amount in some of those cantons. In the other cantons, the minimum tax is based on different indicators of economic importance, such as sales, capital or real-estate property.
82. Other special corporate arrangements also apply for particular types of company. Public enterprises are exempt from CIT and net worth taxes. Income from holdings in

other companies is also generally exempt from the CIT. In addition to being exempt from CIT, qualifying holding companies are taxed at preferential rates on their net worth. Twenty-three cantons also grant exemptions or tax rebates to new corporations for a maximum of ten years. Finally, all the cantons offer preferential treatment to companies that are located in Switzerland but carry out nearly all of their activities abroad.

83. Food, tap water, medication, newspapers and books, plants and flowers are subject to the lowest rate while hostelry services are taxed at an intermediate rate (in recognition of the fact that the clientele is mainly foreign). Excluded services include the following: health, social security, education, culture, sports events, home rentals, financial transactions and insurance. As providers of these services are unable to claim VAT refunds on their purchases, these services are effectively subject to a reduced rate of VAT.

84. The Federal Council and tax administration wanted that change also for personal income taxation, but a majority of cantons were opposed.

85. There are two schedules of standard deductions for the federal income tax, one for taxes collected every year and the other for taxes collected every two years.

86. Müller (1998).

87. Notaries must be used as intermediaries for all aspects of administering the inheritance tax in Bern. On an inheritance of SF 200 000, their fees, which are regulated, amount to about 50 per cent of the death duties levied.

88. According to Swiss law, a tax fraud is committed on the one hand by the use of false documents, such as the presentation of a false balance sheet or falsified invoices to the tax administrations, aiming at the evasion of taxes. On the other hand, any fraudulent deception of the tax authorities with the aim to evade taxes also represents a tax fraud. In addition, according to court practice, Swiss tax authorities have access to bank information seized by the examining magistrate in any criminal procedure. This access is not restricted to information related to persons accused but also includes information with respect to third persons.

89. As an illustration, consider a taxpayer with SF 100 000 gross income and SF 300 000 net wealth. Suppose he earns SF 2 000 income on SF 50 000 wealth that he could "forget" to report. If he does not report that wealth and its income, he loses the withholding tax of SF 700 (35 per cent of SF 2 000). If he reports truthfully, he pays income tax at a marginal rate of about 30 per cent on SF 2 000 and wealth tax at a marginal rate of about 0.4 per cent on SF 50 000 (average across the cantons) for a total of SF 800.

90. Some of the differences in wealth can be explained by geographical factors: the richest cantons are densely populated and located on the central Plateau while the poorest cantons are sparsely populated and located in the mountains.

91. Feld, Kirchgässner, and Pommerehne (1996).

92. Feld, Kirchgässner and Pommerehne (1996) perform such an estimation with data for the 137 largest Swiss municipalities gathering 46 per cent of total population, with data for 1990. Kirchgässner and Pommerehne, did the same with 1987 data.

93. The models have some unsatisfactory features: the authors could not use the tax rates that actually apply to each category of taxpayer, R2 never exceed 50 per cent, and Jarque-Bera tests reject the hypothesis of normal residuals. When the outliers

are excluded to amend that problem, the R2 drop and the influence of tax differentials shrinks.

94. There are a number of limitations to measuring the progressiveness of the tax system in this way that should be borne in mind. First, at higher income levels, families find themselves increasingly in a position different from the APW assumptions because the take-up of tax deductions and exemptions not included in the tax equations gains progressively in significance. Second, social security contributions – which are in the APW tax equations – are not paid by large groups of the taxpayer population in many countries, notably pensioners, the self-employed and benefit recipients. Third, the share of personal income tax in the tax mix and consequently its impact on the redistributive potential of the tax system as a whole varies strongly from one country to another.

95. For Federal Income Tax covering the 1993/94 tax period, the top 50 per cent of income earners declared 72 per cent of income (less if the incomes of taxpayers with incomes below the tax-exempt limit are taken into account) and paid 93 per cent of personal income tax. At the level of the cantons for the 1995/96 tax period, the top 50 per cent of taxpayers contributed 86 per cent of total income tax revenue.

96. The wedge will have no effect on employment if labour supply does not depend on real wage rates (*i.e.* perfectly inelastic labour supply with respect to after-tax real wage rates) or will increase employment if labour supply is inversely related to after-tax real wage rates. Such cases are theoretical curiosities rather than empirical characteristics of labour-supply functions.

97. This includes income taxes, social security contributions, private pension contributions (which should not be considered as taxes) and cash transfers.

98. OECD (1996), *Economic Survey of Switzerland*, p. 60.

99. These use the King-Fullerton method, as adapted by the OECD and described in OECD (1991). They are fairly comparable to the estimates reported in OECD (1991) and in Gordon and Tchilinguirian (1998). The main difference is that the present estimates include wealth taxes, while the earlier ones do not.

100. It also indicates that the cantons tend to extend relatively few tax credits for this type of investment via their normal tax regime. Of course, such credits or other types of tax expenditure with more specific targets (*e.g.* designed to encourage investment in particular sectors or by particular types of firms may exist outside the normal tax regime). These numbers are meant to apply to a manufacturing company to which "normal" tax treatment is accorded.

101. The range of overall average marginal effective tax rates for the OECD as a whole is several percentage points of marginal effective taxation.

102. See Carey *et al.* (1999) for a technical description of the average effective tax rate calculation and a discussion of the limitations of this approach.

103. AETRs for capital based on gross operating surplus provide a more reliable international comparison than estimates based on net operating surplus because this avoids the effects of the large differences in assumed asset service lives in different countries' National Accounts. Asset service lives are assumed to be much shorter in Switzerland than in most other countries. This large difference cannot be easily explained with reference to structural factors. It has a major impact on the estimated capital tax base and therefore on the effective tax rate as well, as can be seen from comparing the estimates based on gross and net operating surplus. As the estimates

based on gross operating surplus exclude depreciation costs, they overestimate the size of the tax base and hence underestimate the effective tax rate on capital.

104. The first pillar provides a universal flat-rate pension financed on a pay-as-you-go basis.

105. Earnings-related contributions (at an average rate of 17 per cent) are compulsory on annual salaries up to SF 71 640.

106. Results of the principal research on this question in the American context are discussed in Mitchell and Moore (1997). Two groups of researchers have dominated the debate in recent years. One group, Poterba, Venti and Wise (1993, 1996*a*, 1996*b*, 1996*c*), concludes that between 45 and 66 cents per dollar contributed to an Individual Retirement Account (IRA) represents new saving, with the balance coming from the tax subsidy and reallocation of existing saving. The other, Gale and Scholz (1994) and Engen, Gale and Scholz (1996), concludes that most IRA and 401(k) contributions are financed through tax savings and the shifting of taxed vehicles to tax-favoured vehicles. Bernheim (1996) and Hubbard and Skinner (1996) review both sets of studies. These authors conclude that the effect of these tax incentives on net saving probably lies between the estimates of the two sets of studies.

107. In principle, regulation could also affect the asset allocation choices of pension funds, especially as Switzerland adopts a prescriptive approach rather than a "prudent man" approach to this regulation. These regulations set the following upper limits to asset allocations: 30 per cent on domestic share, 55 per cent on property and 30 per cent on foreign currency assets (OECD 1998*c*, p. 370). However, none of these limits has been binding in recent years.

108. The Commission also recommended that:
- the adjustment to the value of annuities to exclude the part that represents a return of capital be raised from 40 per cent to 60 per cent (based on average life expectancy and an annuity that lasts for ten years);
- single-premium life insurance policies could only retain all of their tax privileges for contracts signed before the assured reached 60 years old.

109. The following OECD countries, in addition to Switzerland, have personal net wealth taxes: France; Finland; Iceland; Mexico (on businesses controlled by individuals); Netherlands; Norway; Spain; Sweden.

110. Losses may be carried forward for up to seven years.

111. The absence of imputation credits does not discourage investment out of retained earnings because taxes on dividends must be paid even if the investment does not occur and the earnings are instead distributed. Investment out of retained earnings just defers the payment of taxes on dividends. By contrast, investment financed by new equity creates a new liability to dividend taxation at some point in the future.

112. The absence of a personal capital gains tax would complement these arrangements as this ensures that retained earnings are not subject to double taxation.

113. Thalmann, P. (1997).

114. See: http://www.wto.org/wto/environ/environm.htm.

115. The application of the stamp tax to certain financial transactions is another vulnerable capital tax. Switzerland has been gradually lowering the stamp tax, no doubt due to competitive pressures from abroad.

116. The tax may not be lower than the sum of *i)* ordinary income tax computed on gross income from certain Swiss sources and foreign income for which the taxpayers has

applied for a refund, reduction or exemption for foreign taxes under a tax treaty and *ii)* the ordinary net wealth tax computed on gross wealth. Swiss nationals who have been absent from Switzerland for at least ten years can also apply to be taxed under this regime, but it only applies for a year after their return.

117. As it turns out, few non-resident investors ask to qualify for the reduced treaty rates. For some non-residents, this is because they are unwilling to provide the required papers.

118. The income that a non-resident may seek to avoid declaring to his resident country may include the capital deposited in the account, as well as the income earned on that capital.

119. Swiss intermediary banks never act as principals.

120. A variant on this tax treatment is for the so-called "domiciliary company". These are companies that are involved only in such activities as managing intellectual property, personal fortunes, some financial operations, and billing. Normally, such companies are supposed to be foreign-controlled and should have no personnel or offices in Switzerland. In the limited number of cantons where such a tax statute still exists, these companies benefit from a complete exoneration from cantonal taxes. Under the tax harmonisation law, all cantons must eliminate this status from their tax codes by the end of the year 2000. Source: Coopers and Lybrand (1995).

121. Holding companies are almost completely exempt form cantonal income tax if they qualify. To qualify for this status, a company's participation in other companies or the income derived therefrom must represent at least two-thirds of the company's assets or income. Also the holding company's main purpose must be the long-term management of participation and it must have no commercial activities in Switzerland. If these conditions apply, the company is only subject to normal income taxation on income from immovable property in Switzerland. Source: *European Tax Handbook* (1998).

122. These reasons include: *i)* a moderate overall tax burden; *ii)* high tax morale and perceived legitimacy of government tax and spending programmes; *iii)* a structure of capital taxation that does not rely heavily on taxation of financial income but that skewed toward corporate and property taxation; *iv)* extensive use of withholding taxes when individual financial income is subject to tax.

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*Annex I*

Table A1. **Main features of Switzerland's tax system**

| Tax | Main features | Share in total tax revenues in 1996 (%)[1] |
|---|---|---|
| **Personal income tax** | Federal, cantonal and municipal taxation of world-wide income. Levied by the cantons or the communes. The cantons keep 30% of the proceeds of the federal tax. Taxable income includes capital and labour income, pensions and benefits, ancillary incomes, lottery prizes. Deductions from taxable income are allowed for expenses incurred in earning the taxed incomes, for interest paid and for social reasons (spouse earned income, children and other dependants, single parents, insurance premiums). No dividend tax credit, no tax on capital gains from mobile assets unless they belong to a (personal) business. Net income from sole proprietorships (self-employed) and partnerships is part of the income of their owners; loss carry forward for up to 7 years. | **47.9** |
| Federal income tax | Taxation either every second year on the average income of the previous two years or every year on current income. Incomes of husband and wife are aggregated, but distinct schedules apply for single and married taxpayers. Progressive schedules with maximum average rate of 11.5% set in the Constitution, no coefficients. Schedules and allowed deductions are indexed whenever the CPI increased by 7%. | 7.7 |
| Cantonal and municipal taxes | The schedule is generally set in cantonal law; yearly cantonal and municipal taxes are obtained by applying a coefficient. Most cantons apply distinct schedules to single and married taxpayers and grant different personal exemptions. Special tax on lottery prizes in 11 cantons. Special tax on real-estate capital gains. Some cantons subject those gains to ordinary income taxation when the property belongs to a business. | 40.2 |
| **Social security contributions**[2] | Contributions to old-age/survivor insurance, disability insurance, unemployment insurance and military compensation insurance add up to about 15% of an employees' wage and 11% of a self-employed person's income; there are ceilings on taxable income only for unemployment and disability insurance. Contributions are shared between employer and employee and retained at source by employer. Pension plans are mandatory for all regular employees and paid for on the same basis. Sickness insurance is provided by private schemes with government subsidies. | |

Table A1. **Main features of Switzerland's tax system** *(cont.)*

| Tax | Main features | Share in total tax revenues in 1996 (%)[1] |
|---|---|---|
| **Taxes on personal property and wealth** | | **9.0** |
| | Cantonal and municipal taxes on total world-wide wealth minus total debt and foreign located real estate. Progressive schedules apply, with high tax-exempt thresholds; the burden never exceeds 1% of net wealth. | 4.0 |
| | Inheritance and gift tax: great diversity, one canton levying no tax, some exempting the surviving spouse; schedules are generally progressive and depend on family link. | 1.4 |
| | Tax on sale of real-estate property: generally proportional tax, between 1% and 3% of sales price, levied at location of property. | 0.9 |
| | Property tax in 13 cantons: proportional tax, between 0.03% and 0.3% of market value of property (land and buildings) without deduction of debt, levied at its location, generally by municipality. | 0.8 |
| | Federal and cantonal taxes on motor vehicles. Federal tax is 4% of price. Cantonal taxes depend on different characteristics of cars. | 1.9 |
| **Corporate taxes** | | **10.9** |
| Profits | Federal, cantonal and municipal taxes on current yearly world-wide profits of corporations (some cantons still apply different timing). | 8.6 |
| Net worth | Cantonal and municipal taxes on net worth of corporations. Levied by the cantons, which keep 30% of the proceeds of the federal tax. Profit and net worth taxes are deductible from tax base, depreciation must conform to usual business practice. Dividends and to a large extent capital gains from participation in other corporations are exempt in order to avoid multiple taxation. | 2.3 |
| Federal tax[3] | Proportional tax of 8.5% (since 1998), no coefficients apply; maximum possible rate of 9.8% is set in the Constitution. | 3.7 |
| Cantonal and municipal taxes | Graduated schedules, with rates depending generally on the ratio of profit to net worth, except in seven cantons with proportional schedules. Full or partial exemption of new corporations for up to ten years. 15 cantons levy a minimum tax based on sales, real estate holdings or invested capital. Property tax, generally the same as for personal real estate; in some cantons there is an additional tax on buildings not used by the corporation. | 7.2 |

Table A1. **Main features of Switzerland's tax system** *(cont.)*

| Tax | Main features | Share in total tax revenues in 1996 (%)[1] |
|---|---|---|
| **Consumption and excise taxes** | | **22.9** |
| | Federal VAT (6.5%, 2% on necessities, 3% for hostelry)[4]. | 15.4 |
| | Federal taxes on tobacco, liquors (the revenues of those two taxes are reserved for old-age/survivor and disability insurance), and beer. | 1.9 |
| | Federal tax on mineral oils; the greatest part of its revenues is reserved for road construction. | 5.6 |
| **Other taxes** | | **9.3** |
| | Federal withholding tax on capital income and lottery prizes (35%), refundable to resident taxpayers and partly to foreigners under tax treaties; also federal withholding tax on pension benefits and life annuities (15%) and capital payments for death or retirement (8%) paid to Swiss insurance holders, generally replaced by reporting. | 4.2 |
| | Federal stamp duty: 1% on new shares, 0.12% per year of validity for new bonds, 0.15% on the trade of Swiss assets, 0.3% on the trade of foreign assets, and 5% on insurance premiums for liability and real insurance contracts. Cantonal stamp duties on legal documents in 6 cantons. No overlap with federal tax. | 2.5 |
| | Tax for exemption of military duty: levied by the cantons (which keep 20%) levied at 2% on total net income but lowered when days of duty were accomplished. | |
| | Federal customs duties, based on weight. | 1.7 |
| | Road taxes: federal annual tax on trucks and buses, and federal highway stamp. | 0.6 |

1. Computed from: Administration fédérale des contributions. *Finances publiques en Suisse 1996*, Bern 1998.
2. Social security contributions are not counted as part of tax revenue in this table.
3. Fundamental changes were enacted on 1 January 1998. In particular, the federal tax on corporate net worth was repealed. Its revenue in 1996 is included in the statistics.
4. The standard rate of VAT rose to 7.5 per cent on 1 January 1999 and the preferential rates on necessities and hostelry rose to 2.3 per cent and 3.5 per cent, respectively.

*Annex II*

# Calendar of main economic events

## 1997

### January

The second step of unemployment insurance reform comes into force, introducing *inter alia* a shift away from passive income support to active labour market programmes (the "activation principle"). The move implies a prolongation of the duration of benefit eligibility from the previous 170, 250 or 400 workdays – depending on the unemployment contributions paid – to a uniform 520 workdays. To make the new system work, a network of 150 regional placement offices and 1 850 posts of job counsellors are to be created during 1997.

A decree comes into force which allows, during the next two years, to subsidise through the unemployment insurance fund up to half the previous wage of an employee retiring early, to make way for the recruitment of an unemployed, who was out of work for at least six months.

The tenth revision of the old-age and survivors' insurance scheme (AVS) comes into force which, among other things, will give both men and women the right to an individual pension and grant pension credits to women who stop work to care for children or elderly relatives. It also raises the retirement age for women from 62 to 63 in 2001 and from 63 to 64 in 2004.

Switzerland and the United States agree to open their public procurement markets further to each other's companies. The agreement goes beyond the World Trade Organisation accord on public procurement to which both countries are signatories.

The stamp duty on repurchase transactions is abolished, which lays the basis for a domestic repo market.

### March

The Federal Council submits to Parliament a policy package (*programme de relance conjoncturel*) which brings forward infrastructure maintenance expenditures of SF 561 million, most of which will be spent in 1998. The federal government expects the programme to induce additional spending of more than SF 1 billion by lower levels of government and the private sector. In addition, the package introduces a flat-rate federal profit tax of 8.5 per cent (replacing the variable profit tax rate dependent on profitability), the abolition of the corporate tax on capital, the alleviation of the taxation of holding companies, and the re-introduction of the stamp duty on life insurance premiums. The estimated net effect of the tax measures on the Confederation's budget is a revenue reduction of SF 170 million

per year. Moreover, the federal government proposes to moderate the restraint on foreign investment in business real estate imposed by the *Lex Friedrich*.

**April**

The Parliament approves the federal expenditure programme for infrastructure investment but defers the decision on the proposed tax measures.

**May**

The Federal Council proposes to raise the standard VAT rate by 1 percentage point to 7.5 per cent, the rate for hotel services by 0.5 per cent to 3.5 per cent and the reduced rate for a number of goods and services by 0.3 per cent to 2.3 per cent as from 1 January 1999. The rise in VAT is to finance increased expenditures for the old age and survivors pension scheme and the invalidity insurance fund as a result of population ageing.

The Federal Council proposes a constitutional amendment which formally confirms the independence of the Swiss National Bank (SNB) and which defines the maintenance of price stability as SNB's primary objective. Another important aim of the amendment is to formally dissociate the Swiss franc from gold. A referendum on these issues is expected in early 2000.

**July**

The Federal Council decides to curb soaring health insurance contributions, *inter alia*, by raising the obligatory payments patients must make towards the costs of their treatment from SF 150 to SF 230 per annum.

**October**

The Federal Council proposes a continuation of its policy of temporary wage cuts for federal government personnel, which is estimated to result in annual savings of SF 50 million. In this context, it has again been decided not to adjust government wages and pensions to inflation in 1998.

**December**

The Parliament adopts the federal budget for 1998, which provides for a deficit of SF 7.6 billion (about 2 per cent of GDP). The budget includes a one-off payment of SF 1.85 billion for restructuring the Swiss Federal Railways.

Both Houses of Parliament approve a constitutional amendment to reduce the federal budget deficit to no more than 2 per cent of revenue (around SF 1 billion) in 2001, or in 2003 at the latest, in case of a recession. If the target is not met, the federal government must propose a package of urgent measures to attain it. Parliament can choose a different package but it must have the same effect. The issue has to be put to a referendum in 1998.

In view of the absence of inflation pressures, the National Bank announces that it will continue its easier monetary policy in 1998, and allow the monetary base to exceed its medium-term target of 1 per cent annual growth.

## 1998

### January

The new federal law on business taxation comes into force, replacing the progressive federal taxation of profits by a flat rate and easing the tax burden on holding companies and on small and medium-sized enterprises.

The reform of the PTT (Poste, Télégraphe et Télécommunications) comes into effect, which separates the Post Office and telecom business into two autonomous enterprises. The reformed Post Office becomes an independent institution with more entrepreneurial freedom than hitherto.

The telecommunications law comes into effect, which provides for a full opening of the telecommunications market and the partial privatisation of Swisscom, the former telecommunications arm of the Swiss PTT in parallel with the EU's telecoms market.

### March

The federal government approves the creation of seven "*Hautes Écoles Spécialisées*" or "*Fachhochschulen*".

### April

The new market for repurchase transactions begins its trade.

### June

Swiss voters approve the constitutional amendment proposed by the Federal Council to balance the federal budget by 2001.

The Council of States follows the decision of the National Council of April 1998 to inscribe the right to strike and to lock out in the Constitution. However, secondary strikes in support of other workers will be illegal. The right to strike will not be extended to some groups of workers, including civil servants.

The National Council votes in favour of a tax on non-renewable energy.

### July

A new tax on heating oil with high sulphur content comes into effect. Revenues will be returned to the population through a reduction in health insurance premiums.

### September

Swiss voters reject a referendum proposal by a trade union not to raise the retirement age for women from 62 to 64 by 2005.

### November

Swiss voters approve in a referendum the construction of two new transalpine rail tunnels and related improvements in the domestic rail network.

The federal government proposal to reform the labour law is approved by referendum. The reform will make the treatment of men and women generally equal with respect to work time, which *inter alia* implies that it will cancel the ban on night work for women.

**December**

With no inflation pressures in sight, the National Bank announces the continuation of its easy monetary policy in 1999, to support the economic recovery.

The Parliament approves the federal budget for 1999, which projects a deficit of SF 3.9 billion (about 1 per cent of GDP).

After four years of negotiations, the federal government reaches an agreement with the EU on seven key areas (land transport, air transport, free movement of labour, agriculture, research and development, public procurement, and technical trade barriers), which is designed to minimise the economic cost for Switzerland from its non-participation in the EEA agreement.

## 1999

**January**

The general VAT rate is raised by 1 percentage point to 7.5 per cent, the rate for hotel services by 0.5 percentage point to 3.5 per cent and the reduced rate for a number of goods and services by 0.3 percentage point to 2.3 per cent.

The reform of the railway system comes into effect, which transforms the Swiss Federal Railways into a state-owned company with greater freedom to act as a business. The reform also gives other companies access to the railway network for merchandise and non-regular passenger transport.

The new law on agriculture comes into effect, which substantially reduces direct government intervention in the market for agricultural products by phasing out all price guarantees. The reform puts more emphasis on the protection of the environment, linking direct income payments progressively to ecological production methods during a transition period of five years.

**April**

The National Bank lowers the discount rate from 1 per cent to ½ per cent in the wake of the refinancing rate cut by the European Central Bank.

**May**

Based on the new law on agriculture, producer prices and processing margins for dairy products cease to be guaranteed by the federal government. The subsequent income losses of farmers will be partly offset by direct payments.

*STATISTICAL ANNEX AND STRUCTURAL INDICATORS*

OECD 1999

Table A. **Selected background statistics**

| | Average 1989-98 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **A. Percentage change from previous year at constant 1990 prices** | | | | | | | | | | | |
| Private consumption | 1.0 | 2.3 | 1.2 | 1.6 | 0.1 | –0.9 | 1.0 | 0.6 | 0.7 | 1.3 | 1.8 |
| Gross fixed capital formation | 0.8 | 5.3 | 3.8 | –2.9 | –6.6 | –2.7 | 6.5 | 1.8 | –2.4 | 1.5 | 3.8 |
| Construction | 0.0 | 7.3 | 1.3 | –3.4 | –0.8 | –1.0 | 7.0 | –3.8 | –6.2 | –1.5 | 0.7 |
| Machinery and equipment | 1.9 | 2.9 | 7.1 | –2.3 | –13.4 | –5.0 | 5.8 | 9.7 | 2.5 | 4.9 | 7.1 |
| GDP | 1.2 | 4.3 | 3.7 | –0.8 | –0.1 | –0.5 | 0.5 | 0.5 | 0.3 | 1.7 | 2.1 |
| GDP deflator | 2.3 | 3.1 | 4.3 | 6.0 | 2.7 | 2.7 | 1.6 | 1.1 | 0.4 | –0.1 | 1.1 |
| Industrial production | 1.8 | 1.5 | 4.8 | 0.5 | –1.0 | –1.8 | 4.3 | 2.0 | 0.0 | 4.7 | 3.6 |
| Employment | 0.7 | 2.7 | 3.2 | 1.9 | –1.6 | –0.8 | –0.3 | 0.3 | 0.3 | –0.3 | 1.2 |
| Compensation of employees (current prices) | 3.5 | 6.9 | 8.9 | 7.3 | 3.4 | 0.9 | 1.1 | 2.4 | 0.4 | 2.0 | 2.0 |
| Productivity (real GDP/employment) | 0.9 | 2.0 | 1.1 | –0.9 | 2.2 | 0.3 | 1.0 | 0.5 | 0.1 | 1.9 | 0.8 |
| Unit labour cost (compensation/real GDP) | 2.4 | 2.5 | 4.9 | 8.2 | 3.5 | 1.4 | 0.5 | 1.8 | 0.4 | 0.3 | –0.1 |
| **B. Percentage ratios** | | | | | | | | | | | |
| Gross fixed capital formation as per cent of GDP | 22.7 | 27.4 | 27.1 | 25.5 | 23.0 | 21.6 | 22.0 | 21.4 | 20.2 | 19.6 | 19.7 |
| Stockbuilding as a per cent of GDP | 0.2 | 1.0 | 1.2 | –0.1 | –1.0 | –0.9 | –0.9 | 0.0 | 0.1 | 0.6 | 2.4 |
| Foreign balance as per cent of GDP | 3.1 | –0.2 | 0.5 | 1.4 | 3.5 | 4.9 | 4.5 | 4.0 | 4.1 | 4.4 | 3.6 |
| Compensation of employees as per cent of GDP | 60.4 | 59.4 | 59.8 | 61.0 | 61.5 | 60.7 | 60.1 | 60.5 | 60.3 | 60.6 | 59.9 |
| Direct taxes as per cent of household disposable income | 18.2 | 17.9 | 18.5 | 17.7 | 18.1 | 17.6 | 18.9 | 17.8 | 18.5 | 18.3 | 18.7 |
| Households saving as per cent of disposable income | 9.7 | 9.3 | 10.3 | 10.5 | 10.1 | 10.8 | 9.1 | 9.6 | 8.8 | 9.3 | 9.0 |
| Unemployment as per cent of total labour force | 3.2 | 0.5 | 0.5 | 1.1 | 2.5 | 4.5 | 4.7 | 4.2 | 4.7 | 5.2 | 3.9 |
| **C. Other indicator** | | | | | | | | | | | |
| Current balance (US$ billion) | 16.6 | 7.0 | 8.6 | 10.6 | 15.1 | 19.5 | 17.5 | 21.3 | 22.0 | 22.8 | 21.5 |

*Source:* Office fédéral du développement économique et de l'emploi; Office fédéral de la statistique; Banque nationale suisse, *Bulletin mensuel* and OECD, *National Accounts*.

Table B. **Gross national product**

Million Swiss francs, current prices

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Private consumption | 160 258 | 168 762 | 179 587 | 193 426 | 201 854 | 206 777 | 211 182 | 216 053 | 219 780 | 223 904 | 228 511 |
| Public consumption | 38 180 | 41 762 | 46 442 | 50 865 | 53 359 | 53 473 | 54 849 | 55 0 25 | 56 643 | 56 213 | 56 443 |
| Gross fixed asset formation | 72 974 | 80 454 | 85 854 | 84 974 | 78 627 | 75 420 | 78 717 | 77 644 | 73 884 | 72 888 | 75 400 |
| Change in stocks | 113 | 2 910 | 3 787 | –192 | –3 452 | –3 122 | –3 274 | 47 | 486 | 2 347 | 9 201 |
| **Total domestic demand** | 271 525 | 293 888 | 315 670 | 329 073 | 330 388 | 332 548 | 341 473 | 348 769 | 350 793 | 355 351 | 369 555 |
| Exports of goods and services | 97 991 | 110 507 | 115 048 | 116 717 | 122 166 | 125 301 | 127 004 | 127 501 | 131 467 | 147 690 | 152 544 |
| *less:* Imports of goods and services | 96 790 | 111 079 | 113 415 | 112 129 | 110 191 | 108 050 | 111 015 | 112 942 | 116 427 | 131 452 | 138 664 |
| **Gross domestic product at market prices** | 272 726 | 293 316 | 317 303 | 333 661 | 342 363 | 349 799 | 357 463 | 363 329 | 365 833 | 371 590 | 383 435 |
| Factor income from abroad | 31 492 | 39 570 | 42 254 | 42 082 | 39 785 | 40 051 | 39 901 | 40 548 | 43 920 | 52 508 | .. |
| *less:* factor income paid abroad | 16 954 | 24 759 | 28 656 | 27 767 | 26 357 | 24 655 | 27 329 | 24 818 | 26 874 | 30 746 | .. |
| **Gross national income at market prices (= Gross national product at market prices)** | 287 264 | 308 127 | 330 901 | 347 975 | 355 792 | 365 195 | 370 034 | 379 058 | 382 879 | 393 351 | .. |

*Source:* Office fédéral du développemnt économique et de l'emploi; Office fédéral de la statistique; OECD, *National Accounts*.

Table C. **Gross national product**

Million Swiss francs, 1990 prices

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Private consumption | 173 512 | 177 499 | 179 587 | 182 529 | 182 786 | 181 114 | 182 932 | 183 990 | 185 204 | 187 617 | 191 022 |
| Public consumption | 41 807 | 44 049 | 46 442 | 48 080 | 48 438 | 48 370 | 49 354 | 49 320 | 50 322 | 50 649 | 50 942 |
| Gross fixed asset formation | 78 514 | 82 681 | 85 854 | 83 400 | 77 899 | 75 797 | 80 755 | 82 187 | 80 253 | 81 450 | 84 546 |
| Change in stocks | –1 911 | –445 | 3 787 | –376 | –3 982 | –3 305 | –3 002 | 116 | 1 139 | 1 382 | 7 459 |
| **Total domestic demand** | 291 922 | 303 784 | 315 670 | 313 633 | 305 141 | 301 976 | 310 040 | 315 613 | 316 917 | 321 098 | 333 969 |
| Exports of goods and services | 105 665 | 112 659 | 115 048 | 112 688 | 116 051 | 117 766 | 119 835 | 121 769 | 124 783 | 135 986 | 141 555 |
| *less:* Imports of goods and services | 104 456 | 110 589 | 113 415 | 111 557 | 106 826 | 106 890 | 115 356 | 121 279 | 124 589 | 134 655 | 146 471 |
| **Gross domestic product at market prices** | 293 131 | 305 854 | 317 303 | 314 764 | 314 366 | 312 852 | 314 518 | 316 104 | 317 111 | 322 429 | 329 053 |
| Factor income from abroad | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . |
| *less:* factor income paid abroad | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . |
| **Gross national income at market prices (= Gross national product at market prices)** | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . | . . |

*Source:* Office fédéral du développement économique et de l'emploi; Office fédéral de la statistique and OECD, *National Accounts.*

Table D. **Producer and import prices**

May 1993[1] = 100

| | 1995 | 1996 | 1997 | 1998 | 1998 Q1 | 1998 Q2 | 1998 Q3 | 1998 Q4 | 1999 Q1 |
|---|---|---|---|---|---|---|---|---|---|
| **Producer price index** | | | | | | | | | |
| **Total** | 99.4 | 97.6 | 96.9 | 95.7 | 96.3 | 95.9 | 95.7 | 94.9 | 94.5 |
| Agricultural products | 95.2 | 89.4 | 88.4 | 85.4 | 87.1 | 85.7 | 85.5 | 83.4 | 81.2 |
| Manufacturing | 99.4 | 97.9 | 97.2 | 96.1 | 96.6 | 96.3 | 96.0 | 95.4 | 95.0 |
| Food and tobacco | 99.0 | 98.2 | 98.2 | 97.7 | 97.8 | 98.0 | 97.9 | 97.1 | 98.0 |
| Textiles and clothing | 100.9 | 99.3 | 100.1 | 100.8 | 100.9 | 100.8 | 100.8 | 100.5 | 100.5 |
| Paper and paper products | 101.4 | 98.3 | 95.4 | 90.5 | 93.3 | 90.3 | 89.3 | 89.1 | 88.7 |
| Petroleum products | 87.8 | 96.6 | 103.2 | 89.4 | 94.4 | 91.6 | 87.2 | 84.2 | 82.1 |
| Chemicals | 92.2 | 89.1 | 86.1 | 83.3 | 84.0 | 83.6 | 83.3 | 82.4 | 81.0 |
| Machinery | 100.0 | 100.3 | 99.9 | 100.1 | 99.8 | 100.2 | 100.2 | 100.1 | 100.1 |
| Transport equipment | 102.8 | 102.3 | 101.3 | 101.2 | 101.3 | 101.3 | 101.3 | 101.1 | 101.1 |
| Furniture and other articles | 104.5 | 102.9 | 102.9 | 102.3 | 103.6 | 103.3 | 102.2 | 100.2 | 100.9 |
| Electricity and gas | 103.0 | 103.5 | 103.2 | 102.9 | 102.9 | 102.9 | 102.9 | 102.7 | 102.4 |
| Domestic use | 100.0 | 98.1 | 97.5 | 96.1 | 96.9 | 96.3 | 96.0 | 95.2 | 94.7 |
| Exports | 98.4 | 96.8 | 96.0 | 95.1 | 95.4 | 95.3 | 95.2 | 94.5 | 94.0 |
| Raw materials | 95.1 | 90.4 | 89.8 | 85.2 | 87.6 | 87.3 | 84.7 | 81.4 | 78.7 |
| Semi-finished goods | 99.4 | 96.9 | 95.9 | 94.2 | 95.0 | 94.4 | 94.1 | 93.3 | 92.5 |
| Consumer goods | 100.2 | 99.6 | 99.6 | 99.3 | 99.4 | 99.3 | 99.4 | 99.0 | 99.4 |
| Capital goods | 99.6 | 99.2 | 98.8 | 98.8 | 98.8 | 98.9 | 98.9 | 98.7 | 98.7 |
| **Import price index, goods** | 100.0 | 96.3 | 98.2 | 96.1 | 97.4 | 97.2 | 96.1 | 93.6 | 92.5 |
| Raw materials | 121.2 | 109.9 | 130.9 | 121.1 | 128.5 | 126.5 | 121.0 | 108.4 | 104.7 |
| Semi-finished goods | 105.7 | 99.8 | 102.7 | 100.6 | 102.5 | 102.3 | 100.6 | 97.1 | 94.9 |
| Consumer goods | 97.3 | 97.3 | 98.2 | 96.5 | 97.2 | 96.9 | 96.5 | 95.6 | 95.5 |
| Capital goods | 91.3 | 87.0 | 84.4 | 83.4 | 83.5 | 84.1 | 83.5 | 82.5 | 82.3 |
| **Total supply price index** | 99.6 | 97.2 | 97.3 | 95.8 | 96.6 | 96.3 | 95.8 | 94.5 | 93.9 |
| Raw materials | 104.7 | 97.7 | 105.1 | 98.5 | 102.8 | 101.8 | 98.1 | 91.4 | 88.3 |
| Semi-finished goods | 100.6 | 97.5 | 97.2 | 95.5 | 96.5 | 96.0 | 95.4 | 94.0 | 92.9 |
| Consumer goods | 99.0 | 98.7 | 99.0 | 98.2 | 98.5 | 98.3 | 98.2 | 97.6 | 97.8 |
| Capital goods | 96.4 | 94.5 | 93.3 | 92.9 | 92.9 | 93.2 | 93.0 | 92.5 | 92.4 |

1. As from May 1993, the wholesale price index has been replaced by the total supply price index, which is the weighted average of the producer price and the import price indices.

*Source:* Office fédéral de la statistique, *Indice des prix à la production et à l'importation.*

Table E. **Money supply**

Million SF, yearly average

| | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|
| **1985 Definition** | | | | | | | | | | |
| Monetary base | 29 910 | 28 934 | 29 247 | 29 056 | 29 498 | 30 070 | 30 129 | 31 185 | 32 691 | 33 665 |
| Monetary supply | | | | | | | | | | |
| M1 | 77 032 | 73 775 | 74 687 | 74 626 | 81 596 | 86 436 | 90 510 | 101 271 | 113 215 | 122 966 |
| M2 | 179 037 | 202 338 | 208 883 | 209 995 | 193 321 | 189 726 | 187 809 | 189 238 | 200 510 | 203 287 |
| M3 | 353 721 | 362 145 | 373 670 | 384 063 | 402 290 | 425 431 | 437 986 | 470 499 | 496 555 | 501 918 |
| **1995 Definition** | | | | | | | | | | |
| Monetary base | 29 910 | 28 934 | 29 247 | 29 056 | 29 498 | 30 070 | 30 129 | 31 185 | 32 691 | 33 665 |
| Monetary supply | | | | | | | | | | |
| M1 | 117 535 | 111 570 | 113 744 | 116 011 | 128 212 | 135 419 | 144 651 | 161 557 | 177 794 | 191 653 |
| M2 | 247 180 | 227 520 | 231 370 | 237 861 | 276 199 | 304 417 | 319 383 | 357 517 | 380 885 | 392 551 |
| M3 | 349 184 | 356 083 | 365 565 | 373 230 | 387 923 | 407 706 | 416 683 | 445 484 | 468 179 | 472 870 |

*Source:* Banque nationale suisse, *Bulletin mensuel.*

## Table F. **Interest rates and capital markets**

Million Swiss francs and percentages

| | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 | 1997 | | | | 1998 | | | | 1999 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 |
| **Interest rates (average for the period)** | | | | | | | | | | | | | | | |
| Official discont rate (end of period) | 4.0 | 3.5 | 1.5 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| Three-month deposits (Zurich) | 4.3 | 3.5 | 2.4 | 1.4 | 1.0 | 0.9 | 1.2 | 0.9 | 0.9 | 1.2 | 0.6 | 1.1 | 1.1 | 0.7 | 0.6 |
| Government bond yield | 4.6 | 4.9 | 4.6 | 4.0 | 3.4 | 2.8 | 3.5 | 3.3 | 3.3 | 3.3 | 2.8 | 3.0 | 2.8 | 2.4 | 2.6 |
| Saving deposits of cantonal banks | 4.4 | 3.4 | 3.1 | 2.4 | 1.8 | 1.4 | 2.1 | 1.9 | 1.7 | 1.6 | 1.5 | 1.4 | 1.4 | 1.4 | 1.3 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Euro-dollar, three-month | 3.2 | 4.6 | 5.9 | 5.4 | 5.6 | 5.4 | 5.4 | 5.7 | 5.6 | 5.7 | 5.5 | 5.6 | 5.5 | 5.2 | 4.9 |
| **Capital market** | | | | | | | | | | | | | | | |
| Foreign bonds | 45 121 | 36 872 | 37 630 | 36 039 | 34 453 | 41 496 | 9 526 | 7 680 | 10 142 | 7 106 | 10 652 | 11 543 | 12 606 | 6 696 | 13 157 |
| Domestic bonds[1] | 34 909 | 25 157 | 24 102 | 23 785 | 24 673 | 30 444 | 6 346 | 6 272 | 5 892 | 6 162 | 11 512 | 7 509 | 6 267 | 5 156 | 7 396 |
| Public market issues[2] | 32 757 | 24 793 | 23 781 | 23 285 | 24 343 | 30 275 | 6 275 | 6 214 | 5 866 | 5 989 | 11 510 | 7 490 | 6 152 | 5 123 | 7 396 |
| *of which:* | | | | | | | | | | | | | | | |
| Government | 15 793 | 11 166 | 8 574 | 9 681 | 7 861 | 11 315 | 3 194 | 2 031 | 1 029 | 1 607 | 4 969 | 4 011 | 1 249 | 1 086 | 2 277 |
| Financial institutions[3] | 11 786 | 10 530 | 10 805 | 10 320 | 10 581 | 13 731 | 2 025 | 2 737 | 2 299 | 3 520 | 3 200 | 3 003 | 3 947 | 3 582 | 4 501 |
| Other private | 5 178 | 3 097 | 4 402 | 3 283 | 5 901 | 5 229 | 1 056 | 1 445 | 2 538 | 862 | 3 341 | 476 | 957 | 455 | 618 |
| Shares | | | | | | | | | | | | | | | |
| Public market issues | 3 286 | 2 406 | 1 829 | 1 514 | 2 890 | 11 344 | 95 | 1 169 | 1 060 | 567 | 92 | 1 810 | 715 | 8 727 | 759 |
| Redemptions | 30 205 | 32 743 | 28 948 | 41 951 | 32 910 | 35 800 | 8 976 | 7 450 | 8 200 | 8 284 | 8 959 | 6 694 | 9 495 | 10 652 | 12 928 |

1. 1999 Q1 figures include only public market issues.
2. According to date of payments.
3. Including holdings companies.

*Source:* Banque nationale suisse, *Bulletin mensuel;* OECD, *Main Economic Indicators* and OECD Secretariat.

Table G. **Foreign trade by area**

Million US dollar

| | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Exports f.o.b. | | | | | | | | | |
| **OECD countries** | 42 020 | 52 197 | 50 579 | 53 840 | 50 844 | 56 652 | 66 138 | 64 670 | 61 755 | 65 857 |
| EU | 32 526 | 40 768 | 39 683 | 42 372 | 39 426 | 43 496 | 50 737 | 48 445 | 45 449 | 49 158 |
| *of which:* Germany | 10 718 | 14 046 | 14 580 | 15 357 | 14 483 | 16 408 | 19 377 | 18 070 | 17 016 | 18 215 |
| Other OECD | 9 494 | 11 429 | 10 896 | 11 468 | 11 418 | 13 155 | 15 401 | 16 225 | 16 306 | 16 699 |
| *of which:* United States | 4 547 | 5 023 | 4 988 | 5 538 | 5 558 | 6 341 | 6 955 | 7 390 | 7 867 | 8 563 |
| **Non-OECD countries** | 9 491 | 11 336 | 10 751 | 11 691 | 12 290 | 13 413 | 15 248 | 15 093 | 14 326 | 12 985 |
| CEEC | 176 | 157 | 106 | 144 | 302 | 407 | 581 | 536 | 543 | 609 |
| OPEC | 1 542 | 1 928 | 2 020 | 2 304 | 2 400 | 2 052 | 2 471 | 2 174 | 2 139 | 1 969 |
| Other | 7 774 | 9 251 | 8 625 | 9 243 | 9 589 | 10 953 | 12 196 | 12 383 | 11 644 | 10 407 |
| **Total** | 51 511 | 63 533 | 61 330 | 65 531 | 63 135 | 70 065 | 81 386 | 79 764 | 76 081 | 78 842 |
| | Imports c.i.f. | | | | | | | | | |
| **OECD countries** | 52 914 | 63 254 | 59 962 | 60 039 | 55 510 | 61 982 | 73 304 | 71 418 | 68 789 | 72 275 |
| EU | 45 099 | 54 388 | 50 738 | 51 555 | 47 923 | 53 766 | 63 764 | 61 805 | 58 447 | 61 528 |
| *of which:* Germany | 19 589 | 23 492 | 21 711 | 21 962 | 19 794 | 22 232 | 26 939 | 24 555 | 22 934 | 24 420 |
| Other OECD | 7 815 | 8 866 | 9 224 | 8 484 | 7 587 | 8 216 | 9 540 | 9 614 | 10 343 | 10 746 |
| *of which:* United States | 3 717 | 4 262 | 4 862 | 4 172 | 3 777 | 4 170 | 4 942 | 5 539 | 5 965 | 5 680 |
| **Non-OECD countries** | 5 285 | 6 293 | 6 309 | 5 626 | 5 283 | 5 729 | 6 599 | 6 788 | 7 064 | 7 993 |
| CEEC | 42 | 33 | 27 | 29 | 115 | 155 | 253 | 252 | 223 | 303 |
| OPEC | 318 | 842 | 900 | 643 | 711 | 818 | 945 | 1 146 | 1 222 | 920 |
| Other | 4 925 | 5 418 | 5 382 | 4 954 | 4 457 | 4 756 | 5 401 | 5 390 | 5 619 | 6 770 |
| **Total** | 58 199 | 69 546 | 66 271 | 65 665 | 60 794 | 67 711 | 79 903 | 78 207 | 75 854 | 80 268 |

*Source:* OECD, *Foreign Trade Statistics,* Series A.

Table H. **Foreign trade by commodity group**

Million US dollars

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Exports f.o.b. | | | | | | | | | |
| 0. Food and live animals | 1 145 | 1 143 | 1 380 | 1 394 | 1 495 | 1 499 | 1 618 | 1 866 | 1 777 | 1 615 |
| 1. Beverages and tobacco | 258 | 255 | 352 | 377 | 416 | 396 | 513 | 545 | 534 | 480 |
| 2. Crude materials, inedible, except fuels | 680 | 693 | 741 | 685 | 721 | 677 | 795 | 983 | 814 | 843 |
| 3. Mineral fuels, lubricants and related materials | 34 | 22 | 45 | 92 | 71 | 86 | 92 | 82 | 95 | 180 |
| 4. Animal and vegetable oils, fats and waxes | 18 | 20 | 23 | 23 | 22 | 20 | 18 | 20 | 20 | 19 |
| 5. Chemicals and related products, n.e.s. | 11 016 | 11 228 | 13 769 | 13 890 | 15 770 | 16 096 | 17 951 | 21 236 | 21 680 | 21 011 |
| 6. Manufactured goods | 10 025 | 10 136 | 12 068 | 11 332 | 11 617 | 11 102 | 11 654 | 12 823 | 12 091 | 11 606 |
| 7. Machinery and transport equipment | 15 904 | 15 643 | 19 652 | 19 062 | 20 058 | 18 759 | 20 807 | 25 673 | 25 342 | 23 643 |
| 8. Miscellaneous manufactured articles | 10 763 | 11 828 | 15 149 | 14 318 | 15 447 | 15 703 | 16 738 | 18 112 | 17 308 | 16 253 |
| 9. Commodities and transactions, n.e.s. | 101 | 108 | 98 | 61 | 63 | 204 | 89 | 301 | 85 | 150 |
| **Total** | 49 945 | 51 076 | 63 276 | 61 234 | 65 678 | 64 543 | 70 276 | 81 641 | 79 747 | 75 799 |
| | Imports c.i.f. | | | | | | | | | |
| 0. Food and live animals | 3 045 | 2 867 | 3 353 | 3 393 | 3 464 | 3 328 | 3 681 | 3 985 | 3 919 | 3 583 |
| 1. Beverages and tobacco | 715 | 726 | 891 | 884 | 833 | 785 | 864 | 1 007 | 1 071 | 991 |
| 2. Crude materials, inedible, except fuels | 1 776 | 1 833 | 2 019 | 1 750 | 1 739 | 1 567 | 1 768 | 2 069 | 1 782 | 1 651 |
| 3. Mineral fuels, lubricants and related materials | 2 066 | 2 299 | 3 155 | 3 080 | 2 813 | 2 423 | 2 280 | 2 317 | 2 770 | 3 403 |
| 4. Animal and vegetable oils, fats and waxes | 56 | 54 | 69 | 63 | 74 | 78 | 99 | 111 | 109 | 102 |
| 5. Chemicals and related products, n.e.s. | 6 727 | 6 875 | 8 108 | 7 900 | 8 629 | 8 644 | 9 766 | 11 721 | 11 552 | 12 071 |
| 6. Manufactured goods | 11 980 | 12 920 | 14 773 | 13 458 | 13 002 | 12 593 | 13 451 | 15 270 | 14 303 | 14 136 |
| 7. Machinery and transport equipment | 18 201 | 18 264 | 21 988 | 21 139 | 20 229 | 18 335 | 20 899 | 26 795 | 26 162 | 23 853 |
| 8. Miscellaneous manufactured articles | 11 710 | 12 265 | 15 037 | 14 584 | 14 791 | 14 066 | 14 885 | 16 676 | 16 484 | 15 641 |
| 9. Commodities and transactions, n.e.s. | 61 | 86 | 260 | 230 | 119 | 182 | 221 | 199 | 72 | 153 |
| **Total** | 56 335 | 58 189 | 69 651 | 66 480 | 65 692 | 62 000 | 67 915 | 80 152 | 78 224 | 75 583 |

*Source:* OECD, *Foreign Trade Statistics,* Series C.

Table I. **Balance of payments**

Billion Swiss francs

| | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997[1] | 1998[1] |
|---|---|---|---|---|---|---|---|---|---|---|
| **Current account balance** | 11.5 | 12.0 | 15.2 | 21.3 | 28.8 | 23.9 | 25.2 | 27.2 | 33.1 | 31.2 |
| Merchandise trade balance | –12.1 | –10.0 | –8.5 | –1.4 | 2.4 | 2.2 | 1.0 | 1.1 | –0.5 | –2.7 |
| Non-factor services balance | 13.1 | 13.0 | 14.7 | 15.1 | 16.8 | 15.6 | 15.2 | 15.4 | 18.8 | 18.7 |
| Factor services balance | 13.2 | 12.1 | 12.7 | 11.7 | 13.5 | 10.7 | 13.9 | 15.6 | 19.7 | 19.9 |
| Net unrequited transfers | –2.8 | –3.2 | –3.7 | –4.2 | –4.0 | –4.7 | –5.0 | –5.0 | –4.9 | –4.8 |
| **Capital account balance** | –12.3 | –17.2 | –17.1 | –23.4 | –30.5 | –22.4 | –14.0 | –34.9 | –32.2 | . . |
| Foreign direct investment | –10.6 | –2.4 | –5.6 | –7.9 | –13.1 | –10.2 | –11.8 | –16.3 | –14.7 | . . |
| Abroad | –13.0 | –10.0 | –9.4 | –8.5 | –13.0 | –14.8 | –14.4 | –19.8 | –21.0 | . . |
| Into Switzerland | 2.4 | 7.6 | 3.8 | 0.6 | –0.1 | 4.6 | 2.6 | 3.5 | 6.4 | . . |
| Portofolio investment | –4.9 | –1.6 | –17.2 | –8.6 | –26.4 | –24.8 | –4.6 | –11.8 | –17.3 | . . |
| Abroad | –15.4 | –0.8 | –25.2 | –13.6 | –44.8 | –26.1 | –10.5 | –27.7 | –30.4 | . . |
| Into Switzerland | 10.4 | –0.8 | 8.0 | 5.0 | 18.5 | 1.2 | 5.9 | 15.9 | 13.1 | . . |
| Banking sector | 2.5 | –10.4 | 8.0 | –8.7 | 14.1 | 14.8 | –1.1 | –17.9 | –0.8 | . . |
| *of which:* Increase in credit claims | 11.2 | –8.3 | 2.5 | –6.8 | –3.6 | –26.3 | –11.2 | –74.5 | –74.3 | . . |
| Increase in credit liabilities | 13.6 | 16.7 | 2.7 | –1.0 | 6.0 | 41.4 | 1.9 | 60.7 | 74.8 | . . |
| Net increase in fiduciary funds | –23.9 | –16.4 | 3.1 | 0.4 | 14.0 | –0.5 | 7.6 | –3.0 | –3.1 | . . |
| Enterprises | –3.5 | –1.5 | –0.8 | 0.1 | 1.0 | –2.0 | 5.2 | 2.7 | 0.3 | . . |
| *of which:* Increase in claims | –5.4 | –4.3 | 0.3 | –4.3 | 0.8 | –11.9 | 5.8 | –7.2 | 4.3 | . . |
| Increase in liabilities | 1.9 | 2.8 | –1.1 | 4.4 | 0.2 | 9.8 | –0.6 | 9.9 | –4.0 | . . |
| Other private sector | 4.8 | –1.8 | –1.1 | 2.1 | –6.0 | –0.1 | –2.0 | 8.3 | –0.1 | . . |
| Other public sector | –0.6 | –0.3 | –0.4 | –0.2 | –0.1 | –0.1 | 0.2 | 0.0 | 0.5 | . . |
| **Changes in national bank reserves**[2] | –3.9 | 2.3 | –3.0 | –6.9 | –1.4 | 0.9 | 3.7 | –7.8 | –4.9 | . . |
| **Revaluation of national bank reserves**[2] | 1.4 | –4.0 | 1.7 | 0.7 | 0.8 | –2.3 | –3.5 | 4.4 | 1.8 | . . |
| **Errors and omissions** | 3.2 | 7.6 | 3.2 | 8.4 | 2.5 | 0.2 | –11.2 | 11.3 | 2.3 | . . |

1. Provisional data.
2. The sign minus (–) indicates an increase.

*Source:* Banque nationale suisse, *Bulletin mensuel.*

Table J. **Labour market: structural and institutional characteristics**

| | 1975 | 1980 | 1985 | 1990 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
|---|---|---|---|---|---|---|---|---|---|---|
| Labour force (thousand) | 3 129 | 3 172 | 3 382 | 3 841 | 3 960 | 3 941 | 3 937 | 3 967 | 3 969 | 3 985[1] |
| *of which:* Foreigners[2] end-August | 754 | 706 | 756 | 955 | 950 | 947 | 939 | 911 | 874 | 863 |
| Participation rate[3] (per cent) | | | | | | | | | | |
| Total | . . | . . | 62.7 | 68.6 | 68.7 | 68.2 | 67.9 | 68.1 | 68.0 | 67.9[1] |
| Male | . . | . . | 81.3 | 86.2 | 84.5 | 83.4 | 83.0 | 82.4 | 82.1 | 81.5[1] |
| Female | . . | . . | 45.2 | 52.0 | 54.0 | 53.9 | 53.7 | 54.8 | 54.8 | 55.2[1] |
| Employment/population aged 15-64 | 74.5 | 74.3 | 74.8 | 77.6 | 80.6 | 79.8 | 79.8 | 79.9 | 79.6 | 80.4[1] |
| Employment by sector (per cent of total) | | | | | | | | | | |
| Agriculture | 7.6 | 6.9 | 6.1 | 4.2 | 4.3 | 4.1 | 4.3 | 4.5 | 4.6 | 4.6 |
| Industry | 42.2 | 38.1 | 35.6 | 32.2 | 28.8 | 28.8 | 29.2 | 28 | 26.7 | 26.3 |
| Services | 50.1 | 55 | 58.3 | 63.6 | 66.9 | 67 | 66.5 | 67.5 | 68.6 | 69.1 |
| Unemployed (thousand) | 10.2 | 6.2 | 30.3 | 18.1 | 163.1 | 171.0 | 153.3 | 168.6 | 188.3 | 139.7 |
| Unemployed foreigners (thousand) | . . | . . | . . | 7.6 | 63.5 | 68.3 | 64.7 | 75.9 | 86.6 | 65.9 |
| Unemployment rate | 0.3 | 0.2 | 1.0 | 0.5 | 4.5 | 4.7 | 4.2 | 4.7 | 5.2 | 3.9 |
| Vacancies:[4] | | | | | | | | | | |
| Full-time | 2 813 | 12 312 | 7 875 | 16 711 | 5 370 | 4 776 | 4 898 | 5 040 | 8 489 | 12 093 |
| Part-time | | | 647 | 550 | 373 | 449 | 503 | 566 | 949 | 1 234 |
| Number of days not worked (lock-out and strikes) | 1 733 | 5 178 | 662 | 4 090 | – | 14 380 | 351 | 7 266 | 435 | . . |

1. Average of Q1 and Q2 1998 or provisional figures.
2. Includes established foreign workers, workers on a one-year contract and seasonal and border workers.
3. Labour force as a percentage of the corresponding population aged 15 and over.
4. The decomposition between full and part-time begins in 1983, so for 1975 and 1980 it is total vacancies.

*Source:* Office fédéral de la statistique, *Annuaire statistique de la Suisse*, 1999; Département fédéral de l'économie publique, *La vie économique;* and OECD, *Labour Force Statistics.*

Table K. **The structure of taxation**

Per cent of GDP

| | 1980 | 1985 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|---|---|---|---|
| Tax receipts | 29.1 | 30.8 | 30.9 | 30.6 | 31.2 | 32.2 | 33.0 | 33.6 | 34.6 | 34.6 |
| Personal income tax | 10.4 | 10.7 | 10.7 | 10.6 | 10.9 | 10.7 | 11.3 | 10.7 | 11.1 | 10.8 |
| Corporate tax | 1.7 | 1.8 | 2.1 | 2.0 | 2.0 | 1.9 | 1.9 | 1.9 | 1.9 | 2.0 |
| Social security tax | 9.0 | 9.9 | 10.0 | 10.3 | 10.6 | 11.8 | 11.9 | 12.4 | 13.0 | 13.1 |
| Property taxes | 2.1 | 2.5 | 2.4 | 2.2 | 2.3 | 2.4 | 2.5 | 2.4 | 2.4 | 2.6 |
| Tax on goods and services | 5.9 | 5.8 | 5.7 | 5.6 | 5.4 | 5.4 | 5.4 | 6.2 | 6.2 | 6.1 |
| *Memorandum:* | | | | | | | | | | |
| Income tax as a per cent of total tax | 41.4 | 40.9 | 41.4 | 41.1 | 41.6 | 39.0 | 40.1 | 37.4 | 37.6 | 37.0 |

*Source:* OECD, *Revenue Statistics of OECD Member Countries.*

Table L. **Interest rate margins in banking**[1]

Per cent of balance sheet total

| | 1981 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|
| Large banks | 1.14 | 1.36 | 1.37 | 1.24 | 1.52 | 1.68 | 1.92 | 1.44 | 1.22 | 1.13 |
| Cantonal banks | 1.04 | 1.13 | 1.13 | 1.22 | 1.34 | 1.34 | 1.48 | 1.19 | 1.22 | 1.27 |
| Regional and saving banks | 1.25 | 1.21 | 1.20 | 1.32 | 1.48 | 1.51 | 1.51 | 1.20 | 1.44 | 1.54 |
| Loans associations and agricultural credit co-operatives | 0.98 | 0.95 | 0.96 | 1.04 | 1.13 | 1.14 | 1.19 | 1.06 | 1.17 | 1.34 |
| Other Swiss and foreign banks | 2.00 | 1.98 | 2.05 | 2.01 | 2.11 | 2.13 | 2.48 | 1.78 | 1.90 | 1.57 |

1. Interest received less interest paid.

*Source:* OECD, *Bank Profitability, Financial Statements of Banks,* 1998.

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